UM-4 field equations

The complete set of coupled field equations of UM-4: Einstein gravity in a bulk warped in all directions of the fold, dirac16complex and the gauge fields of SU(3), SU(2) and U(1) in the bulk, the Higgs doublet with the ad hoc vacuum and three generations of quarks and leptons with three right-handed neutrinos on the fold

Underlying theory: Patrick L. Nash Derived and verified by Claude (Anthropic)

Field equations of UM-4, 2026-10-25

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1 The fold: definition and examples

Definition. The *fold* is the seven-dimensional surface of the author's eight-dimensional spacetime on which the cosine of the angle $z = 6Hx_0$ of the hidden direction vanishes:

$$\mathcal{F} = \big\{X = (x_0, x_1, \dots, x_7) : \ \cos(6Hx_0) = 0\big\}, \qquad \text{in one period } 0 < z < \pi \text{ of the metric: } z = \frac{\pi}{2}, \quad x_0 = \frac{\pi}{12H} \tag{FD-1}$$

Symbols in (FD-1).

- $\mathcal{F}$: The fold: the seven-dimensional surface $\cos(6Hx_0) = 0$ of the eight-dimensional spacetime (FD-1), in one period $0 < z < \pi$ the surface $z = \pi/2$, $x_0 = \pi/(12H)$, where $\sigma = 0$; $|_{\mathcal{F}}$ means restricted to the fold. Status: DERIVED.

- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_{\mathrm{F}} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant of UM-4, $\Lambda = -21H^2$: the value for which the static block is a vacuum of Einstein gravity (its Einstein tensor is $21H^2$ in every direction). The author writes the pure number Λ_{N} in $0 = -\Lambda_{\mathrm{N}} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$, so that $\Lambda = -\Lambda_{\mathrm{N}} H^2$. Unit: GeV^2. Value: $-21H^2$; the author's original $-10H^2$; UM-3: $-7H^2$. Status: AUTHOR (the term), DERIVED (the value).
- Λ_{N}: The author's pure number in the vacuum equations of his notebook, $\Lambda = -\Lambda_{\mathrm{N}} H^2$: his original value 10; 21 in UM-4 (Einstein gravity), 7 in UM-3. Unit: dimensionless. Value: 21 (the author's original 10). Status: AUTHOR.
- $w_{(p)}$: The Lovelock weights of the author's Einstein-Lovelock action, $w_{(1)} = 1$ and $w_{(2)}$, $w_{(3)}$ of the Gauss-Bonnet and the cubic densities. UM-4 sets $w_{(2)} = w_{(3)} = 0$ (Einstein gravity): the only weights for which gravitons propagate on the light cone on every background, not only around the static block. The author's original weights $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$ (Dirac_Einstein-Lovelock_4+4_Nash.nb, In[134]) keep the gravitons on the light cone only around the static block (with $\Lambda = -161H^2/16$); UM-3 used $(1, 1/20, 1/360)$. Unit: dimensionless. Value: $(1, 0, 0)$ in UM-4; the author's original $(1, 1/32, 1/1152)$. Status: AUTHOR (the form), DERIVED (the values that the order of 2026-10-25 selects).

$X = \{x_0, \ldots, x_7\}$ are the author's coordinates: x_0 the hidden space-like direction, x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 the three extra times; H is the author's constant (an inverse length). The metric of UM-4 depends on x_0 only through $\cot^2 z$ and $(\sin^2 z)^{1/6}$, so it repeats with the period $\pi/(6H)$ of x_0, and every period $0 < z < \pi$ contains exactly one surface of $\mathcal{F}$, at $z = \pi/2$: $\mathcal{F}$ is the union of these copies, one per period (example 2), and "the fold" without a qualifier is the copy in the period that contains our world. A point of the fold is labelled by the seven coordinates $x_1, \ldots, x_7$, and the metric restricted to the fold ($\mathrm{d}x_0 = 0$, $z = \pi/2$) is

$$g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N\big|_{\mathcal{F}} = \mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{d}x_4^2 - \mathrm{e}^{-2a_4} b_4^2 \big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big) \tag{FD-2}$$

Symbols in (FD-2).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- $\mathcal{F}$: The fold: the seven-dimensional surface $\cos(6Hx_0) = 0$ of the eight-dimensional spacetime (FD-1), in one period $0 < z < \pi$ the surface $z = \pi/2$, $x_0 = \pi/(12H)$, where $\sigma = 0$; $|_{\mathcal{F}}$ means restricted to the fold. Status:

DERIVED.

- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.

with three space-like and four time-like directions; a_4 and b_4 are the author's two functions of the time x_4. Notation: ln is the natural logarithm, e Euler's number, $\exp(y) = \mathrm{e}^y$, sin, cos and $\cot = \cos / \sin$ the trigonometric functions, sinh and tanh the hyperbolic functions, $|y|$ the absolute value; $f(0^+)$ and $f(0^-)$ are the limits of $f(\sigma)$ as σ approaches 0 from above and from below.

Why this surface is called a fold. Measured by the proper distance along the hidden direction,

$$\sigma = \mathrm{sgn}\Big(\frac{\pi}{2} - z\Big)\frac{\ln\sin z}{6H}, \qquad \mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0, \qquad \sin z = \mathrm{e}^{-6H|\sigma|} \tag{FD-3}$$

Symbols in (FD-3).

- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no bound. Status: STANDARD.

(the fold is $\sigma = 0$; the half-period $0 < z < \pi/2$ is $\sigma < 0$, the half-period $\pi/2 < z < \pi$ is $\sigma > 0$; the two ends of the period, the tips $z \to 0$ and $z \to \pi$, lie at $\sigma \to -\infty$ and $\sigma \to +\infty$), the metric of UM-4 is

$$g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N = \mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|}\Big[\mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{d}x_4^2 - \mathrm{e}^{-2a_4}b_4^2\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big)\Big] \tag{FD-4}$$

Symbols in (FD-4).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\dots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \dots$): it runs over $x_0, x_1, \dots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \dots$), running over $x_0, \dots, x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X = \{x_0, \dots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \dots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

It depends on σ only through $|\sigma|$. The function $\mathcal{W}(\sigma) = \mathrm{e}^{-2H|\sigma|}$, which multiplies the squared steps of all seven directions of the fold (ordinary space, the time and the extra times), is the warp factor of the metric (its square root $\mathrm{e}^{-H|\sigma|}$ multiplies the lengths). The time x_4 is warped like the other directions (the author's order of 2026-10-25, 13:15 PDT; in UM-3 it was not), so that every surface $\sigma = \text{const}$ has the light cone of the fold. Four properties make the surface $\sigma = 0$ a fold, and together they single it out within the period:

1. *It is a mirror.* The reflection $\sigma \to -\sigma$ (that is $z \to \pi - z$, or $x_0 \to \pi/(6H) - x_0$) maps the metric to itself, and the fold is the only surface of the period that the reflection leaves in place: the two halves of the period are mirror images of each other, joined along the fold, as the two halves of a folded sheet of paper are joined along the crease.
2. *The warp factor is largest on it.* The warp factor $(\sin^2 z)^{1/6} = \mathrm{e}^{-2H|\sigma|}$ equals 1 on the fold and decreases on both sides.
3. *The metric has a kink on it.* The metric is continuous across the fold, but its first derivative along σ jumps. The extrinsic curvature of the surfaces $\sigma = \text{const}$ changes sign across the fold,

$$K^a{}_b = \frac{1}{2}\,g^{ac}\,\partial_\sigma g_{cb} = -H\,\mathrm{sgn}(\sigma)\,\delta^a{}_b \quad \text{(all seven directions of the fold)} \tag{FD-5}$$

Symbols in (FD-5).

- $K^a{}_b$: The extrinsic curvature of a surface of constant σ of the hidden direction, $K^a{}_b = \frac{1}{2}g^{ac}\partial_\sigma g_{cb}$ for a, b along the surface (FD-5): $-H\,\mathrm{sgn}(\sigma)$ along all seven directions of the fold for the author's form of the metric of

UM-4; $K = K^c{}_c$ its trace. Unit: GeV (1/length). Status: DERIVED.

- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1, \ldots, 7\}$; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- c: A frame index (as a). Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+, +, +, +, -, -, -, -$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- σ: The label σ in ∂_σ: the derivative with respect to the proper hidden coordinate σ. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1 M_2 \cdots}_{N_1 N_2 \cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- η_{ab}: The flat metric of the frame, $\eta = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$ (eta4488 of 4+4.nb); η^{ab} is its inverse (the same matrix). Unit: dimensionless. Status: AUTHOR.
- $\hat{g}_{ab}$: The metric of the seven-dimensional fold (FD-2) in its coordinates $x_1, \ldots, x_7$ (or in its frame), $\hat{g}^{ab}$ its inverse; $\hat{h}_{\mu\nu}$ is its part along the four observable directions. Unit: dimensionless. Status: AUTHOR.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- x: The coordinates of the fold: x_1, x_2, x_3 of ordinary space, x_4 the time, x_5, x_6, x_7 the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.

(a, b frame indices along the fold); it jumps by $-2H$ in every direction of the fold. The field equations of gravity of UM-4, $\mathcal{E}_{(1)}{}^M{}_N + \Lambda\delta^M_N = \kappa\mathcal{T}^M{}_N$ (Einstein gravity: the author's Einstein-Lovelock equations with the weights of the Gauss-Bonnet and cubic terms set to zero; $\mathcal{E}_{(1)}$ the Einstein tensor, $\Lambda = -21H^2$ the cosmological constant, $\kappa = 8\pi G_8$ the gravitational coupling of eight dimensions, $\mathcal{T}^M{}_N$ the energy-momentum tensor, δ^M_N the Kronecker delta), are linear in the second σ-derivatives of the metric, so the jump puts a Dirac delta function $\delta_\mathrm{D}(\sigma)$ into them: the metric solves them only if the fold carries a thin source $\mathcal{S}^a{}_b$, an energy-momentum tensor concentrated on $\sigma = 0$, given by the integral of the equations across the fold (the junction condition of Israel),

$$\kappa\,\mathcal{S}^a{}_b = \int_{0^-}^{0^+} \mathrm{d}\sigma\,\mathcal{E}_{(1)}{}^a{}_b = -\Big(\big(K^a{}_b\big|_{0^+} - K^a{}_b\big|_{0^-}\big) - \delta^a{}_b\big(K\big|_{0^+} - K\big|_{0^-}\big)\Big) \tag{FD-11}$$

Symbols in (FD-11).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{S}^a{}_b$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^a{}_b = \int_{0^-}^{0^+} \mathrm{d}\sigma\,\mathcal{E}_{(1)}{}^a{}_b$ (the junction condition of Israel); $\mathcal{S}^4{}_4$ is minus its energy density, $\mathcal{S}^1{}_1$ and $\mathcal{S}^5{}_5$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa\mathcal{S}^a{}_b = -12H\delta^a{}_b$ for every a_4, b_4 of the author's form. Status: DERIVED.

- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1, \ldots, 7\}$; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p, $\mathcal{E}_{(p)}{}^{M}{}_{N} = -2^{-(p+1)}\,\delta^{MP_1\cdots P_{2p}}_{NQ_1\cdots Q_{2p}}\,R^{Q_1Q_2}{}_{P_1P_2}\cdots R^{Q_{2p-1}Q_{2p}}{}_{P_{2p-1}P_{2p}}$ (with both indices down it is $1/\sqrt{|g|}$ times the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to g^{MN}; $\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor, $\mathcal{E}_{(3)}$ the cubic Lovelock tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number (UM-4 keeps only $p = 1$); in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.
- $K^a{}_b$: The extrinsic curvature of a surface of constant σ of the hidden direction, $K^a{}_b = \frac{1}{2}g^{ac}\partial_\sigma g_{cb}$ for a, b along the surface (FD-5): $-H\,\mathrm{sgn}(\sigma)$ along all seven directions of the fold for the author's form of the metric of UM-4; $K = K^c{}_c$ its trace. Unit: GeV (1/length). Status: DERIVED.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1M_2\cdots}_{N_1N_2\cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or −1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- G_8: The eight-dimensional Newton constant ($\kappa = 8\pi G_8$). Unit: GeV^{-6}. Status: AUTHOR.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor $R^M{}_{NPQ}$ (in the convention in which the commutator of two covariant derivatives, acting on a vector, is the Riemann tensor, with its last two indices those of the two derivatives, contracted with the vector), with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the scalar curvature $R = g^{MN}R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: $1/\mathrm{length}^2$. Status: STANDARD.
- $\mathcal{L}_{(p)}$: The Lovelock density of order p, $\mathcal{L}_{(p)} = 2^{-p}\,\delta^{P_1\cdots P_{2p}}_{Q_1\cdots Q_{2p}}\,R^{Q_1Q_2}{}_{P_1P_2}\cdots R^{Q_{2p-1}Q_{2p}}{}_{P_{2p-1}P_{2p}}$, a scalar built from p curvature tensors: $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.

($K = K^c{}_c$ the trace of the extrinsic curvature), with $\mathcal{S}^4{}_4 = -$(its energy density), $\mathcal{S}^1{}_1 = \mathcal{S}^2{}_2 = \mathcal{S}^3{}_3$ its pressure along ordinary space and $\mathcal{S}^5{}_5 = \mathcal{S}^6{}_6 = \mathcal{S}^7{}_7$ its pressure along the extra times; $\mathcal{S}$ is $\mathcal{T}$ integrated across the fold, an energy per unit volume of the six other directions of the fold (unit GeV^7). For the static block (a_4 and b_4 constant) and, to the first order in the curvature of the fold divided by H^2, for every fold

$$\begin{aligned} &\text{static block:}\quad \kappa\mathcal{S}^a{}_b = -12H\,\delta^a{}_b \quad \text{(a pure tension)}, \qquad \kappa\lambda_{\mathrm{F}} = 12H, \\ &\text{any fold:}\quad \kappa\mathcal{S}^a{}_b = -12H\,\delta^a{}_b + \frac{2}{5H}\,\hat{G}^a{}_b + \ldots \end{aligned} \tag{FD-12}$$

Symbols in (FD-12).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{S}^a{}_b$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^a{}_b = \int_{0^-}^{0^+}\mathrm{d}\sigma\,\mathcal{E}_{(1)}{}^a{}_b$ (the junction condition of Israel); $\mathcal{S}^4{}_4$ is minus its energy density, $\mathcal{S}^1{}_1$ and $\mathcal{S}^5{}_5$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa\mathcal{S}^a{}_b = -12H\delta^a{}_b$ for every a_4, b_4 of the author's form. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; in the tensors of the

fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1, \ldots, 7\}$; a repeated index is summed. Status: STANDARD.

- b: A frame index (as a). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1 M_2 \cdots}_{N_1 N_2 \cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or −1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- λ_F: The tension that the static block requires of the fold, $\kappa\lambda_\mathrm{F} = 12H$: the total vacuum energy of the seven-dimensional fold, made of the tension of the fold fluid and the vacuum energy of the fields of the fold (both have the pressure minus their energy density along ordinary space and along the extra times), per unit volume of the six other directions of the fold (eight-dimensional normalisation); in the four-dimensional normalisation it is $\lambda_\mathrm{F} V_*$, and the excess of the total over $\lambda_\mathrm{F} V_*$ is the dark energy ρ_Λ. Unit: GeV^7. Value: $12H/\kappa$; $\lambda_\mathrm{F} V_* = 30H^2 M_\mathrm{red}^2 = 1.1 \times 10^{45}$ GeV^4 for $H = 2.5$ TeV. Status: DERIVED.
- $\hat{G}^\mu{}_\nu$: The Einstein tensor of the metric $\hat{g}_{ab}$ of the fold (FD-2), with frame indices along its seven directions; with four-dimensional indices, $\hat{G}^\mu{}_\nu$, its components along the four observable directions, which for frozen extra times are those of the Einstein tensor of the four-dimensional metric $\hat{h}_{\mu\nu}$. Unit: $1/\mathrm{length}^2$. Status: STANDARD.
- M_red: The reduced Planck mass, $M_\mathrm{red} = (8\pi G_\mathrm{N})^{-1/2}$. Unit: GeV. Value: 2.435×10^{18} GeV. Status: STANDARD.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4} b_4)^3$ (with $V_\mathrm{t} = \int \mathrm{d}x_5\, \mathrm{d}x_6\, \mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4} b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- ρ_Λ: The vacuum energy density of the fold above its static tension, the dark energy of UM-4: $\rho_\Lambda = \rho_\mathrm{ff} + \rho_\mathrm{vac} - \lambda_\mathrm{F} V_*$ (four-dimensional normalisation). Unit: GeV^4. Value: $0.685\,\rho_\mathrm{c}^{(0)} = 2.5 \times 10^{-47}$ GeV^4 (Planck 2018). Status: DERIVED.
- G_N: Newton's constant of four-dimensional gravity (the gravity of our world, on the fold), $G_\mathrm{N} = 1/M_\mathrm{Pl}^2$ with the Planck mass $M_\mathrm{Pl} = 1.221 \times 10^{19}$ GeV. Unit: GeV^{-2}. Value: 6.708×10^{-39} GeV^{-2}. Status: STANDARD.
- V_t: The coordinate volume of the (compact) extra times, $V_\mathrm{t} = \int \mathrm{d}x_5\, \mathrm{d}x_6\, \mathrm{d}x_7$. Unit: length^3. Status: ASSUMED.
- ρ_c: The critical density, the energy density of a spatially flat fold at the expansion rate H_4; $\rho_\mathrm{c}^{(0)}$ its value today. Unit: GeV^4. Value: $\rho_\mathrm{c}^{(0)} = 1.0537 \times 10^{-5}\, h_{100}^2$ $\mathrm{GeV/cm}^3$. Status: STANDARD.
- ρ_ff: The energy density of the fold fluid. Unit: GeV^4. Value: $\lambda_\mathrm{F} V_* + \rho_\Lambda - \rho_\mathrm{vac}$: together with the vacuum energy ρ_vac of the fields of the fold it makes up the tension $\lambda_\mathrm{F} V_*$ that the static block requires plus the dark energy ρ_Λ. Status: ASSUMED (the fluid), DERIVED (the value).
- ρ_vac: The vacuum energy of the fields of the fold: the minimum of their potential (for the Standard Model the minimum of the Higgs potential with its loop corrections), a tension of the seven-dimensional fold. Unit: GeV^4. Status: STANDARD.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\, h_{100}$ km/s/Mpc with $h_{100} = 0.674$ (Planck 2018): 67.4 km/s/Mpc $= 1.44 \times 10^{-42}$ GeV. Status: STANDARD.
- M_Pl: The Planck mass, $M_\mathrm{Pl} = G_\mathrm{N}^{-1/2}$. Unit: GeV. Value: 1.221×10^{19} GeV. Status: STANDARD.
- h_{100}: The present expansion rate of the fold in units of 100 km/s/Mpc. Unit: dimensionless. Value: 0.674 (Planck 2018). Status: STANDARD.

($\hat{G}^a{}_b$ the Einstein tensor of the metric FD-2 of the fold): the tension plus matter that obeys the Einstein equations of seven dimensions, $\hat{G}^a{}_b = \kappa_7 \mathtt{t}^a{}_b$ with $\kappa_7 = 5\kappa H/2$ ($\mathtt{t}^a{}_b$ the matter of the fold), which for frozen extra times are the Friedmann equations of general relativity. The fold is therefore not only a place but a physical object: a thin layer that carries energy and momentum (a brane). 4. *The author's coordinate x_0 degenerates on it.* The metric component $g_{x_0 x_0} = \cot^2(6H x_0)$ vanishes on the fold and nowhere else in the period; near the fold the proper distance

grows with the square of the coordinate distance,

$$|\sigma| = 3H\Big(x_0 - \frac{\pi}{12H}\Big)^2\Big[1 + 6H^2\Big(x_0 - \frac{\pi}{12H}\Big)^2 + \dots\Big] \tag{FD-6}$$

Symbols in (FD-6).

- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

so the two coordinate values on either side of $x_0 = \pi/(12H)$ at the same coordinate distance lie at the same proper distance from the fold: the coordinate line of x_0 is folded onto the line of proper distances (a fold map, example 5).

What lives on the fold. In UM-4 the Higgs doublet (with the author's ad hoc vacuum, a time-dependent choice of its vacuum value), three generations of quarks and leptons, three right-handed neutrinos and the fold fluid (its tension and the pressure along the extra times that keeps their volume fixed) live on the fold: they are fields at $\sigma = 0$, functions of $x_1, \dots, x_4$ (no field depends on the extra times). The gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ live in the bulk; their massless modes are bound to the fold, so that the photon and the gluons of the four-dimensional world are waves of the bulk that the fold holds, and photons that leave the fold travel through the bulk at the speed of light. dirac16complex (the author's spinor field of eight dimensions) and gravity live in the bulk $\sigma \neq 0$ as well as on the fold; the states χ of dirac16complex that the fold holds (in UM-4 long-lived resonances) are the part of it near the fold, and the portals (couplings between dirac16complex and the fields of the fold) act at $\sigma = 0$. Integrated over the extra times, the fold is the four-dimensional universe that we observe, with the metric $\mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in $x_1, \dots, x_4$. In UM-4 files "the fold" means this surface; the examples below use the word for other surfaces with the same properties. The word is not used in its other senses: a set of equal energy levels is called degenerate, and a growth by the factor e is called an e-folding.

Examples of folds that are meaningful for UM-4.

1. *Our fold.* The fold $z = \pi/2$ of the period that contains our world (FD-1): the seven-dimensional world-volume of the quarks, leptons and the Higgs field, and after the integration over the extra times the four-dimensional universe that we observe.
2. *The copies of the fold.* Because the metric repeats with the period $\pi/(6H)$, every surface $x_0 = \pi/(12H) +$ (an integer multiple of $\pi/(6H)$) is a copy of the fold, one per period. Neighbouring copies are separated by a tip (z a multiple of π), where the warp factor $(\sin^2 z)^{1/6}$ vanishes and $g_{x_0x_0} = \cot^2 z$ diverges, and which lies at an infinite proper distance from both: in proper distance every copy has its own complete line $-\infty < \sigma < \infty$. A compact direction of length L would give every field a tower of particles with the masses $2\pi n/L$, $n = 1, 2, \dots$ (a Kaluza-Klein tower); because the copies are infinitely far apart, the period $\pi/(6H)$ of x_0 produces no such tower with the spacing $12H$: every field of the bulk has instead a continuum of modes that starts at zero mass, and the fold holds its massless mode.
3. *A folded sheet of paper.* The crease of a folded sheet is a fold in this sense: the two halves are mirror images (property 1) joined along the crease, the sheet is continuous there and its direction changes abruptly (property 3). In UM-4 the direction across the crease is the hidden direction σ, the seven directions along the crease are $x_1, \dots, x_7$, and the crease is the fold.
4. *A folded function.* The warp factor $\mathrm{e}^{-2H|\sigma|}$ is the simplest folded function: continuous, mirror symmetric, largest at $\sigma = 0$, with a jump of its derivative,

$$\frac{\mathrm{d}}{\mathrm{d}\sigma}\mathrm{e}^{-2H|\sigma|} = -2H\,\mathrm{sgn}(\sigma)\,\mathrm{e}^{-2H|\sigma|}, \qquad \frac{\mathrm{d}^2}{\mathrm{d}\sigma^2}\mathrm{e}^{-2H|\sigma|} = 4H^2\,\mathrm{e}^{-2H|\sigma|} - 4H\,\delta_\mathrm{D}(\sigma) \tag{FD-7}$$

Symbols in (FD-7).

- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- δ_D: The Dirac delta function of σ: zero for $\sigma \neq 0$, with the integral 1 over any range that contains $\sigma = 0$; the integral of $\delta_\mathrm{D}(\sigma)$ times a smooth function of σ is the value of that function at $\sigma = 0$. Unit: 1/length (for $\delta_\mathrm{D}(\sigma)$). Status: STANDARD.

whose delta function is the origin of the thin source of property 3. Every metric function of FD-4 depends on σ through $|\sigma|$ and is folded at $\sigma = 0$ in this sense. 5. *A fold map.* In the mathematics of smooth maps the fold (Whitney's fold singularity) is the map $u \mapsto u^2$ of a line onto a half-line: it folds the line at $u = 0$, two points $\pm u$ land on the same point, and the derivative of the map vanishes on the fold. Near the fold the map from the author's coordinate x_0 to the distance $|\sigma|$ from the fold is of this kind (FD-6), and $g_{x_0x_0} = \cot^2 z$, the square of the derivative $\mathrm{d}\sigma/\mathrm{d}x_0$, vanishes there. 6. *The brane of Randall and Sundrum.* The five-dimensional metric

$$\mathrm{d}y_\mathrm{RS}^2 + \mathrm{e}^{-2k_\mathrm{RS}|y_\mathrm{RS}|}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2 - \mathrm{d}x_4^2\big), \qquad \lambda_\mathrm{RS} = \frac{6k_\mathrm{RS}}{\kappa_5}, \qquad \Lambda_\mathrm{RS} = -\frac{6k_\mathrm{RS}^2}{\kappa_5} \tag{FD-8}$$

Symbols in (FD-8).

- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- y_RS: The proper coordinate across the brane of the five-dimensional model of Randall and Sundrum (FD-8, an example, not UM-4). Unit: length. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- k_RS: The inverse curvature length of the bulk of the model of Randall and Sundrum: the warp factor of the brane is $\mathrm{e}^{-2k_\mathrm{RS}|y_\mathrm{RS}|}$ (FD-8, an example, not UM-4). Unit: GeV (1/length). Status: STANDARD.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- λ_RS: The tension (energy per unit volume of ordinary space) of the brane of the model of Randall and Sundrum, $\lambda_\mathrm{RS} = 6k_\mathrm{RS}/\kappa_5$ (FD-8, an example, not UM-4). Unit: GeV^4. Status: STANDARD.
- κ_5: The gravitational coupling of five dimensions in the model of Randall and Sundrum, 8π times the Newton constant of five dimensions (FD-8, an example, not UM-4). Unit: GeV^{-3}. Status: STANDARD.
- Λ_RS: The vacuum energy density of the bulk of the model of Randall and Sundrum, $\Lambda_\mathrm{RS} = -6k_\mathrm{RS}^2/\kappa_5$ (FD-8, an example, not UM-4). Unit: GeV^5. Status: STANDARD.

has a fold at $y_\mathrm{RS} = 0$ with the properties 1 to 3 (its coordinate y_RS is already the proper distance, so property 4 does not arise); Einstein's equations require there a brane of the tension λ_RS (Israel's junction condition) and in the bulk the vacuum energy Λ_RS, and the bulk is otherwise empty. The fold of UM-4 is of the same kind, with every direction of the fold warped by the same factor as in the model of Randall and Sundrum: its static block is a space of constant curvature $-H^2$ in all eight directions, a vacuum of Einstein gravity with $\Lambda = -21H^2$, with a fold of pure tension (FD-12), and the graviton is bound to the fold as there. It differs in that the fold has seven dimensions, three of them extra times, in that the space is not cut at the brane (Randall and Sundrum identify the two sides of their brane, one physical side; the fold of UM-4 has two distinct sides, example 7), and in that the gauge fields live in the bulk as well. 7. *A particle held by a thin layer.* A particle held by an attractive thin layer has a wave

function that falls off exponentially with the distance on both sides, and the layer makes its derivative jump (for the first-order Dirac equation, the wave function itself). In UM-4 the states χ of dirac16complex near the fold are of this kind only near the fold: at rest and without their self-interaction (the field is $\Psi = \mathrm{e}^{(7/2)H|\sigma|}\chi(\sigma)\mathrm{e}^{-\mathrm{i}\omega x_4}$)

$$\begin{aligned}&\chi(\sigma) = \mathrm{e}^{-q_\mathrm{b}|\sigma|}\chi(0^+)\ \ (0 < \sigma \ll \sigma_\mathrm{t}), \qquad \chi(\sigma) = \mathrm{e}^{-q_\mathrm{b}|\sigma|}\chi(0^-)\ \ (0 < -\sigma \ll \sigma_\mathrm{t}),\\ &\chi(0^+) = \exp\Big(\gamma^{(0)}\big(\mu_\mathrm{L}P_\mathrm{L} + \mu_\mathrm{R}P_\mathrm{R}\big)\Big)\chi(0^-), \qquad \omega_0 = \omega_\mathrm{r} - \frac{\mathrm{i}}{2}\,\Gamma_\chi, \qquad \omega_\mathrm{r} \to \omega_\mathrm{b} = \sqrt{m_\Psi^2 - q_\mathrm{b}^2}\end{aligned} \tag{FD-9}$$

Symbols in (FD-9).

- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{\frac{7}{2}H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x} - \omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of the level of dirac16complex at the fold (a long-lived resonance) and $\bar{\chi}$ its antiparticle. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- q_b: The inverse width of a state of dirac16complex at the fold: inside the turning point its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.
- σ_t: The turning point of a state of dirac16complex of the frequency ω: the depth at which the warped frequency equals the bulk mass, $|\omega|\mathrm{e}^{H\sigma_\mathrm{t}} = m_\Psi$; beyond it every solution oscillates. Unit: length. Status: DERIVED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- μ_X, μ_L, μ_R: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex, set by the mean fields of the Standard Model at the fold: the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_\mathrm{L}P_\mathrm{L} + \mu_\mathrm{R}P_\mathrm{R})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- P_R: The author's right-handed chiral projector, $P_\mathrm{R} = (\mathbf{1} + \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- ω_0: The energy at rest (zero momentum along ordinary space and along the extra times) of a state of dirac16complex at the fold; complex for a resonance, $\omega_0 = \omega_\mathrm{r} - \frac{\mathrm{i}}{2}\Gamma_\chi$. Unit: GeV. Status: DERIVED.
- ω_r: The mass of the particle χ: the real part of the energy at rest of the resonance of dirac16complex at the fold, $\omega_\mathrm{r} = \omega_\mathrm{b}(1 - H/(2q_\mathrm{b})) + \ldots$. Unit: GeV. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Γ_χ: The width of the particle χ at rest: the rate of its decay into the waves of dirac16complex in the bulk. Unit: GeV. Value: below $1/t_0 = 1.5 \times 10^{-42}$ GeV for dark matter. Status: DERIVED.
- ω_b: The energy at zero momentum of the state of dirac16complex at the fold in the limit $H \to 0$ (the bound state of UM-3): the particle χ of UM-4 has the mass $\omega_\mathrm{r} = \omega_\mathrm{b}(1 - H/(2q_\mathrm{b})) + \ldots$, equal to ω_b up to the relative correction $H/(2q_\mathrm{b})$. Unit: GeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- Γ_8: The chirality matrix of eight dimensions, $\Gamma_8 = T_0T_1\cdots T_7$ (the author's T16^A[8]) $= \mathrm{diag}(-\mathbf{1}_8, \mathbf{1}_8)$; $\Gamma_8^2 = \mathbf{1}$; it anticommutes with every T_a. Unit: dimensionless. Status: AUTHOR.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- $\mathbf{1}$, $\mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.
- $\mathbf{e}_a{}^M$, $\mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN} = \eta_{bc}\mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\,\mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- $\vec{k}$: The (comoving) momentum along ordinary space, $\vec{k}^2$ its square. Unit: GeV. Status: STANDARD.
- $\vec{x}$: The position in ordinary space, $\vec{x} = (x_1, x_2, x_3)$. Unit: length. Status: AUTHOR.
- t_0: The age of the universe. Unit: time. Value: 13.8 Gyr. Status: STANDARD.
- σ_{16}: The matrix of the Dirac adjoint, $\sigma_{16} = T_0T_1T_2T_3$, real and symmetric; $\sigma_{16}T_a$ is real and antisymmetric. Unit: dimensionless. Status: AUTHOR.

- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- o: The label of a single-particle state of dirac16complex: a Kohn-Sham orbital, or a mode of the quantised field; a sum over o runs over the orbitals or the modes. Status: STANDARD.

where μ_L, μ_R are the strengths of the fold for the left- and right-handed halves of dirac16complex (set by the fields of the fold), $q_\mathrm{b} = -m_\Psi(\mu_\mathrm{L}+\mu_\mathrm{R})\tanh\varsigma/(2\varsigma)$, $\varsigma = \sqrt{\mu_\mathrm{L}\mu_\mathrm{R}}$ ($q_\mathrm{b} = -m_\Psi\mu_\mathrm{L}/2$ for $\mu_\mathrm{R}=0$), and ω_r tends to the energy ω_b of the bound state of UM-3 when $H/q_\mathrm{b}\to 0$. Because the time is warped, the energy ω of the field enters its equation multiplied by $\mathrm{e}^{H|\sigma|}$: beyond the turning point σ_t ($\omega_\mathrm{r}\mathrm{e}^{H\sigma_\mathrm{t}} = m_\Psi$) the profile oscillates as a wave that leaves the fold, so the state is a resonance of the width Γ_χ, which falls like $\mathrm{e}^{-2F(\upsilon_\mathrm{b})m_\Psi/H}$ with $F(\upsilon) = \ln((1+\sqrt{1-\upsilon^2})/\upsilon) - \sqrt{1-\upsilon^2}$ and $\upsilon_\mathrm{b} = \omega_\mathrm{b}/m_\Psi$ (the tunnelling through the region between the fold and σ_t). The profile is not a function of $|\sigma|$: it jumps at the fold, $\chi(0^+)\neq\chi(0^-)$. 8. *The fold as a mirror for dirac16complex.* The reflection across the fold acts on the profiles $\chi(\sigma)$ in two ways,

$$(\mathsf{R}_1\chi)(\sigma) = T_0T_1T_2T_3\,\chi(-\sigma), \qquad (\mathsf{R}_2\chi)(\sigma) = T_0T_5T_6T_7\,\chi(-\sigma) \tag{FD-10}$$

Symbols in (FD-10).

- R_1: The reflection across the fold acting on a profile of dirac16complex together with the reflection of ordinary space, $(\mathsf{R}_1\chi)(\sigma) = T_0T_1T_2T_3\,\chi(-\sigma)$ (FD-10): a symmetry of dirac16complex, of gravity and of the mean field of the gas of the particles χ; the states at the fold of Krein charge -1 are even, those of Krein charge $+1$ odd. Unit: dimensionless. Status: DERIVED.
- R_2: The reflection across the fold acting on a profile of dirac16complex together with the reflection of the extra times, $(\mathsf{R}_2\chi)(\sigma) = T_0T_5T_6T_7\,\chi(-\sigma)$ (FD-10): it maps the states at the fold of one Krein charge onto those of the other. Unit: dimensionless. Status: DERIVED.
- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{\frac{7}{2}H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x}-\omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of the level of dirac16complex at the fold (a long-lived resonance) and $\bar\chi$ its antiparticle. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2-z)\ln(\sin z)/(6H)$ on one period $0<z<\pi$; $\sigma=0$ is the fold $z=\pi/2$, $\sigma\to\mp\infty$ the tips $z\to 0,\pi$; one period of x_0 is the whole line $-\infty<\sigma<\infty$. Unit: length. Status: DERIVED.
- T_a, T^a: The author's eight real 16×16 Dirac matrices T16ˆA[a] of 4+4.nb ($a=0,\ldots,7$; $T_0,\ldots,T_7$), with $\{T_a,T_b\} = 2\eta_{ab}\mathbf{1}$; $T^a = \eta^{ab}T_b$. Unit: dimensionless. Status: AUTHOR.

(T_a the author's Dirac matrices; for states at rest R_1 is the reflection across the fold combined with the reflection of ordinary space, R_2 the reflection across the fold combined with the reflection of the extra times). The Krein charge of a state is its share of the conserved charge of dirac16complex, which is indefinite: a state can carry the charge -1 or $+1$. R_1 is a symmetry of dirac16complex, of gravity and of a mean field of a gas of the particles χ that is the same on both sides of the fold: the four resonances of Krein charge -1 (the particles χ) are even under it and the four of Krein charge $+1$ are odd, so such a mean field cannot mix the two signs of the indefinite charge. R_2 maps the states of one Krein charge onto those of the other and makes the two sets degenerate without the self-interaction. 9. *The planes of Horava and Witten.* In M-theory (the eleven-dimensional theory whose limits include the superstring theories) on an interval (a circle whose points are identified with their mirror images), the two ten-dimensional planes left in place by the reflection are mirror surfaces (property 1); each carries one E_8 gauge group while gravity fills the bulk. As in UM-4, gravity fills the bulk; unlike UM-4, the gauge fields live on the mirror planes (in UM-4 they live in the bulk), and the two sides of each plane are identified (one physical side).

What is not a fold.

- *The tips* $z=0$ and $z=\pi$ (mod π): the mirror exchanges them within a period, but there the warp factor vanishes, $g_{x_0x_0}$ diverges, and they lie at an infinite proper distance; they carry no thin source.
- *A smooth maximum of a warp factor*, such as $1/\cosh^2(H\sigma)$: mirror symmetric and largest at $\sigma=0$, but without a kink and without a thin source.

2 How to read this document

2.1 What this document contains

This document states the complete set of coupled field equations of the universe model UM-4, derives every one of them from one action, and explains every term. It is self-contained: every equation it uses is written in it, and every symbol of every equation is defined right after the equation. The sections are:

- **Why UM-4** (this section): what was physically incorrect in UM-3, what the author ordered on 2026-10-25 at 13:15 PDT, and how UM-4 meets the order.
- **Where the fields live** (this section): the metric of UM-4 has a fold, the seven-dimensional surface $z = \pi/2$, that carries energy; the Higgs doublet, the quarks and the leptons live on the fold, gravity, dirac16complex and the gauge fields in the bulk.
- **The action** (IN-1 to IN-16): every term of the action, written out and explained; why the gravity of UM-4 is Einstein gravity (FE-68 to FE-70); the gravitational waves of the fold (FE-63), the gravity that the fields of the fold induce (FE-64) and the fold fluid (FE-65).
- **Gravity** (FE-1 to FE-11): the Einstein equations in the bulk, the junction condition at the fold, the energy-momentum tensor of every field and the conservation laws.
- **The matter fields** (FE-12 to FE-26): the field equations of dirac16complex, of the gauge fields in the bulk, of the Higgs doublet and of the quarks and leptons, and the equations of the ad hoc Higgs vacuum.
- **The background** (FE-27 to FE-35, FE-66, FE-71, FE-72): the metric of UM-4, its exact curvature, the static block of constant curvature, the exact bulk equations of the author's form of the metric, the bulk metric and the equations of an expanding fold, the volume of the extra times.
- **Propagation in the bulk** (FE-73 to FE-84): the modes of every field of the bulk, the transverse spin-1 photon, the gravitational wave of spin 2 with its + and × polarisations, the propagators from one point of the fold to another, Newton's and Coulomb's laws, the W and Z bosons, and the causal structure.
- **dirac16complex: modes and Kohn-Sham equations** (FE-36 to FE-43, FE-67, FE-85, FE-86): the equation of the hidden direction, its exact solutions, the resonances at the fold, the Kohn-Sham equations of a gas of them, the Krein norm and its quantisation.
- **The Standard Model on the fold** (FE-44 to FE-47): the four-dimensional couplings, the masses with the ad hoc vacuum, the mixing matrices and the four-dimensional couplings of the particles χ of dirac16complex.
- **The three neutrino flavours** (FE-48 to FE-53): the flavour evolution in vacuum, in matter, in dense neutrino gases (supernovae) and in the early Universe.
- **The cosmological perturbations** (FE-54 to FE-58): the metric of the fold and the Einstein-Boltzmann equations of the cosmic microwave background (CMB), the cosmic neutrino background (CNB) and the baryon acoustic oscillations (BAO).
- **The thermal history** (FE-59 to FE-62, FE-87, FE-88): the expansion of the fold, the degrees of freedom, the Boltzmann equations of the particles χ and of the right-handed neutrinos, and the primordial era.
- **The coupled system**: which field appears in which equation, and how the system is solved.
- **What was verified**: the checks of the computations behind every derived statement.

2.2 Why UM-4

In UM-3 the time x_4 was not warped: its metric was $\mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|}\big(\mathrm{e}^{2a_4}\,\mathrm{d}\vec{x}^2 - \mathrm{e}^{-2a_4}b_4^2\,\mathrm{d}\vec{x}_5^2\big) - \mathrm{d}x_4^2$ (with $\mathrm{d}\vec{x}^2 = \mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2$ and $\mathrm{d}\vec{x}_5^2 = \mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2$), so that at the depth σ the light cone of the bulk was $\mathrm{e}^{H|\sigma|}$ times wider than that of the fold: whatever propagated in the bulk outran light on the fold (bulk scalars 1.22 times, bulk photons 1.41 times the speed of light on the fold; gravitons could be made luminal on the fold only by removing them from the bulk). On 2026-10-25 at 13:15 PDT the author ordered:

> "Since in the UM-3 model anything that propagates in the bulk outruns light on the fold, and bulk scalars and photons faster than light, my model is physically incorrect. You must find a new metric model in which x4 is warped like the other directions, both photons and gravitons actually propagate at c in the bulk, photon waves are transverse spin-1 phenomena, Gravitational waves (ripples in spacetime) are spin-2 phenomena (because they originate from the rank-2 stress-energy tensor) and possess the following Polarization Modes: the spin-2 nature of gravitational waves manifests physically as characteristic"plus" (+) and "cross" (×) polarization modes that stretch and squeeze space perpendicularly. Find the best model and re-calculate and re-document; however keep UM-3 and create a new UM-4, placed in new folders."

UM-4 meets the order as follows (UM-3 is kept unchanged in its folder `universe_model_3`).

- **The metric.** The time is warped like the other six directions of the fold (FE-27, FD-4): the author's metric with the single change $g_{x_4x_4} = -(\sin^2 z)^{1/6}$ (UM-3: -1). Every slice $\sigma = \text{const}$ then has the light cone of the fold (FE-30). This is the smallest change of the author's metric that meets the order, and it keeps everything else of the metric: the coordinates, the period, the fold, the functions a_4 and b_4.
- **The gravity.** Among the author's Einstein-Lovelock theories, only Einstein gravity (the weights of the Gauss-Bonnet and of the cubic Lovelock terms zero) lets gravitons propagate at the speed of light on every background, not only around one (FE-68 to FE-70); with it the static block is a vacuum for $\Lambda = -21H^2$. UM-4 therefore uses Einstein gravity with $\Lambda = -21H^2$ (IN-2). The author's original weights $(1, 1/32, 1/1152)$ would also give gravitons at the speed of light around the static block (with $\Lambda = -161H^2/16$), but not around an expanding fold.
- **The fields of the bulk.** Gravity, dirac16complex and the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ live in the bulk; their massless modes are bound to the fold and travel at the speed of light at every depth and every wavelength, their massive modes are slower (FE-73 to FE-78). The photon is a transverse wave of helicity ± 1 (FE-75); the gravitational wave of the four-dimensional world is a transverse traceless wave of helicity ± 2 with the two polarisations $+$ and $\times$, which stretch and squeeze a ring of test masses along perpendicular directions (FE-76). Nothing that crosses the bulk arrives earlier than light on the fold (FE-83, FE-84).
- **The consequences.** The fold is a seven-dimensional brane of pure tension with $\kappa\lambda_{\mathrm{F}} = 12H$, and to the first order in its curvature it obeys the Einstein equations of seven dimensions (FE-71), which with frozen extra times are the Friedmann equations of general relativity. Newton's and Coulomb's laws get corrections of relative size $0.34/(Hr)^5$ and $0.42/(Hr)^3$, and the W and Z bosons a width into the bulk (FE-81, FE-82), which bound H from below by about 2.5 TeV. The states of dirac16complex at the fold, bound in UM-3, are resonances in UM-4 (FE-86): they live longer than the universe only if m_Ψ/H exceeds about 10^3, so that the particles χ of dark matter are heavier than about 2×10^6 GeV and cannot be a thermal relic (FE-61); bound to the fold by the Higgs portal they cannot live that long and escape direct detection at the same time, so the dark matter of UM-4 needs a binding of χ to the fold that does not couple it to nucleons (OPEN).

2.3 The status labels

Every equation and every result carries one of the labels

- **AUTHOR**: from the author (his notebooks `4+4.nb` and `Dirac_Einstein-Lovelock_4+4_Nash.nb`, and his orders of 2026-10-24 and 2026-10-25: the coordinates, the metric with its change of 2026-10-25 13:15 (the time warped), the Einstein-Lovelock action, the Dirac matrices, the chiral projectors, the list of fields, the Higgs potential and the ad hoc vacuum);
- **RECORD**: from the record of this repository (the Lagrangian of dirac16complex with its canonical spin connection and its contact self-interaction, the Kohn-Sham treatment);
- **STANDARD**: textbook physics;
- **ASSUMED**: a choice made in UM-4 without derivation;
- **INVENTED**: a new interaction (the portals of dirac16complex);
- **DERIVED**: computed and verified in UM-4 (the verification is stated);
- **ESTIMATE**: an order-of-magnitude evaluation;
- **OPEN**: not settled.

2.4 Every equation is followed by its symbols

Every display equation is followed by the list "Symbols in (FE-n)" (or "Symbols in this equation" for a display without a tag) of EVERY symbol that appears in it: every quantity, field, coupling, constant and operator, every index letter with the values it takes, every mark (prime, dagger, transpose, star), the imaginary unit, Euler's number and the differential. Each item gives the meaning, the unit (natural units, in which the reduced Planck constant $\hbar$ and the speed of light c are 1), the value where one exists, and the status. A symbol that one of these meanings uses is itself listed in this document. A field equation carries the tag FE-1 to FE-88 and a term of the action the tag IN-1 to IN-16; the numbering is that of the inventory of UM-4 and is the same in every UM-4 document. The equations that UM-4 shares with UM-3 (the matter fields on the fold, the neutrinos, the cosmological perturbations, the thermal history) keep the numbers they have in UM-3; the equations of gravity, of the background and of dirac16complex have the numbers of the UM-3 equations on the same subjects, and FE-73 to FE-88 are new. Displays without a tag are steps of a derivation.

2.5 Coordinates, signature and units

The author's coordinates are $X = \{x_0, x_1, \ldots, x_7\}$, always with the hidden direction first: x_0 is the hidden space-like coordinate, on which the metric depends periodically with the period $\pi/(6H)$ through the angle $z = 6Hx_0$; x_1, x_2, x_3 are ordinary space, x_4 is the time, and x_5, x_6, x_7 are three extra time-like coordinates. The signature is $(+,+,+,+,-,-,-,-)$ in this order; the four observable coordinates $x_1, \ldots, x_4$ have the signature $(+,+,+,-)$. H is the author's constant (an inverse length); this document keeps it as a free parameter, bounded from below by the propagation of the gauge fields in the bulk (FE-82). Units are natural: the reduced Planck constant $\hbar$ and the speed of light c are 1; in eight dimensions the field dirac16complex has the dimension $\mathrm{GeV}^{7/2}$ and the gravitational coupling $\kappa = 8\pi G_8$ the dimension GeV^{-6}. A prime on a function of the time x_4 alone is its derivative $\mathrm{d}/\mathrm{d}x_4$ (in the cosmological perturbation equations, and only there, a dot is the derivative with respect to the conformal time). The proper coordinate of the hidden direction is σ (FD-3, FE-28): $\sigma = 0$ is the fold $z = \pi/2$.

2.6 The fields of UM-4 and where they live

The author orders a model universe with the following coupled fields (his order of 2026-10-24, items 1 to 10, and his order of 2026-10-25 13:15 for the photons and gravitons); UM-4, as UM-3, adds the right-handed quark singlets, the Yukawa couplings and three right-handed neutrinos (accepted by the author on 2026-10-24 at 17:32), the fold fluid and the induced gravity of the fold (ASSUMED and ESTIMATE, explained below).

field	symbol	what it is	lives in	it couples to	field equations
Einstein gravity	g_{MN}	the metric of eight dimensions	bulk and fold	every field	FE-1 to FE-11, FE-27 to FE-35, FE-54, FE-59, FE-63, FE-66, FE-68 to FE-72, FE-76, FE-79 to FE-81, FE-87
dirac16complex	Ψ	the author's 16-component complex spinor field; its left- and right-handed halves Ψ_L, Ψ_R	the bulk	gravity (spin connection), itself, every field of the fold at the fold (IN-16)	FE-6, FE-12 to FE-14, FE-36 to FE-43, FE-47, FE-55, FE-61, FE-85, FE-86
gluons	G^A	the gauge field of $SU(3)_C$	the bulk; its massless mode is bound to the fold	gravity, quarks, Ψ_L, Ψ_R (at the fold)	FE-8, FE-16, FE-74, FE-79
gauge fields of $SU(2)_L$ and $U(1)_Y$	W^1, W^2, W^3, B	the electroweak gauge fields	the bulk; their massless modes are bound to the fold	gravity, Φ, the fermions, Ψ_L, Ψ_R (at the fold)	FE-8, FE-17, FE-18, FE-45, FE-74, FE-75, FE-82
Higgs doublet	Φ	the scalar doublet with $V = \mu^2\Phi^\dagger\Phi + \lambda(\Phi^\dagger\Phi)^2$, expanded about the ad hoc vacuum $\Phi = (\mathcal{A}, v + h)/\sqrt{2}$	the fold	gravity, W, B, the fermions (Yukawa), Ψ_L only	FE-7, FE-15, FE-26, FE-45, FE-49
three left-handed lepton doublets	ℓ_i	$(\nu_{\mathrm{L}i}, e_{\mathrm{L}i})$, $i = 1, 2, 3$	the fold	gravity, W, B, Φ, Ψ_L and Ψ_R	FE-9, FE-22, FE-48 to FE-53

field	symbol	what it is	lives in	it couples to	field equations
right-handed electron, muon, tau	$e_{\mathrm{R}i}$	the charged-lepton singlets	the fold	gravity, B, Φ, Ψ_{L}, Ψ_{R}	FE-9, FE-23
three right-handed neutrinos	$\nu_{\mathrm{R}i}$	gauge singlets; Dirac neutrino masses	the fold	gravity, Φ, Ψ_{L}, Ψ_{R}	FE-9, FE-24, FE-62
three quark generations	Q_i, $u_{\mathrm{R}i}$, $d_{\mathrm{R}i}$	the quark doublets and singlets	the fold	gravity, gluons, W, B, Φ, Ψ_{L}, Ψ_{R}	FE-9, FE-19 to FE-21, FE-46
fold fluid	$\mathcal{T}_{\mathrm{ff}}$	the tension of the fold and the pressure along the extra times that holds their volume fixed: the part of the thin source at the fold not supplied by the other fields	the fold	gravity	FE-4, FE-5, FE-30, FE-59, FE-65, FE-72
induced gravity of the fold	κ_{ind}	an Einstein-Hilbert term of the fold induced by the quantum fluctuations of its fields	the fold	gravity	FE-5, FE-59, FE-63, FE-64

Why the gauge fields live in the bulk (AUTHOR, by the order). The order requires that photons propagate in the bulk at the speed of light; the photon is the massless mode of the gauge fields of $SU(2)_L \times U(1)_Y$, so UM-4 places all gauge fields of the Standard Model in the bulk. In the metric of UM-4 every field of the bulk has a massless mode bound to the fold that travels at the speed of light at every depth (FE-73 to FE-78), so the four-dimensional photon and gluons are these modes, and a photon that leaves the fold travels through the bulk at the speed of light. The price is a continuum of massive modes of every gauge field, which modify Coulomb's law at short distances, add contact interactions between the currents of the fold and give the W and Z bosons a width into the bulk (FE-80 to FE-82): observation then requires $H \gtrsim 2.5$ TeV (DERIVED, ESTIMATE). The alternative, the gauge fields on the fold (as in UM-3), would keep the Standard Model on the fold without these corrections, but its photon could not propagate in the bulk at all: a gauge field confined to the fold has no wave along σ, so it does not meet the order that photons propagate in the bulk at the speed of light, and UM-4 does not take it.

Why the Higgs doublet, the quarks and the leptons live on the fold (ASSUMED). In UM-4 the light cone of every slice of the bulk is that of the fold, so the argument of UM-3 for placing the Standard Model on the fold (fields with different profiles in the bulk would have different limiting speeds) no longer applies. The fermions could be fields of the bulk with massless modes bound to the fold (a kink mass gives each a normalisable zero mode, FE-77), but then their Yukawa couplings to the Higgs doublet would be set by the overlaps of their profiles, and the flavour structure of three generations by the profiles of the eighteen multiplets; UM-4 keeps the Higgs doublet, the quarks and the leptons on the fold, where their couplings and their flavour physics are those of the Standard Model with three Dirac neutrinos (the simplest choice that reproduces the measured flavour physics). dirac16complex is the author's field of eight dimensions and lives in the bulk; it couples to the fields of the fold at the fold. Status: ASSUMED (the place of the Higgs doublet and the fermions), AUTHOR (the place of gravity and of the photon, by the order), DERIVED (the speeds of the fields of the bulk, `um4_bulk.py`, B1 to B8).

Why the fold carries a source (DERIVED). The warp factor $\mathrm{e}^{-2H|\sigma|}$ has a kink at the fold. The Einstein equations, which are linear in the second derivatives of the metric along σ, then contain a delta function at $\sigma = 0$: the metric is a solution only if the fold carries a thin source $\mathcal{S}^a{}_b$ (FE-4). For the author's form of the metric $\kappa\mathcal{S}^a{}_b = -12H\delta^a{}_b$ in every direction of the fold, a pure tension; on an expanding fold the tension plus matter that obeys the Einstein equations of seven dimensions to the first order in the curvature of the fold (FE-30, FE-71; DERIVED, `um4_geometry.py`, G7; `um4_fold.py`, F2). This source is the fold fluid together with the fields of the

fold, the particles χ of dirac16complex and the portals.

2.7 Conventions

- Indices. M, N, P, Q (with numbered copies $M_1, M_2, \ldots$): coordinate indices of eight dimensions; a, b, c, d: frame indices of eight dimensions (the frame direction (0) is the σ direction); μ, ν, ρ: coordinate indices of the four observable directions $x_1, \ldots, x_4$; $A, B, C = 1, \ldots, 8$: colour (adjoint) indices; $I, J, K = 1, 2, 3$: weak-isospin (adjoint) indices; $i, j = 1, 2, 3$: generations (and the neutrino mass states); $\alpha, \beta = e, \mu, \tau$: lepton flavours; $p = 1, 2, 3$: the Lovelock order; $l = 0, 1, 2, \ldots$: the multipole index of the cosmological perturbations; $X = \mathrm{L}, \mathrm{R}$: the two halves of dirac16complex; $f \in \{Q, u, d, \ell, e, \nu\}$: the six kinds of Standard-Model multiplets. A repeated index is summed over its range.
- Two kinds of Dirac matrices. The author's real 16×16 matrices T_a (IN-5) act on dirac16complex; the four-dimensional complex 4×4 matrices $\hat{\gamma}^\mu$ (IN-14) act on the Standard-Model fermions. They are never multiplied with each other; the couplings between dirac16complex and the Standard Model are products of Lorentz tensors.
- Normalisation of the fold. No field depends on the extra times x_5, x_6, x_7 (ASSUMED: a dependence on them gives tachyonic four-dimensional modes, FE-38). Integrated over them, the extra times give their physical volume on the fold, V_*; the fields of the fold are normalised as four-dimensional fields, the couplings are their measured four-dimensional values, and the fold action carries the factor $1/V_*$ (IN-8). The gauge fields of the bulk are normalised in eight dimensions (IN-7); their zero modes have the measured four-dimensional couplings (FE-44).
- A word or a particle name in a subscript is upright (m_W, ν_R, Ψ_L); an italic letter in a subscript is an index. e is Euler's number, i the imaginary unit, d the differential; δ is the Kronecker delta and δ_D the Dirac delta function.

3 The action

All field equations of UM-4 follow from one action by varying it with respect to every field. This section writes out every term of the action (IN-1 to IN-16) and says in plain words what it couples. The Standard-Model part has exactly the gauge group, the fermion content (three generations, each with a right-handed neutrino), the Higgs potential and the gauge couplings of the Standard Model; its gauge fields live in the bulk, its Higgs doublet and its fermions on the fold. dirac16complex is a singlet of the Standard-Model gauge group; its left-handed and right-handed halves couple to every Standard-Model field at the fold through gauge-invariant portals with their own coupling constants, and only its left-handed half couples to the Higgs doublet.

3.1 The sum of all terms

$$\begin{aligned} S_\mathrm{tot} &= S_\mathrm{EL} + S_\Psi + S_\mathrm{gauge} + S_\mathrm{fold}, \\ S_\mathrm{fold} &= \int \mathrm{d}^8 X \, \sqrt{|g|} \, \delta_\mathrm{D}(\sigma) \, \frac{1}{V_*} \left(\mathcal{L}_\mathrm{IG} + \mathcal{L}_\mathrm{ff} + \mathcal{L}_\mathrm{H} + \mathcal{L}_\mathrm{f} + \mathcal{L}_\mathrm{Y} + \mathcal{L}_\mathrm{P} \right) \end{aligned} \tag{IN-1}$$

Symbols in (IN-1).

- S_tot: The action of UM-4 (the integral of the Lagrangian density over spacetime); its variation with respect to every field gives the field equations. Unit: dimensionless (in units in which the reduced Planck constant is 1). Status: ASSUMED.
- S_EL: The Einstein-Lovelock action of gravity in eight dimensions; in UM-4 the Einstein-Hilbert action with the cosmological constant. Unit: dimensionless. Status: AUTHOR.
- S_Ψ: The action of dirac16complex in the bulk (kinetic term with the canonical spin connection, mass term, contact self-interaction). Unit: dimensionless. Status: RECORD.
- S_gauge: The action of the gauge fields in the bulk. Unit: dimensionless. Status: STANDARD.
- S_fold: The action of the fold: the Standard Model, the portals of dirac16complex, the fold fluid and the Einstein-Hilbert term that the fields of the fold induce, all at $\sigma = 0$. Unit: dimensionless. Status: ASSUMED.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.

- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- δ_D: The Dirac delta function of σ: zero for $\sigma \neq 0$, with the integral 1 over any range that contains $\sigma = 0$; the integral of $\delta_\mathrm{D}(\sigma)$ times a smooth function of σ is the value of that function at $\sigma = 0$. Unit: 1/length (for $\delta_\mathrm{D}(\sigma)$). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\mathcal{L}_\mathrm{IG}$: The Lagrangian of the Einstein-Hilbert term of the four-dimensional metric of the fold that the quantum fluctuations of the fields of the fold induce; it adds to the Einstein-Hilbert term that the graviton of the bulk gives the fold and does not change the speed of gravitational waves. Unit: GeV^4. Status: ESTIMATE.
- $\mathcal{L}_\mathrm{ff}$: The Lagrangian of the fold fluid (only its energy-momentum tensor $\mathcal{T}_\mathrm{ff}$ enters the equations). Unit: GeV^4. Status: ASSUMED.
- $\mathcal{L}_\mathrm{H}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.
- $\mathcal{L}_\mathrm{f}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_\mathrm{Y}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_\mathrm{P}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.

The action has four parts: gravity in eight dimensions (S_EL, IN-2, IN-3), the field dirac16complex in the bulk (S_Ψ, IN-4 to IN-6), the gauge fields in the bulk (S_gauge, IN-7) and the fold (S_fold, IN-8 to IN-16). The Dirac delta $\delta_\mathrm{D}(\sigma)$ of the proper hidden coordinate σ confines the terms of the fold to $\sigma = 0$; the factor $1/V_*$ turns the integral over the extra times into the normalisation of four-dimensional fields (IN-8). On the fold live: the gravity that its fields induce ($\mathcal{L}_\mathrm{IG}$), the fold fluid ($\mathcal{L}_\mathrm{ff}$), the Higgs doublet ($\mathcal{L}_\mathrm{H}$), the quarks and leptons ($\mathcal{L}_\mathrm{f}$), their Yukawa couplings ($\mathcal{L}_\mathrm{Y}$) and the portals of dirac16complex ($\mathcal{L}_\mathrm{P}$); the gauge fields enter the fold through their values at $\sigma = 0$ in the covariant derivatives of the fields of the fold (IN-13). Varying S_tot with respect to the metric gives the gravitational equations (FE-1, FE-4); with respect to $\bar{\Psi}$ the equation of dirac16complex (FE-12, FE-13); with respect to the gauge fields the Yang-Mills equations of the bulk with the currents of the fold as their source (FE-16 to FE-18); with respect to $\Phi^\dagger$ the Higgs equation (FE-15); with respect to every fermion its Dirac equation (FE-19 to FE-24). Status: ASSUMED (the choice and the place of the terms; the content of each term is stated with its own label).

3.2 Gravity

$$S_\mathrm{EL} = \frac{1}{2\kappa}\int \mathrm{d}^8X\,\sqrt{|g|}\,\Big(\sum_{p=1}^{3}\alpha_{(p)}\,\mathcal{L}_{(p)} - 2\Lambda\Big) = \frac{1}{2\kappa}\int \mathrm{d}^8X\,\sqrt{|g|}\,\big(R - 2\Lambda\big), \qquad \kappa = 8\pi G_8,$$
$$\alpha_{(p)} = w_{(p)}\,H^{2-2p}, \qquad w_{(1)} = 1, \quad w_{(2)} = w_{(3)} = 0, \qquad \Lambda = -21H^2 \tag{IN-2}$$

Symbols in (IN-2).

- S_EL: The Einstein-Lovelock action of gravity in eight dimensions; in UM-4 the Einstein-Hilbert action with the cosmological constant. Unit: dimensionless. Status: AUTHOR.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\dots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- p: The Lovelock order $p=1,2,3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number (UM-4 keeps only $p=1$); in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)}=w_{(p)}H^{2-2p}$: in UM-4 $\alpha_{(1)}=1$ and $\alpha_{(2)}=\alpha_{(3)}=0$ (Einstein gravity). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{L}_{(p)}$: The Lovelock density of order p, $\mathcal{L}_{(p)}=2^{-p}\,\delta^{P_1\cdots P_{2p}}_{Q_1\cdots Q_{2p}}\,R^{Q_1Q_2}{}_{P_1P_2}\cdots R^{Q_{2p-1}Q_{2p}}{}_{P_{2p-1}P_{2p}}$, a scalar built from p curvature tensors: $\mathcal{L}_{(1)}=R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant of UM-4, $\Lambda=-21H^2$: the value for which the static block is a vacuum of Einstein gravity (its Einstein tensor is $21H^2$ in every direction). The author writes the pure number Λ_{N} in $0=-\Lambda_{\mathrm{N}}+\sum_p H^{-2p}w_{(p)}\mathcal{E}_{(p)}$, so that $\Lambda=-\Lambda_{\mathrm{N}}H^2$. Unit: GeV^2. Value: $-21H^2$; the author's original $-10H^2$; UM-3: $-7H^2$. Status: AUTHOR (the term), DERIVED (the value).
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor $R^M{}_{NPQ}$ (in the convention in which the commutator of two covariant derivatives, acting on a vector, is the Riemann tensor, with its last two indices those of the two derivatives, contracted with the vector), with two indices the Ricci tensor $R_{MN}=R^P{}_{MPN}$, without indices the scalar curvature $R=g^{MN}R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a,b). Unit: $1/\mathrm{length}^2$. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_8: The eight-dimensional Newton constant ($\kappa=8\pi G_8$). Unit: GeV^{-6}. Status: AUTHOR.
- $w_{(p)}$: The Lovelock weights of the author's Einstein-Lovelock action, $w_{(1)}=1$ and $w_{(2)}$, $w_{(3)}$ of the Gauss-Bonnet and the cubic densities. UM-4 sets $w_{(2)}=w_{(3)}=0$ (Einstein gravity): the only weights for which gravitons propagate on the light cone on every background, not only around the static block. The author's original weights $w_{(2)}=1/32$, $w_{(3)}=1/1152$ (Dirac_Einstein-Lovelock_4+4_Nash.nb, In[134]) keep the gravitons on the light cone only around the static block (with $\Lambda=-161H^2/16$); UM-3 used $(1,1/20,1/360)$. Unit: dimensionless. Value: $(1,0,0)$ in UM-4; the author's original $(1,1/32,1/1152)$. Status: AUTHOR (the form), DERIVED (the values that the order of 2026-10-25 selects).
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda=-21H^2$ and the tension $\kappa\lambda_{\mathrm{F}}=12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H\gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.

Gravity is the author's Einstein-Lovelock theory in eight dimensions with the weights of the Gauss-Bonnet density $\mathcal{L}_{(2)}$ and of the cubic Lovelock density $\mathcal{L}_{(3)}$ set to zero: Einstein gravity, whose density $\mathcal{L}_{(1)}=R$ is the scalar curvature, with the cosmological constant $\Lambda=-21H^2$ that makes the static block a vacuum (FE-70). The author writes the vacuum equations as $0=-\Lambda_{\mathrm{N}}+H^{-2}w_{(1)}\mathcal{E}_{(1)}+H^{-4}w_{(2)}\mathcal{E}_{(2)}+H^{-6}w_{(3)}\mathcal{E}_{(3)}$ (`Dirac_Einstein-Lovelock_4+4_Nash.nb`, In[134]); multiplied by H^2 this is FE-1 without sources and with $\Lambda=-\Lambda_{\mathrm{N}}H^2$, here $\Lambda_{\mathrm{N}}=21$. His original values were $w_{(2)}=1/32$, $w_{(3)}=1/1152$ and $\Lambda_{\mathrm{N}}=10$; UM-3 used $w_{(2)}=1/20$, $w_{(3)}=1/360$, $\Lambda_{\mathrm{N}}=7$. UM-4 uses Einstein gravity because it is the only choice of the weights for which gravitons propagate at the speed of light in the bulk on every background (FE-69), as the order of 2026-10-25 13:15 requires. Gravity couples to every field through the metric that appears in every other term of the action (in $\sqrt{|g|}$, in the contractions of indices and in the spin connection). Status: AUTHOR (the action), DERIVED (the weights and Λ that the order selects, FE-68 to FE-70).

$$\begin{aligned}&\mathcal{L}_{(p)}=\frac{1}{2^p}\,\delta^{M_1N_1\cdots M_pN_p}_{P_1Q_1\cdots P_pQ_p}\,R^{P_1Q_1}{}_{M_1N_1}\cdots R^{P_pQ_p}{}_{M_pN_p},\\&\mathcal{L}_{(1)}=R,\qquad \mathcal{L}_{(2)}=R^2-4R_{MN}R^{MN}+R_{MNPQ}R^{MNPQ}\end{aligned}\tag{IN-3}$$

Symbols in (IN-3).

- $\mathcal{L}_{(p)}$: The Lovelock density of order p, $\mathcal{L}_{(p)}=2^{-p}\,\delta^{P_1\cdots P_{2p}}_{Q_1\cdots Q_{2p}}\,R^{Q_1Q_2}{}_{P_1P_2}\cdots R^{Q_{2p-1}Q_{2p}}{}_{P_{2p-1}P_{2p}}$, a scalar built from p curvature tensors: $\mathcal{L}_{(1)}=R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- p: The Lovelock order $p=1,2,3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number (UM-4 keeps

only $p=1$); in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.

- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1 M_2 \cdots}_{N_1 N_2 \cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor $R^M{}_{NPQ}$ (in the convention in which the commutator of two covariant derivatives, acting on a vector, is the Riemann tensor, with its last two indices those of the two derivatives, contracted with the vector), with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the scalar curvature $R = g^{MN} R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: $1/\mathrm{length}^2$. Status: STANDARD.

The Lovelock densities are built with the generalised Kronecker delta from p Riemann tensors. Their key property (Lovelock's theorem, STANDARD) is that their metric variation contains no derivative of the metric higher than the second, so that the field equations are second-order differential equations, linear in the second derivatives along any one direction (quasi-linearity): a thin source at the fold produces a finite jump of first derivatives, the junction condition FE-4. UM-4 uses only $\mathcal{L}_{(1)}$; the others appear in FE-68 to FE-70, which show why. Status: STANDARD.

3.3 Why Einstein gravity

$$\sum_{p=1}^{3} \alpha_{(p)}\, \mathcal{E}_{(p)}{}^a{}_b + \Lambda\, \delta^a{}_b = \mathsf{P}(\mathsf{y})\, H^2\, \delta^a{}_b, \qquad \mathsf{P}(\mathsf{y}) = 21\mathsf{y} - 420 w_{(2)} \mathsf{y}^2 + 2520 w_{(3)} \mathsf{y}^3 + \frac{\Lambda}{H^2},$$

$$\begin{aligned} &\text{Einstein } (w_{(2)} = w_{(3)} = 0,\ \Lambda = -21H^2): \quad \mathsf{P} = 21(\mathsf{y}-1); \\ &\text{the author's original weights } \Big(\frac{1}{32}, \frac{1}{1152}\Big),\ \Lambda = -\frac{161}{16} H^2: \quad \mathsf{P} = \frac{7}{16}(\mathsf{y}-1)\big(5\mathsf{y}^2 - 25\mathsf{y} + 23\big); \\ &\text{UM-3's weights } \Big(\frac{1}{20}, \frac{1}{360}\Big),\ \Lambda = -7H^2: \quad \mathsf{P} = 7(\mathsf{y}-1)^3 \end{aligned} \tag{FE-70}$$

Symbols in (FE-70).

- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number (UM-4 keeps only $p = 1$); in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order p, $\alpha_{(p)} = w_{(p)} H^{2-2p}$: in UM-4 $\alpha_{(1)} = 1$ and $\alpha_{(2)} = \alpha_{(3)} = 0$ (Einstein gravity). Unit: GeV^{2-2p}. Status: AUTHOR.
- $\mathcal{E}_{(p)}{}^M{}_N$: The Lovelock tensor of order p, $\mathcal{E}_{(p)}{}^M{}_N = -2^{-(p+1)}\, \delta^{M P_1 \cdots P_{2p}}_{N Q_1 \cdots Q_{2p}}\, R^{Q_1 Q_2}{}_{P_1 P_2} \cdots R^{Q_{2p-1} Q_{2p}}{}_{P_{2p-1} P_{2p}}$ (with both indices down it is $1/\sqrt{|g|}$ times the variation of $\sqrt{|g|}\, \mathcal{L}_{(p)}$ with respect to g^{MN}; $\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor, $\mathcal{E}_{(3)}$ the cubic Lovelock tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1, \ldots, 7\}$; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- Λ: The eight-dimensional cosmological constant of UM-4, $\Lambda = -21H^2$: the value for which the static block is a vacuum of Einstein gravity (its Einstein tensor is $21H^2$ in every direction). The author writes the pure

number Λ_{N} in $0 = -\Lambda_{\mathrm{N}} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$, so that $\Lambda = -\Lambda_{\mathrm{N}} H^2$. Unit: GeV^2. Value: $-21H^2$; the author's original $-10H^2$; UM-3: $-7H^2$. Status: AUTHOR (the term), DERIVED (the value).

- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1 M_2 \cdots}_{N_1 N_2 \cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or −1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- $\mathsf{P}(\mathsf{y})$: The vacuum polynomial of the static blocks of curvature $-\mathsf{y}H^2$: the left-hand side of the bulk equations divided by H^2, the same in every direction, $\mathsf{P}(\mathsf{y}) = 21\mathsf{y} - 420 w_{(2)} \mathsf{y}^2 + 2520 w_{(3)} \mathsf{y}^3 + \Lambda/H^2$; $\mathsf{P}'(1)$ its derivative at $\mathsf{y} = 1$, which sets the strength of the graviton around the static block relative to Einstein gravity ($\mathsf{P}'(1)/21$). Unit: dimensionless. Value: $21(\mathsf{y} - 1)$ for Einstein gravity with $\Lambda = -21H^2$. Status: DERIVED.
- y: The curvature of a static block of constant curvature $-\mathsf{y}H^2$ in all eight directions, in units of $-H^2$ (its warp is $\mathrm{e}^{-2\sqrt{\mathsf{y}}H|\sigma|}$ in all seven directions of the fold): $\mathsf{y} = 1$ is the static block of UM-4. Unit: dimensionless. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_{\mathrm{F}} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- $w_{(p)}$: The Lovelock weights of the author's Einstein-Lovelock action, $w_{(1)} = 1$ and $w_{(2)}$, $w_{(3)}$ of the Gauss-Bonnet and the cubic densities. UM-4 sets $w_{(2)} = w_{(3)} = 0$ (Einstein gravity): the only weights for which gravitons propagate on the light cone on every background, not only around the static block. The author's original weights $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$ (Dirac_Einstein-Lovelock_4+4_Nash.nb, In[134]) keep the gravitons on the light cone only around the static block (with $\Lambda = -161H^2/16$); UM-3 used $(1, 1/20, 1/360)$. Unit: dimensionless. Value: $(1, 0, 0)$ in UM-4; the author's original $(1, 1/32, 1/1152)$. Status: AUTHOR (the form), DERIVED (the values that the order of 2026-10-25 selects).

The vacua of the static family: on the static blocks of curvature $-\mathsf{y}H^2$ in all eight directions (the warp $\mathrm{e}^{-2\sqrt{\mathsf{y}}H|\sigma|}$ in all seven directions of the fold, FE-32 with $H \to \sqrt{\mathsf{y}}H$) the left-hand side of the bulk equations is the same in every direction, $\mathsf{P}(\mathsf{y})H^2$; in a space of constant curvature $-K$ in $n = 8$ dimensions the Lovelock tensor of order p is $\mathcal{E}_{(p)}{}^a{}_b = -\frac{1}{2}(-K)^p \frac{(n-1)!}{(n-1-2p)!} \delta^a{}_b$, which gives $21K$, $-420K^2$ and $2520K^3$. The static block of UM-4 ($\mathsf{y} = 1$) is a vacuum if $\mathsf{P}(1) = 0$, and a non-degenerate one if $\mathsf{P}'(1) \neq 0$, $\mathsf{P}'(1) = 21 - 840 w_{(2)} + 7560 w_{(3)}$. For Einstein gravity $\Lambda = -21H^2$ and $\mathsf{y} = 1$ is the only root, with $\mathsf{P}'(1) = 21$. With the author's original weights the cosmological constant would have to be $-161H^2/16$ (not his $-10H^2$) and the polynomial has the roots $\mathsf{y} = 1$, 1.2155 and 3.7845, with $\mathsf{P}'(1) = 21/16$. UM-3's weights make $\mathsf{y} = 1$ a triple root: a degenerate vacuum without linear gravitons. Status: DERIVED (`um4_geometry.py`, G4; `um4_gravity.py`, L1).

$$\frac{\mathsf{P}'(1)}{42}\Big[\mathrm{e}^{2H|\sigma|}\Big(\frac{\partial^2 h_{\mathrm{gw}}}{\partial x_4^2} - \frac{\partial^2 h_{\mathrm{gw}}}{\partial x_3^2}\Big) - \frac{\partial^2 h_{\mathrm{gw}}}{\partial \sigma^2} + 7H\,\mathrm{sgn}(\sigma)\,\frac{\partial h_{\mathrm{gw}}}{\partial \sigma}\Big] = 0, \qquad \mathsf{P}'(1) = 21 - 840 w_{(2)} + 7560 w_{(3)} \quad \text{(FE-68)}$$

Symbols in (FE-68).

- $\mathsf{P}(\mathsf{y})$: The vacuum polynomial of the static blocks of curvature $-\mathsf{y}H^2$: the left-hand side of the bulk equations divided by H^2, the same in every direction, $\mathsf{P}(\mathsf{y}) = 21\mathsf{y} - 420 w_{(2)} \mathsf{y}^2 + 2520 w_{(3)} \mathsf{y}^3 + \Lambda/H^2$; $\mathsf{P}'(1)$ its derivative at $\mathsf{y} = 1$, which sets the strength of the graviton around the static block relative to Einstein gravity ($\mathsf{P}'(1)/21$). Unit: dimensionless. Value: $21(\mathsf{y} - 1)$ for Einstein gravity with $\Lambda = -21H^2$. Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_{\mathrm{F}} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the

fold). Status: STANDARD.

- h_{gw}: The amplitude of a gravitational wave of the + polarisation travelling along x_3: the metric components along x_1 and x_2 multiplied by $\mathrm{e}^{h_{\mathrm{gw}}}$ and $\mathrm{e}^{-h_{\mathrm{gw}}}$; a function of σ, x_3 and x_4 in the bulk, and on the fold the gravitational wave of the four-dimensional world, to the first order its plus polarisation h_+. Unit: dimensionless. Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- $w_{(p)}$: The Lovelock weights of the author's Einstein-Lovelock action, $w_{(1)} = 1$ and $w_{(2)}$, $w_{(3)}$ of the Gauss-Bonnet and the cubic densities. UM-4 sets $w_{(2)} = w_{(3)} = 0$ (Einstein gravity): the only weights for which gravitons propagate on the light cone on every background, not only around the static block. The author's original weights $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$ (Dirac_Einstein-Lovelock_4+4_Nash.nb, In[134]) keep the gravitons on the light cone only around the static block (with $\Lambda = -161H^2/16$); UM-3 used $(1, 1/20, 1/360)$. Unit: dimensionless. Value: $(1, 0, 0)$ in UM-4; the author's original $(1, 1/32, 1/1152)$. Status: AUTHOR (the form), DERIVED (the values that the order of 2026-10-25 selects).

The graviton around the static block of UM-4, for any weights: a transverse traceless wave of the + polarisation travelling along x_3 multiplies the metric components along x_1 and x_2 by $\mathrm{e}^{h_{\mathrm{gw}}}$ and $\mathrm{e}^{-h_{\mathrm{gw}}}$ (h_{gw} a function of σ, x_3 and x_4); FE-68 is the bulk equation (FE-1) to the first order in h_{gw}, half the difference of its components along x_1 and x_2, away from the fold. The time and ordinary space enter only through $\mathrm{e}^{2H|\sigma|}(\partial^2/\partial x_4^2 - \partial^2/\partial x_3^2)$: at every depth the wave has the light cone of the fold, for every choice of the weights, with the strength $\mathsf{P}'(1)/21$ relative to Einstein gravity. The + polarisation along the extra times obeys the same equation. A degenerate choice ($\mathsf{P}'(1) = 0$, UM-3's weights) gives no linear graviton, and $\mathsf{P}'(1) < 0$ a ghost. Around the static block, therefore, the order does not single out Einstein gravity; FE-69 does. Status: DERIVED (`um4_gravity.py`, L2).

$$
\begin{aligned}
&\mathrm{d}\sigma^2 + \mathrm{e}^{-2\mathfrak{p}H\sigma}\,\mathrm{d}\vec{x}^2 - \mathrm{e}^{-2\mathfrak{q}H\sigma}\,\mathrm{d}x_4^2 - \mathrm{e}^{-2\mathfrak{r}H\sigma}\,\mathrm{d}\vec{x}_5^2 \quad (\sigma > 0)\,:\\
&c_{\mathrm{x}} - c_{\mathrm{t}} = -4\,(\mathfrak{p}-\mathfrak{q})\big(18\mathfrak{r}^2 w_{(3)} - w_{(2)}\big)\big(\mathfrak{p}+\mathfrak{q}+3\mathfrak{r}\big),\\
&c_\sigma - c_{\mathrm{t}} = -4\big[w_{(2)}\big(\mathfrak{p}\mathfrak{q} - \mathfrak{p}^2 + 3\mathfrak{q}\mathfrak{r} - 3\mathfrak{r}^2\big) + 18 w_{(3)}\mathfrak{r}^2\big(\mathfrak{p}^2 - 3\mathfrak{p}\mathfrak{q} + 2\mathfrak{p}\mathfrak{r} - \mathfrak{q}\mathfrak{r} + \mathfrak{r}^2\big)\big]
\end{aligned}
\tag{FE-69}
$$

Symbols in (FE-69).

- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $\mathfrak{p}$, $\mathfrak{q}$, $\mathfrak{r}$: The rates of the warps of ordinary space ($\mathfrak{p}$), of the time ($\mathfrak{q}$) and of the extra times ($\mathfrak{r}$) of a static warped background, in units of H. Unit: dimensionless. Value: $\mathfrak{p} = \mathfrak{q} = \mathfrak{r} = 1$ for the static block of UM-4; $\mathfrak{p} = \mathfrak{r} = 1$, $\mathfrak{q} = 0$ for that of UM-3. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_{\mathrm{F}} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- $\vec{x}$: The position in ordinary space, $\vec{x} = (x_1, x_2, x_3)$. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\vec{x}_5$: The position along the extra times, $\vec{x}_5 = (x_5, x_6, x_7)$; $\mathrm{d}\vec{x}_5^2 = \mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2$. Unit: length. Status: AUTHOR.
- c_{x}: The coefficient of the second derivative along the proper length of x_3 in the same equation; the wave moves along x_3 with the local speed $\sqrt{c_{\mathrm{x}}/c_{\mathrm{t}}}$ in units of the local speed of light. Unit: dimensionless. Value: 1 for Einstein gravity on every background; $1 - 24w_{(2)} + 72w_{(3)}$ at the static block of UM-3 ($c_{\mathrm{x}}/c_{\mathrm{t}} = 5$ with the author's original weights). Status: DERIVED.
- c_{t}: The coefficient of the second derivative along the proper time in the linearised equation of a gravitational wave of the + polarisation on a static warped background, normalised so that Einstein gravity gives 1. Unit: dimensionless. Value: 1 for Einstein gravity on every background; $1 - 40w_{(2)} + 360w_{(3)}$ at the static block of

UM-3. Status: DERIVED.

- c_σ: The coefficient of the second derivative along σ in the linearised equation of a gravitational wave on a static warped background, normalised so that Einstein gravity gives 1; the wave moves along σ with the local speed $\sqrt{c_\sigma/c_\mathrm{t}}$ in units of the local speed of light. Unit: dimensionless. Value: 1 for Einstein gravity on every background. Status: DERIVED.
- $w_{(p)}$: The Lovelock weights of the author's Einstein-Lovelock action, $w_{(1)} = 1$ and $w_{(2)}$, $w_{(3)}$ of the Gauss-Bonnet and the cubic densities. UM-4 sets $w_{(2)} = w_{(3)} = 0$ (Einstein gravity): the only weights for which gravitons propagate on the light cone on every background, not only around the static block. The author's original weights $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$ (Dirac_Einstein-Lovelock_4+4_Nash.nb, In[134]) keep the gravitons on the light cone only around the static block (with $\Lambda = -161H^2/16$); UM-3 used $(1, 1/20, 1/360)$. Unit: dimensionless. Value: $(1, 0, 0)$ in UM-4; the author's original $(1, 1/32, 1/1152)$. Status: AUTHOR (the form), DERIVED (the values that the order of 2026-10-25 selects).

The gravitons on backgrounds that are not of constant curvature: the static warped metrics of the first line ($\mathfrak{p}$, $\mathfrak{q}$, $\mathfrak{r}$ constant; $\mathfrak{p} = \mathfrak{q} = \mathfrak{r} = 1$ is the static block of UM-4, $\mathfrak{p} = \mathfrak{r} = 1$, $\mathfrak{q} = 0$ that of UM-3), and the + polarisation along x_1, x_2 travelling along x_3. Its linearised equation contains the second derivatives in the frame with the coefficients c_t (of the second derivative along the proper time), c_x (along the proper length of x_3) and c_σ (along σ), normalised so that Einstein gravity gives 1; the wave moves along x_3 with the local speed $\sqrt{c_\mathrm{x}/c_\mathrm{t}}$ and along σ with $\sqrt{c_\sigma/c_\mathrm{t}}$ in units of the local speed of light. Einstein gravity gives $c_\mathrm{t} = c_\mathrm{x} = c_\sigma = 1$ on every such background. The Gauss-Bonnet and cubic terms give the deviations of FE-69, which vanish for all $\mathfrak{p}$, $\mathfrak{q}$, $\mathfrak{r}$ only for $w_{(2)} = w_{(3)} = 0$. A curved fold (an expanding universe, a star, a gravitational wave of finite amplitude) makes the warps of the different directions differ, so with the Gauss-Bonnet or cubic terms gravitons would leave the light cone there. At the static block of UM-3 the same coefficients are $c_\mathrm{t} = 1 - 40w_{(2)} + 360w_{(3)}$ and $c_\mathrm{x} = 1 - 24w_{(2)} + 72w_{(3)}$, with the author's original weights $c_\mathrm{x}/c_\mathrm{t} = 5$: the bulk gravitons of UM-3 that were $\sqrt{5}$ times faster than light. Status: DERIVED (`um4_gravity.py`, L3 and L4).

The gravity of UM-4 is therefore Einstein gravity with $\Lambda = -21H^2$: the static block is its unique vacuum ($\mathsf{P}'(1) = 21 > 0$), gravitons propagate on the light cone on every background, and the fold is a pure tension $\kappa\lambda_\mathrm{F} = 12H$ (FE-30). The author's original weights would also make the static block a vacuum (with $\Lambda = -161H^2/16$) and give gravitons at the speed of light around it with 1/16 of the Einstein strength, but on every background that is not of constant curvature their gravitons would leave the light cone; UM-4 does not use them. Status: DERIVED (`um4_gravity.py`, L4).

3.4 The field dirac16complex

$$S_\Psi = \int \mathrm{d}^8X\,\sqrt{|g|}\,\Big[\frac{1}{2}\big(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi\big) - m_\Psi\,\bar\Psi\Psi - \frac{\lambda_\Psi}{2}\big(\bar\Psi\Psi\big)^2\Big] \tag{IN-4}$$

Symbols in (IN-4).

- S_Ψ: The action of dirac16complex in the bulk (kinetic term with the canonical spin connection, mass term, contact self-interaction). Unit: dimensionless. Status: RECORD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

- $D_M,\ D_\mu$: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- λ_Ψ: The coupling of the contact self-interaction of dirac16complex. Unit: GeV^{-6}. Status: RECORD.

dirac16complex Ψ is the author's complex spinor field of eight dimensions with 16 components. Its Lagrangian is that of the record: the kinetic term with the canonical spin connection (in the symmetric form, which is real without a factor i for the author's real matrices, IN-5), the mass term and a contact self-interaction. It couples to gravity through $\sqrt{|g|}$, γ^M and the spin connection Ω_M inside D_M (IN-6), and to itself through λ_Ψ. $\bar\Psi\Psi = \bar\Psi_\mathrm{L}\Psi_\mathrm{L} + \bar\Psi_\mathrm{R}\Psi_\mathrm{R}$ (the mass term does not mix the halves, IN-5), while the kinetic term mixes them (it contains one Dirac matrix). dirac16complex is a singlet of the gauge group: it has no gauge coupling in the bulk. Status: RECORD.

$$\begin{aligned}
&\{T_a, T_b\} = 2\eta_{ab}\mathbf{1}, \qquad \eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1), \qquad T^a = \eta^{ab}T_b, \qquad \gamma^M = \mathbf{e}_a{}^M\, T^a,\\
&\bar\Psi = \Psi^\dagger\sigma_{16}, \qquad \sigma_{16} = T_0T_1T_2T_3, \qquad \Gamma_8 = T_0T_1T_2T_3T_4T_5T_6T_7,\\
&P_\mathrm{L} = \frac{\mathbf{1}-\Gamma_8}{2}, \qquad P_\mathrm{R} = \frac{\mathbf{1}+\Gamma_8}{2}, \qquad \Psi_\mathrm{L} = P_\mathrm{L}\Psi, \qquad \Psi_\mathrm{R} = P_\mathrm{R}\Psi
\end{aligned} \tag{IN-5}$$

Symbols in (IN-5).

- $T_a,\ T^a$: The author's eight real 16×16 Dirac matrices T16^A[a] of 4+4.nb ($a = 0,\dots,7$; $T_0,\dots,T_7$), with $\{T_a,T_b\} = 2\eta_{ab}\mathbf{1}$; $T^a = \eta^{ab}T_b$. Unit: dimensionless. Status: AUTHOR.
- a: A frame (orthonormal) index running over $0,1,\dots,7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$; on the fold ($a,b\in\{1,\dots,7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat g_{ab}$ and the tensors built from it) a coordinate index of the fold, $a\in\{1,\dots,7\}$; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- η_{ab}: The flat metric of the frame, $\eta = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$ (eta4488 of 4+4.nb); η^{ab} is its inverse (the same matrix). Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1},\ \mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0,x_1,\dots,x_7$ for a tensor of eight dimensions, $x_1,\dots,x_7$ for a tensor of the fold, $x_1,\dots,x_4$ for a four-dimensional tensor). Status: STANDARD.
- $\gamma^M,\ \gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2,\dots$): it runs over $x_0,x_1,\dots,x_7$; a repeated index is summed. Status: STANDARD.
- $\mathbf{e}_a{}^M,\ \mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN} = \eta_{bc}\mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\,\mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- $\Psi,\ \bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- σ_{16}: The matrix of the Dirac adjoint, $\sigma_{16} = T_0T_1T_2T_3$, real and symmetric; $\sigma_{16}T_a$ is real and antisymmetric. Unit: dimensionless. Status: AUTHOR.
- Γ_8: The chirality matrix of eight dimensions, $\Gamma_8 = T_0T_1\cdots T_7$ (the author's T16^A[8]) $= \mathrm{diag}(-\mathbf{1}_8,\mathbf{1}_8)$; $\Gamma_8^2 = \mathbf{1}$; it anticommutes with every T_a. Unit: dimensionless. Status: AUTHOR.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1}-\Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- P_R: The author's right-handed chiral projector, $P_\mathrm{R} = (\mathbf{1}+\Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- Ψ_L: The left-handed half of dirac16complex, $\Psi_\mathrm{L} = P_\mathrm{L}\Psi$ (the upper eight components in the basis of 4+4.nb). Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.

- Ψ_{R}: The right-handed half of dirac16complex, $\Psi_{\mathrm{R}} = P_{\mathrm{R}}\Psi$ (the lower eight components). Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.

T_a ($a = 0, \ldots, 7$) are the author's eight real 16×16 Dirac matrices `T16^A[a]` of `4+4.nb`, and Γ_8 is the author's chirality matrix `T16^A[8]` $= \mathrm{diag}(-\mathbf{1}_8, \mathbf{1}_8)$; P_{L} and P_{R} are the author's chiral projectors (the Clifford relations, the reality of the matrices and the projectors are checked with the author's matrices in `um4_dirac_matrices.py`, 55 checks; the facts used here are stated in this paragraph). The Dirac adjoint uses $\sigma_{16} = T_0 T_1 T_2 T_3$: σ_{16} is real and symmetric, $\sigma_{16} T_a$ is real and antisymmetric, and σ_{16} commutes with Γ_8 (it is a product of four Dirac matrices), so $\overline{P_X \Psi} = \bar{\Psi} P_X$ and every bilinear with an even number of Dirac matrices joins a half with itself, every bilinear with an odd number joins Ψ_{L} with Ψ_{R} (DERIVED, `um4_fields.py`, checks F1 and F2). Status: AUTHOR (matrices, projectors), RECORD (adjoint).

$$
\begin{aligned}
&D_M \Psi = \partial_M \Psi + \Omega_M \Psi, \qquad D_M \bar{\Psi} = \partial_M \bar{\Psi} - \bar{\Psi}\, \Omega_M, \qquad \Omega_M = \frac{1}{2}\, \varpi_{Mab}\, \Sigma^{ab}, \qquad \Sigma^{ab} = \frac{1}{4}\big[T^a, T^b\big], \\
&\varpi_{Mab} = \mathbf{e}_a{}^N \big(\partial_M \mathbf{e}_{bN} - \Gamma^P{}_{MN}\, \mathbf{e}_{bP} \big)
\end{aligned}
\tag{IN-6}
$$

Symbols in (IN-6).

- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M \bar{\Psi} = \partial_M \bar{\Psi} - \bar{\Psi} \Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2 \epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- Ω_M: The spin connection acting on dirac16complex, $\Omega_M = \frac{1}{2} \varpi_{Mab} \Sigma^{ab}$. Unit: 1/length. Status: RECORD.
- ϖ_{Mab}: The canonical (Levi-Civita) spin connection of the vielbein, antisymmetric in a, b. Unit: 1/length. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1, \ldots, 7\}$; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- Σ^{ab}, $\Sigma^{\mu\nu}$: The generators of Spin(4,4) on dirac16complex, $\Sigma^{ab} = \frac{1}{4}[T^a, T^b]$ (the scaled commutators of the author's matrices); $\Sigma^{\mu\nu} = \mathbf{e}_a{}^\mu \mathbf{e}_b{}^\nu \Sigma^{ab}$ along the fold. Unit: dimensionless. Status: AUTHOR.
- T_a, T^a: The author's eight real 16×16 Dirac matrices T16^A[a] of 4+4.nb ($a = 0, \ldots, 7$; $T_0, \ldots, T_7$), with $\{T_a, T_b\} = 2\eta_{ab} \mathbf{1}$; $T^a = \eta^{ab} T_b$. Unit: dimensionless. Status: AUTHOR.
- $\mathbf{e}_a{}^M$, $\mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\, \mathbf{e}^a{}_M \mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN} = \eta_{bc} \mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\, \mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\Gamma^P{}_{MN}$: The Christoffel symbols of the metric g_{MN} (the Levi-Civita connection). Unit: 1/length. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

The canonical spin connection ϖ_{Mab} of the vielbein (antisymmetric in a, b) acts on the spinor through the generators Σ^{ab} of Spin(4,4). With it the Dirac matrices are covariantly constant ($D_M \gamma^N = 0$), the kinetic term of IN-4 is a scalar and the field equation FE-12 is covariant. For the metric of UM-4 $\gamma^M \Omega_M$ is never zero (FE-14). Status: RECORD (the canonical construction), DERIVED (the vielbein postulate for the metric of UM-4, `um4_geometry.py`, G9).

3.5 The gauge fields in the bulk

$$S_{\text{gauge}} = \int \mathrm{d}^8X\,\sqrt{|g|}\,\mathcal{L}_{\text{gauge}}, \qquad \mathcal{L}_{\text{gauge}} = -\frac{1}{4\mathtt{g}_3^2}\,G^A_{MN}G^{AMN} - \frac{1}{4\mathtt{g}_2^2}\,W^I_{MN}W^{IMN} - \frac{1}{4\mathtt{g}_1^2}\,B_{MN}B^{MN},$$
$$G^A_{MN} = \partial_M G^A_N - \partial_N G^A_M + f^{ABC}G^B_M G^C_N, \qquad W^I_{MN} = \partial_M W^I_N - \partial_N W^I_M + \epsilon^{IJK}W^J_M W^K_N, \tag{IN-7}$$
$$B_{MN} = \partial_M B_N - \partial_N B_M$$

Symbols in (IN-7).

- S_{gauge}: The action of the gauge fields in the bulk. Unit: dimensionless. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- $\mathcal{L}_{\text{gauge}}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ in the bulk. Unit: GeV^8. Status: STANDARD.
- $\mathtt{g}_1$, $\mathtt{g}_2$, $\mathtt{g}_3$: The gauge couplings of eight dimensions of $U(1)_Y$, $SU(2)_L$ and $SU(3)_C$: the coefficients of the kinetic terms of the gauge fields of the bulk in their geometric normalisation; the four-dimensional couplings are $g_k^2 = 3H\mathtt{g}_k^2/(2V_*)$. Unit: GeV^{-2} ($\mathtt{g}^2$: length^4). Status: ASSUMED.
- G^A_M, G^A_μ, G^A_{MN}: The gluon field of $SU(3)_C$ ($A = 1, \ldots, 8$): in the bulk G^A_M, a field of eight dimensions in its geometric normalisation (the covariant derivative $\partial - \mathrm{i}G$), and on the fold the canonically normalised four-dimensional field $G^A_\mu(x)$ of its massless mode, $g_3G^A_\mu(x) = G^A_\mu(\sigma = 0, x)$; G^A_{MN} and $G^A_{\mu\nu}$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- W^I_M, W^I_μ, W^I_{MN}: The gauge field of $SU(2)_L$ ($I = 1, 2, 3$): in the bulk W^I_M (geometric normalisation), on the fold the canonically normalised four-dimensional field $W^I_\mu(x)$ of its massless mode, $g_2W^I_\mu(x) = W^I_\mu(\sigma = 0, x)$; W^I_{MN}, $W^I_{\mu\nu}$ their field strengths; $W^\pm_\mu = (W^1_\mu \mp \mathrm{i}W^2_\mu)/\sqrt{2}$. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- B_M, B_μ, B_{MN}: The hypercharge gauge field of $U(1)_Y$: in the bulk B_M (geometric normalisation), on the fold the canonically normalised four-dimensional field $B_\mu(x)$ of its massless mode, $g_1B_\mu(x) = B_\mu(\sigma = 0, x)$; B_{MN}, $B_{\mu\nu}$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- f^{ABC}: The structure constants of $SU(3)$, $[t^A, t^B] = \mathrm{i}f^{ABC}t^C$. Unit: dimensionless. Status: STANDARD.
- B: A colour (adjoint SU(3)) index (as A). Status: STANDARD.
- C: A colour (adjoint SU(3)) index (as A). Status: STANDARD.
- ϵ^{IJK}: The Levi-Civita symbol of three indices (the structure constants of $SU(2)$), $\epsilon^{123} = 1$. Unit: dimensionless. Status: STANDARD.
- J: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.
- K: A weak-isospin (adjoint SU(2)) index (as I). Status: STANDARD.

The gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ in the bulk of eight dimensions, with the couplings $\mathtt{g}_3$, $\mathtt{g}_2$, $\mathtt{g}_1$ of eight dimensions in front of the kinetic terms (the fields normalised geometrically, so that the covariant derivative is $\partial - \mathrm{i}G$; their dimension is that of an inverse length, and $\mathtt{g}^2$ has the dimension of a length to the fourth power). The order of 2026-10-25 13:15 places the photon in the bulk; the photon is a combination of W^3 and B, so the whole electroweak gauge field is a field of the bulk, and UM-4 places the gluons there as well (all gauge fields of the Standard Model in

one place). Their massless modes are bound to the fold with the profile 1 along σ and have the four-dimensional couplings $g_k^2 = 3H\mathsf{g}_k^2/(2V_*)$ (FE-44, FE-74), the measured couplings of the Standard Model. The gauge fields have no components along the extra times and no dependence on them (ASSUMED: such components would be scalars of wrong sign, FE-38), and no component along σ in the gauge of FE-74 (a component B_σ has no normalisable zero mode). They couple to the fields of the fold through their values at $\sigma = 0$ (IN-13) and to dirac16complex only through the portals at the fold (IN-16). Status: AUTHOR (the photon in the bulk, by the order), ASSUMED (all gauge fields in the bulk; no components along the extra times), STANDARD (the Yang-Mills action).

3.6 The fold

$$\int \mathrm{d}^8X\,\sqrt{|g|}\,\delta_\mathrm{D}(\sigma)\,\frac{1}{V_*}\,\mathbb{F} = \int \mathrm{d}^4x\,\sqrt{-\hat{h}}\,\mathcal{Z}\,\mathbb{F}\big|_{\sigma=0}, \qquad \mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4,$$
$$\hat{h}_{\mu\nu} = \mathrm{diag}\big(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1\big), \qquad \sqrt{-\hat{h}} = \mathrm{e}^{3a_4}, \qquad \mathcal{Z} = \frac{V_\mathrm{t}\big(\mathrm{e}^{-a_4}b_4\big)^3}{V_*}, \qquad V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7 \tag{IN-8}$$

Symbols in (IN-8).

- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- δ_D: The Dirac delta function of σ: zero for $\sigma \neq 0$, with the integral 1 over any range that contains $\sigma = 0$; the integral of $\delta_\mathrm{D}(\sigma)$ times a smooth function of σ is the value of that function at $\sigma = 0$. Unit: 1/length (for $\delta_\mathrm{D}(\sigma)$). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\mathbb{F}$: Any function of the fields at the fold (used to state a rule of integration). Status: STANDARD.
- x: The coordinates of the fold: x_1, x_2, x_3 of ordinary space, x_4 the time, x_5, x_6, x_7 the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.

- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold and the kinetic terms of the massless modes of the fields of the bulk; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- V_t: The coordinate volume of the (compact) extra times, $V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$. Unit: length^3. Status: ASSUMED.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

The fold is the seven-dimensional surface $\sigma = 0$ ($z = \pi/2$) with the coordinates $x_1, \ldots, x_7$; at $\sigma = 0$ the metric of UM-4 is that of the fold (FD-2), the same as in UM-3. No field depends on the extra times (ASSUMED, FE-38), so the integral over them gives their coordinate volume V_t (the extra times are compact) times their scale factors $(\mathrm{e}^{-a_4}b_4)^3$ at the fold. V_* is the physical volume of the extra times on the fold in the late universe, where $\mathrm{e}^{-a_4}b_4$ is constant (FE-72); with this normalisation $\mathcal{Z} = 1$ in the whole late universe, the fields of the fold are four-dimensional fields in the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold, and their couplings are the measured four-dimensional ones. $\mathcal{Z}(x_4)$ is the factor by which the extra times rescale every term of the fold when $\mathrm{e}^{-a_4}b_4$ changes, and with it the couplings of the fold (FE-44). $\mathbb{F}$ stands for any function of the fields on the fold. Status: ASSUMED (the place of the fields), DERIVED (the reduction).

$$\mathcal{L}_\mathrm{IG} = \frac{\hat{R}}{2\kappa_\mathrm{ind}} \tag{IN-9}$$

Symbols in (IN-9).

- $\mathcal{L}_\mathrm{IG}$: The Lagrangian of the Einstein-Hilbert term of the four-dimensional metric of the fold that the quantum fluctuations of the fields of the fold induce; it adds to the Einstein-Hilbert term that the graviton of the bulk gives the fold and does not change the speed of gravitational waves. Unit: GeV^4. Status: ESTIMATE.
- $\hat{R}$: The curvature of the metric $\hat{g}_{ab}$ of the fold (FD-2): with four frame indices its Riemann tensor, with two its Ricci tensor $\hat{R}_{ab}$, without indices its scalar curvature $\hat{R}$; for frozen extra times the scalar curvature equals that of the four-dimensional metric $\hat{h}_{\mu\nu}$. Unit: 1/length^2. Status: STANDARD.
- κ_ind: The coupling of the Einstein-Hilbert term $\hat{R}/(2\kappa_\mathrm{ind})$ that the quantum fluctuations of the fields of the fold may induce, $\kappa_\mathrm{ind} = 8\pi G_\mathrm{ind}$; it adds to the gravity of the fold, $1/(8\pi G_\mathrm{N}) = V_*/\kappa_7 + 1/\kappa_\mathrm{ind}$. Unit: GeV^{-2}. Value: not fixed by UM-4; negligible for a cutoff of the order of H. Status: ESTIMATE.

The Einstein-Hilbert term of the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold ($\hat{R}$ its scalar curvature) that the quantum fluctuations of the fields of the fold induce (FE-64), with the coupling $\kappa_\mathrm{ind} = 8\pi G_\mathrm{ind}$. It adds $1/\kappa_\mathrm{ind}$ to the coefficient V_*/κ_7 of the Einstein-Hilbert term that the bulk gives the fold (FE-63, FE-71), so that four-dimensional gravity has $1/(8\pi G_\mathrm{N}) = V_*/\kappa_7 + 1/\kappa_\mathrm{ind}$ (FE-59), and it has the same light cone, so that it leaves the speed of gravitational waves equal to the speed of light (FE-63). Its strength is not fixed by UM-4 (FE-64); for a cutoff of the order of H it is negligible. Status: ESTIMATE.

$$\frac{1}{2}\Big(\frac{\partial^2 h_\mathrm{gw}}{\partial x_4^2} + 3a_4'\,\frac{\partial h_\mathrm{gw}}{\partial x_4} - \mathrm{e}^{-2a_4}\,\frac{\partial^2 h_\mathrm{gw}}{\partial x_3^2}\Big) = \kappa_7\,\Pi_\mathrm{gw}, \qquad \kappa_7 = \frac{5}{2}\,\kappa H, \qquad c_\mathrm{g} = 1 \tag{FE-63}$$

Symbols in (FE-63).

- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- h_gw: The amplitude of a gravitational wave of the + polarisation travelling along x_3: the metric components along x_1 and x_2 multiplied by $\mathrm{e}^{h_\mathrm{gw}}$ and $\mathrm{e}^{-h_\mathrm{gw}}$; a function of σ, x_3 and x_4 in the bulk, and on the fold the gravitational wave of the four-dimensional world, to the first order its plus polarisation h_+. Unit: dimensionless. Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.

- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- κ_7: The gravitational coupling of the fold: to the first order in its curvature over H^2 the fold obeys the Einstein equations of seven dimensions $\hat{G}^a{}_b = \kappa_7 \mathsf{t}^a{}_b$ with $\kappa_7 = 5\kappa H/2$, the coupling of the massless graviton of the bulk bound to the fold; four-dimensional gravity on the fold has $8\pi G_\mathrm{N} = \kappa_7/V_*$ (plus the induced term, which is negligible). Unit: GeV^{-5}. Value: $5\kappa H/2$. Status: DERIVED.
- Π_gw: The transverse traceless stress of the matter of the fold that drives the gravitational wave h_gw (half the difference of its components along x_1 and x_2), integrated across the fold (eight-dimensional normalisation). Unit: GeV^7. Status: STANDARD.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- c_g: The speed of gravitational waves on the fold, in units of the speed of light. Unit: dimensionless. Value: 1 at every wavelength for the four-dimensional graviton on the static block and on every fold with frozen extra times, and for the massless graviton of the bulk at every depth; measured $-3\times10^{-15} \le c_\mathrm{g} - 1 \le 7\times10^{-16}$ (GW170817). Status: DERIVED.

The gravitational wave of the fold: a wave of the + polarisation travelling along x_3 on a fold with frozen extra times (any $a_4(x_4)$, the static block included). The left-hand side is half the difference of the components along x_1 and x_2 of the Einstein tensor of the metric of the fold to the first order in h_gw (computed exactly), and the right-hand side is the same difference of κ_7 times the matter of the fold, Π_gw its transverse traceless stress: by FE-71 the fold obeys the Einstein equations of seven dimensions with $\kappa_7 = 5\kappa H/2$, the same coupling as in its Friedmann equations. FE-63 is the wave equation of four-dimensional general relativity with the friction $3a_4'$ of the expansion: the gravitational waves of the four-dimensional world travel at the speed of light at every wavelength, and their Planck mass equals that of the expansion. In the bulk the same wave is the zero mode of FE-76: h_gw independent of σ solves the bulk equation FE-68 (with $\mathsf{P}'(1) = 21$) and the junction condition, so the wave of the fold is a wave of the bulk at the speed of light that the fold holds. The gravitational waves of the neutron-star merger GW170817 arrived together with its light, $-3\times10^{-15} \le c_\mathrm{g} - 1 \le 7\times10^{-16}$ for the distance 26 Mpc (B. P. Abbott et al., Astrophys. J. Lett. 848 (2017) L13): UM-4 agrees with it for every value of H, at every order of the bulk (the light cone of every slice is that of the fold). The induced term IN-9 adds to the coefficient of the same operator and keeps $c_\mathrm{g} = 1$. Status: DERIVED (`um4_foldsources.py`, FS2; `um4_bulk.py`, B4 and B5; `um4_fold.py`, F2).

$$\frac{1}{16\pi G_\mathrm{ind}} = \frac{N_\mathrm{ind}\,\Lambda_\mathrm{c}^2}{192\pi^2}, \qquad N_\mathrm{ind} = \sum_{\text{real scalars}} \big(1 - 6\xi\big) + N_\mathrm{Weyl} - 4N_\mathrm{gauge} = 52 - 24\xi_\mathrm{H} \tag{FE-64}$$

Symbols in (FE-64).

- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_ind: The Newton constant of the induced Einstein-Hilbert term of the fold, $\kappa_\mathrm{ind} = 8\pi G_\mathrm{ind}$. Unit: GeV^{-2}. Status: ESTIMATE.
- N_ind: The weighted number of the fields of the fold that induce the Einstein-Hilbert term: each real scalar counts $1 - 6\xi$, each Weyl fermion 1, each gauge field -4 (with its ghost fields). Unit: dimensionless. Value: 52 for the fields of the fold of UM-4 (the four real scalars of the Higgs doublet and 48 Weyl fermions; the gauge fields live in the bulk) with $\xi_\mathrm{H} = 0$; $52 - 24\xi_\mathrm{H}$ with the curvature coupling ξ_H of an extension. Status: DERIVED.
- Λ_c: The cutoff of the proper-time integral of the one-loop effective action of the fields of the fold: the highest energy of the quantum fluctuations that induce gravity. Unit: GeV. Value: taken of the order of H (2.5 TeV); the induced term alone would give the measured Newton constant for $\Lambda_\mathrm{c} = \pi M_\mathrm{red}\sqrt{96/52} = 1.0\times10^{19}$ GeV. Status: ESTIMATE.

- ξ: The coupling of a real scalar field to the scalar curvature $\hat{R}$ of the fold: the term $-\frac{1}{2}\xi\hat{R}$ times the square of the field in its Lagrangian ($\xi = 1/6$ is the conformal coupling). Unit: dimensionless. Status: STANDARD.
- N_{Weyl}: The number of Weyl (two-component) fermion fields of the fold. Unit: dimensionless. Value: 48 (three generations of quarks and leptons with three right-handed neutrinos). Status: STANDARD.
- N_{gauge}: The number of gauge fields of the fold. Unit: dimensionless. Value: 0 in UM-4 (the gauge fields live in the bulk); 12 (eight gluons, three W, one B) if they lived on the fold. Status: STANDARD.
- ξ_{H}: The coupling of the Higgs doublet to the scalar curvature of the fold, the term $-\xi_{\text{H}}\hat{R}\,\Phi^\dagger\Phi$ in the Lagrangian of the fold: an extension of UM-4, whose Higgs Lagrangian has no such term ($\xi_{\text{H}} = 0$). Unit: dimensionless. Value: 0 in UM-4; about 1.6×10^4 if the Higgs field drove inflation (ESTIMATE, an extension of the action). Status: ESTIMATE (an extension of the action).

The Einstein-Hilbert term that the quantum fluctuations of the fields of the fold induce (Sakharov's induced gravity, A. D. Sakharov, Dokl. Akad. Nauk SSSR 177 (1967) 70), evaluated at one loop with the heat kernel and the proper-time cutoff Λ_{c} (the coefficient of the scalar curvature in the heat kernel of a Laplace-type operator $-(\nabla^2 + E)$ is the trace of $E + R/6$; D. V. Vassilevich, Phys. Rep. 388 (2003) 279): each real scalar with the curvature coupling ξ counts $1 - 6\xi$, each Weyl fermion 1, each gauge field of the fold -4 with its ghost fields (S. N. Solodukhin, arXiv:1502.03758). In UM-4 the gauge fields live in the bulk, so the fold carries the four real scalars of the Higgs doublet and the 48 Weyl fermions of three generations with three right-handed neutrinos, and no gauge field ($N_{\text{gauge}} = 0$), so $N_{\text{ind}} = 4(1 - 6\xi_{\text{H}}) + 48$. The Higgs Lagrangian of UM-4 (IN-11) has no coupling to the curvature of the fold, so $\xi_{\text{H}} = 0$ and $N_{\text{ind}} = 52$. For a cutoff Λ_{c} of the order of H (2.5 TeV) the induced term is a part 6×10^{-32} of M_{red}^2 (M_{red} the reduced Planck mass): four-dimensional gravity is the gravity of the bulk (FE-59). The same loops give the fold a vacuum energy of about $|N_{\text{B}} - N_{\text{F}}|\Lambda_{\text{c}}^4/(64\pi^2)$ ($N_{\text{B}} = 4$ bosonic and $N_{\text{F}} = 96$ fermionic degrees of freedom), a tension that adds to λ_{F} (FE-65). Status: ESTIMATE (one loop and cutoff-dependent; `um4_foldsources.py`, FS3).

$$\mathcal{T}_{\text{ff}}{}^{a}{}_{b} = \text{diag}\big(p_{\text{ff}}, p_{\text{ff}}, p_{\text{ff}}, -\rho_{\text{ff}}, q_{\text{ff}}, q_{\text{ff}}, q_{\text{ff}}\big) \tag{IN-10}$$

Symbols in (IN-10).

- $\mathcal{T}_{\text{ff}}{}^{a}{}_{b}$: The energy-momentum tensor of the fold fluid (frame components of the fold, four-dimensional normalisation). Unit: GeV^4. Status: ASSUMED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \text{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1, \ldots, 7\}$; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- p_{ff}: The pressure of the fold fluid along ordinary space. Unit: GeV^4. Value: $-\rho_{\text{ff}}$ (a tension). Status: ASSUMED (the fluid), DERIVED (the value).
- ρ_{ff}: The energy density of the fold fluid. Unit: GeV^4. Value: $\lambda_{\text{F}}V_* + \rho_\Lambda - \rho_{\text{vac}}$: together with the vacuum energy ρ_{vac} of the fields of the fold it makes up the tension $\lambda_{\text{F}}V_*$ that the static block requires plus the dark energy ρ_Λ. Status: ASSUMED (the fluid), DERIVED (the value).
- q_{ff}: The pressure of the fold fluid along the extra times: minus its energy density (its tension) plus the pressure q_{hold} that holds the volume of the extra times fixed. Unit: GeV^4. Value: $-\rho_{\text{ff}} + q_{\text{hold}}$. Status: ASSUMED (the fluid), DERIVED (the value).

The fold fluid: a medium at rest on the fold (the frame indices a, b run over the seven directions $1, \ldots, 7$ of the fold), with the energy density ρ_{ff}, the pressure p_{ff} along ordinary space and the pressure q_{ff} along the extra times (four-dimensional normalisation, IN-8). The kink of the metric at the fold requires a thin source (FE-4); the fold fluid is the part of it that the fields of the fold, the particles χ of dirac16complex and the portals do not supply. It has two parts (FE-65): a tension, a vacuum energy of the seven-dimensional fold like the tension of the brane of Randall and Sundrum, which the static block requires; and a pressure q_{hold} along the extra times, which holds their volume fixed. Without q_{hold} the volume of the extra times on the fold changes whenever the matter of the fold is not radiation (FE-72), and with it Newton's constant, the squares of the gauge and Yukawa couplings and the quartic coupling of the Higgs doublet: in a matter era the extra times would expand as ordinary space and Newton's

constant would fall at three times the Hubble rate, which lunar laser ranging excludes by a factor of about 8500. UM-4 therefore assumes that the fold fluid supplies $q_{\rm hold}$. Its microscopic origin is OPEN: it acts as a potential for the volume of the extra times, as the stabilisation of the sizes of compact extra dimensions does in other theories. Status: ASSUMED (its existence), DERIVED (its values, FE-65).

$$\begin{aligned} &p_{\rm ff} = -\rho_{\rm ff}, \qquad q_{\rm ff} = -\rho_{\rm ff} + q_{\rm hold}, \qquad \kappa\lambda_{\rm F} = 12H \quad \big(\lambda_{\rm F} V_* = 30H^2 M_{\rm red}^2\big), \\ &q_{\rm hold} = -\frac{\rho - 3\mathcal{P}}{2} - q_{\rm SM} - q_\chi - q_{\rm P}: \qquad \text{radiation } 0, \quad \text{dust } -\frac{\rho}{2}, \quad \text{a vacuum energy } -\rho \end{aligned} \tag{FE-65}$$

Symbols in (FE-65).

- $p_{\rm ff}$: The pressure of the fold fluid along ordinary space. Unit: GeV^4. Value: $-\rho_{\rm ff}$ (a tension). Status: ASSUMED (the fluid), DERIVED (the value).
- $\rho_{\rm ff}$: The energy density of the fold fluid. Unit: GeV^4. Value: $\lambda_{\rm F} V_* + \rho_\Lambda - \rho_{\rm vac}$: together with the vacuum energy $\rho_{\rm vac}$ of the fields of the fold it makes up the tension $\lambda_{\rm F} V_*$ that the static block requires plus the dark energy ρ_Λ. Status: ASSUMED (the fluid), DERIVED (the value).
- $q_{\rm ff}$: The pressure of the fold fluid along the extra times: minus its energy density (its tension) plus the pressure $q_{\rm hold}$ that holds the volume of the extra times fixed. Unit: GeV^4. Value: $-\rho_{\rm ff} + q_{\rm hold}$. Status: ASSUMED (the fluid), DERIVED (the value).
- $q_{\rm hold}$: The pressure along the extra times that the fold fluid adds to its tension to hold the volume of the extra times fixed (four-dimensional normalisation). Unit: GeV^4. Value: $-(\rho - 3\mathcal{P})/2 - q_{\rm SM} - q_\chi - q_{\rm P}$: 0 for radiation, $-\rho/2$ for dust, $-\rho$ for a vacuum energy; today $-0.842\,\rho_{\rm c}^{(0)}$ if the baryons have no pressure along the extra times (ASSUMED). Status: DERIVED (the value), ASSUMED (that the fold fluid supplies it).
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\lambda_{\rm F}$: The tension that the static block requires of the fold, $\kappa\lambda_{\rm F} = 12H$: the total vacuum energy of the seven-dimensional fold, made of the tension of the fold fluid and the vacuum energy of the fields of the fold (both have the pressure minus their energy density along ordinary space and along the extra times), per unit volume of the six other directions of the fold (eight-dimensional normalisation); in the four-dimensional normalisation it is $\lambda_{\rm F} V_*$, and the excess of the total over $\lambda_{\rm F} V_*$ is the dark energy ρ_Λ. Unit: GeV^7. Value: $12H/\kappa$; $\lambda_{\rm F} V_* = 30H^2 M_{\rm red}^2 = 1.1 \times 10^{45}$ GeV^4 for $H = 2.5$ TeV. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_{\rm F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\rm t}(\mathrm{e}^{-a_4} b_4)^3$ (with $V_{\rm t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4} b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $M_{\rm red}$: The reduced Planck mass, $M_{\rm red} = (8\pi G_{\rm N})^{-1/2}$. Unit: GeV. Value: 2.435×10^{18} GeV. Status: STANDARD.
- ρ: The total energy density of the matter of the fold (all species); $\Delta\rho$ its perturbation. Unit: GeV^4. Status: STANDARD.
- $\mathcal{P}$: The total pressure of the matter of the fold; $\Delta\mathcal{P}$ its perturbation. Unit: GeV^4. Status: STANDARD.
- $q_{\rm SM}$: The pressure of the Standard Model along the extra times at the fold: the mean on-shell Lagrangian density of the Standard-Model fields (the extra-time component of the energy-momentum tensor of every term of the fold is its Lagrangian, because the fields of the fold do not depend on the extra times). Unit: GeV^4. Status: DERIVED (for radiation and free fields), ASSUMED (for the baryons).
- q_χ: The pressure of the particles χ of dirac16complex along the extra times, integrated across the fold (their on-shell Lagrangian, the self-interaction energy per volume; zero for a dilute gas). Unit: GeV^4. Status: DERIVED.
- $q_{\rm P}$: The pressure of the portals along the extra times: their mean on-shell Lagrangian density. Unit: GeV^4. Status: DERIVED.

The fold fluid of UM-4 with frozen extra times. Its energy density and its pressure along ordinary space are those of a tension. The static block requires the total tension $\kappa\lambda_{\rm F} = 12H$ (FD-12, FE-30), which in the four-dimensional normalisation is $\lambda_{\rm F} V_* = 30H^2 M_{\rm red}^2$ (FE-71): the tension of the fold fluid together with the vacuum energy of the fields of the fold (their potential at the minimum), which is a tension of the same kind, because the fields do not depend on the extra times and their pressure along the extra times is their on-shell Lagrangian (FE-5). The excess of the total tension over $\lambda_{\rm F}$ is the vacuum energy ρ_Λ with which the fold expands, the dark energy of UM-4 (FE-59; its

value needs the tension tuned to one part in 10^{92} for $H = 2.5$ TeV, the cosmological-constant problem of the brane of Randall and Sundrum, OPEN). When the fold expands with frozen extra times, the component of its equations along the extra times requires of the matter above λ_{F} the pressure $-(\rho - 3\mathcal{P})/2$ along the extra times (ρ, $\mathcal{P}$ its energy density and its pressure along ordinary space; FE-72). The pressure of a species of the fold along the extra times is its mean on-shell Lagrangian density (FE-5): radiation has none, and supplies what is required ($\rho = 3\mathcal{P}$); dust has none and needs $-\rho/2$ (point particles at rest in the extra times have no stress along them; for the baryons this is ASSUMED, because the on-shell Lagrangian of the quark and gluon fields inside a nucleon is not computed); a vacuum energy has $-\rho$ and needs -2ρ. The particles χ of dirac16complex have no pressure along the extra times in a dilute gas (the on-shell Lagrangian of a free Dirac field vanishes; the contact term gives $+(\lambda_\Psi/2)(\bar\Psi\Psi)^2$, negligible in a dilute gas). q_{hold} makes up the difference. Status: DERIVED (`um4_foldsources.py`, FS4 and FS5; `um4_fold.py`, F4 and F5; `um4_background.py`, BG2 and BG3), ASSUMED (the pressure of the baryons along the extra times; that the fold fluid supplies q_{hold}), OPEN (what supplies it, and the tuning of the tension).

3.7 The Higgs doublet with the ad hoc vacuum

$$\mathcal{L}_{\mathrm{H}} = -\hat h^{\mu\nu}\big(D_\mu\Phi\big)^\dagger D_\nu\Phi - V(\Phi), \qquad V(\Phi) = \mu^2\,\Phi^\dagger\Phi + \lambda\big(\Phi^\dagger\Phi\big)^2, \qquad \lambda > 0, \quad \mu^2 < 0 \tag{IN-11}$$

Symbols in (IN-11).

- $\mathcal{L}_{\mathrm{H}}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.
- $\hat h_{\mu\nu}$: The four-dimensional metric of the fold, $\hat h_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat h^{\mu\nu}$ its inverse, $\hat h$ its determinant, $\sqrt{-\hat h} = \mathrm{e}^{3a_4}$; $\hat h^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $V(\Phi)$: The Higgs potential, $V = \mu^2\Phi^\dagger\Phi + \lambda(\Phi^\dagger\Phi)^2$. Unit: GeV^4. Status: AUTHOR.
- μ^2: The mass parameter of the Higgs potential, $\mu^2 < 0$ (it enters only as μ^2). Unit: GeV^2. Value: $\mu^2 = -\lambda v^2 = -m_h^2/2 = -(88.6\ \mathrm{GeV})^2$. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.

The Higgs doublet Φ (two complex components, hypercharge 1/2) with the author's potential. In the signature $(+,+,+,-)$ of the fold the kinetic term carries a minus sign, so that the energy density is positive. Status: AUTHOR (the potential), STANDARD (the rest).

$$\Phi = \frac{1}{\sqrt2}\begin{pmatrix}\mathcal{A}(x_4)\\ v + h\end{pmatrix}, \qquad \mathcal{A}(x_4) = v_{\mathcal{A}}\,\mathrm{e}^{-2a_{40}x_4}, \qquad a_{40} > 0, \quad x_4 \geq 0, \qquad v^2 = -\frac{\mu^2}{\lambda} \tag{IN-12}$$

Symbols in (IN-12).

- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_{\mathcal{A}}\mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt2\,G_{\mathrm{F}})^{-1/2}$ with the Fermi constant $G_{\mathrm{F}} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.

- h: The Higgs field: the fluctuation of the lower component of the Higgs doublet about v. Unit: GeV. Value: $m_h = 125.25$ GeV. Status: STANDARD.
- $v_{\mathcal{A}}$: The amplitude of almightyAdHoc at $x_4 = 0$; the author's almightyAdHoc, $\mathrm{e}^{-2a_{40}x_4}$ (written Exp[-2 a40 x4] in his notebook), has $v_{\mathcal{A}} = 1$ in the units of his notebook. Unit: GeV. Value: 1 in the units of the author's notebook; UM-4 keeps it free (the examples take $v_{\mathcal{A}} = v$ and $10v$). Status: AUTHOR (the form), ASSUMED (a free amplitude).
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_{40}: The decay constant of almightyAdHoc, $a_{40} > 0$: $\mathcal{A}$ falls by e in the time $1/(2a_{40})$. Unit: GeV (1/length). Status: AUTHOR.
- μ^2: The mass parameter of the Higgs potential, $\mu^2 < 0$ (it enters only as μ^2). Unit: GeV^2. Value: $\mu^2 = -\lambda v^2 = -m_h^2/2 = -(88.6\ \mathrm{GeV})^2$. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.

The author's ad hoc vacuum: the upper component of the Higgs doublet is the prescribed function almightyAdHoc $\mathcal{A}(x_4) = \exp(-2a_{40}x_4)$ (the author writes it with the amplitude $v_{\mathcal{A}} = 1$ in the unit of the field; UM-4 keeps $v_{\mathcal{A}}$ to show where it enters), the lower component is the vacuum expectation value v plus the Higgs field h. The Goldstone fields are absorbed by the gauge choice (unitary gauge). $\mathcal{A}$ decays with the rate $2a_{40}$ from the beginning of the time $x_4 = 0$: it acts only in the primordial era. The ad hoc vacuum is not an exact solution of the Higgs equation; the residuals are FE-26. Status: AUTHOR.

3.8 The covariant derivative of the fields of the fold

$$D_\mu = \partial_\mu + \hat{\omega}_\mu - \mathrm{i}g_3\, G_\mu^A\, t^A - \mathrm{i}g_2\, W_\mu^I\, \frac{\tau^I}{2} - \mathrm{i}g_1\, Y_f\, B_\mu,$$
$$g_3\, G_\mu^A(x) = G_\mu^A(\sigma = 0, x), \qquad g_2\, W_\mu^I(x) = W_\mu^I(\sigma = 0, x), \qquad g_1\, B_\mu(x) = B_\mu(\sigma = 0, x) \tag{IN-13}$$

Symbols in (IN-13).

- $D_M,\ D_\mu$: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G_\mu^B X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- $\hat{\omega}_\mu$: The spin connection of the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold acting on four-dimensional spinors. Unit: 1/length. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- g_3: The $SU(3)_C$ gauge coupling of the four-dimensional world (the strong coupling g_s): the coupling of the massless mode of the gluon field of the bulk, $g_3^2 = 3H\mathsf{g}_3^2/(2V_*)$. Unit: dimensionless. Value: $g_3(m_Z) = 1.2177$ ($g_3^2/(4\pi) = 0.1180$). Status: STANDARD.
- $G_M^A,\ G_\mu^A,\ G_{MN}^A$: The gluon field of $SU(3)_C$ ($A = 1, \ldots, 8$): in the bulk G_M^A, a field of eight dimensions in its geometric normalisation (the covariant derivative $\partial - \mathrm{i}G$), and on the fold the canonically normalised four-dimensional field $G_\mu^A(x)$ of its massless mode, $g_3 G_\mu^A(x) = G_\mu^A(\sigma = 0, x)$; G_{MN}^A and $G_{\mu\nu}^A$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- t^A: The generators of $SU(3)$ on colour triplets, t^A: half the Gell-Mann matrices, with $\mathrm{tr}(t^A t^B) = \delta^{AB}/2$. Unit: dimensionless. Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling of the four-dimensional world (usually g): the coupling of the massless mode of the $SU(2)_L$ field of the bulk, $g_2^2 = 3H\mathsf{g}_2^2/(2V_*)$. Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- $W_M^I,\ W_\mu^I,\ W_{MN}^I$: The gauge field of $SU(2)_L$ ($I = 1, 2, 3$): in the bulk W_M^I (geometric normalisation), on the fold the canonically normalised four-dimensional field $W_\mu^I(x)$ of its massless mode, $g_2 W_\mu^I(x) = W_\mu^I(\sigma = 0, x)$;

W^I_{MN}, $W^I_{\mu\nu}$ their field strengths; $W^\pm_\mu = (W^1_\mu \mp \mathrm{i} W^2_\mu)/\sqrt{2}$. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.

- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak doublets (τ^1, τ^2, τ^3; τ^2 is the second one, not a square). Unit: dimensionless. Status: STANDARD.
- g_1: The $U(1)_Y$ gauge coupling of the four-dimensional world (usually g'; not normalised for grand unification): the coupling of the massless mode of the hypercharge field of the bulk, $g_1^2 = 3H\mathtt{g}_1^2/(2V_*)$. Unit: dimensionless. Value: $g_1(m_Z) = 0.3574$. Status: STANDARD.
- Y_f: The hypercharge of a field, the charge of its coupling $g_1 Y_f B_\mu$ to the hypercharge field in the covariant derivative (1/6, 2/3, −1/3, −1/2, −1, 0 for $Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}$, 1/2 for Φ). Unit: dimensionless. Status: STANDARD.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- B_M, B_μ, B_{MN}: The hypercharge gauge field of $U(1)_Y$: in the bulk B_M (geometric normalisation), on the fold the canonically normalised four-dimensional field $B_\mu(x)$ of its massless mode, $g_1 B_\mu(x) = B_\mu(\sigma = 0, x)$; B_{MN}, $B_{\mu\nu}$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- x: The coordinates of the fold: x_1, x_2, x_3 of ordinary space, x_4 the time, x_5, x_6, x_7 the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.

The gauge fields act on the fields of the fold through their values at the fold, $\sigma = 0$ (the second line: the field of the bulk in its geometric normalisation, IN-7, at $\sigma = 0$ equals the coupling times the canonically normalised field of four dimensions). For the massless modes of the bulk, whose profile along σ is 1 (FE-74), the canonical four-dimensional fields G^A_μ, W^I_μ, B_μ have the kinetic terms $-\frac{1}{4}G^A_{\mu\nu}G^{A\mu\nu}$ and so on, with the four-dimensional couplings $g_k^2 = 3H\mathtt{g}_k^2/(2V_*)$ (FE-44): the gauge theory of the Standard Model on the fold. The massive modes of the bulk enter the same covariant derivative with their values at the fold (FE-79). The covariant derivative of a field acts with the part that fits the field: t^A on colour triplets, $\tau^I/2$ on weak doublets, the hypercharge Y_f of the field, and the four-dimensional spin connection $\hat\omega_\mu$ on spinors. The hypercharges (STANDARD) are $Y = 1/6$ for Q, 2/3 for u_R, −1/3 for d_R, −1/2 for ℓ, −1 for e_R, 0 for ν_R and 1/2 for Φ. Status: STANDARD (the covariant derivative), DERIVED (the normalisation of the zero modes, `um4_bulk.py`, B2).

3.9 The quarks and leptons

$$\mathcal{L}_\mathrm{f} = \sum_\psi \mathrm{i}\,\bar\psi\,\hat\gamma^\mu D_\mu \psi, \qquad \psi \in \{Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}\},$$
$$\{\hat\gamma^\mu, \hat\gamma^\nu\} = -2\hat h^{\mu\nu}\mathbf{1}, \qquad \bar\psi = \psi^\dagger\beta, \qquad \beta = \hat\gamma^{(4)}, \qquad \hat\gamma_5 = \mathrm{i}\,\hat\gamma^{(4)}\hat\gamma^{(1)}\hat\gamma^{(2)}\hat\gamma^{(3)},$$
$$\hat P_\mathrm{L} = \frac{\mathbf{1} - \hat\gamma_5}{2}, \quad \hat P_\mathrm{R} = \frac{\mathbf{1} + \hat\gamma_5}{2}, \qquad \hat P_\mathrm{L} Q = Q, \quad \hat P_\mathrm{L}\ell = \ell, \quad \hat P_\mathrm{R} u_\mathrm{R} = u_\mathrm{R}, \quad \hat P_\mathrm{R} d_\mathrm{R} = d_\mathrm{R}, \quad \hat P_\mathrm{R} e_\mathrm{R} = e_\mathrm{R}, \quad \hat P_\mathrm{R}\nu_\mathrm{R} = \nu_\mathrm{R} \tag{IN-14}$$

Symbols in (IN-14).

- $\mathcal{L}_\mathrm{f}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- ψ: The label of a Standard-Model fermion field in a sum: ψ runs over $Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}$. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ψ, $\bar\psi$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar\psi = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\hat\gamma^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat\gamma^{(a)}$ along the frame directions of the fold and $\hat\gamma^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat\gamma^\mu, \hat\gamma^\nu\} = -2\hat h^{\mu\nu}\mathbf{1}$; $\hat\gamma_\mu = \hat h_{\mu\nu}\hat\gamma^\nu$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+, +, +, -$); a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the

fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2 \epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.

- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- u_R: The right-handed up-type quarks (up, charm and top; singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- d_R: The right-handed down-type quarks (down, strange and bottom; hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- e_R: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- ν_R: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathbf{1}$, $\mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- β: The matrix of the four-dimensional Dirac adjoint, $\beta = \hat{\gamma}^{(4)}$ (Hermitian, $\beta^2 = \mathbf{1}$). Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}_5$: The four-dimensional chirality matrix, $\hat{\gamma}_5 = \mathrm{i}\hat{\gamma}^{(4)}\hat{\gamma}^{(1)}\hat{\gamma}^{(2)}\hat{\gamma}^{(3)}$, $\hat{\gamma}_5^2 = \mathbf{1}$. Unit: dimensionless. Status: STANDARD.
- $\hat{P}_\mathrm{L}$: The four-dimensional left-handed projector, $\hat{P}_\mathrm{L} = (\mathbf{1} - \hat{\gamma}_5)/2$. Unit: dimensionless. Status: STANDARD.
- $\hat{P}_\mathrm{R}$: The four-dimensional right-handed projector, $\hat{P}_\mathrm{R} = (\mathbf{1} + \hat{\gamma}_5)/2$. Unit: dimensionless. Status: STANDARD.

Every Standard-Model fermion is a four-dimensional Weyl field on the fold, written as a four-component spinor with a fixed chirality: the quark doublets Q and the lepton doublets ℓ are left-handed, the singlets u_R, d_R, e_R, ν_R right-handed. Each symbol stands for the three generations (a vector in generation space) and, for quarks, the three colours. $\hat{\gamma}^\mu = \hat{\mathbf{e}}_a{}^\mu \hat{\gamma}^{(a)}$ ($\hat{\mathbf{e}}_a{}^\mu$ the inverse frame field of the fold) are complex 4×4 Dirac matrices of the fold (with the frame matrices $\hat{\gamma}^{(a)}$, $a = 1, \ldots, 4$, $(\hat{\gamma}^{(4)})^2 = \mathbf{1}$, $(\hat{\gamma}^{(1)})^2 = (\hat{\gamma}^{(2)})^2 = (\hat{\gamma}^{(3)})^2 = -\mathbf{1}$); $\beta = \hat{\gamma}^{(4)}$ is Hermitian and $(\hat{\gamma}^\mu)^\dagger = \beta\hat{\gamma}^\mu\beta$. With these conventions the free Dirac equation is $(-\mathrm{i}\hat{\gamma}^\mu\partial_\mu + m)\psi = 0$ and $\bar{\psi}\beta\psi$-type energy densities are positive. Status: STANDARD.

$$\mathcal{L}_\mathrm{Y} = -\big(\bar{Q}\,\mathcal{Y}^u\,\tilde{\Phi}\,u_\mathrm{R} + \bar{Q}\,\mathcal{Y}^d\,\Phi\,d_\mathrm{R} + \bar{\ell}\,\mathcal{Y}^e\,\Phi\,e_\mathrm{R} + \bar{\ell}\,\mathcal{Y}^\nu\,\tilde{\Phi}\,\nu_\mathrm{R}\big) + \mathrm{h.c.}, \qquad \tilde{\Phi} = \mathrm{i}\tau^2\Phi^* \tag{IN-15}$$

Symbols in (IN-15).

- $\mathcal{L}_\mathrm{Y}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- $\tilde{\Phi}$: The conjugate Higgs doublet, $\tilde{\Phi} = \mathrm{i}\tau^2\Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- u_R: The right-handed up-type quarks (up, charm and top; singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- d_R: The right-handed down-type quarks (down, strange and bottom; hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.

- e_{R}: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- ν_{R}: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- h.c.: The Hermitian conjugate of the preceding terms. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- τ^I: The Pauli matrices acting on weak doublets (τ^1, τ^2, τ^3; τ^2 is the second one, not a square). Unit: dimensionless. Status: STANDARD.
- *: Complex conjugation of a number, or of each entry of a matrix or column. Status: STANDARD.

The Yukawa couplings, with the complex 3×3 matrices $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$ in generation space; $\tilde{\Phi}$ is the conjugate doublet (hypercharge $-1/2$). In the vacuum they give the quarks, the charged leptons and the neutrinos their Dirac masses (FE-46); with three right-handed neutrinos the neutrino masses are Dirac masses and the lepton mixing is that of the PMNS matrix. No Majorana mass of ν_{R} is included (lepton number is conserved). Status: STANDARD (with ν_{R}: accepted by the author, 2026-10-24 17:32).

3.10 The portals of dirac16complex

$$\begin{aligned}\mathcal{L}_{\mathrm{P}} = & -c_\Phi \left(\Phi^\dagger \Phi\right) \bar{\Psi} P_{\mathrm{L}} \Psi - \sum_X \Big[\frac{1}{4} \big(c_X^G \, G^A_{\mu\nu} G^{A\mu\nu} + c_X^W \, W^I_{\mu\nu} W^{I\mu\nu} + c_X^B \, B_{\mu\nu} B^{\mu\nu} \big) \, \bar{\Psi} P_X \Psi \\ & + \frac{d_X}{2} \, B_{\mu\nu} \, \mathcal{D}_X^{\mu\nu} + \sum_f c_X^f \, J_f^\mu \, j_{X\mu} \Big], \qquad \text{(IN-16)} \\ \mathcal{D}_X^{\mu\nu} = & \, \mathrm{i} \, \bar{\Psi} \, \Sigma^{\mu\nu} P_X \Psi, \qquad j_X^\mu = \mathrm{i} \big(\bar{\Psi} P_X D^\mu \Psi - (D^\mu \bar{\Psi}) P_X \Psi \big), \qquad J_f^\mu = \bar{\psi}_f \, \hat{\gamma}^\mu \psi_f\end{aligned}$$

Symbols in (IN-16).

- $\mathcal{L}_{\mathrm{P}}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_{L}: The author's left-handed chiral projector, $P_{\mathrm{L}} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- G^A_M, G^A_μ, G^A_{MN}: The gluon field of $SU(3)_C$ ($A = 1, \ldots, 8$): in the bulk G^A_M, a field of eight dimensions in its geometric normalisation (the covariant derivative $\partial - \mathrm{i}G$), and on the fold the canonically normalised four-dimensional field $G^A_\mu(x)$ of its massless mode, $g_3 G^A_\mu(x) = G^A_\mu(\sigma = 0, x)$; G^A_{MN} and $G^A_{\mu\nu}$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+, +, +, -$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- c_X^W: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- W^I_M, W^I_μ, W^I_{MN}: The gauge field of $SU(2)_L$ ($I = 1, 2, 3$): in the bulk W^I_M (geometric normalisation), on the fold the canonically normalised four-dimensional field $W^I_\mu(x)$ of its massless mode, $g_2 W^I_\mu(x) = W^I_\mu(\sigma = 0, x)$; W^I_{MN}, $W^I_{\mu\nu}$ their field strengths; $W^\pm_\mu = (W^1_\mu \mp \mathrm{i} W^2_\mu)/\sqrt{2}$. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- B_M, B_μ, B_{MN}: The hypercharge gauge field of $U(1)_Y$: in the bulk B_M (geometric normalisation), on the fold the canonically normalised four-dimensional field $B_\mu(x)$ of its massless mode, $g_1 B_\mu(x) = B_\mu(\sigma = 0, x)$; B_{MN}, $B_{\mu\nu}$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.

- P_X: The chiral projector of the half X: $P_X = P_\mathrm{L}$ for $X = \mathrm{L}$, P_R for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- $\mathcal{D}_X^{\mu\nu}$: The dipole (antisymmetric tensor) density of the half X of dirac16complex at the fold, $\mathcal{D}_X^{\mu\nu} = \mathrm{i}\bar{\Psi}\Sigma^{\mu\nu}P_X\Psi$ (real, DERIVED). Unit: GeV^7. Status: DERIVED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- J_f^μ: The fermion-number current of the Standard-Model multiplets of kind f, $J_f^\mu = \bar{\psi}_f\hat{\gamma}^\mu\psi_f$ summed over the three generations and the gauge components; with other labels the currents that are the sources of the gauge fields. Unit: GeV^3. Status: STANDARD.
- j_X^μ: The convective current of the half X of dirac16complex, $j_X^\mu = \mathrm{i}(\bar{\Psi}P_X D^\mu\Psi - (D^\mu\bar{\Psi})P_X\Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^8. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Σ^{ab}, $\Sigma^{\mu\nu}$: The generators of Spin(4,4) on dirac16complex, $\Sigma^{ab} = \frac{1}{4}[T^a, T^b]$ (the scaled commutators of the author's matrices); $\Sigma^{\mu\nu} = \mathbf{e}_a{}^\mu\mathbf{e}_b{}^\nu\Sigma^{ab}$ along the fold. Unit: dimensionless. Status: AUTHOR.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G_\mu^B X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- ψ, $\bar{\psi}$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar{\psi} = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat{\gamma}^{(a)}$ along the frame directions of the fold and $\hat{\gamma}^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.

The new interactions of UM-4 (INVENTED; the same as in UM-3), the order of the author that the left-handed and the right-handed halves of dirac16complex each couple to every Standard-Model field with its own coupling, and that only the left-handed half couples to the Higgs doublet. Ψ is the value of dirac16complex at the fold, and the field strengths are those of the four-dimensional gauge fields at the fold (IN-13). Every term is a product of a gauge-invariant Standard-Model operator and a Lorentz-tensor bilinear of one half X of dirac16complex, so every term is invariant under $SU(3)_C \times SU(2)_L \times U(1)_Y$ and under the Lorentz transformations of the fold; none changes the flavour of a quark or a lepton (DERIVED by construction):

- the Higgs portal: the scalar density $\bar{\Psi}P_\mathrm{L}\Psi$ of the left-handed half times $\Phi^\dagger\Phi$, with the coupling c_Φ; the right-handed half has no Higgs coupling;
- the gauge-kinetic portals: the scalar density of each half times the gauge-invariant kinetic densities of the gluons, of the W fields and of the B field, with the six couplings c_X^G, c_X^W, c_X^B: they make the gauge couplings at the fold depend on the density of dirac16complex at the fold and couple the particles χ to gluons, photons and weak bosons;
- the dipole portal: the antisymmetric tensor $\mathcal{D}_X^{\mu\nu}$ of each half times the hypercharge field strength (the only Standard-Model field strength that is gauge invariant by itself), with the couplings d_X: an electric and magnetic dipole moment of dirac16complex with respect to the photon and the Z boson;
- the current portals: the convective current j_X^μ of each half (the only real vector current that joins a half with itself: $\bar{\Psi}\gamma^\mu P_X\Psi$ joins Ψ_L with Ψ_R and is not real, DERIVED, `um4_fields.py`, F1) times the fermion-number current J_f^μ of each kind f of Standard-Model multiplet, with the twelve couplings c_X^f; J_f^μ sums over the three generations and over the gauge components, so the portal is flavour blind.

The 21 couplings are c_Φ, c_L^G, c_R^G, c_L^W, c_R^W, c_L^B, c_R^B, d_L, d_R and c_L^f, c_R^f for $f = Q, u, d, \ell, e, \nu$. Every term is bilinear in Ψ: the action is invariant under a change of the phase of Ψ by a constant (the charge of dirac16complex is conserved), and baryon and lepton numbers are conserved. Status: INVENTED.

4 Gravity

4.1 The Einstein equations in the bulk

$$\mathcal{E}_{(1)}{}^{M}{}_{N} + \Lambda\,\delta^{M}_{N} = \kappa\left(\mathcal{T}_{\Psi}{}^{M}{}_{N} + \mathcal{T}_{\text{gauge}}{}^{M}{}_{N}\right) \qquad (\sigma \neq 0) \tag{FE-1}$$

Symbols in (FE-1).

- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p, $\mathcal{E}_{(p)}{}^{M}{}_{N} = -2^{-(p+1)}\,\delta^{MP_1\cdots P_{2p}}_{NQ_1\cdots Q_{2p}}\,R^{Q_1Q_2}{}_{P_1P_2}\cdots R^{Q_{2p-1}Q_{2p}}{}_{P_{2p-1}P_{2p}}$ (with both indices down it is $1/\sqrt{|g|}$ times the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to g^{MN}; $\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor, $\mathcal{E}_{(3)}$ the cubic Lovelock tensor); with frame indices its frame components. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number (UM-4 keeps only $p = 1$); in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Λ: The eight-dimensional cosmological constant of UM-4, $\Lambda = -21H^2$: the value for which the static block is a vacuum of Einstein gravity (its Einstein tensor is $21H^2$ in every direction). The author writes the pure number Λ_{N} in $0 = -\Lambda_{\text{N}} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$, so that $\Lambda = -\Lambda_{\text{N}} H^2$. Unit: GeV^2. Value: $-21H^2$; the author's original $-10H^2$; UM-3: $-7H^2$. Status: AUTHOR (the term), DERIVED (the value).
- δ: The Kronecker delta: with two indices $\delta^{M}_{N} = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\text{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1M_2\cdots}_{N_1N_2\cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or −1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_{\Psi}$: The energy-momentum tensor of dirac16complex: $-2/\sqrt{|g|}$ times the variation of the action S_{Ψ} of dirac16complex with respect to g^{MN}. Unit: GeV^8. Status: STANDARD.
- $\mathcal{T}_{\text{gauge}}$: The energy-momentum tensor of the gauge fields of the bulk. Unit: GeV^8. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \text{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

The variation of S_{tot} with respect to the metric, away from the fold. The left-hand side is the Einstein tensor $\mathcal{E}_{(1)}{}^{M}{}_{N}$ (FE-2) plus the cosmological constant $\Lambda = -21H^2$ (IN-2); the right-hand side is κ times the energy-momentum tensors of the fields of the bulk, dirac16complex (FE-6) and the gauge fields (FE-8). With the energy-momentum tensor defined by $\mathcal{T}_{MN} = -(2/\sqrt{|g|})\,\delta S_{\text{matter}}/\delta g^{MN}$ (S_{matter} the action of the field), the time-time component $\mathcal{T}_{x_4x_4} = -\mathcal{T}^{x_4}{}_{x_4}$ is the energy density seen by an observer at rest. Status: AUTHOR (the theory, with the weights of the order), STANDARD (the variation).

$$\begin{aligned}
\mathcal{E}_{(p)}{}^{M}{}_{N} &= -\frac{1}{2^{p+1}}\,\delta^{MM_1N_1\cdots M_pN_p}_{NP_1Q_1\cdots P_pQ_p}\,R^{P_1Q_1}{}_{M_1N_1}\cdots R^{P_pQ_p}{}_{M_pN_p},\\
\mathcal{E}_{(1)}{}^{M}{}_{N} &= R^{M}{}_{N} - \frac{1}{2}\,\delta^{M}_{N}\,R,\\
\mathcal{E}_{(2)}{}^{M}{}_{N} &= 2\big(R\,R^{M}{}_{N} - 2R^{M}{}_{P}R^{P}{}_{N} - 2R^{MP}{}_{NQ}R^{Q}{}_{P} + R^{MPQS}R_{NPQS}\big) - \frac{1}{2}\,\delta^{M}_{N}\,\mathcal{L}_{(2)}
\end{aligned} \tag{FE-2}$$

Symbols in (FE-2).

- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p, $\mathcal{E}_{(p)}{}^{M}{}_{N} = -2^{-(p+1)}\,\delta^{MP_1\cdots P_{2p}}_{NQ_1\cdots Q_{2p}}\,R^{Q_1Q_2}{}_{P_1P_2}\cdots R^{Q_{2p-1}Q_{2p}}{}_{P_{2p-1}P_{2p}}$ (with both indices down it is $1/\sqrt{|g|}$ times the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to g^{MN}; $\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor, $\mathcal{E}_{(3)}$ the cubic Lovelock tensor); with frame indices its frame components. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number (UM-4 keeps only $p = 1$); in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.

- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise δ_{XL} is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1 M_2 \cdots}_{N_1 N_2 \cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Q: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor $R^M{}_{NPQ}$ (in the convention in which the commutator of two covariant derivatives, acting on a vector, is the Riemann tensor, with its last two indices those of the two derivatives, contracted with the vector), with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the scalar curvature $R = g^{MN} R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: $1/\mathrm{length}^2$. Status: STANDARD.
- S: A coordinate index of eight dimensions, running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathcal{L}_{(p)}$: The Lovelock density of order p, $\mathcal{L}_{(p)} = 2^{-p}\, \delta^{P_1 \cdots P_{2p}}_{Q_1 \cdots Q_{2p}}\, R^{Q_1 Q_2}{}_{P_1 P_2} \cdots R^{Q_{2p-1} Q_{2p}}{}_{P_{2p-1} P_{2p}}$, a scalar built from p curvature tensors: $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.

The Lovelock tensor of order p: the variation of $\sqrt{|g|}\, \mathcal{L}_{(p)}$ with respect to the metric. For $p = 1$ it is the Einstein tensor (the second line), which is the left-hand side of FE-1; for $p = 2$ the Gauss-Bonnet tensor (written out in the third line); the cubic tensor $\mathcal{E}_{(3)}$ is the first line with $p = 3$. The tensors of order 2 and 3 enter only the comparison FE-68 to FE-70. UM-4 evaluates the first line exactly by enumerating the non-zero frame components of the curvature (`um4_lovelock.py`); the second line is an independent check of it (DERIVED, `um4_geometry.py`, G2 and G5). Status: STANDARD.

$$\mathcal{E}_{(p)}{}^M{}_M = -\frac{8 - 2p}{2}\, \mathcal{L}_{(p)}, \qquad \nabla_M\, \mathcal{E}_{(p)}{}^M{}_N = 0 \tag{FE-3}$$

Symbols in (FE-3).

- $\mathcal{E}_{(p)}{}^M{}_N$: The Lovelock tensor of order p, $\mathcal{E}_{(p)}{}^M{}_N = -2^{-(p+1)}\, \delta^{M P_1 \cdots P_{2p}}_{N Q_1 \cdots Q_{2p}}\, R^{Q_1 Q_2}{}_{P_1 P_2} \cdots R^{Q_{2p-1} Q_{2p}}{}_{P_{2p-1} P_{2p}}$ (with both indices down it is $1/\sqrt{|g|}$ times the variation of $\sqrt{|g|}\, \mathcal{L}_{(p)}$ with respect to g^{MN}; $\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor, $\mathcal{E}_{(3)}$ the cubic Lovelock tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number (UM-4 keeps only $p = 1$); in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathcal{L}_{(p)}$: The Lovelock density of order p, $\mathcal{L}_{(p)} = 2^{-p}\, \delta^{P_1 \cdots P_{2p}}_{Q_1 \cdots Q_{2p}}\, R^{Q_1 Q_2}{}_{P_1 P_2} \cdots R^{Q_{2p-1} Q_{2p}}{}_{P_{2p-1} P_{2p}}$, a scalar built from p curvature tensors: $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- ∇: The covariant derivative of a tensor with the Levi-Civita connection of the metric g_{MN} (Christoffel symbols $\Gamma^P{}_{MN}$). Status: STANDARD.

Two identities of every Lovelock tensor in eight dimensions: its trace is a multiple of the Lovelock density, and its covariant divergence vanishes (the contracted Bianchi identity). The second makes the energy-momentum tensor of the sources conserved (FE-11). Status: STANDARD.

4.2 The junction condition at the fold

$$\kappa\,\mathcal{S}^{a}{}_{b} = \lim_{\varepsilon\to 0}\int_{-\varepsilon}^{\varepsilon} \mathrm{d}\sigma\,\mathcal{E}_{(1)}{}^{a}{}_{b} = -\Big(\big(K^{a}{}_{b}\big|_{0^+} - K^{a}{}_{b}\big|_{0^-}\big) - \delta^{a}{}_{b}\big(K\big|_{0^+} - K\big|_{0^-}\big)\Big), \qquad K = K^{c}{}_{c} \tag{FE-4}$$

Symbols in (FE-4).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{S}^{a}{}_{b}$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^{a}{}_{b} = \int_{0^-}^{0^+}\mathrm{d}\sigma\,\mathcal{E}_{(1)}{}^{a}{}_{b}$ (the junction condition of Israel); $\mathcal{S}^{4}{}_{4}$ is minus its energy density, $\mathcal{S}^{1}{}_{1}$ and $\mathcal{S}^{5}{}_{5}$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^{7}. Value: $\kappa\mathcal{S}^{a}{}_{b} = -12H\delta^{a}{}_{b}$ for every a_4, b_4 of the author's form. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$; on the fold ($a, b \in \{1,\ldots,7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1,\ldots,7\}$; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- ε: The half-thickness of a layer around the fold, taken to zero. Unit: length. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order p, $\mathcal{E}_{(p)}{}^{M}{}_{N} = -2^{-(p+1)}\,\delta^{MP_1\cdots P_{2p}}_{NQ_1\cdots Q_{2p}}\,R^{Q_1Q_2}{}_{P_1P_2}\cdots R^{Q_{2p-1}Q_{2p}}{}_{P_{2p-1}P_{2p}}$ (with both indices down it is $1/\sqrt{|g|}$ times the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to g^{MN}; $\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor, $\mathcal{E}_{(3)}$ the cubic Lovelock tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number (UM-4 keeps only $p = 1$); in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.
- $K^{a}{}_{b}$: The extrinsic curvature of a surface of constant σ of the hidden direction, $K^{a}{}_{b} = \frac{1}{2}g^{ac}\partial_\sigma g_{cb}$ for a, b along the surface (FD-5): $-H\,\mathrm{sgn}(\sigma)$ along all seven directions of the fold for the author's form of the metric of UM-4; $K = K^{c}{}_{c}$ its trace. Unit: GeV (1/length). Status: DERIVED.
- δ: The Kronecker delta: with two indices $\delta^{M}_{N} = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1M_2\cdots}_{N_1N_2\cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- c: A frame index (as a). Status: STANDARD.

The field equations at the fold: the junction condition of Israel (W. Israel, Nuovo Cimento B 44 (1966) 1). The metric of UM-4 has a kink at $\sigma = 0$: its first σ-derivatives jump at the fold, so its second derivatives contain $\delta_{\mathrm{D}}(\sigma)$. The Einstein tensor is linear in the second σ-derivatives, so FE-1 extended across the fold reads $\mathcal{E}_{(1)} + \Lambda = \kappa(\mathcal{T}_\Psi + \mathcal{T}_{\mathrm{gauge}} + \mathcal{S}\,\delta_{\mathrm{D}}(\sigma))$: the fold carries the thin source $\mathcal{S}^{a}{}_{b}$ (a, b along the fold; an energy per unit volume of the six other directions of the fold, $\mathcal{S}^{4}{}_{4}$ minus its energy density, $\mathcal{S}^{1}{}_{1}$ and $\mathcal{S}^{5}{}_{5}$ its pressures along ordinary space and along the extra times), equal to the integral of the Einstein tensor across the fold, which is the jump of the extrinsic curvature $K^{a}{}_{b}$ of the slices $\sigma = \mathrm{const}$ (FE-30). The (σ,σ) and (σ, a) components contain no second σ-derivative: they carry no thin part and hold on each side separately (the constraints of the bulk). Status: STANDARD (DERIVED for the metric of UM-4: `um4_geometry.py`, G7; `um4_fold.py`, F2).

$$\mathcal{S}^{a}{}_{b} = \frac{1}{V_*}\Big(\mathcal{T}_{\mathrm{H}}{}^{a}{}_{b} + \mathcal{T}_{\mathrm{f}}{}^{a}{}_{b} + \mathcal{T}_{\mathrm{P}}{}^{a}{}_{b} + \mathcal{T}_{\mathrm{ff}}{}^{a}{}_{b} + \mathcal{T}_{\mathrm{IG}}{}^{a}{}_{b}\Big), \qquad \mathcal{T}_{\mathrm{IG}}{}^{\mu}{}_{\nu} = -\frac{1}{\kappa_{\mathrm{ind}}}\,\hat{G}^{\mu}{}_{\nu}, \qquad \mathcal{T}^{r}{}_{r} = \mathcal{L}_{\mathrm{fold}} \tag{FE-5}$$

Symbols in (FE-5).

- $\mathcal{S}^{a}{}_{b}$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^{a}{}_{b} = \int_{0^-}^{0^+}\mathrm{d}\sigma\,\mathcal{E}_{(1)}{}^{a}{}_{b}$ (the junction condition of Israel); $\mathcal{S}^{4}{}_{4}$ is minus its energy density, $\mathcal{S}^{1}{}_{1}$ and $\mathcal{S}^{5}{}_{5}$ its

pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa\mathcal{S}^a{}_b = -12H\delta^a{}_b$ for every a_4, b_4 of the author's form. Status: DERIVED.

- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1, \ldots, 7\}$; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\mathcal{T}_\mathrm{H}$: The energy-momentum tensor of the Higgs doublet (four-dimensional normalisation). Unit: GeV^4. Status: STANDARD.
- $\mathcal{T}_\mathrm{f}$: The energy-momentum tensor of the quarks and leptons. Unit: GeV^4. Status: STANDARD.
- $\mathcal{T}_\mathrm{P}$: The energy-momentum tensor of the portals of dirac16complex. Unit: GeV^4. Status: DERIVED.
- $\mathcal{T}_\mathrm{ff}{}^a{}_b$: The energy-momentum tensor of the fold fluid (frame components of the fold, four-dimensional normalisation). Unit: GeV^4. Status: ASSUMED.
- $\mathcal{T}_\mathrm{IG}$: The energy-momentum tensor of the induced Einstein-Hilbert term of the fold, $\mathcal{T}_\mathrm{IG}{}^\mu{}_\nu = -\hat{G}^\mu{}_\nu/\kappa_\mathrm{ind}$. Unit: GeV^4. Status: ESTIMATE.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+, +, +, -$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- κ_ind: The coupling of the Einstein-Hilbert term $\hat{R}/(2\kappa_\mathrm{ind})$ that the quantum fluctuations of the fields of the fold may induce, $\kappa_\mathrm{ind} = 8\pi G_\mathrm{ind}$; it adds to the gravity of the fold, $1/(8\pi G_\mathrm{N}) = V_*/\kappa_7 + 1/\kappa_\mathrm{ind}$. Unit: GeV^{-2}. Value: not fixed by UM-4; negligible for a cutoff of the order of H. Status: ESTIMATE.
- $\hat{G}^\mu{}_\nu$: The Einstein tensor of the metric $\hat{g}_{ab}$ of the fold (FD-2), with frame indices along its seven directions; with four-dimensional indices, $\hat{G}^\mu{}_\nu$, its components along the four observable directions, which for frozen extra times are those of the Einstein tensor of the four-dimensional metric $\hat{h}_{\mu\nu}$. Unit: $1/\mathrm{length}^2$. Status: STANDARD.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- $\mathcal{T}$: The energy-momentum tensor of any one term of the fold (in $\mathcal{T}^r{}_r = \mathcal{L}_\mathrm{fold}$). Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_\mathrm{fold}$: The Lagrangian of any one term of the fold (four-dimensional normalisation). Unit: GeV^4. Status: STANDARD.

The thin source of the fold is the energy-momentum tensor of everything that lives on the fold, divided by V_* (IN-8): the Higgs doublet (FE-7), the quarks and leptons (FE-9), the portals (FE-10), the fold fluid (IN-10) and the induced gravity (IN-9), whose energy-momentum tensor is $-1/\kappa_\mathrm{ind}$ times the Einstein tensor $\hat{G}^\mu{}_\nu$ of the metric of the fold; the particles χ of dirac16complex enter through the jump of its profile at the fold. The gauge fields live in the bulk: their energy-momentum tensor is finite at the fold and has no thin part (their field strengths jump there, FE-16 to FE-18, but do not diverge). Because the fields of the fold do not depend on the extra times and have no components along them, the extra times enter only through $\sqrt{|g|}$: for every term of the fold the component along an extra time is the Lagrangian of the term, $\mathcal{T}^r{}_r = \mathcal{L}_\mathrm{fold}$ (DERIVED: $\delta\sqrt{|g|} = -\frac{1}{2}\sqrt{|g|}g_{rr}\delta g^{rr}$). Combining FE-4 and FE-5 gives the equations of the fold (FE-71). Status: DERIVED.

4.3 The energy-momentum tensors

$$\mathcal{T}_{\Psi\,MN} = -\frac{1}{4}\Big(\bar{\Psi}\gamma_M D_N\Psi + \bar{\Psi}\gamma_N D_M\Psi - (D_M\bar{\Psi})\gamma_N\Psi - (D_N\bar{\Psi})\gamma_M\Psi\Big) + g_{MN}\,\mathcal{L}_\Psi \tag{FE-6}$$

Symbols in (FE-6).

- $\mathcal{T}_\Psi$: The energy-momentum tensor of dirac16complex: $-2/\sqrt{|g|}$ times the variation of the action S_Ψ of dirac16complex with respect to g^{MN}. Unit: GeV^8. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- $\mathcal{L}_\Psi$: The Lagrangian density of dirac16complex: the square bracket in its action S_Ψ. Unit: GeV^8. Status: RECORD.

The symmetric energy-momentum tensor of dirac16complex, from the variation of the vielbein in IN-4 ($\mathcal{L}_\Psi$ is the Lagrangian in the square bracket of IN-4; the variation of the spin connection adds only the divergence of the spin current, which drops out of the symmetric part). On solutions of FE-12, $\mathcal{L}_\Psi = \frac{\lambda_\Psi}{2}(\bar{\Psi}\Psi)^2$. For a stationary state $\Psi \propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ of the free field the energy density is $\mathcal{T}_{\Psi\, x_4 x_4} = -\omega\,\mathrm{i}\bar{\Psi}\gamma_{x_4}\Psi$: its sign is the product of the sign of the frequency and of the Krein density (FE-43). Status: STANDARD (DERIVED for the energy density).

$$\mathcal{T}_{\mathrm{H}\mu\nu} = (D_\mu\Phi)^\dagger D_\nu\Phi + (D_\nu\Phi)^\dagger D_\mu\Phi + \hat{h}_{\mu\nu}\,\mathcal{L}_\mathrm{H} \tag{FE-7}$$

Symbols in (FE-7).

- $\mathcal{T}_\mathrm{H}$: The energy-momentum tensor of the Higgs doublet (four-dimensional normalisation). Unit: GeV^4. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{L}_\mathrm{H}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.

The energy-momentum tensor of the Higgs doublet (four-dimensional normalisation): its time-time component is $|D_4\Phi|^2 + \hat{h}^{ij}(D_i\Phi)^\dagger D_j\Phi + V \geq V$. Status: STANDARD.

$$\mathcal{T}_{\mathrm{gauge}\,MN} = \frac{1}{\mathrm{g}_3^2}\,G^A_{MP}G^A{}_N{}^P + \frac{1}{\mathrm{g}_2^2}\,W^I_{MP}W^I{}_N{}^P + \frac{1}{\mathrm{g}_1^2}\,B_{MP}B_N{}^P + g_{MN}\,\mathcal{L}_\mathrm{gauge} \tag{FE-8}$$

Symbols in (FE-8).

- $\mathcal{T}_\mathrm{gauge}$: The energy-momentum tensor of the gauge fields of the bulk. Unit: GeV^8. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- g_1, g_2, g_3: The gauge couplings of eight dimensions of $U(1)_Y$, $SU(2)_L$ and $SU(3)_C$: the coefficients of the kinetic terms of the gauge fields of the bulk in their geometric normalisation; the four-dimensional couplings are $g_k^2 = 3H\mathsf{g}_k^2/(2V_*)$. Unit: GeV^{-2} (g^2: length^4). Status: ASSUMED.
- G^A_M, G^A_μ, G^A_{MN}: The gluon field of $SU(3)_C$ ($A = 1, \ldots, 8$): in the bulk G^A_M, a field of eight dimensions in its geometric normalisation (the covariant derivative $\partial - \mathrm{i}G$), and on the fold the canonically normalised four-dimensional field $G^A_\mu(x)$ of its massless mode, $g_3 G^A_\mu(x) = G^A_\mu(\sigma = 0, x)$; G^A_{MN} and $G^A_{\mu\nu}$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- P: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- W^I_M, W^I_μ, W^I_{MN}: The gauge field of $SU(2)_L$ ($I = 1, 2, 3$): in the bulk W^I_M (geometric normalisation), on the fold the canonically normalised four-dimensional field $W^I_\mu(x)$ of its massless mode, $g_2 W^I_\mu(x) = W^I_\mu(\sigma = 0, x)$; W^I_{MN}, $W^I_{\mu\nu}$ their field strengths; $W^\pm_\mu = (W^1_\mu \mp \mathrm{i}W^2_\mu)/\sqrt{2}$. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- B_M, B_μ, B_{MN}: The hypercharge gauge field of $U(1)_Y$: in the bulk B_M (geometric normalisation), on the fold the canonically normalised four-dimensional field $B_\mu(x)$ of its massless mode, $g_1 B_\mu(x) = B_\mu(\sigma = 0, x)$; B_{MN}, $B_{\mu\nu}$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- $\mathcal{L}_{\text{gauge}}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ in the bulk. Unit: GeV^8. Status: STANDARD.

The energy-momentum tensor of the gauge fields in the bulk; for each field its energy density is $(\vec{E}^2 + \vec{B}^2)/(2\mathsf{g}^2)$ in terms of its electric and magnetic components $\vec{E}$ and $\vec{B}$ in the frame (seven-dimensional vectors along the fold and along σ). For the massless modes integrated over σ and over the extra times it is the energy-momentum tensor of the gauge fields of the Standard Model on the fold, $G^A_{\mu\rho} G^{A\,\nu\rho} + W^I_{\mu\rho} W^{I\,\nu\rho} + B_{\mu\rho} B_\nu{}^\rho + \hat{h}_{\mu\nu}\mathcal{L}_4$ (the canonical fields of IN-13; $\mathcal{L}_4$ their Lagrangian). Status: STANDARD.

$$\mathcal{T}_{\mathrm{f}\,\mu\nu} = -\frac{\mathrm{i}}{4}\sum_\psi \Big(\bar\psi\hat\gamma_\mu D_\nu\psi + \bar\psi\hat\gamma_\nu D_\mu\psi - (D_\mu\bar\psi)\hat\gamma_\nu\psi - (D_\nu\bar\psi)\hat\gamma_\mu\psi\Big) + \hat{h}_{\mu\nu}\big(\mathcal{L}_{\mathrm{f}} + \mathcal{L}_{\mathrm{Y}}\big) \tag{FE-9}$$

Symbols in (FE-9).

- $\mathcal{T}_{\mathrm{f}}$: The energy-momentum tensor of the quarks and leptons. Unit: GeV^4. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ψ: The label of a Standard-Model fermion field in a sum: ψ runs over $Q, u_{\mathrm{R}}, d_{\mathrm{R}}, \ell, e_{\mathrm{R}}, \nu_{\mathrm{R}}$. Status: STANDARD.
- ψ, $\bar\psi$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar\psi = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\hat\gamma^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat\gamma^{(a)}$ along the frame directions of the fold and $\hat\gamma^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat\gamma^\mu, \hat\gamma^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat\gamma_\mu = \hat{h}_{\mu\nu}\hat\gamma^\nu$. Unit: dimensionless. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.

- $\mathcal{L}_\mathrm{f}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_\mathrm{Y}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.

The energy-momentum tensor of the quarks and leptons (with the kinetic term in its symmetric form). For a free particle of energy E at rest in the frame of the fold its time-time component is $E\,\psi^\dagger\psi > 0$. Status: STANDARD.

$$\begin{aligned}\mathcal{T}_{\mathrm{P}\mu\nu} = \sum_X \Big[&\big(c_X^G\, G^A_{\mu\rho} G^A{}_\nu{}^\rho + c_X^W\, W^I_{\mu\rho} W^I{}_\nu{}^\rho + c_X^B\, B_{\mu\rho} B_\nu{}^\rho\big)\,\bar\Psi P_X \Psi \\ &+ \frac{d_X}{2}\big(B_{\mu\rho}\,\mathcal{D}_{X\nu}{}^\rho + B_{\nu\rho}\,\mathcal{D}_{X\mu}{}^\rho\big) + \sum_f \frac{c_X^f}{2}\big(J_{f\mu}\,j_{X\nu} + J_{f\nu}\,j_{X\mu}\big)\Big] + \hat h_{\mu\nu}\,\mathcal{L}_\mathrm{P}\end{aligned} \tag{FE-10}$$

Symbols in (FE-10).

- $\mathcal{T}_\mathrm{P}$: The energy-momentum tensor of the portals of dirac16complex. Unit: GeV^4. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- G^A_M, G^A_μ, G^A_{MN}: The gluon field of $SU(3)_C$ ($A = 1, \ldots, 8$): in the bulk G^A_M, a field of eight dimensions in its geometric normalisation (the covariant derivative $\partial - \mathrm{i}G$), and on the fold the canonically normalised four-dimensional field $G^A_\mu(x)$ of its massless mode, $g_3 G^A_\mu(x) = G^A_\mu(\sigma = 0, x)$; G^A_{MN} and $G^A_{\mu\nu}$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- ρ: A four-dimensional coordinate index (as μ). Status: STANDARD.
- c_X^W: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- W^I_M, W^I_μ, W^I_{MN}: The gauge field of $SU(2)_L$ ($I = 1, 2, 3$): in the bulk W^I_M (geometric normalisation), on the fold the canonically normalised four-dimensional field $W^I_\mu(x)$ of its massless mode, $g_2 W^I_\mu(x) = W^I_\mu(\sigma = 0, x)$; W^I_{MN}, $W^I_{\mu\nu}$ their field strengths; $W^\pm_\mu = (W^1_\mu \mp \mathrm{i}W^2_\mu)/\sqrt{2}$. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- B_M, B_μ, B_{MN}: The hypercharge gauge field of $U(1)_Y$: in the bulk B_M (geometric normalisation), on the fold the canonically normalised four-dimensional field $B_\mu(x)$ of its massless mode, $g_1 B_\mu(x) = B_\mu(\sigma = 0, x)$; B_{MN}, $B_{\mu\nu}$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_\mathrm{L}$ for $X = \mathrm{L}$, P_R for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- $\mathcal{D}_X^{\mu\nu}$: The dipole (antisymmetric tensor) density of the half X of dirac16complex at the fold, $\mathcal{D}_X^{\mu\nu} = \mathrm{i}\bar\Psi\Sigma^{\mu\nu}P_X\Psi$ (real, DERIVED). Unit: GeV^7. Status: DERIVED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- J^μ_f: The fermion-number current of the Standard-Model multiplets of kind f, $J^\mu_f = \bar\psi_f\hat\gamma^\mu\psi_f$ summed over the three generations and the gauge components; with other labels the currents that are the sources of the gauge fields. Unit: GeV^3. Status: STANDARD.
- j^μ_X: The convective current of the half X of dirac16complex, $j^\mu_X = \mathrm{i}(\bar\Psi P_X D^\mu\Psi - (D^\mu\bar\Psi)P_X\Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^8. Status: DERIVED.

- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \dots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{L}_\mathrm{P}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.

The energy-momentum tensor of the portals (from the variation of IN-16 with respect to the metric of the fold, which enters through the raised indices, through $\hat{\gamma}^\mu$ in J^μ_f and through $\sqrt{-\hat{h}}$); the Higgs portal enters only through $\hat{h}_{\mu\nu}\mathcal{L}_\mathrm{P}$; the field strengths are those of the canonical four-dimensional fields at the fold (IN-13). It is a source of the fold. Status: DERIVED.

4.4 The conservation laws

$$\begin{aligned} &\nabla_M \left(\mathcal{T}_\Psi{}^M{}_N + \mathcal{T}_\mathrm{gauge}{}^M{}_N\right) = 0 \qquad (\sigma \neq 0), \\ &\hat{\nabla}_a \, \mathcal{S}^a{}_b = -\Big(\mathcal{T}_\mathrm{bulk}{}^{(0)}{}_b\big|_{\sigma=0^+} - \mathcal{T}_\mathrm{bulk}{}^{(0)}{}_b\big|_{\sigma=0^-}\Big), \qquad \mathcal{T}_\mathrm{bulk} = \mathcal{T}_\Psi + \mathcal{T}_\mathrm{gauge} \end{aligned} \tag{FE-11}$$

Symbols in (FE-11).

- ∇: The covariant derivative of a tensor with the Levi-Civita connection of the metric g_{MN} (Christoffel symbols $\Gamma^P{}_{MN}$). Status: STANDARD.
- $\mathcal{T}_\Psi$: The energy-momentum tensor of dirac16complex: $-2/\sqrt{|g|}$ times the variation of the action S_Ψ of dirac16complex with respect to g^{MN}. Unit: GeV^8. Status: STANDARD.
- $\mathcal{T}_\mathrm{gauge}$: The energy-momentum tensor of the gauge fields of the bulk. Unit: GeV^8. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \dots$): it runs over $x_0, x_1, \dots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \dots$), running over $x_0, \dots, x_7$; a repeated index is summed. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z| \, \mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z) \ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $\hat{\nabla}$: The covariant derivative of the fold: with the Levi-Civita connection of the metric of the fold (for four-dimensional tensors of its metric $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the observable coordinates $x_1, \dots, x_4$; for $\hat{\nabla}_a \mathcal{S}^a{}_b$ of the induced metric of the seven-dimensional fold). Status: STANDARD.
- $\mathcal{S}^a{}_b$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^a{}_b = \int_{0^-}^{0^+} \mathrm{d}\sigma \, \mathcal{E}_{(1)}{}^a{}_b$ (the junction condition of Israel); $\mathcal{S}^4{}_4$ is minus its energy density, $\mathcal{S}^1{}_1$ and $\mathcal{S}^5{}_5$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa\mathcal{S}^a{}_b = -12H\delta^a{}_b$ for every a_4, b_4 of the author's form. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \dots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \dots, 7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1, \dots, 7\}$; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- $\mathcal{T}_\mathrm{bulk}$: The energy-momentum tensor of all sources of the bulk, dirac16complex and the gauge fields, $\mathcal{T}_\mathrm{bulk} = \mathcal{T}_\Psi + \mathcal{T}_\mathrm{gauge}$; $\mathcal{T}_\mathrm{bulk}{}^{(0)}{}_b$ is its flux along σ (frame components). Unit: GeV^8. Status: STANDARD.

The energy and momentum of the bulk are conserved away from the fold (FE-3 and FE-1). At the fold the thin source exchanges energy and momentum with the bulk: its covariant divergence along the fold equals minus the jump of the flux $\mathcal{T}_\mathrm{bulk}{}^{(0)}{}_b$ of the sources of the bulk along σ (the frame component $(0), b$). In UM-4 two kinds of flux leave the fold: the waves of dirac16complex by which the particles χ decay into the bulk (FE-86), and the massive modes of the gauge fields that the currents of the fold emit (the width of the W and Z bosons into the bulk, FE-82). Without them (the static block, a fold with frozen extra times and no such emission) the matter of the fold is conserved by itself. Status: DERIVED.

5 The matter fields

5.1 dirac16complex

$$\gamma^M D_M \Psi - m_\Psi \Psi - \lambda_\Psi \big(\bar\Psi\Psi\big)\Psi = 0 \qquad (\sigma \neq 0) \tag{FE-12}$$

Symbols in (FE-12).

- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- λ_Ψ: The coupling of the contact self-interaction of dirac16complex. Unit: GeV^{-6}. Status: RECORD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

The field equation of dirac16complex in the bulk: the variation of IN-4 with respect to $\bar\Psi$ (the second half of the symmetric kinetic term, integrated by parts with $D_M\gamma^M = 0$, doubles the first). With the author's real matrices the operator $\gamma^M D_M - m_\Psi$ is real: the field equation does not contain i, and a real solution exists for every complex one. Status: RECORD.

$$\begin{aligned}
&\Psi\big|_{\sigma=0^+} = \exp\Big(\frac{1}{V_*}\,\gamma^{(0)}\,\mathbb{M}\Big)\,\Psi\big|_{\sigma=0^-},\\
&\mathbb{M} = c_\Phi\big(\Phi^\dagger\Phi\big)P_\mathrm{L} + \sum_X \Big[\frac{1}{4}\big(c^G_X\,G^A_{\mu\nu}G^{A\mu\nu} + c^W_X\,W^I_{\mu\nu}W^{I\mu\nu} + c^B_X\,B_{\mu\nu}B^{\mu\nu}\big)P_X + \frac{d_X}{2}\,B_{\mu\nu}\,\mathrm{i}\Sigma^{\mu\nu}P_X\\
&\qquad + \sum_f c^f_X\big(2\mathrm{i}\,J^\mu_f P_X D_\mu + \mathrm{i}(\hat\nabla_\mu J^\mu_f)P_X\big)\Big]
\end{aligned} \tag{FE-13}$$

Symbols in (FE-13).

- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- $\mathbb{M}$: The fold operator of dirac16complex: the 16×16 matrix (with a derivative along the fold) built from the Standard-Model fields at the fold by varying the portals with respect to $\bar\Psi$. Unit: GeV^{-3} (so that $\mathbb{M}/V_*$ is dimensionless). Status: DERIVED.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.

- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- P_{L}: The author's left-handed chiral projector, $P_{\mathrm{L}} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- G_M^A, G_μ^A, G_{MN}^A: The gluon field of $SU(3)_C$ ($A = 1, \ldots, 8$): in the bulk G_M^A, a field of eight dimensions in its geometric normalisation (the covariant derivative $\partial - \mathrm{i}G$), and on the fold the canonically normalised four-dimensional field $G_\mu^A(x)$ of its massless mode, $g_3 G_\mu^A(x) = G_\mu^A(\sigma = 0, x)$; G_{MN}^A and $G_{\mu\nu}^A$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+, +, +, -$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- c_X^W: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- W_M^I, W_μ^I, W_{MN}^I: The gauge field of $SU(2)_L$ ($I = 1, 2, 3$): in the bulk W_M^I (geometric normalisation), on the fold the canonically normalised four-dimensional field $W_\mu^I(x)$ of its massless mode, $g_2 W_\mu^I(x) = W_\mu^I(\sigma = 0, x)$; W_{MN}^I, $W_{\mu\nu}^I$ their field strengths; $W_\mu^\pm = (W_\mu^1 \mp \mathrm{i} W_\mu^2)/\sqrt{2}$. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- B_M, B_μ, B_{MN}: The hypercharge gauge field of $U(1)_Y$: in the bulk B_M (geometric normalisation), on the fold the canonically normalised four-dimensional field $B_\mu(x)$ of its massless mode, $g_1 B_\mu(x) = B_\mu(\sigma = 0, x)$; B_{MN}, $B_{\mu\nu}$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_{\mathrm{L}}$ for $X = \mathrm{L}$, P_{R} for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Σ^{ab}, $\Sigma^{\mu\nu}$: The generators of Spin(4,4) on dirac16complex, $\Sigma^{ab} = \frac{1}{4}[T^a, T^b]$ (the scaled commutators of the author's matrices); $\Sigma^{\mu\nu} = \mathbf{e}_a{}^\mu \mathbf{e}_b{}^\nu \Sigma^{ab}$ along the fold. Unit: dimensionless. Status: AUTHOR.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- J_f^μ: The fermion-number current of the Standard-Model multiplets of kind f, $J_f^\mu = \bar{\psi}_f \hat{\gamma}^\mu \psi_f$ summed over the three generations and the gauge components; with other labels the currents that are the sources of the gauge fields. Unit: GeV^3. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M \bar{\Psi} = \partial_M \bar{\Psi} - \bar{\Psi} \Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G_\mu^B X^C$ (likewise for $SU(2)$ with $g_2 \epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\hat{\nabla}$: The covariant derivative of the fold: with the Levi-Civita connection of the metric of the fold (for four-dimensional tensors of its metric $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the observable coordinates $x_1, \ldots, x_4$; for $\hat{\nabla}_a \mathcal{S}^a{}_b$ of the induced metric of the seven-dimensional fold). Status: STANDARD.

The equation of dirac16complex at the fold: the variation of the portals IN-16 with respect to $\bar{\Psi}$ adds $-\delta_{\mathrm{D}}(\sigma)\, \mathbb{M}\Psi/V_*$ to FE-12. $\mathbb{M}$ is the fold operator, a 16×16 matrix (with a derivative along the fold from the current portals) built from the fields of the fold and the gauge fields at the fold. Integrating FE-12 with this term across a fold of vanishing thickness gives the matching condition in the first line: the values of dirac16complex on the two sides of the fold differ by the exponential of $\gamma^{(0)}\mathbb{M}/V_*$ (the exact solution of $\partial_\sigma \Psi = \gamma^{(0)} \mathbb{M}\, \delta_{\mathrm{D}}(\sigma) \Psi / V_*$ in a layer in which the fold term dominates; $(\gamma^{(0)})^2 = \mathbf{1}$). The Higgs term acts on the left-handed half only. Status: DERIVED (from INVENTED terms).

$$\gamma^M \Omega_M = -\frac{7}{2}\, H\, \mathrm{sgn}(\sigma)\, \gamma^{(0)} + \frac{3}{2}\, \frac{b_4'}{b_4}\, \mathrm{e}^{H|\sigma|}\, \gamma^{(4)} \neq 0 \tag{FE-14}$$

Symbols in (FE-14).

- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- Ω_M: The spin connection acting on dirac16complex, $\Omega_M = \frac{1}{2}\varpi_{Mab}\Sigma^{ab}$. Unit: 1/length. Status: RECORD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.

The coupling of dirac16complex to gravity through the canonical spin connection, for the author's form of the metric of UM-4: the contraction $\gamma^M\Omega_M$ that enters the Dirac operator. The first term comes from the warp of the seven directions of the fold (each contributes $\frac{1}{2}\partial_\sigma$ of the logarithm of its squared scale factor, $-H\,\mathrm{sgn}(\sigma)$; in UM-3 six directions gave $-3H$), the second from the time dependence of their scale factors (a_4 cancels between ordinary space and the extra times; the factor $\mathrm{e}^{H|\sigma|}$ is the inverse of the warp of the time). The first term never vanishes ($H > 0$, and it only changes sign at the fold); the second is non-zero whenever b_4 depends on x_4. Hence the coupling of dirac16complex to gravity through the spin connection is never identically zero (DERIVED, `um4_geometry.py`, G9, with the author's matrices; the vielbein postulate $D_M\gamma^N = 0$ holds, G9). The first term is removed exactly by the factor $\mathrm{e}^{(7/2)H|\sigma|}$ of FE-36. Status: DERIVED.

5.2 The Higgs doublet

$$\begin{aligned} &\frac{1}{\sqrt{-\hat{h}}\,\mathcal{Z}}\, D_\mu\big(\sqrt{-\hat{h}}\,\mathcal{Z}\,\hat{h}^{\mu\nu}D_\nu\Phi\big) - \mu^2\Phi - 2\lambda\big(\Phi^\dagger\Phi\big)\Phi - c_\Phi\,\bar{\Psi}P_\mathrm{L}\Psi\,\Phi = \Theta, \\ &\Theta = \begin{pmatrix} \bar{d}_\mathrm{R}(\mathcal{Y}^d)^\dagger u_\mathrm{L} + \bar{e}_\mathrm{R}(\mathcal{Y}^e)^\dagger \nu_\mathrm{L} - \bar{d}_\mathrm{L}\mathcal{Y}^u u_\mathrm{R} - \bar{e}_\mathrm{L}\mathcal{Y}^\nu \nu_\mathrm{R} \\ \bar{d}_\mathrm{R}(\mathcal{Y}^d)^\dagger d_\mathrm{L} + \bar{e}_\mathrm{R}(\mathcal{Y}^e)^\dagger e_\mathrm{L} + \bar{u}_\mathrm{L}\mathcal{Y}^u u_\mathrm{R} + \bar{\nu}_\mathrm{L}\mathcal{Y}^\nu \nu_\mathrm{R} \end{pmatrix} \end{aligned} \tag{FE-15}$$

Symbols in (FE-15).

- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold and the kinetic terms of the massless modes of the fields of the bulk; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.

- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- μ^2: The mass parameter of the Higgs potential, $\mu^2 < 0$ (it enters only as μ^2). Unit: GeV^2. Value: $\mu^2 = -\lambda v^2 = -m_h^2/2 = -(88.6\ \mathrm{GeV})^2$. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- Θ: The Yukawa source of the Higgs equation, $\Theta = -\partial\mathcal{L}_\mathrm{Y}/\partial\Phi^\dagger$ (a doublet of fermion bilinears). Unit: GeV^3. Status: STANDARD.
- d_R: The right-handed down-type quarks (down, strange and bottom; hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- u_L: The upper (up-type, left-handed) components of the quark doublets. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- e_R: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- ν_L: The upper (neutral, left-handed) components of the lepton doublets: the left-handed neutrinos. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- d_L: The lower (down-type, left-handed) components of the quark doublets. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- u_R: The right-handed up-type quarks (up, charm and top; singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- ν_R: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- e_L: The lower (charged, left-handed) components of the lepton doublets. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.

The field equation of the Higgs doublet on the fold: the variation of IN-11, IN-15 and IN-16 with respect to $\Phi^\dagger$, with the gauge fields of the bulk at the fold in D_μ (IN-13). The factor $\mathcal{Z}$ (IN-8) is 1 in the late universe. The Higgs portal adds the term $-c_\Phi\bar\Psi P_\mathrm{L}\Psi\,\Phi$: the scalar density of the left-handed half of dirac16complex at the fold shifts the mass parameter of the Higgs field. Θ is the Yukawa source, $-\partial\mathcal{L}_\mathrm{Y}/\partial\Phi^\dagger$, written with the components $Q = (u_\mathrm{L}, d_\mathrm{L})$ and $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ of the doublets (generation indices contracted with the Yukawa matrices). Status: STANDARD (AUTHOR potential; INVENTED portal term).

5.3 The gauge fields

$$\frac{1}{\sqrt{|g|}}\,D_M\big(\sqrt{|g|}\,G^{AMN}\big) = 0 \qquad (\sigma \neq 0),$$
$$\frac{1}{\mathrm{g}_3^2}\Big(G^{A\sigma\nu}\big|_{0^+} - G^{A\sigma\nu}\big|_{0^-}\Big) = -\frac{1}{V_*}\Big(J_\mathrm{c}^{A\nu} + \frac{1}{g_3}\,D_\mu\big[(\mathcal{K}_G - 1)G^{A\mu\nu}\big]\Big), \qquad \mathcal{K}_G = 1 + \sum_X c_X^G\,\bar\Psi P_X\Psi, \tag{FE-16}$$
$$J_\mathrm{c}^{A\nu} = \bar Q\,\hat\gamma^\nu t^A Q + \bar u_\mathrm{R}\,\hat\gamma^\nu t^A u_\mathrm{R} + \bar d_\mathrm{R}\,\hat\gamma^\nu t^A d_\mathrm{R}$$

Symbols in (FE-16).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G_\mu^B X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.

- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- G^A_M, G^A_μ, G^A_{MN}: The gluon field of $SU(3)_C$ ($A = 1, \ldots, 8$): in the bulk G^A_M, a field of eight dimensions in its geometric normalisation (the covariant derivative $\partial - \mathrm{i}G$), and on the fold the canonically normalised four-dimensional field $G^A_\mu(x)$ of its massless mode, $g_3 G^A_\mu(x) = G^A_\mu(\sigma = 0, x)$; G^A_{MN} and $G^A_{\mu\nu}$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- g_1, g_2, g_3: The gauge couplings of eight dimensions of $U(1)_Y$, $SU(2)_L$ and $SU(3)_C$: the coefficients of the kinetic terms of the gauge fields of the bulk in their geometric normalisation; the four-dimensional couplings are $g_k^2 = 3H\mathsf{g}_k^2/(2V_*)$. Unit: GeV^{-2} (g^2: length^4). Status: ASSUMED.
- σ: The label σ in ∂_σ: the derivative with respect to the proper hidden coordinate σ. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $J_\mathrm{c}^{A\nu}$: The colour current of the quarks. Unit: GeV^3. Status: STANDARD.
- g_3: The $SU(3)_C$ gauge coupling of the four-dimensional world (the strong coupling g_s): the coupling of the massless mode of the gluon field of the bulk, $g_3^2 = 3H\mathsf{g}_3^2/(2V_*)$. Unit: dimensionless. Value: $g_3(m_Z) = 1.2177$ ($g_3^2/(4\pi) = 0.1180$). Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- $\mathcal{K}_G$, $\mathcal{K}_W$, $\mathcal{K}_B$: The portal factors of the gauge kinetic terms, $\mathcal{K}_G = 1 + \sum_X c_X^G \bar\Psi P_X \Psi$ (likewise for W and B): the gauge couplings at the fold are divided by the square root of the portal factor. Unit: dimensionless. Value: 1 where dirac16complex vanishes. Status: DERIVED.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_\mathrm{L}$ for $X = \mathrm{L}$, P_R for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- Q, $\bar Q$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\hat\gamma^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat\gamma^{(a)}$ along the frame directions of the fold and $\hat\gamma^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat\gamma^\mu, \hat\gamma^\nu\} = -2\hat h^{\mu\nu}\mathbf{1}$; $\hat\gamma_\mu = \hat h_{\mu\nu}\hat\gamma^\nu$. Unit: dimensionless. Status: STANDARD.
- t^A: The generators of $SU(3)$ on colour triplets, t^A: half the Gell-Mann matrices, with $\mathrm{tr}(t^A t^B) = \delta^{AB}/2$. Unit: dimensionless. Status: STANDARD.
- u_R: The right-handed up-type quarks (up, charm and top; singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- d_R: The right-handed down-type quarks (down, strange and bottom; hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.

The Yang-Mills equation of the gluons in the bulk (first line; D_M with the gauge connection in the adjoint representation) and at the fold (second line): the variation of IN-7, IN-14 and IN-16 with respect to the gluon field of the bulk. The quarks on the fold couple to the value of the field at $\sigma = 0$ (IN-13), which puts their colour current $J_\mathrm{c}^{A\nu}$, times $\delta_\mathrm{D}(\sigma)/V_*$, on the right-hand side of the bulk equation; integrated across the fold it makes the component $G^{A\sigma\nu}$ of the field strength jump by $-\mathsf{g}_3^2 J_\mathrm{c}^{A\nu}/V_*$ (convention of the signature: $\nabla_M F^{MN} = -J^N$ for a Lagrangian $-\frac{1}{4}F^2 + A_M J^M$). The gluon-kinetic portals add a kinetic term on the fold ($\mathcal{K}_G - 1$, with the canonical four-dimensional field strength $G^{A\mu\nu}$ of IN-13 and the covariant derivative of the fold). For the massless mode (the profile 1, FE-74) the integral of the first line over σ with the weight of the metric, together with the jump, gives the Yang-Mills equation of the fold, $\hat\nabla_\mu(\mathcal{K}_G G^{A\mu\nu}) + g_3 f^{ABC} G^B_\mu \mathcal{K}_G G^{C\mu\nu} = -g_3 J_\mathrm{c}^{A\nu}$ (in UM-3 the field equation of the

gluons on the fold), with $g_3^2 = 3H\mathtt{g}_3^2/(2V_*)$; the massive modes add the corrections of FE-79 and FE-80. Status: STANDARD (DERIVED for the reduction to the fold, `um4_bulk.py`, B2; INVENTED portal factor).

$$\begin{aligned}
&\frac{1}{\sqrt{|g|}}\,D_M\big(\sqrt{|g|}\,W^{IMN}\big) = 0 \qquad (\sigma \neq 0),\\
&\frac{1}{\mathtt{g}_2^2}\Big(W^{I\sigma\nu}\big|_{0^+} - W^{I\sigma\nu}\big|_{0^-}\Big) = -\frac{1}{V_*}\Big(J_{\mathrm{w}}^{I\nu} + J_{\Phi}^{I\nu} + \frac{1}{g_2}\,D_\mu\big[(\mathcal{K}_W - 1)W^{I\mu\nu}\big]\Big),\\
&\mathcal{K}_W = 1 + \sum_X c_X^W\,\bar\Psi P_X \Psi, \qquad J_{\mathrm{w}}^{I\nu} = \bar Q\,\hat\gamma^\nu\frac{\tau^I}{2}Q + \bar\ell\,\hat\gamma^\nu\frac{\tau^I}{2}\ell, \qquad J_{\Phi}^{I\nu} = \mathrm{i}\Big[(D^\nu\Phi)^\dagger\frac{\tau^I}{2}\Phi - \Phi^\dagger\frac{\tau^I}{2}D^\nu\Phi\Big]
\end{aligned} \tag{FE-17}$$

Symbols in (FE-17).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G_\mu^B X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- W_M^I, W_μ^I, W_{MN}^I: The gauge field of $SU(2)_L$ ($I = 1,2,3$): in the bulk W_M^I (geometric normalisation), on the fold the canonically normalised four-dimensional field $W_\mu^I(x)$ of its massless mode, $g_2W_\mu^I(x) = W_\mu^I(\sigma = 0, x)$; W_{MN}^I, $W_{\mu\nu}^I$ their field strengths; $W_\mu^\pm = (W_\mu^1 \mp \mathrm{i}W_\mu^2)/\sqrt{2}$. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1,2,3$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0,\ldots,x_7$; a repeated index is summed. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $\mathtt{g}_1$, $\mathtt{g}_2$, $\mathtt{g}_3$: The gauge couplings of eight dimensions of $U(1)_Y$, $SU(2)_L$ and $SU(3)_C$: the coefficients of the kinetic terms of the gauge fields of the bulk in their geometric normalisation; the four-dimensional couplings are $g_k^2 = 3H\mathtt{g}_k^2/(2V_*)$. Unit: GeV^{-2} ($\mathtt{g}^2$: length^4). Status: ASSUMED.
- σ: The label σ in ∂_σ: the derivative with respect to the proper hidden coordinate σ. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_{\mathrm{t}} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $J_{\mathrm{w}}^{I\nu}$: The weak-isospin current of the left-handed fermion doublets. Unit: GeV^3. Status: STANDARD.
- $J_\Phi^{I\nu}$: The weak-isospin current of the Higgs doublet (it contains the mass terms of the W fields); $J_\Phi^{2\,x_4}$ its isospin charge density with $I = 2$. Unit: GeV^3. Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling of the four-dimensional world (usually g): the coupling of the massless mode of the $SU(2)_L$ field of the bulk, $g_2^2 = 3H\mathtt{g}_2^2/(2V_*)$. Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- $\mathcal{K}_G$, $\mathcal{K}_W$, $\mathcal{K}_B$: The portal factors of the gauge kinetic terms, $\mathcal{K}_G = 1 + \sum_X c_X^G\bar\Psi P_X\Psi$ (likewise for W and B): the gauge couplings at the fold are divided by the square root of the portal factor. Unit: dimensionless. Value: 1 where dirac16complex vanishes. Status: DERIVED.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^W: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_{\mathrm{L}}$ for $X = \mathrm{L}$, P_{R} for $X = \mathrm{R}$. Unit: dimensionless. Status:

AUTHOR.

- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_{\mathrm{L}}, d_{\mathrm{L}})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\hat{\gamma}^{\mu}$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat{\gamma}^{(a)}$ along the frame directions of the fold and $\hat{\gamma}^{\mu}$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat{\gamma}^{\mu}, \hat{\gamma}^{\nu}\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_{\mu} = \hat{h}_{\mu\nu}\hat{\gamma}^{\nu}$. Unit: dimensionless. Status: STANDARD.
- τ^{I}: The Pauli matrices acting on weak doublets (τ^1, τ^2, τ^3; τ^2 is the second one, not a square). Unit: dimensionless. Status: STANDARD.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_{\mathrm{L}}, e_{\mathrm{L}})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.

The Yang-Mills equation of the $SU(2)_L$ fields in the bulk and at the fold. The sources on the fold are the weak-isospin current of the left-handed doublets and the isospin current of the Higgs doublet, which contains the mass terms of the W fields: in the vacuum $J^{I\nu}_{\Phi}$ contains $-g_2(v^2 + \mathcal{A}^2)W^{I\nu}/4$, a mass term of the field of the bulk concentrated on the fold, which makes the W boson a resonance of the bulk (FE-82). For the massless mode the reduction gives the field equation of UM-3 on the fold, $\hat{\nabla}_{\mu}(\mathcal{K}_W W^{I\mu\nu}) + \cdots = -g_2(J^{I\nu}_{\mathrm{w}} + J^{I\nu}_{\Phi})$, with $g_2^2 = 3H\mathsf{g}_2^2/(2V_*)$. Status: STANDARD (INVENTED portal factor).

$$
\begin{aligned}
&\frac{1}{\sqrt{|g|}}\,\partial_M\big(\sqrt{|g|}\,B^{MN}\big) = 0 \qquad (\sigma \neq 0),\\
&\frac{1}{\mathsf{g}_1^2}\Big(B^{\sigma\nu}\big|_{0^+} - B^{\sigma\nu}\big|_{0^-}\Big) = -\frac{1}{V_*}\Big(J^{\nu}_{Y} + \frac{1}{g_1}\,\partial_{\mu}\Big[\big(\mathcal{K}_B - 1\big)B^{\mu\nu} + \sum_X d_X\,\mathcal{D}^{\mu\nu}_X\Big]\Big), \qquad \mathcal{K}_B = 1 + \sum_X c^B_X\,\bar{\Psi}P_X\Psi, \qquad \text{(FE-18)}\\
&J^{\nu}_{Y} = \sum_{\psi} Y_{\psi}\,\bar{\psi}\hat{\gamma}^{\nu}\psi + \frac{\mathrm{i}}{2}\big[(D^{\nu}\Phi)^{\dagger}\Phi - \Phi^{\dagger}D^{\nu}\Phi\big]
\end{aligned}
$$

Symbols in (FE-18).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_{σ} with respect to σ; ∂_M and ∂_{μ} with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- B_M, B_{μ}, B_{MN}: The hypercharge gauge field of $U(1)_Y$: in the bulk B_M (geometric normalisation), on the fold the canonically normalised four-dimensional field $B_{\mu}(x)$ of its massless mode, $g_1 B_{\mu}(x) = B_{\mu}(\sigma = 0, x)$; B_{MN}, $B_{\mu\nu}$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- g_1, g_2, g_3: The gauge couplings of eight dimensions of $U(1)_Y$, $SU(2)_L$ and $SU(3)_C$: the coefficients of the kinetic terms of the gauge fields of the bulk in their geometric normalisation; the four-dimensional couplings are $g_k^2 = 3H\mathsf{g}_k^2/(2V_*)$. Unit: GeV^{-2} (g^2: length^4). Status: ASSUMED.
- σ: The label σ in ∂_{σ}: the derivative with respect to the proper hidden coordinate σ. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_{\mathrm{t}} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the

eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- J_Y^ν: The hypercharge current of the fermions and of the Higgs doublet. Unit: GeV^3. Status: STANDARD.
- g_1: The $U(1)_Y$ gauge coupling of the four-dimensional world (usually g'; not normalised for grand unification): the coupling of the massless mode of the hypercharge field of the bulk, $g_1^2 = 3H\mathsf{g}_1^2/(2V_*)$. Unit: dimensionless. Value: $g_1(m_Z) = 0.3574$. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- $\mathcal{K}_G$, $\mathcal{K}_W$, $\mathcal{K}_B$: The portal factors of the gauge kinetic terms, $\mathcal{K}_G = 1 + \sum_X c_X^G \bar\Psi P_X \Psi$ (likewise for W and B): the gauge couplings at the fold are divided by the square root of the portal factor. Unit: dimensionless. Value: 1 where dirac16complex vanishes. Status: DERIVED.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- $\mathcal{D}_X^{\mu\nu}$: The dipole (antisymmetric tensor) density of the half X of dirac16complex at the fold, $\mathcal{D}_X^{\mu\nu} = \mathrm{i}\bar\Psi \Sigma^{\mu\nu} P_X \Psi$ (real, DERIVED). Unit: GeV^7. Status: DERIVED.
- c_X^B: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: GeV$^{7/2}$. Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_\mathrm{L}$ for $X = \mathrm{L}$, P_R for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- ψ: The label of a Standard-Model fermion field in a sum: ψ runs over $Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}$. Status: STANDARD.
- Y_f: The hypercharge of a field, the charge of its coupling $g_1 Y_f B_\mu$ to the hypercharge field in the covariant derivative (1/6, 2/3, $-1/3$, $-1/2$, -1, 0 for $Q, u_\mathrm{R}, d_\mathrm{R}, \ell, e_\mathrm{R}, \nu_\mathrm{R}$, 1/2 for Φ). Unit: dimensionless. Status: STANDARD.
- ψ, $\bar\psi$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar\psi = \psi^\dagger\beta$; ψ_f the fields of the kind f. Unit: GeV$^{3/2}$. Status: STANDARD.
- $\hat\gamma^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat\gamma^{(a)}$ along the frame directions of the fold and $\hat\gamma^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat\gamma^\mu, \hat\gamma^\nu\} = -2\hat h^{\mu\nu}\mathbf{1}$; $\hat\gamma_\mu = \hat h_{\mu\nu}\hat\gamma^\nu$. Unit: dimensionless. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G_\mu^B X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.

The Maxwell equation of the hypercharge field in the bulk and at the fold. The dipole portal adds the dipole density of dirac16complex at the fold to the field strength, as a polarisation adds to the electric field in a medium: dirac16complex at the fold carries electric and magnetic dipole moments. The source on the fold is the hypercharge current of the fermions and of the Higgs doublet. The photon is the combination of W^3 and B that the Higgs vacuum leaves massless (FE-45); in the bulk it obeys the free Maxwell equations of the first lines of FE-17 and FE-18, so a photon that leaves the fold travels through the bulk at the speed of light (FE-74). Status: STANDARD (INVENTED portal terms).

5.4 The quarks and leptons

$$\mathrm{i}\hat\gamma^\mu\big(D_\mu + \mathrm{i}\,\mathcal{V}_\mu^Q\big)Q = \mathcal{Y}^u\tilde\Phi\,u_\mathrm{R} + \mathcal{Y}^d\Phi\,d_\mathrm{R}, \qquad \mathcal{V}_\mu^f = \sum_X c_X^f\, j_{X\mu} \tag{FE-19}$$

Symbols in (FE-19).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.

- $\hat{\gamma}^{\mu}$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat{\gamma}^{(a)}$ along the frame directions of the fold and $\hat{\gamma}^{\mu}$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat{\gamma}^{\mu},\hat{\gamma}^{\nu}\}=-2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_{\mu}=\hat{h}_{\mu\nu}\hat{\gamma}^{\nu}$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1,x_2,x_3,x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_{μ}: The covariant derivative: on dirac16complex $D_M=\partial_M+\Omega_M$ (with $D_M\bar{\Psi}=\partial_M\bar{\Psi}-\bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_{μ} (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^{\mu}=\hat{h}^{\mu\nu}D_{\nu}$; on an adjoint field $(D_{\mu}X)^{A}=\partial_{\mu}X^{A}+g_3f^{ABC}G^{B}_{\mu}X^{C}$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^{f}_{\mu}$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^{f}_{\mu}=\sum_X c^{f}_{X}j_{X\mu}$ ($\mathcal{V}^{Q},\mathcal{V}^{u},\ldots,\mathcal{V}^{\nu}$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q=(u_{\mathrm{L}},d_{\mathrm{L}})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^{u}$, $\mathcal{Y}^{d}$, $\mathcal{Y}^{e}$, $\mathcal{Y}^{\nu}$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- $\tilde{\Phi}$: The conjugate Higgs doublet, $\tilde{\Phi}=\mathrm{i}\tau^2\Phi^{*}$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- u_{R}: The right-handed up-type quarks (up, charm and top; singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- d_{R}: The right-handed down-type quarks (down, strange and bottom; hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- X: The label of a half of dirac16complex: $X\in\{\mathrm{L},\mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c^{f}_{X}: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c^{ℓ}_{X} and c^{ν}_{X} are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- j^{μ}_{X}: The convective current of the half X of dirac16complex, $j^{\mu}_{X}=\mathrm{i}(\bar{\Psi}P_XD^{\mu}\Psi-(D^{\mu}\bar{\Psi})P_X\Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^{8}. Status: DERIVED.
- f: The label of a kind of Standard-Model multiplet: $f\in\{Q,u,d,\ell,e,\nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.

The Dirac equation of the quark doublets: the variation of IN-14, IN-15 and IN-16 with respect to $\bar{Q}$. The current portals act on every Standard-Model fermion of kind f as a vector potential $\mathcal{V}^{f}_{\mu}$ built from the convective currents of the two halves of dirac16complex at the fold; it is the same for the three generations (flavour blind). Status: STANDARD (INVENTED potential).

$$\mathrm{i}\hat{\gamma}^{\mu}\big(D_{\mu}+\mathrm{i}\,\mathcal{V}^{u}_{\mu}\big)u_{\mathrm{R}}=(\mathcal{Y}^{u})^{\dagger}\tilde{\Phi}^{\dagger}Q \tag{FE-20}$$

Symbols in (FE-20).

- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}^{\mu}$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat{\gamma}^{(a)}$ along the frame directions of the fold and $\hat{\gamma}^{\mu}$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat{\gamma}^{\mu},\hat{\gamma}^{\nu}\}=-2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_{\mu}=\hat{h}_{\mu\nu}\hat{\gamma}^{\nu}$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1,x_2,x_3,x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_{μ}: The covariant derivative: on dirac16complex $D_M=\partial_M+\Omega_M$ (with $D_M\bar{\Psi}=\partial_M\bar{\Psi}-\bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_{μ} (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^{\mu}=\hat{h}^{\mu\nu}D_{\nu}$; on an adjoint field $(D_{\mu}X)^{A}=\partial_{\mu}X^{A}+g_3f^{ABC}G^{B}_{\mu}X^{C}$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^{f}_{\mu}$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^{f}_{\mu}=\sum_X c^{f}_{X}j_{X\mu}$ ($\mathcal{V}^{Q},\mathcal{V}^{u},\ldots,\mathcal{V}^{\nu}$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.

- u_{R}: The right-handed up-type quarks (up, charm and top; singlets of $SU(2)_L$, hypercharge 2/3). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $\tilde{\Phi}$: The conjugate Higgs doublet, $\tilde{\Phi} = \mathrm{i}\tau^2\Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_{\mathrm{L}}, d_{\mathrm{L}})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.

$$\mathrm{i}\hat{\gamma}^\mu\big(D_\mu + \mathrm{i}\,\mathcal{V}^d_\mu\big)d_{\mathrm{R}} = (\mathcal{Y}^d)^\dagger\Phi^\dagger Q \tag{FE-21}$$

Symbols in (FE-21).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat{\gamma}^{(a)}$ along the frame directions of the fold and $\hat{\gamma}^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^f_\mu$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_\mu = \sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- d_{R}: The right-handed down-type quarks (down, strange and bottom; hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- Q, $\bar{Q}$: The left-handed quark doublets of the three generations, $Q = (u_{\mathrm{L}}, d_{\mathrm{L}})$ (colour triplets, hypercharge 1/6). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.

$$\mathrm{i}\hat{\gamma}^\mu\big(D_\mu + \mathrm{i}\,\mathcal{V}^\ell_\mu\big)\ell = \mathcal{Y}^e\Phi\, e_{\mathrm{R}} + \mathcal{Y}^\nu\tilde{\Phi}\,\nu_{\mathrm{R}} \tag{FE-22}$$

Symbols in (FE-22).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat{\gamma}^{(a)}$ along the frame directions of the fold and $\hat{\gamma}^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^f_\mu$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_\mu = \sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.

- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_{\mathrm{L}}, e_{\mathrm{L}})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge $1/2$), a field of the fold. Unit: GeV. Status: AUTHOR.
- e_{R}: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\tilde{\Phi}$: The conjugate Higgs doublet, $\tilde{\Phi} = \mathrm{i}\tau^2\Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- ν_{R}: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.

$$\mathrm{i}\hat{\gamma}^\mu\big(D_\mu + \mathrm{i}\,\mathcal{V}^e_\mu\big)e_{\mathrm{R}} = (\mathcal{Y}^e)^\dagger\Phi^\dagger\ell \tag{FE-23}$$

Symbols in (FE-23).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat{\gamma}^{(a)}$ along the frame directions of the fold and $\hat{\gamma}^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^f_\mu$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_\mu = \sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- e_{R}: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge $1/2$), a field of the fold. Unit: GeV. Status: AUTHOR.
- ℓ, $\bar{\ell}$: The left-handed lepton doublets of the three generations, $\ell = (\nu_{\mathrm{L}}, e_{\mathrm{L}})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.

$$\mathrm{i}\hat{\gamma}^\mu\big(D_\mu + \mathrm{i}\,\mathcal{V}^\nu_\mu\big)\nu_{\mathrm{R}} = (\mathcal{Y}^\nu)^\dagger\tilde{\Phi}^\dagger\ell \tag{FE-24}$$

Symbols in (FE-24).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat{\gamma}^{(a)}$ along the frame directions of the fold and $\hat{\gamma}^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- $\mathcal{V}^f_\mu$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_\mu = \sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- ν_{R}: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.

- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $\tilde\Phi$: The conjugate Higgs doublet, $\tilde\Phi = \mathrm{i}\tau^2\Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- ℓ, $\bar\ell$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.

The Dirac equations of the up-quark singlets, the down-quark singlets, the lepton doublets, the charged-lepton singlets and the right-handed neutrinos (FE-20 to FE-24), each the variation of IN-14, IN-15 and IN-16 with respect to the conjugate field. Each fermion sees the gauge fields of the bulk at the fold (in D_μ), the Higgs doublet (through the Yukawa matrices) and dirac16complex (through $\mathcal{V}^f_\mu$). The right-handed neutrinos see no gauge field; they couple to the Standard Model only through the neutrino Yukawa matrix, which gives the neutrinos their Dirac masses. Status: STANDARD (INVENTED potentials).

$$\begin{aligned}\hat\nabla_\mu J^\mu_\ell &= -\hat\nabla_\mu\big(J^\mu_e + J^\mu_\nu\big) = -\mathrm{i}\big(\bar\ell\,\mathcal{Y}^e\Phi\,e_\mathrm{R} + \bar\ell\,\mathcal{Y}^\nu\tilde\Phi\,\nu_\mathrm{R}\big) + \mathrm{h.c.},\\ \hat\nabla_\mu J^\mu_Q &= -\hat\nabla_\mu\big(J^\mu_u + J^\mu_d\big) = -\mathrm{i}\big(\bar Q\,\mathcal{Y}^u\tilde\Phi\,u_\mathrm{R} + \bar Q\,\mathcal{Y}^d\Phi\,d_\mathrm{R}\big) + \mathrm{h.c.}\end{aligned} \tag{FE-25}$$

Symbols in (FE-25).

- $\hat\nabla$: The covariant derivative of the fold: with the Levi-Civita connection of the metric of the fold (for four-dimensional tensors of its metric $\hat h_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the observable coordinates $x_1, \ldots, x_4$; for $\hat\nabla_a \mathcal{S}^a{}_b$ of the induced metric of the seven-dimensional fold). Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- J^μ_f: The fermion-number current of the Standard-Model multiplets of kind f, $J^\mu_f = \bar\psi_f\hat\gamma^\mu\psi_f$ summed over the three generations and the gauge components; with other labels the currents that are the sources of the gauge fields. Unit: GeV^3. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ℓ, $\bar\ell$: The left-handed lepton doublets of the three generations, $\ell = (\nu_\mathrm{L}, e_\mathrm{L})$ (hypercharge $-1/2$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge $1/2$), a field of the fold. Unit: GeV. Status: AUTHOR.
- e_R: The right-handed electron, muon and tau (singlets, hypercharge -1). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- $\tilde\Phi$: The conjugate Higgs doublet, $\tilde\Phi = \mathrm{i}\tau^2\Phi^*$ (hypercharge $-1/2$). Unit: GeV. Status: STANDARD.
- ν_R: The three right-handed neutrinos (gauge singlets). Unit: $\mathrm{GeV}^{3/2}$. Status: ASSUMED.
- h.c.: The Hermitian conjugate of the preceding terms. Status: STANDARD.
- Q, $\bar Q$: The left-handed quark doublets of the three generations, $Q = (u_\mathrm{L}, d_\mathrm{L})$ (colour triplets, hypercharge $1/6$). Unit: $\mathrm{GeV}^{3/2}$. Status: AUTHOR.
- u_R: The right-handed up-type quarks (up, charm and top; singlets of $SU(2)_L$, hypercharge $2/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- d_R: The right-handed down-type quarks (down, strange and bottom; hypercharge $-1/3$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.

The divergences of the fermion-number currents of the six kinds of multiplets (DERIVED from FE-19 to FE-24): the Yukawa couplings turn doublets into singlets, so the current of each kind is not conserved by itself, while the lepton number $J_\ell + J_e + J_\nu$ and the baryon number $(J_Q + J_u + J_d)/3$ are conserved. The divergences enter the fold operator $\mathbb{M}$ (FE-13). Status: DERIVED.

5.5 The ad hoc Higgs vacuum

$$\begin{aligned}&\mathcal{R}_\mathcal{A} = \mathcal{A}'' + 3\frac{b_4'}{b_4}\mathcal{A}' + \lambda\mathcal{A}^3 + c_\Phi\,\bar\Psi P_\mathrm{L}\Psi\,\mathcal{A} = \Big(4a_{40}^2 - 6a_{40}\frac{b_4'}{b_4}\Big)\mathcal{A} + \lambda\mathcal{A}^3 + c_\Phi\,\bar\Psi P_\mathrm{L}\Psi\,\mathcal{A},\\ &\mathcal{R}_v = \lambda\mathcal{A}^2 v + c_\Phi\,\bar\Psi P_\mathrm{L}\Psi\,v, \qquad J_\Phi^{2\,x_4} = a_{40}\,v\,\mathcal{A}, \qquad J_\Phi^{1\,x_4} = J_\Phi^{3\,x_4} = 0\end{aligned} \tag{FE-26}$$

Symbols in (FE-26).

- $\mathcal{R}_\mathcal{A}$, $\mathcal{R}_v$: The residuals of the ad hoc Higgs vacuum in the upper ($\mathcal{R}_\mathcal{A}$) and lower ($\mathcal{R}_v$) components of the Higgs equation: what the ad hoc vacuum leaves unbalanced. Unit: GeV^3. Status: DERIVED.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_\mathcal{A} \mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4} b_4$ on the fold and $\mathrm{e}^{-a_4} b_4 \, \mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4} b_4$). Unit: dimensionless. Status: AUTHOR.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: GeV$^{7/2}$. Status: AUTHOR.
- P_L: The author's left-handed chiral projector, $P_\mathrm{L} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- a_{40}: The decay constant of almightyAdHoc, $a_{40} > 0$: $\mathcal{A}$ falls by e in the time $1/(2a_{40})$. Unit: GeV (1/length). Status: AUTHOR.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\, G_\mathrm{F})^{-1/2}$ with the Fermi constant $G_\mathrm{F} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- $J_\Phi^{I\nu}$: The weak-isospin current of the Higgs doublet (it contains the mass terms of the W fields); $J_\Phi^{2\,x_4}$ its isospin charge density with $I = 2$. Unit: GeV^3. Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.

The author's ad hoc vacuum IN-12 inserted into the Higgs equation FE-15 (homogeneous fields, no gauge fields, the fold factor $\mathcal{Z} \propto b_4^3 \mathrm{e}^{-3a_4}$ so that the time derivative of $\sqrt{-\hat h}\mathcal{Z}$ gives $3b_4'/b_4$; the metric of the fold is the same in UM-4 as in UM-3). The upper and lower components of FE-15 are not satisfied: $\mathcal{R}_\mathcal{A}$ and $\mathcal{R}_v$ are what remains (the residuals; FE-15 holds exactly if the source Θ is replaced by $\Theta - (\mathcal{R}_\mathcal{A}, \mathcal{R}_v)/\sqrt{2}$). The ad hoc vacuum is an exact solution only in the limit $\mathcal{A} \to 0$. It also carries a weak-isospin charge density $J_\Phi^{2\,x_4} = a_{40} v \mathcal{A}$ (the component along x_4 of the isospin current of FE-17 with $I = 2$), which the jump condition of the W^2 field (FE-17) must balance with a background field $W_{x_4}^2$; with the pure-gauge background $g_2 W_{x_4}^2 = \theta'$ ($\tan(\theta/2) = \mathcal{A}/v$) the charge density vanishes and the Higgs equation reduces to the single radial residual $\mathcal{R}_V = V'' + 3(b_4'/b_4)V' + \lambda \mathcal{A}^2 V + c_\Phi \bar\Psi P_\mathrm{L} \Psi \, V$ of the modulus $V = \sqrt{v^2 + \mathcal{A}^2}$ (DERIVED, `um4_fields.py`, F5; `um4_background.py`, BG5 and BG6). Status: DERIVED.

6 The background

The background is the metric of UM-4 with the two functions $a_4(x_4)$ and $b_4(x_4)$ of the time, on which every other field moves. This section writes out its geometry, the exact equations of the author's form of the metric, the static block, and the bulk metric and the equations of an expanding fold, to the first order in its curvature and exactly.

6.1 The metric of UM-4 and the proper chart

$$
\begin{aligned}
g_{MN} &= \mathrm{diag}\Big(\cot^2 z,\ \mathrm{e}^{2a_4} (\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4} (\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4} (\sin^2 z)^{1/6},\ -(\sin^2 z)^{1/6}, \\
&\qquad -\mathrm{e}^{-2a_4} b_4^2 (\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4} b_4^2 (\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4} b_4^2 (\sin^2 z)^{1/6} \Big), \qquad z = 6Hx_0, \\
\sigma &= \mathrm{sgn}\Big(\frac{\pi}{2} - z\Big) \frac{\ln \sin z}{6H}, \qquad \mathrm{d}\sigma = |\cot z| \, \mathrm{d}x_0, \\
g_{MN} \, \mathrm{d}X^M \mathrm{d}X^N &= \mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|} \Big[\mathrm{e}^{2a_4} \big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2 \big) - \mathrm{d}x_4^2 - \mathrm{e}^{-2a_4} b_4^2 \big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2 \big) \Big]
\end{aligned}
\tag{FE-27}
$$

Symbols in (FE-27).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

The metric of UM-4: the author's metric of the primordial gravitational field (his order of 2026-10-24: the metric of `Dirac_Einstein-Lovelock_4+4_Nash.nb` with $b_4 \to b_4^2$; $(\sin^2 z)^{1/6}$ is the positive branch of `Sin[6 H x0]^(1/3)`, the author's rule of 2026-10-16) with the one change that his order of 2026-10-25 13:15 requires: the time x_4 is warped like the other directions, $g_{x_4x_4} = -(\sin^2 z)^{1/6}$ instead of -1. In the proper chart σ of the hidden direction (second and third lines) the warp factor $\mathrm{e}^{-2H|\sigma|}$ multiplies the whole metric of the fold (FD-2), time included. In the author's chart the metric is smooth but degenerate at the fold $z = \pi/2$ ($\cot z = 0$); the proper distance from the fold is $|\sigma|$, and σ runs from $-\infty$ (the tip $z \to 0$) through 0 (the fold) to $+\infty$ (the tip $z \to \pi$): one period of x_0 is the whole line, and the two halves of a period are mirror images under $z \to \pi - z$ ($\sigma \to -\sigma$). a_4 and b_4 are functions of x_4. Status: AUTHOR (the metric and the order), ASSUMED (the way the order is met: the time gets the warp factor of the other directions, the smallest change of the metric that the order allows), DERIVED (the

chart, `um4_geometry.py`, G1).

$$
\begin{aligned}
&\mathbf{e}^{(0)} = \mathrm{d}\sigma, \quad \mathbf{e}^{(i)} = \mathrm{e}^{a_4 - H|\sigma|}\mathrm{d}x_i, \quad \mathbf{e}^{(4)} = \mathrm{e}^{-H|\sigma|}\mathrm{d}x_4, \quad \mathbf{e}^{(r)} = b_4\,\mathrm{e}^{-a_4 - H|\sigma|}\mathrm{d}x_r, \qquad \sqrt{|g|} = b_4^3\,\mathrm{e}^{-7H|\sigma|},\\
&R^{(a)(b)}{}_{(c)(d)} = \mathrm{e}^{2H|\sigma|}\,\hat{R}^{(a)(b)}{}_{(c)(d)} - H^2\,\delta^{ab}_{cd} \quad \text{(all four along the fold)}, \qquad R^{(0)(a)}{}_{(0)(b)} = -H^2\,\delta^a{}_b,\\
&\hat{R}^{(i)(j)}{}_{(i)(j)} = \mathsf{A}^2H^2, \quad \hat{R}^{(i)(4)}{}_{(i)(4)} = (\mathsf{A}^2 + \mathsf{A}_1)H^2, \quad \hat{R}^{(i)(r)}{}_{(i)(r)} = \mathsf{A}\mathsf{V}H^2,\\
&\hat{R}^{(r)(s)}{}_{(r)(s)} = \mathsf{V}^2H^2, \quad \hat{R}^{(4)(r)}{}_{(4)(r)} = (\mathsf{V}^2 + \mathsf{V}_1)H^2,\\
&\mathsf{A} = \frac{a_4'}{H}, \qquad \mathsf{U} = \frac{b_4'}{Hb_4}, \qquad \mathsf{V} = \mathsf{U} - \mathsf{A}, \qquad \mathsf{A}_1 = \frac{\mathsf{A}'}{H}, \qquad \mathsf{V}_1 = \frac{\mathsf{V}'}{H}
\end{aligned}
\tag{FE-28}
$$

Symbols in (FE-28).

- $\mathbf{e}_a{}^M$, $\mathbf{e}^a{}_M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is its inverse; $\mathbf{e}_{bN} = \eta_{bc}\mathbf{e}^c{}_N$; $\mathbf{e}^{(a)}$ is the frame one-form $\mathbf{e}^a{}_M\,\mathrm{d}X^M$. Unit: dimensionless. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- x: The coordinates of the fold: x_1, x_2, x_3 of ordinary space, x_4 the time, x_5, x_6, x_7 the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- i: A generation index ($i = 1, 2, 3$), which also labels the three neutrino mass states; in a frame component such as $\mathcal{S}^{(i)}{}_{(i)}$ one of the three directions of ordinary space (the three components are equal), and in a tensor of ordinary space such as h_{ij} one of the directions x_1, x_2, x_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor $R^M{}_{NPQ}$ (in the convention in which the commutator of two covariant derivatives, acting on a vector, is the Riemann tensor, with its last two indices those of the two derivatives, contracted with the vector), with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the scalar curvature $R = g^{MN}R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: 1/length^2. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1, \ldots, 7\}$; a repeated index is summed. Status: STANDARD.

- b: A frame index (as a). Status: STANDARD.
- c: A frame index (as a). Status: STANDARD.
- d: A frame index (as a). Status: STANDARD.
- $\hat{R}$: The curvature of the metric $\hat{g}_{ab}$ of the fold (FD-2): with four frame indices its Riemann tensor, with two its Ricci tensor $\hat{R}_{ab}$, without indices its scalar curvature $\hat{R}$; for frozen extra times the scalar curvature equals that of the four-dimensional metric $\hat{h}_{\mu\nu}$. Unit: 1/length^2. Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1 M_2 \cdots}_{N_1 N_2 \cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- j: A second generation or mass-state index, or a second direction of ordinary space (as i). Status: STANDARD.
- s: A second extra-time direction of the fold (as r, different from it). Status: STANDARD.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant); the first-order equations of the fold hold for $\mathsf{A}^2 \ll 1$; 1 at the author's vacuum, which is not a solution of UM-4. Status: DERIVED.
- A_1: The time derivative of A in units of H, $\mathsf{A}_1 = \mathsf{A}'/H = a_4''/H^2$. Unit: dimensionless. Status: DERIVED.
- V: The rate of change of the scale factor $\mathrm{e}^{-a_4} b_4$ of the extra times on the fold in units of H, $\mathsf{V} = \mathsf{U} - \mathsf{A}$; zero with frozen extra times. Unit: dimensionless. Status: DERIVED.
- V_1: The time derivative of V in units of H, $\mathsf{V}_1 = \mathsf{V}'/H$. Unit: dimensionless. Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(H b_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.

The vielbein and the volume element (i, j any two different directions of ordinary space, r, s any two different extra times; x_i and x_r the coordinates of these directions), the exact frame curvature of the metric of UM-4 for any $a_4(x_4)$, $b_4(x_4)$ (second line), and the curvature $\hat{R}$ of the metric of the fold FD-2 in its 21 frame planes (third line), with the dimensionless rates of the background (A the expansion rate of ordinary space in units of H, U the rate of change of b_4, V the rate of the scale factor $\mathrm{e}^{-a_4} b_4$ of the extra times). In every plane of two directions of the fold the curvature of the bulk is that of the fold, enlarged by $\mathrm{e}^{2H|\sigma|}$, minus H^2; in every plane that contains the hidden direction it is $-H^2$; there is no other component (UM-3 had mixed components, because its time was not warped). The metric of UM-4 is a warped product: the bulk is a space of curvature $-H^2$ wherever the fold is flat (DERIVED, `um4_geometry.py`, G2, exact, computed in the author's coordinates on both halves of the period; `um4_background.py`, BG1). Status: DERIVED.

6.2 The exact bulk equations of the author's form of the metric

$$\mathcal{E}_{(1)}{}^{(a)}{}_{(b)} + \Lambda\,\delta^a{}_b = \mathrm{e}^{2H|\sigma|}\,\hat{G}^a{}_b \quad \text{(both along the fold)}, \qquad \mathcal{E}_{(1)}{}^{(0)}{}_{(0)} + \Lambda = -\frac{1}{2}\,\mathrm{e}^{2H|\sigma|}\,\hat{R}, \qquad \mathcal{E}_{(1)}{}^{(0)}{}_{(a)} = 0 \quad \text{(FE-29)}$$

Symbols in (FE-29).

- $\mathcal{E}_{(p)}{}^M{}_N$: The Lovelock tensor of order p, $\mathcal{E}_{(p)}{}^M{}_N = -2^{-(p+1)}\,\delta^{M P_1 \cdots P_{2p}}_{N Q_1 \cdots Q_{2p}}\,R^{Q_1 Q_2}{}_{P_1 P_2} \cdots R^{Q_{2p-1} Q_{2p}}{}_{P_{2p-1} P_{2p}}$ (with both indices down it is $1/\sqrt{|g|}$ times the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to g^{MN}; $\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor, $\mathcal{E}_{(3)}$ the cubic Lovelock tensor); with frame indices its frame components. Unit: 1/length^{2p}. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number (UM-4 keeps only $p = 1$); in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1, \ldots, 7\}$; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- Λ: The eight-dimensional cosmological constant of UM-4, $\Lambda = -21H^2$: the value for which the static block is a vacuum of Einstein gravity (its Einstein tensor is $21H^2$ in every direction). The author writes the pure

number $\Lambda_{\rm N}$ in $0 = -\Lambda_{\rm N} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$, so that $\Lambda = -\Lambda_{\rm N} H^2$. Unit: GeV^2. Value: $-21H^2$; the author's original $-10H^2$; UM-3: $-7H^2$. Status: AUTHOR (the term), DERIVED (the value).

- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\rm L}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1 M_2 \cdots}_{N_1 N_2 \cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or −1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_{\rm F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $\hat{G}^\mu{}_\nu$: The Einstein tensor of the metric $\hat{g}_{ab}$ of the fold (FD-2), with frame indices along its seven directions; with four-dimensional indices, $\hat{G}^\mu{}_\nu$, its components along the four observable directions, which for frozen extra times are those of the Einstein tensor of the four-dimensional metric $\hat{h}_{\mu\nu}$. Unit: 1/length^2. Status: STANDARD.
- $\hat{R}$: The curvature of the metric $\hat{g}_{ab}$ of the fold (FD-2): with four frame indices its Riemann tensor, with two its Ricci tensor $\hat{R}_{ab}$, without indices its scalar curvature $\hat{R}$; for frozen extra times the scalar curvature equals that of the four-dimensional metric $\hat{h}_{\mu\nu}$. Unit: 1/length^2. Status: STANDARD.

The bulk equations FE-1 for the metric of UM-4 with Einstein gravity and $\Lambda = -21H^2$, exactly, for any $a_4(x_4)$, $b_4(x_4)$ ($\hat{G}^a{}_b$ and $\hat{R}$ the Einstein tensor and the scalar curvature of the metric of the fold FD-2). They state what the bulk sources must be: the author's form of the metric is a vacuum of the bulk exactly when the metric of the fold is Ricci flat ($\hat{R}_{ab} = 0$), and with sources only when these grow like $\mathrm{e}^{2H|\sigma|}$ away from the fold. An expanding fold with matter therefore has a bulk metric that is not of the author's form (FE-34). (In UM-3 the left-hand side was a polynomial in the rates, with a vacuum branch for any a_4 with frozen extra times; with Einstein gravity and the time warped there is no such branch, FE-33.) Status: DERIVED (`um4_geometry.py`, G5).

6.3 The fold of the author's form of the metric

$$K^a{}_b = \frac{1}{2}\, g^{ac}\, \partial_\sigma g_{cb} = -H\,\mathrm{sgn}(\sigma)\,\delta^a{}_b, \qquad \kappa\,\mathcal{S}^a{}_b = -12H\,\delta^a{}_b \quad \big(\kappa\lambda_{\rm F} = 12H\big) \tag{FE-30}$$

Symbols in (FE-30).

- $K^a{}_b$: The extrinsic curvature of a surface of constant σ of the hidden direction, $K^a{}_b = \frac{1}{2} g^{ac}\partial_\sigma g_{cb}$ for a, b along the surface (FD-5): $-H\,\mathrm{sgn}(\sigma)$ along all seven directions of the fold for the author's form of the metric of UM-4; $K = K^c{}_c$ its trace. Unit: GeV (1/length). Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1, \ldots, 7\}$; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- c: A frame index (as a). Status: STANDARD.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- σ: The label σ in ∂_σ: the derivative with respect to the proper hidden coordinate σ. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_{\rm F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.

- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1 M_2 \cdots}_{N_1 N_2 \cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{S}^a{}_b$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^a{}_b = \int_{0^-}^{0^+} \mathrm{d}\sigma\, \mathcal{E}_{(1)}{}^a{}_b$ (the junction condition of Israel); $\mathcal{S}^4{}_4$ is minus its energy density, $\mathcal{S}^1{}_1$ and $\mathcal{S}^5{}_5$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa\mathcal{S}^a{}_b = -12H\delta^a{}_b$ for every a_4, b_4 of the author's form. Status: DERIVED.
- λ_F: The tension that the static block requires of the fold, $\kappa\lambda_\mathrm{F} = 12H$: the total vacuum energy of the seven-dimensional fold, made of the tension of the fold fluid and the vacuum energy of the fields of the fold (both have the pressure minus their energy density along ordinary space and along the extra times), per unit volume of the six other directions of the fold (eight-dimensional normalisation); in the four-dimensional normalisation it is $\lambda_\mathrm{F} V_*$, and the excess of the total over $\lambda_\mathrm{F} V_*$ is the dark energy ρ_Λ. Unit: GeV^7. Value: $12H/\kappa$; $\lambda_\mathrm{F} V_* = 30H^2 M_\mathrm{red}^2 = 1.1 \times 10^{45}$ GeV^4 for $H = 2.5$ TeV. Status: DERIVED.

The extrinsic curvature of the slices $\sigma = \mathrm{const}$ in all seven directions of the fold (UM-3: zero along the time) and the thin source of the fold from Israel's condition FE-4: for every $a_4(x_4)$, $b_4(x_4)$ of the author's form a pure tension λ_F with $\kappa\lambda_\mathrm{F} = 12H$, the same in every direction ($p = q = -\rho$). For a fold of n dimensions in a bulk of $n+1$ dimensions with this warp the tension is $\kappa\lambda_\mathrm{F} = 2(n-1)H$, the critical tension that belongs to the bulk energy $\Lambda = -n(n-1)H^2/2$: for $n = 4$ it is the tension $6k_\mathrm{RS}/\kappa_5$ of the brane of Randall and Sundrum (FD-8). Status: DERIVED (`um4_geometry.py`, G7; `um4_foldsources.py`, FS1).

6.4 The author's vacuum

$$a_4 = Hx_4, \quad b_4 = \mathrm{constant}\ (\mathsf{A} = 1,\ \mathsf{U} = 0): \qquad \hat{R}_{ab} \neq 0, \qquad \kappa_7\rho_7 = -3H^2, \quad \kappa_7 P_7 = \kappa_7 q_7 = -3H^2 \qquad \text{(FE-31)}$$

Symbols in (FE-31).

- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4} b_4$ on the fold and $\mathrm{e}^{-a_4} b_4\, \mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4} b_4$). Unit: dimensionless. Status: AUTHOR.
- A: The expansion rate of ordinary space in units of H, $\mathsf{A} = a_4'/H$. Unit: dimensionless. Value: 0 at the static block; $H_4^{(0)}/H$ today ($H_4^{(0)}$ the Hubble constant); the first-order equations of the fold hold for $\mathsf{A}^2 \ll 1$; 1 at the author's vacuum, which is not a solution of UM-4. Status: DERIVED.
- U: The rate of change of b_4 in units of H, $\mathsf{U} = b_4'/(Hb_4)$. Unit: dimensionless. Value: $\mathsf{U} = \mathsf{A}$ with frozen extra times. Status: DERIVED.
- $\hat{R}$: The curvature of the metric $\hat{g}_{ab}$ of the fold (FD-2): with four frame indices its Riemann tensor, with two its Ricci tensor $\hat{R}_{ab}$, without indices its scalar curvature $\hat{R}$; for frozen extra times the scalar curvature equals that of the four-dimensional metric $\hat{h}_{\mu\nu}$. Unit: $1/\mathrm{length}^2$. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1, \ldots, 7\}$; a repeated index is summed. Status: STANDARD.

- b: A frame index (as a). Status: STANDARD.
- κ_7: The gravitational coupling of the fold: to the first order in its curvature over H^2 the fold obeys the Einstein equations of seven dimensions $\hat{G}^a{}_b = \kappa_7 \mathsf{t}^a{}_b$ with $\kappa_7 = 5\kappa H/2$, the coupling of the massless graviton of the bulk bound to the fold; four-dimensional gravity on the fold has $8\pi G_{\mathrm{N}} = \kappa_7/V_*$ (plus the induced term, which is negligible). Unit: GeV^{-5}. Value: $5\kappa H/2$. Status: DERIVED.
- ρ_7: The energy density of the matter of the fold above its tension in the seven-dimensional normalisation (minus the component $\mathsf{t}^{(4)}{}_{(4)}$: an energy per unit volume of the six other directions of the fold); $V_*\rho_7$ in the four-dimensional normalisation. Unit: GeV^7. Status: DERIVED.
- P_7: The pressure along ordinary space of the matter of the fold above its tension in the seven-dimensional normalisation (the component $\mathsf{t}^{(i)}{}_{(i)}$). Unit: GeV^7. Status: DERIVED.
- q_7: The pressure along the extra times of the matter of the fold above its tension in the seven-dimensional normalisation (the component $\mathsf{t}^{(r)}{}_{(r)}$). Unit: GeV^7. Status: DERIVED.

The author's vacuum of the earlier version (a de Sitter expansion of ordinary space at the rate H with constant b_4, so that the extra times contract) is not a solution of UM-4: the metric of its fold is not Ricci flat, so the bulk is not a vacuum of the author's form (FE-29), and as a fold with matter it would need, by the first-order equations FE-71, the energy density $\rho_7 = -3H^2/\kappa_7 < 0$ (a violation of the weak energy condition) with the pressures $P_7 = q_7 = -3H^2/\kappa_7$ along ordinary space and the extra times; its rate $\mathsf{A} = 1$ is moreover outside the regime $\mathsf{A} \ll 1$ in which FE-71 holds. The primordial era of UM-4 is the inflation of FE-87 and FE-88. Status: DERIVED (`um4_primordial.py`, PR0).

6.5 The static block

$$a_4,\ b_4 \text{ constant}: \qquad R^{(a)(b)}{}_{(a)(b)} = -H^2 \text{ in all 28 frame planes}, \qquad \mathcal{E}_{(1)}{}^a{}_b = 21H^2\,\delta^a{}_b, \qquad \Lambda = -21H^2 \quad \text{(FE-32)}$$

Symbols in (FE-32).

- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.
- $R^{MN}{}_{PQ}$, R_{MN}, R: The curvature of the metric g_{MN}: with four indices the Riemann tensor $R^M{}_{NPQ}$ (in the convention in which the commutator of two covariant derivatives, acting on a vector, is the Riemann tensor, with its last two indices those of the two derivatives, contracted with the vector), with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the scalar curvature $R = g^{MN}R_{MN}$; with frame indices $R^{(a)(b)}{}_{(c)(d)}$ its frame components, $R^{(a)(b)}{}_{(a)(b)}$ (no sum) the sectional curvature of the frame plane (a, b). Unit: $1/\mathrm{length}^2$. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1, \ldots, 7\}$; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_{\mathrm{F}} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- $\mathcal{E}_{(p)}{}^M{}_N$: The Lovelock tensor of order p, $\mathcal{E}_{(p)}{}^M{}_N = -2^{-(p+1)}\,\delta^{MP_1\cdots P_{2p}}_{NQ_1\cdots Q_{2p}}\,R^{Q_1Q_2}{}_{P_1P_2}\cdots R^{Q_{2p-1}Q_{2p}}{}_{P_{2p-1}P_{2p}}$ (with both indices down it is $1/\sqrt{|g|}$ times the variation of $\sqrt{|g|}\,\mathcal{L}_{(p)}$ with respect to g^{MN}; $\mathcal{E}_{(1)}$ the Einstein tensor, $\mathcal{E}_{(2)}$ the Gauss-Bonnet tensor, $\mathcal{E}_{(3)}$ the cubic Lovelock tensor); with frame indices its frame components. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- p: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic), as an index and as a number (UM-4 keeps only $p = 1$); in the Kohn-Sham equations also a second label of a Kohn-Sham orbital, with the Kronecker delta of the two labels. Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised

Kronecker delta $\delta^{M_1 M_2 \cdots}_{N_1 N_2 \cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.

- Λ: The eight-dimensional cosmological constant of UM-4, $\Lambda = -21H^2$: the value for which the static block is a vacuum of Einstein gravity (its Einstein tensor is $21H^2$ in every direction). The author writes the pure number Λ_N in $0 = -\Lambda_\mathrm{N} + \sum_p H^{-2p} w_{(p)} \mathcal{E}_{(p)}$, so that $\Lambda = -\Lambda_\mathrm{N} H^2$. Unit: GeV^2. Value: $-21H^2$; the author's original $-10H^2$; UM-3: $-7H^2$. Status: AUTHOR (the term), DERIVED (the value).

The static block (a_4 and b_4 constant; the units of x_1, x_2, x_3 and of x_5, x_6, x_7 can make $a_4 = 0$, $b_4 = 1$): a space of constant curvature $-H^2$ in all eight directions (its Weyl tensor vanishes), on each side of the fold a region of the eight-dimensional anti-de Sitter space of radius $1/H$. Its Einstein tensor is $21H^2$ in every direction, so it is a vacuum of Einstein gravity with $\Lambda = -21H^2$ (in UM-3 the time was not warped, its planes with the time were flat, and the static block was a vacuum only of the Lovelock gravity with UM-3's weights). The fold of the static block is a pure tension (FE-30). Status: DERIVED (`um4_geometry.py`, G3 and G4).

$$\begin{aligned} &\mathrm{e}^{a_4} = x_4^{\mathfrak{k}_1}, \quad \mathrm{e}^{-a_4} b_4 = x_4^{\mathfrak{k}_2} : \quad 3\mathfrak{k}_1 + 3\mathfrak{k}_2 = 1, \quad 3\mathfrak{k}_1^2 + 3\mathfrak{k}_2^2 = 1, \quad \{\mathfrak{k}_1, \mathfrak{k}_2\} = \Big\{\frac{1+\sqrt{5}}{6}, \frac{1-\sqrt{5}}{6}\Big\}; \\ &\mathsf{V} = 0 : \quad \hat{G}^{(4)}{}_{(4)} = -3a_4'^2 \end{aligned} \tag{FE-33}$$

Symbols in (FE-33).

- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4} b_4$ on the fold and $\mathrm{e}^{-a_4} b_4\, \mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4} b_4$). Unit: dimensionless. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\mathfrak{k}_1$, $\mathfrak{k}_2$: The exponents of the Kasner folds (the expanding folds without matter of the author's form of the metric): $\mathrm{e}^{a_4} = x_4^{\mathfrak{k}_1}$ and $\mathrm{e}^{-a_4} b_4 = x_4^{\mathfrak{k}_2}$. Unit: dimensionless. Value: $(1 \pm \sqrt{5})/6$. Status: DERIVED.
- V: The rate of change of the scale factor $\mathrm{e}^{-a_4} b_4$ of the extra times on the fold in units of H, $\mathsf{V} = \mathsf{U} - \mathsf{A}$; zero with frozen extra times. Unit: dimensionless. Status: DERIVED.
- $\hat{G}^\mu{}_\nu$: The Einstein tensor of the metric $\hat{g}_{ab}$ of the fold (FD-2), with frame indices along its seven directions; with four-dimensional indices, $\hat{G}^\mu{}_\nu$, its components along the four observable directions, which for frozen extra times are those of the Einstein tensor of the four-dimensional metric $\hat{h}_{\mu\nu}$. Unit: 1/length^2. Status: STANDARD.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.

The vacua of the bulk of the author's form (FE-29: the Ricci-flat folds): besides the static fold, the two Kasner folds in which ordinary space and the extra times expand and contract with the powers $\mathfrak{k}_1$, $\mathfrak{k}_2$ of the time ($x_4 > 0$). With frozen extra times ($\mathsf{V} = 0$, $\mathrm{e}^{-a_4} b_4$ constant) the energy component of the Einstein tensor of the fold is $-3a_4'^2$, so the only Ricci-flat fold with frozen extra times is static: an expanding fold with frozen extra times needs matter, and its bulk metric is FE-34. With constant rates the only vacuum of the author's form is the static block. Status: DERIVED (`um4_geometry.py`, G6; `um4_background.py`, BG1).

$$\begin{aligned} &\hat{G}^{(i)}{}_{(i)} = -\big(2a_4'' + 3\mathsf{c}'' + 3a_4'^2 + 6a_4'\mathsf{c}' + 6\mathsf{c}'^2\big), \qquad \hat{G}^{(4)}{}_{(4)} = -\big(3a_4'^2 + 9a_4'\mathsf{c}' + 3\mathsf{c}'^2\big), \\ &\hat{G}^{(r)}{}_{(r)} = -\big(3a_4'' + 2\mathsf{c}'' + 6a_4'^2 + 6a_4'\mathsf{c}' + 3\mathsf{c}'^2\big), \qquad \mathsf{c} = \ln\big(\mathrm{e}^{-a_4} b_4\big) \end{aligned} \tag{FE-66}$$

Symbols in (FE-66).

- $\hat{G}^\mu{}_\nu$: The Einstein tensor of the metric $\hat{g}_{ab}$ of the fold (FD-2), with frame indices along its seven directions; with four-dimensional indices, $\hat{G}^\mu{}_\nu$, its components along the four observable directions, which for frozen extra times are those of the Einstein tensor of the four-dimensional metric $\hat{h}_{\mu\nu}$. Unit: 1/length^2. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the three neutrino mass states; in a frame component such as $\mathcal{S}^{(i)}{}_{(i)}$ one of the three directions of ordinary space (the three components are equal), and in a tensor of

ordinary space such as h_{ij} one of the directions x_1, x_2, x_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.

- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- c: The logarithm of the scale factor of the extra times on the fold, $\mathsf{c} = \ln(\mathrm{e}^{-a_4} b_4)$; $\mathsf{c}' = \mathsf{V}H$; frozen extra times have $\mathsf{c}' = 0$. Unit: dimensionless. Status: DERIVED.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4} b_4$ on the fold and $\mathrm{e}^{-a_4} b_4\, \mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4} b_4$). Unit: dimensionless. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.

The Einstein tensor of the metric of the fold FD-2 for any $a_4(x_4)$, $b_4(x_4)$, written with the logarithm c of the scale factor of the extra times ($\mathsf{c}' = \mathsf{V}H$); the off-diagonal components vanish. With the equations of the fold FE-71 it gives the Friedmann equations of the fold and the evolution of the volume of the extra times (FE-72). Status: DERIVED (`um4_background.py`, BG2).

6.6 The bulk metric of an expanding fold

$$
\begin{aligned}
& g_{MN}\, \mathrm{d}X^M \mathrm{d}X^N = \mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|}\, \tilde{g}_{ab}\, \mathrm{d}x^a \mathrm{d}x^b, \\
& \tilde{g}_{ab} = \hat{g}_{ab} - \frac{\mathrm{e}^{2H|\sigma|} - 1}{5H^2} \Big(\hat{R}_{ab} - \frac{\hat{R}}{12}\, \hat{g}_{ab} \Big) + \mathrm{e}^{7H|\sigma|}\, \mathfrak{c}_{ab} + \ldots
\end{aligned}
\tag{FE-34}
$$

Symbols in (FE-34).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\, \mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\, \mathrm{d}^8 X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8 X = \mathrm{d}x_0\, \mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\, \mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z) \ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- $\tilde{g}_{ab}$: The metric of the slice at the depth σ divided by the warp factor $\mathrm{e}^{-2H|\sigma|}$: equal to the metric of the fold at $\sigma = 0$. Unit: dimensionless. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; in the tensors of the

fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1, \ldots, 7\}$; a repeated index is summed. Status: STANDARD.

- b: A frame index (as a). Status: STANDARD.
- x: The coordinates of the fold: x_1, x_2, x_3 of ordinary space, x_4 the time, x_5, x_6, x_7 the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $\hat{g}_{ab}$: The metric of the seven-dimensional fold (FD-2) in its coordinates $x_1, \ldots, x_7$ (or in its frame), $\hat{g}^{ab}$ its inverse; $\hat{h}_{\mu\nu}$ is its part along the four observable directions. Unit: dimensionless. Status: AUTHOR.
- $\hat{R}$: The curvature of the metric $\hat{g}_{ab}$ of the fold (FD-2): with four frame indices its Riemann tensor, with two its Ricci tensor $\hat{R}_{ab}$, without indices its scalar curvature $\hat{R}$; for frozen extra times the scalar curvature equals that of the four-dimensional metric $\hat{h}_{\mu\nu}$. Unit: 1/length^2. Status: STANDARD.
- $\mathfrak{c}_{ab}$: A traceless conserved tensor of the fold that sets the homogeneous part of the bulk metric of an expanding fold: the Weyl tensor of the bulk that a field travelling through the bulk can leave (dark radiation); UM-4 takes $\mathfrak{c}_{ab} = 0$. Unit: dimensionless. Value: 0. Status: ASSUMED.

The bulk metric of a fold with any metric $\hat{g}_{ab}(x)$ of the fold (x^a its seven coordinates), expanded in the number of derivatives along the fold: each derivative counts once, the curvature of the fold twice, so the expansion parameter is the curvature of the fold divided by H^2 (for a homogeneous fold $\mathsf{A}^2 = (a_4'/H)^2$). To the first order the second term solves every component of the bulk equations FE-1 (the diagonal components to the second order in the derivatives, the mixed component (0)(4) to the third), for every $a_4(x_4)$, $b_4(x_4)$, and it vanishes at the fold, so that $\tilde{g}_{ab}(\sigma = 0) = \hat{g}_{ab}$. The third term is the homogeneous solution: a traceless conserved tensor $\mathfrak{c}_{ab}(x)$ of the fold times $\mathrm{e}^{7H|\sigma|}$, the bulk Weyl tensor that a field travelling through the bulk can leave (the dark radiation of the brane of Randall and Sundrum); UM-4 takes $\mathfrak{c}_{ab} = 0$ (ASSUMED: no dark radiation). The dots are the terms of higher order. Status: DERIVED (`um4_fold.py`, F1), ASSUMED ($\mathfrak{c}_{ab} = 0$).

6.7 The equations of the fold

$$\kappa\,\mathcal{S}^a{}_b = -12H\,\delta^a{}_b + \frac{2}{5H}\,\hat{G}^a{}_b + \ldots, \qquad \mathcal{S}^a{}_b = -\lambda_{\mathrm{F}}\,\delta^a{}_b + \mathsf{t}^a{}_b : \qquad \hat{G}^a{}_b = \kappa_7\,\mathsf{t}^a{}_b, \qquad \kappa_7 = \frac{5}{2}\,\kappa H \tag{FE-71}$$

Symbols in (FE-71).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{S}^a{}_b$: The thin source of the fold: the energy-momentum tensor concentrated on the fold, integrated across it, $\kappa\mathcal{S}^a{}_b = \int_{0^-}^{0^+} \mathrm{d}\sigma\,\mathcal{E}_{(1)}{}^a{}_b$ (the junction condition of Israel); $\mathcal{S}^4{}_4$ is minus its energy density, $\mathcal{S}^1{}_1$ and $\mathcal{S}^5{}_5$ its pressures along ordinary space and along the extra times (frame components along the fold, eight-dimensional normalisation: an energy per unit volume of the six other directions of the fold). Unit: GeV^7. Value: $\kappa\mathcal{S}^a{}_b = -12H\delta^a{}_b$ for every a_4, b_4 of the author's form. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1, \ldots, 7\}$; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_{\mathrm{F}} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- δ: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta^{M_1 M_2 \cdots}_{N_1 N_2 \cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or −1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- $\hat{G}^\mu{}_\nu$: The Einstein tensor of the metric $\hat{g}_{ab}$ of the fold (FD-2), with frame indices along its seven directions; with four-dimensional indices, $\hat{G}^\mu{}_\nu$, its components along the four observable directions, which for frozen extra times are those of the Einstein tensor of the four-dimensional metric $\hat{h}_{\mu\nu}$. Unit: 1/length^2. Status: STANDARD.
- λ_{F}: The tension that the static block requires of the fold, $\kappa\lambda_{\mathrm{F}} = 12H$: the total vacuum energy of the seven-dimensional fold, made of the tension of the fold fluid and the vacuum energy of the fields of the fold

(both have the pressure minus their energy density along ordinary space and along the extra times), per unit volume of the six other directions of the fold (eight-dimensional normalisation); in the four-dimensional normalisation it is $\lambda_{\rm F}V_*$, and the excess of the total over $\lambda_{\rm F}V_*$ is the dark energy ρ_Λ. Unit: GeV^7. Value: $12H/\kappa$; $\lambda_{\rm F}V_* = 30H^2M_{\rm red}^2 = 1.1\times10^{45}$ GeV^4 for $H = 2.5$ TeV. Status: DERIVED.

- $\mathtt{t}^a{}_b$, $\mathtt{t}$: The matter of the fold above its tension: the thin source minus the tension, $\mathcal{S}^a{}_b = -\lambda_{\rm F}\delta^a{}_b + \mathtt{t}^a{}_b$ (eight-dimensional normalisation); $\mathtt{t} = \mathtt{t}^c{}_c$ its trace; with frozen extra times $\mathtt{t}^a{}_b = \mathrm{diag}(P_7, P_7, P_7, -\rho_7, q_7, q_7, q_7)$. Unit: GeV^7. Status: DERIVED.
- κ_7: The gravitational coupling of the fold: to the first order in its curvature over H^2 the fold obeys the Einstein equations of seven dimensions $\hat{G}^a{}_b = \kappa_7\mathtt{t}^a{}_b$ with $\kappa_7 = 5\kappa H/2$, the coupling of the massless graviton of the bulk bound to the fold; four-dimensional gravity on the fold has $8\pi G_{\rm N} = \kappa_7/V_*$ (plus the induced term, which is negligible). Unit: GeV^{-5}. Value: $5\kappa H/2$. Status: DERIVED.

The junction condition FE-4 with the bulk metric FE-34: the thin source is the tension of FE-30 plus $2/(5H)$ times the Einstein tensor of the fold, up to terms of the fourth order in the derivatives (computed for the space, time and extra-time components). Written with the tension $\lambda_{\rm F}$ and the matter $\mathtt{t}^a{}_b$ of the fold, the fold obeys the Einstein equations of seven dimensions with the coupling $\kappa_7 = 5\kappa H/2$ and no cosmological constant of its own (for the critical tension; a tension above it gives the fold the cosmological constant $\kappa_7(\lambda - \lambda_{\rm F})$, FE-65). The fold is a seven-dimensional brane on which gravity is Einstein's to the first order in its curvature over H^2, as for the brane of Randall and Sundrum. Integrated over the extra times (volume V_*, four-dimensional densities $\rho = V_*\rho_7$) Newton's constant is $8\pi G_{\rm N} = \kappa_7/V_*$, and the tension in four-dimensional normalisation is $\lambda_{\rm F}V_* = 30H^2M_{\rm red}^2$. Status: DERIVED (`um4_fold.py`, F2 and F5).

$$\begin{aligned}\hat{G}_{ab} &= \kappa_7\,\mathtt{t}_{ab} + \kappa^2\,\Pi_{ab} - \mathfrak{E}_{ab},\\ \Pi_{ab} &= -\frac{1}{4}\,\mathtt{t}_{ac}\mathtt{t}^c{}_b + \frac{1}{24}\,\mathtt{t}\,\mathtt{t}_{ab} + \frac{1}{8}\,\hat{g}_{ab}\,\mathtt{t}_{cd}\mathtt{t}^{cd} - \frac{1}{48}\,\hat{g}_{ab}\,\mathtt{t}^2, \qquad \mathtt{t} = \mathtt{t}^c{}_c\end{aligned} \tag{FE-35}$$

Symbols in (FE-35).

- $\hat{G}^\mu{}_\nu$: The Einstein tensor of the metric $\hat{g}_{ab}$ of the fold (FD-2), with frame indices along its seven directions; with four-dimensional indices, $\hat{G}^\mu{}_\nu$, its components along the four observable directions, which for frozen extra times are those of the Einstein tensor of the four-dimensional metric $\hat{h}_{\mu\nu}$. Unit: 1/length^2. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$; on the fold ($a, b \in \{1,\ldots,7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1,\ldots,7\}$; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- κ_7: The gravitational coupling of the fold: to the first order in its curvature over H^2 the fold obeys the Einstein equations of seven dimensions $\hat{G}^a{}_b = \kappa_7\mathtt{t}^a{}_b$ with $\kappa_7 = 5\kappa H/2$, the coupling of the massless graviton of the bulk bound to the fold; four-dimensional gravity on the fold has $8\pi G_{\rm N} = \kappa_7/V_*$ (plus the induced term, which is negligible). Unit: GeV^{-5}. Value: $5\kappa H/2$. Status: DERIVED.
- $\mathtt{t}^a{}_b$, $\mathtt{t}$: The matter of the fold above its tension: the thin source minus the tension, $\mathcal{S}^a{}_b = -\lambda_{\rm F}\delta^a{}_b + \mathtt{t}^a{}_b$ (eight-dimensional normalisation); $\mathtt{t} = \mathtt{t}^c{}_c$ its trace; with frozen extra times $\mathtt{t}^a{}_b = \mathrm{diag}(P_7, P_7, P_7, -\rho_7, q_7, q_7, q_7)$. Unit: GeV^7. Status: DERIVED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- Π_{ab}: The correction of high energies in the exact equations of the fold, quadratic in the matter $\mathtt{t}_{ab}$ of the fold. Unit: GeV^{14}. Status: DERIVED.
- $\mathfrak{E}_{ab}$: The projection of the Weyl tensor of the bulk onto the fold (its electric part), which enters the exact equations of the fold; zero for a bulk of constant curvature, and not fixed by the fold alone. Unit: 1/length^2. Status: DERIVED.
- c: A frame index (as a). Status: STANDARD.
- d: A frame index (as a). Status: STANDARD.
- $\hat{g}_{ab}$: The metric of the seven-dimensional fold (FD-2) in its coordinates $x_1, \ldots, x_7$ (or in its frame), $\hat{g}^{ab}$ its inverse; $\hat{h}_{\mu\nu}$ is its part along the four observable directions. Unit: dimensionless. Status: AUTHOR.

The exact equations of the fold: Gauss's equation for the fold in the bulk with Israel's condition FE-4 and the tension of FE-30 (the method of T. Shiromizu, K. Maeda and M. Sasaki, Phys. Rev. D 62 (2000) 024012, here in

eight dimensions). The quadratic term Π_{ab} is the correction of high energies, and $\mathfrak{E}_{ab}$ is the projection of the Weyl tensor of the bulk onto the fold (zero for a bulk of constant curvature; not fixed by the fold alone). Relative to $\kappa_7 \mathsf{t}_{ab}$ the term $\kappa^2 \Pi_{ab}$ is of the order $(24/5)\rho/(\lambda_\mathrm{F} V_*) = (12/25)\mathsf{A}^2$ for a fold that expands at the rate $\mathsf{A}H$: the equations of the fold are those of general relativity in seven dimensions whenever the expansion rate is small compared to H. With frozen extra times the conservation of Π_{ab} fails for matter other than vacuum energy, radiation and stiff matter (its divergence is $a_4'\rho^2$ times $-3(w-1)(w+1)(3w-1)/32$ for a fluid with $\mathcal{P} = w\rho$): for dust $\mathfrak{E}_{ab}$ must be non-zero at the order $\rho^2/\lambda_\mathrm{F}$ (its value needs the bulk solution: OPEN). Status: DERIVED (`um4_fold.py`, F3, F5 and F6; the identity verified for a diagonal t_{ab} with independent entries).

6.8 The volume of the extra times

$$\mathsf{c}' = 0: \quad 3a_4'^2 = \kappa_7 \rho_7, \quad -\big(2a_4'' + 3a_4'^2\big) = \kappa_7 P_7, \quad -3\big(a_4'' + 2a_4'^2\big) = \kappa_7 q_7 \quad \Rightarrow \quad q_7 = -\frac{\rho_7 - 3P_7}{2};$$
$$q_7 = P_7 = 0,\ \mathsf{c}' = 0: \ \mathsf{c}'' = \frac{3}{5}\, a_4'^2 \tag{FE-72}$$

Symbols in (FE-72).

- c: The logarithm of the scale factor of the extra times on the fold, $\mathsf{c} = \ln(\mathrm{e}^{-a_4} b_4)$; $\mathsf{c}' = \mathsf{V}H$; frozen extra times have $\mathsf{c}' = 0$. Unit: dimensionless. Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- κ_7: The gravitational coupling of the fold: to the first order in its curvature over H^2 the fold obeys the Einstein equations of seven dimensions $\hat{G}^a{}_b = \kappa_7 \mathsf{t}^a{}_b$ with $\kappa_7 = 5\kappa H/2$, the coupling of the massless graviton of the bulk bound to the fold; four-dimensional gravity on the fold has $8\pi G_\mathrm{N} = \kappa_7/V_*$ (plus the induced term, which is negligible). Unit: GeV^{-5}. Value: $5\kappa H/2$. Status: DERIVED.
- ρ_7: The energy density of the matter of the fold above its tension in the seven-dimensional normalisation (minus the component $\mathsf{t}^{(4)}{}_{(4)}$: an energy per unit volume of the six other directions of the fold); $V_*\rho_7$ in the four-dimensional normalisation. Unit: GeV^7. Status: DERIVED.
- P_7: The pressure along ordinary space of the matter of the fold above its tension in the seven-dimensional normalisation (the component $\mathsf{t}^{(i)}{}_{(i)}$). Unit: GeV^7. Status: DERIVED.
- q_7: The pressure along the extra times of the matter of the fold above its tension in the seven-dimensional normalisation (the component $\mathsf{t}^{(r)}{}_{(r)}$). Unit: GeV^7. Status: DERIVED.

The fold with frozen extra times ($\mathsf{c}' = 0$, FE-66 with FE-71; $\mathsf{t}^a{}_b = \mathrm{diag}(P_7, P_7, P_7, -\rho_7, q_7, q_7, q_7)$): the first two equations are the Friedmann equations of four-dimensional general relativity, and the component along the extra times requires the pressure $q_7 = -(\rho_7 - 3P_7)/2$ along them (the radion condition). Radiation satisfies it by itself; dust ($P_7 = q_7 = 0$) does not: without a pressure along the extra times the volume of the extra times accelerates, $\mathsf{c}'' = 3a_4'^2/5$, and through a matter era the ratio c'/a_4' grows from 0 to 0.75, 0.98 and 1.00 after one, two and three e-folds (the seven-dimensional dust solution, in which the extra times expand as ordinary space); a vacuum energy ($q_7 = P_7 = -\rho_7$) drives it too, $\mathsf{c}'' = 6a_4'^2/5$. Newton's constant $8\pi G_\mathrm{N} = \kappa_7/(V_\mathrm{t} \mathrm{e}^{3\mathsf{c}})$ would then change at the rate $-3\mathsf{c}' = -3H_4$, which lunar laser ranging (the relative rate of change of Newton's constant is $(-5.0 \pm 9.6) \times 10^{-15}$ per year; L. Biskupek, J. Müller and J.-M. Torre, Universe 7 (2021) 34) excludes by a factor of about 8500. The fold fluid supplies the missing pressure (IN-10, FE-65). Status: DERIVED (`um4_fold.py`, F4; `um4_background.py`, BG2 to BG4), ASSUMED (the fold fluid).

7 Propagation in the bulk

In the metric of UM-4 the warp factor $\mathrm{e}^{-2H|\sigma|}$ multiplies the whole metric of the fold, the time included (FE-27). Every slice $\sigma = $ const of the static block is therefore a copy of the flat fold, scaled by $\mathrm{e}^{-H|\sigma|}$ in all seven directions, and its light cone is the light cone of the fold: at every depth a signal of light moves along ordinary space by one unit of x_1, x_2, x_3 per unit of the time x_4, exactly as on the fold. For every field of the bulk the dependence on σ then separates from the dependence on the seven coordinates of the fold, and the field of the fold that remains obeys the Lorentz-invariant equation of seven dimensions with a mass m that its profile along σ fixes. This section writes the equations of the profiles of every field of the bulk on the static block ($a_4 = 0$, $b_4 = 1$ by the choice of the units of $x_1, \ldots, x_7$), the photon and the gravitational wave as waves of four dimensions, the propagators of the bulk from

one point of the fold to another, Newton's and Coulomb's laws, the widths of the W and Z bosons, and the causal structure of the bulk.

7.1 The modes of the fields of the bulk

$$\phi_{\text{bulk}} = u(\sigma)\,\mathrm{e}^{\mathrm{i}(kx_3-\omega x_4)}: \qquad u'' - 7H\,\mathrm{sgn}(\sigma)\,u' + \mathrm{e}^{2H|\sigma|}\,m^2\,u = 0, \qquad m^2 = \omega^2 - k^2,$$
$$u_0 = 1, \quad \int_{-\infty}^{\infty} \mathrm{d}\sigma\,\mathrm{e}^{-5H|\sigma|}\,u_0^2 = \frac{2}{5H} \tag{FE-73}$$

Symbols in (FE-73).

- ϕ_{bulk}: A massless scalar field of the bulk, $\nabla_M\nabla^M\phi_{\text{bulk}} = 0$ (an illustration: each polarisation of the graviton obeys its equation). Unit: GeV^3. Status: STANDARD.
- $u(\sigma)$, u_0: The profile along the hidden direction of a mode of a field of the bulk (a scalar, a gauge field or a polarisation of the graviton): the field is $u(\sigma)$ times a field of the fold; $u_0 = 1$ is the profile of the massless mode (the zero mode); in a propagator the response of the field to a unit source on the fold. Unit: dimensionless (a relative amplitude); length in a propagator. Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- k: A wave number: the momentum of a plane wave along x_3, $\mathrm{e}^{\mathrm{i}(kx_3-\omega x_4)}$; in the cosmological perturbations the comoving wave number of a Fourier mode, $k = |\vec{k}|$. Unit: GeV (1/length). Status: STANDARD.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_{\text{F}} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- m: A mass: m_i the mass of the neutrino mass state i; $\Delta m_{ij}^2 = m_i^2 - m_j^2$ the squared-mass difference of the states i and j (Δm_{21}^2, Δm_{31}^2, Δm_{32}^2); Δm^2 without labels the difference of the two states of a resonance, $m_2^2 - m_1^2$, multiplied by v_{loc}^2/v^2 where this differs from 1; m without a label the mass of a mode of a field of the bulk as the fold sees it, $m^2 = \omega^2 - k^2$; for the W and Z bosons their complex pole mass, $m^2 = m_0^2 - \mathrm{i}m_0\Gamma_{\text{bulk}}$ to the leading order; in the relic abundance of the particles χ their mass. Unit: GeV. Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no bound. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.

A massless scalar field of the bulk, $\nabla_M\nabla^M\phi_{\text{bulk}} = 0$ (an illustration: it is the equation of each polarisation of the graviton, FE-76), for a wave of the frequency ω and the wave number k along x_3, independent of the extra times. The profile $u(\sigma)$ obeys the equation in the middle (on both sides of the fold; the term $-7H\,\mathrm{sgn}(\sigma)u'$ comes from the volume element $\sqrt{|g|} \propto \mathrm{e}^{-7H|\sigma|}$), in which ω and k appear only through $m^2 = \omega^2 - k^2$: the mode is a field of the fold of the mass m, with the Lorentz-invariant dispersion $\omega^2 = k^2 + m^2$, whatever its profile. The mode $m = 0$ has the profile $u_0 = 1$, the same amplitude at every depth; it is normalisable, because its kinetic term carries the weight $\sqrt{|g|}\,|g^{x_4x_4}| \propto \mathrm{e}^{-5H|\sigma|}$ with the finite integral $2/(5H)$: the zero mode is a massless field of the seven-dimensional fold, bound to it, and it travels at the speed of light at every wavelength. The other modes form a continuum of masses from $m = 0$ upwards (no gap) and move slower than light, with the group velocity k/ω. (In UM-3, whose time was not warped, the same equation had ω^2 without and k^2 with the factor $\mathrm{e}^{2H|\sigma|}$, and the zero mode had $\omega^2 = \frac{3}{2}k^2$: 1.22 times the speed of light.) Status: DERIVED (`um4_bulk.py`, B1: the profile equation exactly on both halves; the lowest ω^2 of the operator in a box $|\sigma| < 6/H$ equals k^2 to 4×10^{-12} for $k = 0.1H$ to $2H$; B10: the massive modes, Bessel functions of the warped distance from the fold, a continuum from zero mass).

$$B_\sigma = 0, \quad B_a = u(\sigma)\,\hat{B}_a(x): \qquad u'' - 5H\,\mathrm{sgn}(\sigma)\,u' + \mathrm{e}^{2H|\sigma|}\,m^2\,u = 0, \qquad \eta^{ac}\,\partial_a \hat{B}_{cb} = m^2\,\hat{B}_b,$$
$$u_0 = 1, \quad \int_{-\infty}^{\infty} \mathrm{d}\sigma\,\mathrm{e}^{-3H|\sigma|}\,u_0^2 = \frac{2}{3H} \tag{FE-74}$$

Symbols in (FE-74).

- B_M, B_μ, B_{MN}: The hypercharge gauge field of $U(1)_Y$: in the bulk B_M (geometric normalisation), on the fold the canonically normalised four-dimensional field $B_\mu(x)$ of its massless mode, $g_1 B_\mu(x) = B_\mu(\sigma = 0, x)$; B_{MN}, $B_{\mu\nu}$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- σ: The label σ in ∂_σ: the derivative with respect to the proper hidden coordinate σ. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab} = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat{g}_{ab}$ and the tensors built from it) a coordinate index of the fold, $a \in \{1, \ldots, 7\}$; a repeated index is summed. Status: STANDARD.
- $u(\sigma)$, u_0: The profile along the hidden direction of a mode of a field of the bulk (a scalar, a gauge field or a polarisation of the graviton): the field is $u(\sigma)$ times a field of the fold; $u_0 = 1$ is the profile of the massless mode (the zero mode); in a propagator the response of the field to a unit source on the fold. Unit: dimensionless (a relative amplitude); length in a propagator. Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $\hat{B}_a$, $\hat{B}_{ab}$: The field of the fold of a mode of the hypercharge field of the bulk (of any gauge field of the bulk to the first order in the field), $B_a = u(\sigma)\hat{B}_a(x)$; $\hat{B}_{ab} = \partial_a \hat{B}_b - \partial_b \hat{B}_a$ its field strength. Unit: GeV (field), GeV^2 (field strength). Status: STANDARD.
- x: The coordinates of the fold: x_1, x_2, x_3 of ordinary space, x_4 the time, x_5, x_6, x_7 the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- m: A mass: m_i the mass of the neutrino mass state i; $\Delta m^2_{ij} = m_i^2 - m_j^2$ the squared-mass difference of the states i and j (Δm^2_{21}, Δm^2_{31}, Δm^2_{32}); Δm^2 without labels the difference of the two states of a resonance, $m_2^2 - m_1^2$, multiplied by v_loc^2/v^2 where this differs from 1; m without a label the mass of a mode of a field of the bulk as the fold sees it, $m^2 = \omega^2 - k^2$; for the W and Z bosons their complex pole mass, $m^2 = m_0^2 - \mathrm{i}m_0\Gamma_\mathrm{bulk}$ to the leading order; in the relic abundance of the particles χ their mass. Unit: GeV. Status: STANDARD.
- η_{ab}: The flat metric of the frame, $\eta = \mathrm{diag}(1, 1, 1, 1, -1, -1, -1, -1)$ (eta4488 of 4+4.nb); η^{ab} is its inverse (the same matrix). Unit: dimensionless. Status: AUTHOR.
- c: A frame index (as a). Status: STANDARD.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no bound. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.

The modes of a gauge field of the bulk, written for the hypercharge field B_M (each gauge field of the bulk, to the first order in the field, IN-7), in the gauge $B_\sigma = 0$: the field of the fold $\hat{B}_a(x)$ (x the coordinates of the fold, a, b, c the seven directions of the fold, $\hat{B}_{cb} = \partial_c \hat{B}_b - \partial_b \hat{B}_c$) obeys the Proca equation of the fold with the mass m, and the

profile the equation in the middle (the factor $-5H$ from $\sqrt{|g|}\,g^{\sigma\sigma}g^{bb} \propto \mathrm{e}^{-5H|\sigma|}$). The massless mode has the profile 1 and the weight $\sqrt{|g|}\,g^{aa}g^{bb} \propto \mathrm{e}^{-3H|\sigma|}$ in its kinetic term, whose integral is $2/(3H)$: a massless gauge field of the fold, bound to it, with the coupling of seven dimensions $\mathtt{g}_1^2 \cdot \frac{3}{2}H$ and the four-dimensional coupling $g_1^2 = 3H\mathtt{g}_1^2/(2V_*)$ (FE-44). This is the photon, the gluons and the massless electroweak gauge fields of the Standard Model before the Higgs vacuum gives the W and Z bosons their masses (FE-82). A component B_σ along the hidden direction has no normalisable zero mode (its profile would be $\mathrm{e}^{5H|\sigma|}$), so the four-dimensional world sees no scalar from it. The massive modes ($m > 0$, a continuum without gap) modify Coulomb's law at short distances (FE-81). (In UM-3 the bulk photon had $\omega^2 = 2k^2$: 1.41 times the speed of light.) Status: DERIVED (`um4_bulk.py`, B2: exactly on both halves; the lowest ω^2 in the box equals k^2 to 4×10^{-12}; B10: the continuum of massive modes from zero mass).

$$
\begin{aligned}
&A_\mu = \mathfrak{e}_\mu\, \mathrm{e}^{\mathrm{i}k(x_3-x_4)}: \qquad \mathfrak{e}_3 + \mathfrak{e}_4 = 0, \qquad \mathfrak{e}_\mu \sim \mathfrak{e}_\mu + \beta_\mathrm{g}\, k\,(0,0,1,-1),\\
&\mathfrak{e}_\pm = \frac{1}{\sqrt{2}}\big(1, \pm\mathrm{i}, 0, 0\big) \to \mathrm{e}^{\mp\mathrm{i}\theta_\mathrm{r}}\,\mathfrak{e}_\pm, \qquad \vec{E}\cdot\vec{B} = 0, \quad \vec{E}^2 = \vec{B}^2
\end{aligned}
\tag{FE-75}
$$

Symbols in (FE-75).

- A_μ: The photon field of the fold (canonically normalised): the massless combination of the massless modes of W^1, W^2, W^3 and B that the Higgs vacuum leaves massless; a field of the bulk whose zero mode travels at the speed of light. Unit: GeV. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- $\mathfrak{e}_\mu$, $\mathfrak{e}_\pm$: The polarisation vector of a plane wave of the photon, with components along x_1, x_2, x_3, x_4; $\mathfrak{e}_\pm = (1, \pm\mathrm{i}, 0, 0)/\sqrt{2}$ the circular polarisations of a wave travelling along x_3. Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- k: A wave number: the momentum of a plane wave along x_3, $\mathrm{e}^{\mathrm{i}(kx_3 - \omega x_4)}$; in the cosmological perturbations the comoving wave number of a Fourier mode, $k = |\vec{k}|$. Unit: GeV (1/length). Status: STANDARD.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- β_g: Any constant: the size of a gauge transformation, which adds a multiple of the wave vector to the polarisation vector of the photon. Unit: dimensionless. Status: STANDARD.
- θ_r: The angle of a rotation about the direction of motion of a wave. Unit: dimensionless (radian). Status: STANDARD.
- $\vec{E}$: The electric field of the photon on the fold: its component along the direction x_i of ordinary space is $\partial A_4/\partial x_i - \partial A_i/\partial x_4$ ($i = 1, 2, 3$; A_4 and A_i the components of A_μ); $\vec{E}^2$ its square. Unit: GeV^2. Status: STANDARD.
- $\vec{B}$: The magnetic field of the photon on the fold, the curl of the components A_1, A_2, A_3 of A_μ along ordinary space (its component along x_1 is $\partial A_3/\partial x_2 - \partial A_2/\partial x_3$, and cyclically); $\vec{B}^2$ its square. Unit: GeV^2. Status: STANDARD.

The photon as a wave: the zero mode of FE-74 for the combination A_μ of W^3 and B that the Higgs vacuum leaves massless (FE-45), a plane wave of the wave number k along x_3 with the polarisation vector $\mathfrak{e}_\mu$ (components along $x_1, \ldots, x_4$; independent of the extra times). The wave satisfies Maxwell's equations of the fold because its wave vector is null. The Lorenz condition $\partial^\mu A_\mu = 0$ requires $\mathfrak{e}_3 + \mathfrak{e}_4 = 0$, and adding the gradient of a gauge function (any constant β_g in the second relation) removes the remaining component along the direction of motion: the only physical polarisations are the two transverse ones, along x_1 and along x_2; the longitudinal and the time-like polarisations are pure gauge. Its electric field $\vec{E}$ and its magnetic field $\vec{B}$ are orthogonal to the direction of motion x_3 and to each other and equal in size. Under a rotation by the angle θ_r about the direction of motion the circular polarisations $\mathfrak{e}_\pm$ are multiplied by $\mathrm{e}^{\mp\mathrm{i}\theta_\mathrm{r}}$: the photon has the helicities ± 1; it is a transverse wave of spin 1, as the order of 2026-10-25 13:15 requires, and it travels at the speed of light. A photon that leaves the fold travels through the bulk at the local speed of light, which at every depth is that of the fold (FE-83). The massless vector of seven dimensions has five polarisations; the three that point along the extra times are the components A_r, which UM-4 excludes (IN-7: they would have kinetic terms of the wrong sign). Status: DERIVED (`um4_bulk.py`, B3), ASSUMED (no components along the extra times).

$$\frac{1}{2}\Big[\mathrm{e}^{2H|\sigma|}\Big(\frac{\partial^2 h_{\mathrm{gw}}}{\partial x_4^2}-\frac{\partial^2 h_{\mathrm{gw}}}{\partial x_3^2}\Big)-\frac{\partial^2 h_{\mathrm{gw}}}{\partial\sigma^2}+7H\,\mathrm{sgn}(\sigma)\,\frac{\partial h_{\mathrm{gw}}}{\partial\sigma}\Big]=0: \qquad h_{\mathrm{gw}}=h_+(x_3-x_4),$$
$$\mathsf{h}_{ij}=\begin{pmatrix} h_+ & h_\times & 0\\ h_\times & -h_+ & 0\\ 0 & 0 & 0\end{pmatrix}, \qquad \frac{\Delta L}{L}=\frac{1}{2}\big(h_+\cos 2\phi_{\mathrm{r}}+h_\times\sin 2\phi_{\mathrm{r}}\big), \qquad h_+\pm\mathrm{i}h_\times\to\mathrm{e}^{\mp 2\mathrm{i}\theta_{\mathrm{r}}}\big(h_+\pm\mathrm{i}h_\times\big) \tag{FE-76}$$

Symbols in (FE-76).

- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda=-21H^2$ and the tension $\kappa\lambda_{\mathrm{F}}=12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H\gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma=|\cot z|\,\mathrm{d}x_0$, $\sigma=\mathrm{sgn}(\pi/2-z)\ln(\sin z)/(6H)$ on one period $0<z<\pi$; $\sigma=0$ is the fold $z=\pi/2$, $\sigma\to\mp\infty$ the tips $z\to 0,\pi$; one period of x_0 is the whole line $-\infty<\sigma<\infty$. Unit: length. Status: DERIVED.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- h_{gw}: The amplitude of a gravitational wave of the + polarisation travelling along x_3: the metric components along x_1 and x_2 multiplied by $\mathrm{e}^{h_{\mathrm{gw}}}$ and $\mathrm{e}^{-h_{\mathrm{gw}}}$; a function of σ, x_3 and x_4 in the bulk, and on the fold the gravitational wave of the four-dimensional world, to the first order its plus polarisation h_+. Unit: dimensionless. Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c=1$). Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- h_+: The plus polarisation of a gravitational wave travelling along x_3: $\mathsf{h}_{11}=-\mathsf{h}_{22}=h_+$, a function of x_3-x_4; to the first order the zero mode of h_{gw}. Unit: dimensionless. Status: STANDARD.
- h_{ij}: The transverse traceless perturbation of the metric of ordinary space by a gravitational wave as the four-dimensional world sees it: $\hat{h}_{ij}=\mathrm{e}^{2a_4}(\delta_{ij}+\mathsf{h}_{ij})$ $(i,j=1,2,3)$. Unit: dimensionless. Status: STANDARD.
- i: A generation index $(i=1,2,3)$, which also labels the three neutrino mass states; in a frame component such as $\mathcal{S}^{(i)}{}_{(i)}$ one of the three directions of ordinary space (the three components are equal), and in a tensor of ordinary space such as h_{ij} one of the directions x_1,x_2,x_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- j: A second generation or mass-state index, or a second direction of ordinary space (as i). Status: STANDARD.
- $h_\times$: The cross polarisation of a gravitational wave travelling along x_3: $\mathsf{h}_{12}=\mathsf{h}_{21}=h_\times$, a function of x_3-x_4. Unit: dimensionless. Status: STANDARD.
- Δ: A change, difference or small variation of the quantity that follows (for example an interval of the time, a perturbation of a density, a difference of two squared masses, or an addition to an effective number of species). Status: STANDARD.
- L: A distance between two points of the fold along ordinary space: for a gravitational wave the radius of a ring of free test masses (ΔL the change of the distance of one of them from the centre), for a signal the distance between its emitter and its receiver. Unit: length. Status: STANDARD.
- ϕ_{r}: The angle at which a test mass sits on a ring of test masses in the plane of x_1 and x_2, measured from the x_1 axis. Unit: dimensionless (radian). Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- θ_{r}: The angle of a rotation about the direction of motion of a wave. Unit: dimensionless (radian). Status: STANDARD.

The gravitational wave. First line: the + polarisation h_{gw} of a wave along x_3 in the bulk of the static block (the metric components along x_1 and x_2 multiplied by $\mathrm{e}^{h_{\mathrm{gw}}}$ and $\mathrm{e}^{-h_{\mathrm{gw}}}$), to the first order: Einstein's equation FE-1 gives the equation of the massless scalar FE-73, and the zero mode, independent of σ, is any function of x_3-x_4 (written h_+): it solves the bulk equation on both sides and the junction condition at the fold (its σ-derivative does not jump), so the graviton of the bulk has a massless mode bound to the fold that travels at the speed of light at every wavelength, with the coupling $\kappa_7=5\kappa H/2$ of seven dimensions (FE-63, FE-71). Second line: the wave as the

four-dimensional world sees it, the transverse traceless perturbation h_{ij} of the metric of ordinary space ($i,j=1,2,3$: $\hat{h}_{ij}=\mathrm{e}^{2a_4}(\delta_{ij}+\mathsf{h}_{ij})$) of a wave along x_3. It has the two polarisations h_+ (the "plus" mode, $\mathsf{h}_{11}=-\mathsf{h}_{22}$) and $h_\times$ (the "cross" mode, $\mathsf{h}_{12}=\mathsf{h}_{21}$). A ring of free test masses in the plane of x_1 and x_2, of the radius L, changes the distance L of the mass at the angle ϕ_r (from the x_1 axis) by ΔL: the plus mode stretches the ring along x_1 while it squeezes it along x_2 and then the reverse, the cross mode does the same along the two diagonals, rotated by 45°: they stretch and squeeze space perpendicularly to each other and to the direction of motion. Under a rotation by the angle θ_r about the direction of motion h_+ and $h_\times$ rotate into each other by the angle $2\theta_\mathrm{r}$, so that the circular combinations $h_+\pm\mathrm{i}h_\times$ are multiplied by $\mathrm{e}^{\mp2\mathrm{i}\theta_\mathrm{r}}$: helicity ±2, a wave of spin 2, as it must be for a wave sourced by the rank-2 energy-momentum tensor (FE-63: its source is the transverse traceless stress Π_gw). The massless graviton of seven dimensions has $7\cdot(7-3)/2=14$ polarisations, which the four-dimensional world sees as the graviton (2), three vectors $h_{\mu r}$ (6) and six scalars h_{rs} (6). Matter of the fold that does not depend on the extra times and is the same along the three of them has no stress $\mathcal{T}_{\mu r}$ and only the trace of $\mathcal{T}_{rs}$, so it emits only h_+, $h_\times$ and the trace of h_{rs}, the volume of the extra times, which the fold fluid holds fixed (IN-10, FE-65): the gravitational waves that the matter of the fold emits have only the plus and the cross polarisations, as the order requires and as observed (no other polarisation has been detected). Status: DERIVED (`um4_bulk.py`, B4 and B5; `um4_foldsources.py`, FS2), ASSUMED (that the fold fluid holds the volume of the extra times fixed for waves as well, a stiff modulus).

$$\begin{aligned}&\Psi_\mathrm{k}=\mathrm{e}^{\frac{7}{2}H|\sigma|}\,\mathrm{e}^{-M_\mathrm{k}|\sigma|}\,\chi_0(x),\qquad \gamma^{(0)}\chi_0=-\chi_0,\qquad \gamma^{(a)}\,\partial_a\chi_0=0\quad\text{(along the fold)},\\&\sqrt{|g|}\,\bar\Psi_\mathrm{k}\gamma^{x_4}\Psi_\mathrm{k}\propto\mathrm{e}^{(H-2M_\mathrm{k})|\sigma|}\end{aligned}\qquad\text{(FE-77)}$$

Symbols in (FE-77).

- Ψ_k, $\bar\Psi_\mathrm{k}$: A fermion field of the bulk with the kink mass $M_\mathrm{k}\,\mathrm{sgn}(\sigma)$ (an illustration, not a field of UM-4). Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda=-21H^2$ and the tension $\kappa\lambda_\mathrm{F}=12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H\gtrsim2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma=|\cot z|\,\mathrm{d}x_0$, $\sigma=\mathrm{sgn}(\pi/2-z)\ln(\sin z)/(6H)$ on one period $0<z<\pi$; $\sigma=0$ is the fold $z=\pi/2$, $\sigma\to\mp\infty$ the tips $z\to0,\pi$; one period of x_0 is the whole line $-\infty<\sigma<\infty$. Unit: length. Status: DERIVED.
- M_k: The mass of a fermion of the bulk that changes sign at the fold (a kink mass $M_\mathrm{k}\,\mathrm{sgn}(\sigma)$), used to illustrate the speeds of bulk fermions. Unit: GeV. Status: ASSUMED.
- χ_0: The spinor of the fold of the massless mode of a fermion of the bulk: a solution of the massless Dirac equation of the fold of one chirality along σ. Unit: $\mathrm{GeV}^{7/2}$. Status: DERIVED.
- x: The coordinates of the fold: x_1,x_2,x_3 of ordinary space, x_4 the time, x_5,x_6,x_7 the extra times; a field of the fold is a function of the four observable coordinates $x=(x_1,x_2,x_3,x_4)$, and $\mathrm{d}^4x=\mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M=\mathbf{e}_a{}^MT^a$, $\gamma_M=g_{MN}\gamma^N$; $\gamma^{(a)}=T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- a: A frame (orthonormal) index running over $0,1,\ldots,7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with the frame metric $\eta_{ab}=\mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$; on the fold ($a,b\in\{1,\ldots,7\}$) the frame of the fold; in the tensors of the fold written with its coordinates x^a (its metric $\hat g_{ab}$ and the tensors built from it) a coordinate index of the fold, $a\in\{1,\ldots,7\}$; a repeated index is summed. Status: STANDARD.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c=1$). Status: AUTHOR.

A fermion of the bulk with a mass that changes sign at the fold, $M_\mathrm{k}\,\mathrm{sgn}(\sigma)$ (an illustration: UM-4 has no such field;

it shows that the quarks and leptons could live in the bulk without outrunning light). The factor $\mathrm{e}^{(7/2)H|\sigma|}$ removes the spin connection (FE-14), and the zero mode is the profile $\mathrm{e}^{-M_\mathrm{k}|\sigma|}$ times a spinor $\chi_0(x)$ of the fold of one chirality along σ that obeys the massless Dirac equation of the fold. Its charge density along σ falls as $\mathrm{e}^{(H-2M_\mathrm{k})|\sigma|}$, so it is normalisable for $M_\mathrm{k} > H/2$: a massless fermion of seven dimensions, bound to the fold, at the speed of light (in UM-3 it moved at $2M_\mathrm{k}/(2M_\mathrm{k}-H)$ times the speed of light). Status: DERIVED (`um4_bulk.py`, B6).

$$\mathsf{S} = \frac{1}{2}\,\gamma^{(3)}\gamma^{(4)}: \qquad \big[\mathsf{S},\gamma^{(0)}\big] = \big[\mathsf{S},\Gamma_8\big] = 0, \qquad \big[\mathsf{S},\gamma^{(3)}\big] = -\gamma^{(4)}, \qquad \big[\mathsf{S},\gamma^{(4)}\big] = -\gamma^{(3)} \qquad \Rightarrow \qquad \omega^2 = \omega_0^2 + k^2 \tag{FE-78}$$

Symbols in (FE-78).

- S: The generator of the boosts along x_3 on dirac16complex, $\mathsf{S} = \frac{1}{2}\gamma^{(3)}\gamma^{(4)}$. Unit: dimensionless. Status: DERIVED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- Γ_8: The chirality matrix of eight dimensions, $\Gamma_8 = T_0T_1\cdots T_7$ (the author's T16^A[8]) $= \mathrm{diag}(-\mathbf{1}_8, \mathbf{1}_8)$; $\Gamma_8^2 = \mathbf{1}$; it anticommutes with every T_a. Unit: dimensionless. Status: AUTHOR.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- ω_0: The energy at rest (zero momentum along ordinary space and along the extra times) of a state of dirac16complex at the fold; complex for a resonance, $\omega_0 = \omega_\mathrm{r} - \frac{\mathrm{i}}{2}\Gamma_\chi$. Unit: GeV. Status: DERIVED.
- k: A wave number: the momentum of a plane wave along x_3, $\mathrm{e}^{\mathrm{i}(kx_3-\omega x_4)}$; in the cosmological perturbations the comoving wave number of a Fourier mode, $k = |\vec{k}|$. Unit: GeV (1/length). Status: STANDARD.

dirac16complex is invariant under the boosts of the fold. S is the generator of the boosts along x_3 on dirac16complex, built from the author's matrices ($\gamma^{(a)} = T^a$, $(\gamma^{(3)})^2 = \mathbf{1}$, $(\gamma^{(4)})^2 = -\mathbf{1}$). It commutes with $\gamma^{(0)}$, with the chirality Γ_8 and with the mass term, and it rotates $\gamma^{(3)}$ and $\gamma^{(4)}$ into each other hyperbolically; in the bulk equation (FE-36) the warp multiplies the frequency and the momentum by the same factor $\mathrm{e}^{H|\sigma|}$, so the boost $\mathrm{e}^{\beta_\mathrm{b}\mathsf{S}}$ of the rapidity β_b maps a solution of frequency ω and momentum k along x_3 onto one with $(\omega\cosh\beta_\mathrm{b} - k\sinh\beta_\mathrm{b},\ k\cosh\beta_\mathrm{b} - \omega\sinh\beta_\mathrm{b})$. The fold terms of the vacuum (the Higgs portal and the gluon condensate, FE-39) are scalars of the fold and products of $\gamma^{(0)}$ and the projectors $P_\mathrm{L}, P_\mathrm{R}$, so they are invariant too. Every state of dirac16complex at the fold with the energy ω_0 at rest therefore has $\omega^2 = \omega_0^2 + k^2$ exactly and moves slower than light (in UM-3 the time was not warped, the boosts were not a symmetry, and the bound states had the speed coefficient $c_\mathrm{b}^2 = 1.19$ along ordinary space). Status: DERIVED (`um4_bulk.py`, B7, with the author's matrices; `um4_krein.py`, KR6: the moving states have $\sqrt{\omega_0^2+k^2}$ to 10^{-10}).

7.2 The propagators from one point of the fold to another

$$-\mathrm{e}^{nH|\sigma|}\,\partial_\sigma\big(\mathrm{e}^{-nH|\sigma|}\,\partial_\sigma u\big) + \mathrm{e}^{2H|\sigma|}\,\mathsf{p}^2\,u = \delta_\mathrm{D}(\sigma), \qquad \mathbb{G}_n(\mathsf{p}) = u(0) = \frac{\mathrm{K}_{n/2}(\mathsf{p}/H)}{2\mathsf{p}\,\mathrm{K}_{n/2-1}(\mathsf{p}/H)} \tag{FE-79}$$

Symbols in (FE-79).

- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- n: The label of a field of the bulk in its profile equation: the exponent of the weight $\mathrm{e}^{-nH|\sigma|}$ of its σ-derivatives, $n = 7$ for a scalar and for each polarisation of the graviton, $n = 5$ for a gauge field. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- σ: The label σ in ∂_σ: the derivative with respect to the proper hidden coordinate σ. Status: STANDARD.

- $u(\sigma)$, u_0: The profile along the hidden direction of a mode of a field of the bulk (a scalar, a gauge field or a polarisation of the graviton): the field is $u(\sigma)$ times a field of the fold; $u_0 = 1$ is the profile of the massless mode (the zero mode); in a propagator the response of the field to a unit source on the fold. Unit: dimensionless (a relative amplitude); length in a propagator. Status: DERIVED.
- p: The Euclidean momentum along the fold of a source on the fold: p^2 the square of its seven-dimensional momentum (for a static source spread uniformly over the extra times the square of its momentum along ordinary space); $\mathsf{p} = -\mathrm{i}m$ for a time-like momentum of the mass m with outgoing waves into the bulk. Unit: GeV. Status: STANDARD.
- δ_{D}: The Dirac delta function of σ: zero for $\sigma \neq 0$, with the integral 1 over any range that contains $\sigma = 0$; the integral of $\delta_{\mathrm{D}}(\sigma)$ times a smooth function of σ is the value of that function at $\sigma = 0$. Unit: 1/length (for $\delta_{\mathrm{D}}(\sigma)$). Status: STANDARD.
- $\mathbb{G}_n$, $\mathbb{G}_7$, $\mathbb{G}_5$: The propagator of a massless field of the bulk from one point of the fold to another, in the momentum space of the fold: the value at the fold of its response to a unit source on the fold; $\mathbb{G}_7$ for each polarisation of the graviton and for a scalar, $\mathbb{G}_5$ for a gauge field. Unit: GeV^{-1}. Status: DERIVED.
- K: The modified Bessel function of the second kind (of the order written as its subscript): the solution of the modified Bessel equation that decays exponentially for a large argument; for a half-integer order it is an exponential times a polynomial in the inverse of the argument, which makes the propagators of the bulk rational functions. Unit: dimensionless. Status: STANDARD.

The response of a massless field of the bulk to a unit source on the fold: the profile $u(\sigma)$ of the field for the Euclidean momentum p along the fold (p^2 the square of the seven-dimensional momentum; for a static source spread uniformly over the extra times, the momentum along ordinary space), with the label $n = 7$ for the graviton (each polarisation) and a scalar, and $n = 5$ for a gauge field (FE-73, FE-74). With $\zeta_{\mathrm{w}} = \mathrm{e}^{H|\sigma|}/H$ the solution that decays into the bulk is $\zeta_{\mathrm{w}}^{n/2}\mathrm{K}_{n/2}(\mathsf{p}\zeta_{\mathrm{w}})$, and the jump of u' at the fold fixes its normalisation: $\mathbb{G}_n(\mathsf{p})$ is the propagator of the bulk from one point of the fold to another, in the momentum space of the fold (K_ν the modified Bessel function of the second kind). For small momenta $\mathbb{G}_n \to (n/2-1)H/\mathsf{p}^2$: the massless propagator of the fold with the strength $\frac{5}{2}H$ for the graviton (the coupling $\kappa_7 = \frac{5}{2}\kappa H$) and $\frac{3}{2}H$ for a gauge field ($g_1^2 = \frac{3}{2}H\mathsf{g}_1^2/V_*$). Status: DERIVED (`um4_propagators.py`, P1: the solution and the jump checked with sympy).

$$\begin{aligned}\mathbb{G}_7 &= \frac{H}{2\mathsf{p}^2}\,\frac{x_{\mathrm{p}}^3 + 6x_{\mathrm{p}}^2 + 15x_{\mathrm{p}} + 15}{x_{\mathrm{p}}^2 + 3x_{\mathrm{p}} + 3} = \frac{5H}{2\mathsf{p}^2}\Big(1 + \frac{x_{\mathrm{p}}^2}{15} - \frac{x_{\mathrm{p}}^4}{45} + \frac{x_{\mathrm{p}}^5}{45} + \dots\Big),\\ \mathbb{G}_5 &= \frac{H}{2\mathsf{p}^2}\,\frac{x_{\mathrm{p}}^2 + 3x_{\mathrm{p}} + 3}{x_{\mathrm{p}} + 1} = \frac{3H}{2\mathsf{p}^2}\Big(1 + \frac{x_{\mathrm{p}}^2}{3} - \frac{x_{\mathrm{p}}^3}{3} + \dots\Big), \qquad x_{\mathrm{p}} = \frac{\mathsf{p}}{H}\end{aligned} \tag{FE-80}$$

Symbols in (FE-80).

- $\mathbb{G}_n$, $\mathbb{G}_7$, $\mathbb{G}_5$: The propagator of a massless field of the bulk from one point of the fold to another, in the momentum space of the fold: the value at the fold of its response to a unit source on the fold; $\mathbb{G}_7$ for each polarisation of the graviton and for a scalar, $\mathbb{G}_5$ for a gauge field. Unit: GeV^{-1}. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_{\mathrm{F}} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- p: The Euclidean momentum along the fold of a source on the fold: p^2 the square of its seven-dimensional momentum (for a static source spread uniformly over the extra times the square of its momentum along ordinary space); $\mathsf{p} = -\mathrm{i}m$ for a time-like momentum of the mass m with outgoing waves into the bulk. Unit: GeV. Status: STANDARD.
- x_{p}: The Euclidean momentum along the fold in units of H, $x_{\mathrm{p}} = \mathsf{p}/H$. Unit: dimensionless. Status: DERIVED.

For the half-integer orders of the Bessel functions the propagators are rational functions of $x_{\mathrm{p}} = \mathsf{p}/H$. The expansions for momenta small compared with H show what the massive modes add to the zero mode: the even powers of x_{p} are polynomials in p^2, contact interactions between the sources on the fold (local terms, which act only where the sources overlap); the first odd power (x_{p}^5 for gravity, x_{p}^3 for a gauge field) is not analytic in p^2 and gives a correction of the potential of long range (FE-81). Status: DERIVED (`um4_propagators.py`, P2).

$$\mathcal{V}_7(r) = \frac{5H}{8\pi r}\Big[1 + \frac{16}{15\pi}\,\frac{1}{(Hr)^5} + \dots\Big], \qquad \mathcal{V}_5(r) = \frac{3H}{8\pi r}\Big[1 + \frac{4}{3\pi}\,\frac{1}{(Hr)^3} + \dots\Big] \tag{FE-81}$$

Symbols in (FE-81).

- $\mathcal{V}_7(r)$, $\mathcal{V}_5(r)$: The static potentials between two unit point sources on the fold at the distance r (spread uniformly over the extra times) from the exchange of the graviton ($\mathcal{V}_7$) and of a gauge field ($\mathcal{V}_5$) of the bulk: the three-dimensional Fourier transforms of $\mathbb{G}_7$ and $\mathbb{G}_5$. Unit: GeV^2. Status: DERIVED.
- r: A distance from a centre: the distance from the centre of the star along the path of a neutrino; in Newton's and Coulomb's laws the distance between two point sources on the fold. Unit: length. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

The static potentials between two unit point sources on the fold at the distance r (spread uniformly over the extra times): the three-dimensional Fourier transforms of FE-80. The first terms are the potentials of the zero modes, $1/(4\pi r)$ times the strengths $\frac{5}{2}H$ and $\frac{3}{2}H$; multiplied by the couplings they are Newton's law (with $8\pi G_\mathrm{N} = \kappa_7/V_*$, FE-59) and Coulomb's law. The massive modes of the bulk add to Newton's law the relative correction $0.3395/(Hr)^5$ and to Coulomb's law $0.4244/(Hr)^3$ ($16/(15\pi)$ and $4/(3\pi)$). For gravity this is the scalar part of the exchange; the tensor structure of the massive modes of spin 2 changes its coefficient by a factor of order one (as in the brane of Randall and Sundrum, where the correction is $1/(kr)^2$ in five dimensions). Newton's law, tested by torsion balances down to 52 micrometres (J. G. Lee, E. G. Adelberger, T. S. Cook, S. M. Fleischer and B. R. Heckel, Phys. Rev. Lett. 124 (2020) 101101), holds there to one per cent if $H > 7.7 \times 10^{-3}$ eV; for the bound $H \gtrsim 2.5$ TeV of FE-82 the corrections are 3×10^{-74} at 52 micrometres and 1.4×10^{-27} for Coulomb's law at the Bohr radius: unobservable. Status: DERIVED (`um4_propagators.py`, P3: the asymptotic coefficients exactly, and the numerical Fourier transforms at $Hr = 32$ within 1.2 per cent of them), ESTIMATE (the factor of the tensor structure; the bounds).

$$\Big(1+M_\mathrm{loc}^2\,\mathbb{G}_5(\mathsf{p})\Big)\Big|_{\mathsf{p}=-\mathrm{i}m} = 0: \qquad m^2 = m_0^2\Big(1-\frac{m_0^2}{3H^2}\Big)-\mathrm{i}\,\frac{m_0^5}{3H^3}+\ldots, \qquad m_0^2 = \frac{3}{2}\,HM_\mathrm{loc}^2, \qquad \Gamma_\mathrm{bulk} = \frac{m_0^4}{3H^3} \quad \text{(FE-82)}$$

Symbols in (FE-82).

- M_loc: The mass term that the Higgs vacuum on the fold gives a gauge field of the bulk: $M_\mathrm{loc}^2\delta_\mathrm{D}(\sigma)$ in its profile equation; for the W, $M_\mathrm{loc}^2 = \mathsf{g}_2^2(v^2+\mathcal{A}^2)/(4V_*)$. Unit: GeV$^{1/2}$ (M_loc^2 in GeV). Status: DERIVED.
- $\mathbb{G}_n$, $\mathbb{G}_7$, $\mathbb{G}_5$: The propagator of a massless field of the bulk from one point of the fold to another, in the momentum space of the fold: the value at the fold of its response to a unit source on the fold; $\mathbb{G}_7$ for each polarisation of the graviton and for a scalar, $\mathbb{G}_5$ for a gauge field. Unit: GeV^{-1}. Status: DERIVED.
- p: The Euclidean momentum along the fold of a source on the fold: p^2 the square of its seven-dimensional momentum (for a static source spread uniformly over the extra times the square of its momentum along ordinary space); $\mathsf{p} = -\mathrm{i}m$ for a time-like momentum of the mass m with outgoing waves into the bulk. Unit: GeV. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- m: A mass: m_i the mass of the neutrino mass state i; $\Delta m_{ij}^2 = m_i^2 - m_j^2$ the squared-mass difference of the states i and j (Δm_{21}^2, Δm_{31}^2, Δm_{32}^2); Δm^2 without labels the difference of the two states of a resonance, $m_2^2 - m_1^2$, multiplied by v_loc^2/v^2 where this differs from 1; m without a label the mass of a mode of a field of the bulk as the fold sees it, $m^2 = \omega^2 - k^2$; for the W and Z bosons their complex pole mass, $m^2 = m_0^2 - \mathrm{i}m_0\Gamma_\mathrm{bulk}$ to the leading order; in the relic abundance of the particles χ their mass. Unit: GeV. Status: STANDARD.
- m_0: The mass of the W or the Z boson without the massive modes of the bulk (the mass of the massless mode with the Higgs vacuum), $m_0^2 = \frac{3}{2}HM_\mathrm{loc}^2$; $m_0^2 = g_2^2(v^2+\mathcal{A}^2)/4$ for the W. Unit: GeV. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- Γ_bulk: The width of the W or the Z boson into the bulk (its decay into the massive modes of the gauge fields of the bulk), $\Gamma_\mathrm{bulk} = m_0(m_0/H)^3/3$. Unit: GeV. Value: 1.4 MeV for the Z and 0.9 MeV for the W at $H = 2.5$ TeV. Status: DERIVED.

The W and Z bosons. The Higgs vacuum on the fold gives the gauge fields of the bulk a mass term concentrated on the fold, $M_{\mathrm{loc}}^2\,\delta_{\mathrm{D}}(\sigma)$ in their profile equation (from the isospin current J_Φ of FE-17 and the hypercharge current of FE-18: $M_{\mathrm{loc}}^2 = \mathsf{g}_2^2(v^2+\mathcal{A}^2)/(4V_*)$ for the W), and the propagator of the bulk becomes $\mathbb{G}_5/(1+M_{\mathrm{loc}}^2\mathbb{G}_5)$. Its pole for a time-like momentum ($\mathsf{p} = -\mathrm{i}m$, the continuation with outgoing waves into the bulk) lies at the complex m^2 of the first relation: m_0 is the mass that the zero mode alone would have, $m_0^2 = g_2^2(v^2+\mathcal{A}^2)/4$ for the W and $(g_1^2+g_2^2)(v^2+\mathcal{A}^2)/4$ for the Z (FE-45), and the massive modes shift the mass by $-m_0^4/(6H^2)$ and give the W and the Z the width $\Gamma_{\mathrm{bulk}} = m_0(m_0/H)^3/3$ into the bulk ($m^2 = m_{\mathrm{W}}^2 - \mathrm{i}m_{\mathrm{W}}\Gamma_{\mathrm{bulk}}$): the W and the Z are resonances of the bulk. A Z boson that decays into the bulk leaves the fold unseen and adds to its invisible width. The number of light neutrinos measured at the Z, $N_\nu = 2.9963 \pm 0.0074$ (P. Janot and S. Jadach, Phys. Lett. B 803 (2020) 135319), with the width 167.2 MeV of the Z into one neutrino and its antineutrino, allows at most 1.4 MeV more invisible width (95 per cent, one-sided), so $H > m_{\mathrm{Z}}\big(m_{\mathrm{Z}}/(3\cdot 1.4\ \mathrm{MeV})\big)^{1/3} = 2.5$ TeV (m_{Z} the mass of the Z), the strongest bound on H. The contact term $x_{\mathrm{p}}^2/3$ of $\mathbb{G}_5$ is a contact interaction between the currents of the fold whose strength is the square of the four-dimensional gauge coupling divided by $3H^2$; the limit of ATLAS on such interactions of electrons and muons, written as $4\pi/\Lambda_{\mathrm{CI}}^2$ (Λ_{CI} the scale of a contact interaction), $\Lambda_{\mathrm{CI}} > 26.0$ TeV with destructive interference (JHEP 11 (2020) 005), gives with the photon alone $H > 1.3$ TeV; the shift of the masses changes the ratio $m_{\mathrm{W}}^2/(\cos^2\theta_{\mathrm{W}} m_{\mathrm{Z}}^2)$ (the parameter rho of the electroweak fits, θ_{W} the weak mixing angle) by $\sin^2\theta_{\mathrm{W}} m_{\mathrm{Z}}^2/(3H^2)$, and its measured value 1.00031 ± 0.00019 (Particle Data Group 2024) gives $H > 0.97$ TeV. At $H = 2.5$ TeV the width of the W into the bulk is 0.9 MeV (its measured total width is 2.1 GeV). Status: DERIVED (`um4_propagators.py`, P4: the pole of the rational propagator agrees with the expansion to 10^{-5} in the real part and 10^{-2} in the imaginary part for $m_0/H = 10^{-2}$ to 10^{-3}), ESTIMATE (the bounds, P5: the measured values are inputs).

7.3 The causal structure

$$\tilde\sigma = \mathrm{sgn}(\sigma)\,\frac{\mathrm{e}^{H|\sigma|}-1}{H}: \qquad g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N = \frac{\mathrm{d}\tilde\sigma^2 + \mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2 - \mathrm{d}x_4^2 - \mathrm{d}x_5^2 - \mathrm{d}x_6^2 - \mathrm{d}x_7^2}{\big(1+H|\tilde\sigma|\big)^2}, \qquad \Delta x_4 \geq L \tag{FE-83}$$

Symbols in (FE-83).

- $\tilde\sigma$: The conformal coordinate of the hidden direction on the static block, $\tilde\sigma = \mathrm{sgn}(\sigma)(\mathrm{e}^{H|\sigma|}-1)/H$, in which the static block is conformally flat. Unit: length. Status: DERIVED.
- sgn: The sign function: $+1$ for a positive argument and -1 for a negative one. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_{\mathrm{F}} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.

- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- Δ: A change, difference or small variation of the quantity that follows (for example an interval of the time, a perturbation of a density, a difference of two squared masses, or an addition to an effective number of species). Status: STANDARD.
- L: A distance between two points of the fold along ordinary space: for a gravitational wave the radius of a ring of free test masses (ΔL the change of the distance of one of them from the centre), for a signal the distance between its emitter and its receiver. Unit: length. Status: STANDARD.

The static block in the conformal coordinate $\tilde{\sigma}$ of the hidden direction: the metric is flat space of signature (4, 4) times the factor $(1 + H|\tilde{\sigma}|)^{-2}$. A conformal factor does not change which curves are null, and the null geodesics are the null straight lines of flat space (each with its own parameter). A signal that leaves a point of the fold, travels through the bulk at the local speed of light and arrives at another point of the fold at the distance L along ordinary space needs the time $\Delta x_4 = \int \sqrt{\mathrm{d}x_3^2 + \mathrm{d}\tilde{\sigma}^2} \geq L$, with equality only for the path along the fold: nothing crosses the static bulk faster than light on the fold. (In UM-3 the light cone of the bulk at the depth σ was $\mathrm{e}^{H|\sigma|}$ times wider than that of the fold, and signals through the bulk arrived first.) Status: DERIVED (`um4_causality.py`, C1: the conformal form exactly; the geodesic equation of the conformal metric for a null straight line; 200 sampled bulk paths, all later than the path along the fold).

$$c_{\mathrm{loc}}(\sigma) = 1 - \frac{a_4''}{5H^2}\left(\mathrm{e}^{2H|\sigma|} - 1\right), \qquad \sigma(x_3) = \frac{|a_4''|}{5H}\, x_3\,(L - x_3), \qquad \Delta x_4 = L - \frac{a_4''^2 L^3}{150H^2} \tag{FE-84}$$

Symbols in (FE-84).

- c_{loc}: The coordinate speed of light along x_3 at the depth σ (the rate of change of x_3 with x_4 along a null line at fixed σ), in units of that on the fold: $\mathrm{e}^{H|\sigma|}$ on the static block of UM-3, whose time was not warped, 1 on the static block of UM-4, and on an expanding fold of UM-4 different from 1 in proportion to the acceleration a_4 of the expansion. Unit: dimensionless. Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_{\mathrm{F}} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- L: A distance between two points of the fold along ordinary space: for a gravitational wave the radius of a ring of free test masses (ΔL the change of the distance of one of them from the centre), for a signal the distance between its emitter and its receiver. Unit: length. Status: STANDARD.
- Δ: A change, difference or small variation of the quantity that follows (for example an interval of the time, a perturbation of a density, a difference of two squared masses, or an addition to an effective number of species). Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.

An expanding fold with frozen extra times, with the bulk metric FE-34 to the first order in its curvature: the coordinate speed of light c_{loc} along x_3 at the depth σ, in units of the speed of light on the fold, differs from 1 in proportion to the acceleration a_4'' of the expansion. Where the expansion decelerates ($a_4'' < 0$, the eras of radiation and of matter) the light cone of the bulk is slightly wider than that of the fold, and a signal can gain time by dipping into the bulk; the path that gains the most between two points of the fold at the distance L (for distances small compared with $1/H_4$, $H_4 = a_4'$ the expansion rate) is the parabola of the second relation, and it arrives earlier than light along the fold by $a_4''^2 L^3/(150H^2)$ (the third relation). Relative to the travel time this is

$(1+q_{\rm dec})^2(H_4L)^2(H_4/H)^2/150$, with the deceleration parameter $q_{\rm dec}=-1-a_4''/a_4'^2$; over the Hubble distance today ($q_{\rm dec}=-0.55$, $H_4=1.44\times10^{-42}$ GeV) it is 4.5×10^{-94} for $H=2.5$ TeV (3×10^{-63} even for $H=10^{-3}$ eV): no signal through the bulk outruns light on the fold at any observable level, and on the static fold none at all. Status: DERIVED (`um4_causality.py`, C2 and C3: the speed from the metric of FE-34; the advance by minimising the travel time numerically, 0.997 of the formula).

8 dirac16complex: modes and Kohn-Sham equations

dirac16complex lives in the bulk; its four-dimensional particles are its states at the fold. This section writes the equation of those states on the static block (the late universe; an expanding fold replaces a_4, b_4 by their values at the time), its exact solutions, the conditions at the fold, the resonances that the fold holds, and the Kohn-Sham equations that make the description self-consistent with the self-interaction, with gravity and with the mean fields of the Standard Model.

8.1 The equation of the hidden direction

$$\Psi=\mathrm{e}^{\frac{7}{2}H|\sigma|}\,\chi(\sigma)\,\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x}-\omega x_4)},\qquad \gamma^{(0)}\chi'+\mathrm{e}^{H|\sigma|}\big(-\mathrm{i}\omega\,\gamma^{(4)}+\mathrm{i}\,\vec{k}\cdot\vec{\gamma}\big)\chi-m_\Psi\chi=0\qquad(\sigma\neq0)\qquad\text{(FE-36)}$$

Symbols in (FE-36).

- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi=\Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma=0$. Unit: GeV$^{7/2}$. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda=-21H^2$ and the tension $\kappa\lambda_{\rm F}=12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H\gtrsim2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma=|\cot z|\,\mathrm{d}x_0$, $\sigma=\mathrm{sgn}(\pi/2-z)\ln(\sin z)/(6H)$ on one period $0<z<\pi$; $\sigma=0$ is the fold $z=\pi/2$, $\sigma\to\mp\infty$ the tips $z\to0,\pi$; one period of x_0 is the whole line $-\infty<\sigma<\infty$. Unit: length. Status: DERIVED.
- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi=\mathrm{e}^{\frac{7}{2}H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x}-\omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of the level of dirac16complex at the fold (a long-lived resonance) and $\bar\chi$ its antiparticle. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- $\vec{k}$: The (comoving) momentum along ordinary space, $\vec{k}^2$ its square. Unit: GeV. Status: STANDARD.
- $\vec{x}$: The position in ordinary space, $\vec{x}=(x_1,x_2,x_3)$. Unit: length. Status: AUTHOR.
- ω: A frequency: the energy of a mode $\propto\mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c=1$). Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M=\mathbf{e}_a{}^MT^a$, $\gamma_M=g_{MN}\gamma^N$; $\gamma^{(a)}=T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4'=\mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $\vec{\gamma}$: The three frame Dirac matrices of dirac16complex along ordinary space, $\vec{\gamma}=(\gamma^{(1)},\gamma^{(2)},\gamma^{(3)})$. Unit: dimensionless. Status: AUTHOR.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.

The free field equation FE-12 on the static block (a_4 and b_4 constant, ordinary space and the extra times normalised so that $a_4=0$ and $b_4=1$) for a mode of definite momentum $\vec{k}$ along ordinary space and frequency ω, independent of the extra times. The factor $\mathrm{e}^{(7/2)H|\sigma|}$ removes the spin connection exactly ($\gamma^{(0)}\partial_\sigma\mathrm{e}^{(7/2)H|\sigma|}=\frac{7}{2}H\,\mathrm{sgn}(\sigma)\gamma^{(0)}\mathrm{e}^{(7/2)H|\sigma|}$ cancels the first term of FE-14), so that the profile $\chi(\sigma)$ obeys a Dirac equation of one variable in which the factor $\mathrm{e}^{H|\sigma|}$ (one over the square root of the warp factor) multiplies the whole operator of the fold, the frequency as well as the momentum: the time is warped like ordinary space (in UM-3 the factor was $\mathrm{e}^{3H|\sigma|}$, and $\mathrm{e}^{H|\sigma|}$ multiplied only the momentum). The density of the conserved charge along σ is $\mathrm{e}^{H|\sigma|}\chi^\dagger\mathfrak{B}\chi$ (times b_4^3; FE-43). Status: DERIVED (`um4_geometry.py`, G9, with the author's matrices).

$$\chi' = \mathbb{K}\,\chi, \qquad \mathbb{K} = \gamma^{(0)}\big(m_\Psi + \mathrm{i}\omega\,\mathrm{e}^{H|\sigma|}\gamma^{(4)}\big), \qquad \mathbb{K}^2 = \big(m_\Psi^2 - \omega^2\mathrm{e}^{2H|\sigma|}\big)\mathbf{1},$$
$$q = \sqrt{m_\Psi^2 - \omega^2\mathrm{e}^{2H|\sigma|}}, \qquad |\omega|\,\mathrm{e}^{H\sigma_\mathrm{t}} = m_\Psi \tag{FE-37}$$

Symbols in (FE-37).

- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{\frac{7}{2}H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x}-\omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of the level of dirac16complex at the fold (a long-lived resonance) and $\bar{\chi}$ its antiparticle. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $\mathbb{K}$: The transfer matrix of the profile equation at zero momentum, $\chi' = \mathbb{K}\chi$. Unit: GeV. Status: DERIVED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $\mathbf{1}$, $\mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.
- q: The local decay rate of a profile of dirac16complex along σ at the frequency ω, $q = \sqrt{m_\Psi^2 - \omega^2\mathrm{e}^{2H|\sigma|}}$ (imaginary beyond the turning point, where the profile oscillates). Unit: GeV. Status: DERIVED.
- σ_t: The turning point of a state of dirac16complex of the frequency ω: the depth at which the warped frequency equals the bulk mass, $|\omega|\mathrm{e}^{H\sigma_\mathrm{t}} = m_\Psi$; beyond it every solution oscillates. Unit: length. Status: DERIVED.

At zero momentum along ordinary space ($\vec{k} = 0$) the profile equation is a first-order system whose matrix $\mathbb{K}$ depends on σ through the warped frequency; its square is the local gap. Where $|\omega|\mathrm{e}^{H|\sigma|} < m_\Psi$ the solutions grow or decay locally at the rate $q(\sigma)$; at the turning point σ_t the local gap closes, and beyond it every solution oscillates: the warped frequency exceeds the bulk mass far from the fold for every $\omega \neq 0$. There is therefore no gap in the bulk, no threshold of the continuum other than $\omega = 0$, and no normalisable state of non-zero frequency: the states that UM-3 bound to the fold are resonances in UM-4 (FE-86). The 16 components split into eight copies of two inequivalent problems of two components (the element $\gamma^{(0)}\gamma^{(4)}\Gamma_8$ squares to $\mathbf{1}$ and commutes with $\gamma^{(0)}$, $\gamma^{(4)}$ and Γ_8), in each of which $\gamma^{(0)}$, $\mathrm{i}\gamma^{(0)}\gamma^{(4)}$ and Γ_8 act as the Pauli matrices s_3, is_1 and $\pm\mathsf{s}_1$ (FE-85). With chirality-dependent masses $m_\mathrm{L}P_\mathrm{L} + m_\mathrm{R}P_\mathrm{R}$ in place of m_Ψ the local gap would be $m_\mathrm{L}m_\mathrm{R} - \omega^2\mathrm{e}^{2H|\sigma|}$ (tachyonic for $m_\mathrm{L}m_\mathrm{R} < 0$); UM-4 keeps the record's single bulk mass and puts the difference of the halves into the fold terms. Status: DERIVED (`um4_spectra.py`, U1, with the author's matrices).

$$\mathrm{e}^{2H|\sigma|}\Big(\omega^2 - \mathrm{e}^{-2a_4}\,\vec{k}^2 + \mathrm{e}^{2a_4}\,b_4^{-2}\,\vec{\kappa}^2\Big) = m_\Psi^2 + p_\sigma^2, \qquad \omega^2 = \omega_0^2 + \vec{k}^2 - \vec{\kappa}^2 \quad \big(a_4 = 0,\ b_4 = 1\big) \tag{FE-38}$$

Symbols in (FE-38).

- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free

parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.

- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $\vec{k}$: The (comoving) momentum along ordinary space, $\vec{k}^2$ its square. Unit: GeV. Status: STANDARD.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.
- $\vec{\kappa}$: A momentum along the extra times (three components); the fields of UM-4 have $\vec{\kappa} = 0$. Unit: GeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- p_σ: The momentum along the hidden direction (local wave number of a profile along σ). Unit: GeV. Status: DERIVED.
- ω_0: The energy at rest (zero momentum along ordinary space and along the extra times) of a state of dirac16complex at the fold; complex for a resonance, $\omega_0 = \omega_\mathrm{r} - \frac{\mathrm{i}}{2}\Gamma_\chi$. Unit: GeV. Status: DERIVED.

The local dispersion relation of dirac16complex (WKB: for profiles that vary slowly compared with the warp), with the momentum p_σ along the hidden direction and a momentum $\vec{\kappa}$ along the extra times, and the exact dispersion of every state at the fold with the energy ω_0 at rest. The second relation holds exactly because the boosts along ordinary space (FE-78) and the rotations of the time into an extra time (generated by $\frac{1}{2}\gamma^{(4)}\gamma^{(5)}$, which commutes with $\gamma^{(0)}$, with Γ_8 and with the mass term) are symmetries of the bulk equation and of the fold terms of the vacuum: the warp multiplies the frequency and the momenta alike. A momentum along a time-like direction enters with the sign of the frequency: a state with $|\vec{\kappa}| > \omega_0$ has $\omega^2 < 0$ and grows exponentially in x_4 (a tachyon); it is the rotation of the state at rest and has its width. The same holds for every field, in the bulk as on the fold (the sign comes from $g^{rr} < 0$): a massless field has growing modes for every $|\vec{\kappa}| > |\vec{k}|$. UM-4 therefore keeps only the fields that do not depend on the extra times (ASSUMED); this restriction is consistent (the metric does not depend on x_5, x_6, x_7, so the momentum along the extra times is conserved and a field without it stays without it classically) but not stable (interactions create pairs of opposite momenta, an ESTIMATE: no rate is computed). Compact extra times of the period L_t do not help: the first harmonic stays below one e-folding over the age of the universe only for $L_\mathrm{t} \geq 26.6$ Gpc (2π times the age), but the n-th harmonic grows n times faster, so infinitely many modes grow for every L_t (and a compact time-like direction is itself a closed time-like curve). UM-4 needs a gauge principle that removes the momentum along the extra times of every field. (In UM-3 the time was not warped, the momentum along the extra times carried the factor $\mathrm{e}^{H|\sigma|}$ and the frequency not, and a bound state with $\vec{\kappa} \neq 0$ leaked into the bulk beyond a finite depth.) Status: DERIVED (the dispersion and the instability; `um4_extratimes.py`, EX1 to EX5: $\omega^2 = \omega_0^2 - \vec{\kappa}^2$ and $\omega_0^2 + \vec{k}^2$ computed with the author's matrices for $|\vec{\kappa}|, |\vec{k}| = 0.1$, 0.3, 0.6 m_Ψ to ten digits), ASSUMED (the restriction), OPEN (the gauge principle).

$$\chi = (\omega\zeta_\mathrm{w})^{1/2}\begin{pmatrix}\mathfrak{z}_-(\omega\zeta_\mathrm{w})\\ \mathrm{i}\,\mathfrak{z}_+(\omega\zeta_\mathrm{w})\end{pmatrix} \quad (\sigma > 0), \qquad \chi = (\omega\zeta_\mathrm{w})^{1/2}\begin{pmatrix}-\mathrm{i}\,\mathfrak{z}_+(\omega\zeta_\mathrm{w})\\ \mathfrak{z}_-(\omega\zeta_\mathrm{w})\end{pmatrix} \quad (\sigma < 0), \qquad \zeta_\mathrm{w} = \frac{\mathrm{e}^{H|\sigma|}}{H} \tag{FE-85}$$

Symbols in (FE-85).

- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{\frac{7}{2}H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x} - \omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of the level of dirac16complex at the fold (a long-lived resonance) and $\bar{\chi}$ its antiparticle. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- ζ_w: The warped distance from the fold, $\zeta_\mathrm{w} = \mathrm{e}^{H|\sigma|}/H$ ($1/H$ at the fold): $\omega\zeta_\mathrm{w}$ is the argument of the exact solutions of dirac16complex, and $m\zeta_\mathrm{w}$ that of the massive modes of the mass m of the massless fields of the bulk. Unit: length. Status: DERIVED.
- $\mathfrak{z}_-$, $\mathfrak{z}_+$: Bessel functions of the orders $m_\Psi/H - \frac{1}{2}$ ($\mathfrak{z}_-$) and $m_\Psi/H + \frac{1}{2}$ ($\mathfrak{z}_+$), both of the same kind (of the first kind, of the second kind, or the Hankel functions of the first kind for waves that leave the fold), of the

argument $\omega\zeta_{\mathrm{w}}$. Unit: dimensionless. Status: STANDARD.

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_{\mathrm{F}} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.

The exact solutions of FE-37 in one copy of two components ($\gamma^{(0)} = \mathsf{s}_3$, $\mathrm{i}\gamma^{(0)}\gamma^{(4)} = \mathrm{i}\mathsf{s}_1$): $\mathfrak{z}_-$ and $\mathfrak{z}_+$ are Bessel functions of the orders $m_\Psi/H - \frac{1}{2}$ and $m_\Psi/H + \frac{1}{2}$ of the argument $\omega\zeta_{\mathrm{w}}$ (the same kind for both: of the first kind, of the second kind, or a Hankel function), with ζ_{w} the warped distance from the fold ($\zeta_{\mathrm{w}} = 1/H$ at the fold). For $\omega\zeta_{\mathrm{w}} \ll m_\Psi/H$ (close to the fold) the solutions grow or decay like $\mathrm{e}^{\pm q|\sigma|}$; for $\omega\zeta_{\mathrm{w}} \gg m_\Psi/H$ (beyond the turning point) they are waves, and the outgoing ones, which carry the flux away from the fold, are the Hankel functions of the first kind. Status: DERIVED (`um4_spectra.py`, U2: the solutions checked against the differential equation at three depths).

8.2 The fold and the resonances that it holds

$$\chi(0^+) = \exp\Big(\gamma^{(0)}\big(\mu_{\mathrm{L}}P_{\mathrm{L}} + \mu_{\mathrm{R}}P_{\mathrm{R}}\big)\Big)\,\chi(0^-),$$
$$\mu_X = \frac{1}{V_*}\Big[c_\Phi\,\langle\Phi^\dagger\Phi\rangle\,\delta_{X\mathrm{L}} + \frac{1}{4}\big(c_X^G\langle G^2\rangle + c_X^W\langle W^2\rangle + c_X^B\langle B^2\rangle\big) + 2\omega\sum_f c_X^f\,n_f\Big], \qquad \langle\Phi^\dagger\Phi\rangle = \frac{v^2 + \mathcal{A}^2}{2} \tag{FE-39}$$

Symbols in (FE-39).

- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{\frac{7}{2}H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x} - \omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of the level of dirac16complex at the fold (a long-lived resonance) and $\bar{\chi}$ its antiparticle. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- μ_X, μ_{L}, μ_{R}: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex, set by the mean fields of the Standard Model at the fold: the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_{\mathrm{L}}P_{\mathrm{L}} + \mu_{\mathrm{R}}P_{\mathrm{R}})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- P_{L}: The author's left-handed chiral projector, $P_{\mathrm{L}} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- P_{R}: The author's right-handed chiral projector, $P_{\mathrm{R}} = (\mathbf{1} + \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_{\mathrm{t}} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta_N^M = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised

Kronecker delta $\delta^{M_1 M_2 \cdots}_{N_1 N_2 \cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta^{M_a}_{N_b}$ ($+1$ or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.

- c^G_X: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- $\langle G^2 \rangle$: The mean square of the gluon field strength, $\langle G^A_{\mu\nu} G^{A\mu\nu} \rangle$ (the gluon condensate in the QCD vacuum, a thermal value in the plasma). Unit: GeV^4. Status: STANDARD.
- G^A_M, G^A_μ, G^A_{MN}: The gluon field of $SU(3)_C$ ($A = 1, \ldots, 8$): in the bulk G^A_M, a field of eight dimensions in its geometric normalisation (the covariant derivative $\partial - \mathrm{i}G$), and on the fold the canonically normalised four-dimensional field $G^A_\mu(x)$ of its massless mode, $g_3 G^A_\mu(x) = G^A_\mu(\sigma = 0, x)$; G^A_{MN} and $G^A_{\mu\nu}$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- c^W_X: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- $\langle W^2 \rangle$: The mean square of the $SU(2)$ field strength, $\langle W^I_{\mu\nu} W^{I\mu\nu} \rangle$. Unit: GeV^4. Status: STANDARD.
- W^I_M, W^I_μ, W^I_{MN}: The gauge field of $SU(2)_L$ ($I = 1, 2, 3$): in the bulk W^I_M (geometric normalisation), on the fold the canonically normalised four-dimensional field $W^I_\mu(x)$ of its massless mode, $g_2 W^I_\mu(x) = W^I_\mu(\sigma = 0, x)$; W^I_{MN}, $W^I_{\mu\nu}$ their field strengths; $W^\pm_\mu = (W^1_\mu \mp \mathrm{i} W^2_\mu)/\sqrt{2}$. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- c^B_X: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- $\langle B^2 \rangle$: The mean square of the hypercharge field strength, $\langle B_{\mu\nu} B^{\mu\nu} \rangle$. Unit: GeV^4. Status: STANDARD.
- B_M, B_μ, B_{MN}: The hypercharge gauge field of $U(1)_Y$: in the bulk B_M (geometric normalisation), on the fold the canonically normalised four-dimensional field $B_\mu(x)$ of its massless mode, $g_1 B_\mu(x) = B_\mu(\sigma = 0, x)$; B_{MN}, $B_{\mu\nu}$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- c^f_X: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c^ℓ_X and c^ν_X are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- n_f: The number density of the Standard-Model fermions of kind f (particles minus antiparticles, summed over generations and gauge components), the time component of the mean current $\langle J^\mu_f \rangle$ in the rest frame of the medium. Unit: GeV^3. Status: STANDARD.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\, G_\mathrm{F})^{-1/2}$ with the Fermi constant $G_\mathrm{F} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_\mathcal{A} \mathrm{e}^{-2 a_{40} x_4}$. Unit: GeV. Status: AUTHOR.

The matching condition FE-13 for a mode, with the Standard-Model fields replaced by their mean values in a homogeneous, isotropic medium at rest on the fold (no mean field strength, no mean spatial current; the factor $\mathrm{e}^{(7/2)H|\sigma|}$ is 1 at the fold, so χ and Ψ have the same jump): the fold acts on dirac16complex as a chirality-dependent mass concentrated at $\sigma = 0$, with the dimensionless strengths μ_L and μ_R. The Higgs portal acts on the left-handed half only ($\delta_{X\mathrm{L}} = 1$ for $X = \mathrm{L}$, 0 for $X = \mathrm{R}$); the gauge-kinetic portals act through the mean squares $\langle G^2 \rangle = \langle G^A_{\mu\nu} G^{A\mu\nu} \rangle$ (the gluon condensate of the QCD vacuum, or of the plasma), $\langle W^2 \rangle$, $\langle B^2 \rangle$ of the canonical fields of the fold (IN-13); the current portals act through the number densities n_f of the Standard-Model fermions of kind f (they give an energy-dependent strength). In the vacuum of the late universe only the Higgs term and the gluon condensate contribute. $(\gamma^{(0)}(\mu_\mathrm{L} P_\mathrm{L} + \mu_\mathrm{R} P_\mathrm{R}))^2 = \mu_\mathrm{L}\mu_\mathrm{R}\mathbf{1}$, so the exponential is $\cosh\varsigma + (\sinh\varsigma/\varsigma)\gamma^{(0)}(\mu_\mathrm{L} P_\mathrm{L} + \mu_\mathrm{R} P_\mathrm{R})$ with $\varsigma^2 = \mu_\mathrm{L}\mu_\mathrm{R}$; for $\mu_\mathrm{R} = 0$ it is exactly $1 + \mu_\mathrm{L}\gamma^{(0)} P_\mathrm{L}$. Status: DERIVED (as in UM-3; `um4_spectra.py`, U3, uses it with the author's matrices).

$$
\begin{aligned}
&q_\mathrm{b} = -m_\Psi \left(\mu_\mathrm{L} + \mu_\mathrm{R}\right) \frac{\tanh\varsigma}{2\varsigma}, \qquad \varsigma = \sqrt{\mu_\mathrm{L}\mu_\mathrm{R}}, \qquad \omega_\mathrm{b} = \sqrt{m_\Psi^2 - q_\mathrm{b}^2}, \\
&\omega_\mathrm{r} = \omega_\mathrm{b}\Big(1 - \frac{H}{2q_\mathrm{b}}\Big) + \ldots, \qquad |\chi_\mathrm{b}(\sigma)| \propto \mathrm{e}^{-q_\mathrm{b}|\sigma|} \quad \big(|\sigma| < \sigma_\mathrm{t}\big)
\end{aligned}
\tag{FE-40}
$$

Symbols in (FE-40).

- q_b: The inverse width of a state of dirac16complex at the fold: inside the turning point its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.

- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- μ_X, μ_L, μ_R: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex, set by the mean fields of the Standard Model at the fold: the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_\mathrm{L}P_\mathrm{L}+\mu_\mathrm{R}P_\mathrm{R})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- ς: The auxiliary variable of the bound-state condition, $\varsigma=\sqrt{\mu_\mathrm{L}\mu_\mathrm{R}}$ ($\varsigma_-=\sqrt{-\mu_\mathrm{L}\mu_\mathrm{R}}$ when $\mu_\mathrm{L}\mu_\mathrm{R}<0$). Unit: dimensionless. Status: DERIVED.
- ω_b: The energy at zero momentum of the state of dirac16complex at the fold in the limit $H\to 0$ (the bound state of UM-3): the particle χ of UM-4 has the mass $\omega_\mathrm{r}=\omega_\mathrm{b}(1-H/(2q_\mathrm{b}))+\ldots$, equal to ω_b up to the relative correction $H/(2q_\mathrm{b})$. Unit: GeV. Status: DERIVED.
- ω_r: The mass of the particle χ: the real part of the energy at rest of the resonance of dirac16complex at the fold, $\omega_\mathrm{r}=\omega_\mathrm{b}(1-H/(2q_\mathrm{b}))+\ldots$. Unit: GeV. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda=-21H^2$ and the tension $\kappa\lambda_\mathrm{F}=12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H\gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- χ_b: The profile of a state of dirac16complex at the fold (a resonance): it decays as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$ out to the turning point and continues beyond it as an outgoing wave of exponentially small amplitude. Unit: GeV^2. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma=|\cot z|\,\mathrm{d}x_0$, $\sigma=\mathrm{sgn}(\pi/2-z)\ln(\sin z)/(6H)$ on one period $0<z<\pi$; $\sigma=0$ is the fold $z=\pi/2$, $\sigma\to\mp\infty$ the tips $z\to 0,\pi$; one period of x_0 is the whole line $-\infty<\sigma<\infty$. Unit: length. Status: DERIVED.
- σ_t: The turning point of a state of dirac16complex of the frequency ω: the depth at which the warped frequency equals the bulk mass, $|\omega|\mathrm{e}^{H\sigma_\mathrm{t}}=m_\Psi$; beyond it every solution oscillates. Unit: length. Status: DERIVED.

The states of dirac16complex at the fold at zero momentum. Close to the fold ($|\sigma|\ll 1/H$) the warped frequency is ω itself, the profile equation is that of UM-3 with constant coefficients, and the decaying solutions on the two sides, joined by FE-39, are the bound states of UM-3: they exist if the fold is attractive, $\mu_\mathrm{L}+\mu_\mathrm{R}<0$, with the inverse width $0<q_\mathrm{b}<m_\Psi$ and the energy ω_b (for $\mu_\mathrm{L}\mu_\mathrm{R}<0$, $\tanh(\varsigma)/\varsigma$ means $\tan(\varsigma_-)/\varsigma_-$ with $\varsigma_-=\sqrt{-\mu_\mathrm{L}\mu_\mathrm{R}}$; for the Higgs portal alone, $\mu_\mathrm{R}=0$: $q_\mathrm{b}=-m_\Psi\mu_\mathrm{L}/2$, $\omega_\mathrm{b}=m_\Psi\sqrt{1-\mu_\mathrm{L}^2/4}$). In UM-4 the profile χ_b keeps decaying as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$ out to the turning point σ_t and beyond it continues as an outgoing wave of exponentially small amplitude: the state is a resonance, the particle χ of dirac16complex, with the four-dimensional mass ω_r (its "x_0 mass") and the width of FE-86. The warped frequency lowers the energy: to the first order in H by $\omega_\mathrm{b}H\langle|\sigma|\rangle$, with $\langle|\sigma|\rangle=1/(2q_\mathrm{b})$ the mean distance from the fold of the density $\mathrm{e}^{-2q_\mathrm{b}|\sigma|}$. The resonances are degenerate eight times at $+\omega_\mathrm{r}$ (and eight times at $-\omega_\mathrm{r}$). Status: DERIVED (`um4_spectra.py`, U3: the zeros of the 16×16 matching with outgoing waves on both sides, for $(\mu_\mathrm{L},\mu_\mathrm{R})=(-1,0)$ and $(-0.6,-0.3)$ and $m_\Psi/H=10$ to 40, an eight-dimensional kernel and $\omega_\mathrm{r}\to\omega_\mathrm{b}$ as $H/m_\Psi\to 0$; `um4_kohnsham.py`, K1: $\omega_\mathrm{r}=\omega_\mathrm{b}(1-H/(2q_\mathrm{b}))$ up to $-27H^2/m_\Psi$ at $\mu_\mathrm{L}=-0.6$, $H=10^{-4}m_\Psi$, and the box method against the exact resonance to 2×10^{-9}; K9: the same for the weak binding $q_\mathrm{b}=0.03\,m_\Psi$).

$$\omega_0=\omega_\mathrm{r}-\frac{\mathrm{i}}{2}\,\Gamma_\chi,\qquad \Gamma_\chi\simeq p_\mathrm{w}\,m_\Psi\exp\Big(-2F(\upsilon_\mathrm{b})\,\frac{m_\Psi}{H}\Big),\qquad F(\upsilon)=\ln\frac{1+\sqrt{1-\upsilon^2}}{\upsilon}-\sqrt{1-\upsilon^2},\qquad \upsilon_\mathrm{b}=\frac{\omega_\mathrm{b}}{m_\Psi} \tag{FE-86}$$

Symbols in (FE-86).

- ω_0: The energy at rest (zero momentum along ordinary space and along the extra times) of a state of dirac16complex at the fold; complex for a resonance, $\omega_0=\omega_\mathrm{r}-\frac{\mathrm{i}}{2}\Gamma_\chi$. Unit: GeV. Status: DERIVED.
- ω_r: The mass of the particle χ: the real part of the energy at rest of the resonance of dirac16complex at the fold, $\omega_\mathrm{r}=\omega_\mathrm{b}(1-H/(2q_\mathrm{b}))+\ldots$. Unit: GeV. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- Γ_χ: The width of the particle χ at rest: the rate of its decay into the waves of dirac16complex in the bulk. Unit: GeV. Value: below $1/t_0=1.5\times 10^{-42}$ GeV for dark matter. Status: DERIVED.
- p_w: The prefactor of the width of the particles χ in its tunnelling form $\Gamma_\chi\approx p_\mathrm{w}m_\Psi\mathrm{e}^{-2F(\upsilon_\mathrm{b})m_\Psi/H}$, taken from the computed widths at $m_\Psi/H=40$. Unit: dimensionless. Value: 0.14. Status: ESTIMATE.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.

- $F(\upsilon)$: The tunnelling exponent of the width of the particles χ per unit of m_Ψ/H: the integral of the local decay rate q over the region inside the turning point, divided by m_Ψ/H, as a function of the ratio υ of the energy to the bulk mass. Unit: dimensionless. Value: 0.049 for $\upsilon = 0.866$; $(q_\mathrm{b}/\omega_\mathrm{b})^3/3$ for weak binding. Status: DERIVED.
- υ, υ_b: The argument of the tunnelling exponent F, a ratio of an energy to the bulk mass m_Ψ; $\upsilon_\mathrm{b} = \omega_\mathrm{b}/m_\Psi$ that of the particle χ. Unit: dimensionless. Status: DERIVED.
- ω_b: The energy at zero momentum of the state of dirac16complex at the fold in the limit $H \to 0$ (the bound state of UM-3): the particle χ of UM-4 has the mass $\omega_\mathrm{r} = \omega_\mathrm{b}(1 - H/(2q_\mathrm{b})) + \ldots$, equal to ω_b up to the relative correction $H/(2q_\mathrm{b})$. Unit: GeV. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.

The resonances of dirac16complex at the fold: the solutions of FE-37 with outgoing waves on both sides of the fold (the Hankel functions of FE-85) that satisfy the matching FE-39 exist only at the complex energies ω_0, with the real part ω_r of FE-40 and the width Γ_χ, the rate at which a particle χ at rest decays into the waves of the bulk. The width is the tunnelling through the region $|\sigma| < \sigma_\mathrm{t}$ in which the warped frequency is below the bulk mass: $2\int_0^{\sigma_\mathrm{t}} q\,\mathrm{d}\sigma = 2F(\upsilon_\mathrm{b})\,m_\Psi/H$ ($F(\upsilon) = \int_\upsilon^1 \sqrt{1-t^2}\,\mathrm{d}t/t$, with $t = \omega\mathrm{e}^{H\sigma}/m_\Psi$), so the particles are long-lived only if m_Ψ/H is large; the prefactor $p_\mathrm{w} \approx 0.14$ is taken from the computed widths at $m_\Psi/H = 40$. Particles χ that make up the dark matter today must live longer than the age $t_0 = 13.8$ Gyr of the universe, $\Gamma_\chi < 1/t_0 = 1.5 \times 10^{-42}$ GeV, which requires $m_\Psi/H > \ln(p_\mathrm{w} m_\Psi t_0)/(2F(\upsilon_\mathrm{b}))$: about 1.1×10^3 for $q_\mathrm{b} = m_\Psi/2$ ($\upsilon_\mathrm{b} = 0.866$, $F = 0.049$) and 1.7×10^5 for the weak binding $q_\mathrm{b}/\omega_\mathrm{b} = 0.1$ ($\upsilon_\mathrm{b} = 0.995$, $F = 3.3 \times 10^{-4}$; for weak binding $F \simeq (q_\mathrm{b}/\omega_\mathrm{b})^3/3$). With H at its lower bound of 2.5 TeV (FE-82) the particles χ are then heavier than 2.4×10^6 GeV ($q_\mathrm{b} = m_\Psi/2$) or 4.4×10^8 GeV ($q_\mathrm{b}/\omega_\mathrm{b} = 0.1$); a lifetime of 100 times the age raises these bounds by about 4 per cent. Direct detection demands a far weaker binding by the Higgs portal (its Yukawa coupling, FE-47): the largest $q_\mathrm{b}/\omega_\mathrm{b}$ that the limit of LZ allows falls like $\omega_\mathrm{b}^{-1/4}$, from 0.019 at 10 TeV to 4.9×10^{-3} at 2.4×10^6 GeV and 3×10^{-6} at the Planck mass, and with that binding the particles live longer than the universe only if H is below 2×10^{-3} GeV at 2.4×10^6 GeV and below 2 GeV even at the Planck mass: never for $H > 2.5$ TeV. A particle moving with the momentum $\vec{k}$ has the energy $\sqrt{\omega_0^2 + \vec{k}^2}$ (FE-38) and the width $\Gamma_\chi\omega_\mathrm{r}/\sqrt{\omega_\mathrm{r}^2 + \vec{k}^2}$ (the time dilation of the boost). Status: DERIVED (`um4_spectra.py`, U3 and U4: the slope of $\ln\Gamma_\chi$ in m_Ψ/H between 30 and 40 agrees with $-2F(\upsilon_\mathrm{b})$ within 6 per cent), ESTIMATE (the prefactor and the bounds, U6; the binding that direct detection allows, `um4_thermal.py`, T8).

8.3 The Kohn-Sham equations

$$\big[\gamma^M D_M - m_\Psi - \lambda_\Psi\, n_\mathrm{s} + \lambda_\Psi\,\mathcal{N}\big]\varphi_o = \delta_\mathrm{D}(\sigma)\,\frac{1}{V_*}\,\langle\mathbb{M}\rangle\,\varphi_o,$$

$$\mathcal{N} = \sum_o \mathsf{n}_o\,\varphi_o\bar{\varphi}_o, \qquad n_\mathrm{s} = \mathrm{tr}\,\mathcal{N} = \sum_o \mathsf{n}_o\,\bar{\varphi}_o\varphi_o, \qquad \mathsf{n}_o = \frac{1}{\mathrm{e}^{(E_o-\mu_\Psi)/\mathsf{T}_\Psi}+1},$$

$$\chi_o' = \gamma^{(0)}\Big[\mathrm{i}\omega\,\mathrm{e}^{H|\sigma|}\gamma^{(4)} + m_\Psi + \mathcal{M}(\sigma)\Big]\chi_o, \qquad \mathcal{M} = G_\mathrm{KS}\,\mathrm{e}^{7H|\sigma|}\big(\hat{\rho}\,\mathbf{1} - \hat{\mathcal{N}}\big), \qquad \hat{\rho} = \frac{1}{4}\sum_o \bar{\chi}_o\chi_o, \qquad \hat{\mathcal{N}} = \frac{1}{4}\sum_o \chi_o\bar{\chi}_o \tag{FE-41}$$

Symbols in (FE-41).

- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar{\Psi} = \partial_M\bar{\Psi} - \bar{\Psi}\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat{h}^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- λ_Ψ: The coupling of the contact self-interaction of dirac16complex. Unit: GeV^{-6}. Status: RECORD.

- n_s: The scalar density of dirac16complex in the Kohn-Sham state, $n_\mathrm{s} = \langle\bar\Psi\Psi\rangle = \mathrm{tr}\,\mathcal{N}$. Unit: GeV^7. Status: DERIVED.
- $\mathcal{N}$: The one-body density matrix of dirac16complex, $\mathcal{N} = \sum_o \mathsf{n}_o\varphi_o\bar\varphi_o$ (a 16×16 matrix at each point). Unit: GeV^7. Status: DERIVED.
- φ_o, $\bar\varphi_o$: A Kohn-Sham orbital of dirac16complex (a single-particle solution of the Kohn-Sham equation: the Dirac equation of dirac16complex with the mean field of the gas added to its mass) and its Dirac adjoint. Unit: GeV$^{7/2}$. Status: RECORD.
- o: The label of a single-particle state of dirac16complex: a Kohn-Sham orbital, or a mode of the quantised field; a sum over o runs over the orbitals or the modes. Status: STANDARD.
- δ_D: The Dirac delta function of σ: zero for $\sigma\neq0$, with the integral 1 over any range that contains $\sigma=0$; the integral of $\delta_\mathrm{D}(\sigma)$ times a smooth function of σ is the value of that function at $\sigma=0$. Unit: 1/length (for $\delta_\mathrm{D}(\sigma)$). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2-z)\ln(\sin z)/(6H)$ on one period $0<z<\pi$; $\sigma=0$ is the fold $z=\pi/2$, $\sigma\to\mp\infty$ the tips $z\to0,\pi$; one period of x_0 is the whole line $-\infty<\sigma<\infty$. Unit: length. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_\mathrm{t} = \int\mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $\mathbb{M}$: The fold operator of dirac16complex: the 16×16 matrix (with a derivative along the fold) built from the Standard-Model fields at the fold by varying the portals with respect to $\bar\Psi$. Unit: GeV^{-3} (so that $\mathbb{M}/V_*$ is dimensionless). Status: DERIVED.
- n_o: The occupation of the Kohn-Sham orbital o (between 0 and 1). Unit: dimensionless. Status: RECORD.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- μ_Ψ: The chemical potential of dirac16complex. Unit: GeV. Status: DERIVED.
- T_Ψ: The temperature of dirac16complex. Unit: GeV. Status: DERIVED.
- $\chi_o(\sigma)$, $\bar\chi_o$: The profile of the occupied Kohn-Sham orbital o ($o=1,\ldots,4$, the four positive-energy states of the level at the fold) of a homogeneous gas of the particles χ at rest on the static block: a 16-component spinor function of σ; the orbital of each particle is $\mathrm{e}^{\frac{7}{2}H|\sigma|}\chi_o(\sigma)\mathrm{e}^{-\mathrm{i}\omega x_4}$ divided by the square root of the volume of ordinary space and of V_*; normalised to the Krein charge -1 with the measure of UM-4, $\int\mathrm{d}\sigma\,\mathrm{e}^{H|\sigma|}\chi_o^\dagger\mathfrak{B}\chi_o=-1$; $\bar\chi_o = \chi_o^\dagger\sigma_{16}$ is its Dirac adjoint. Unit: GeV$^{1/2}$. Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- ω: A frequency: the energy of a mode $\propto\mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda=-21H^2$ and the tension $\kappa\lambda_\mathrm{F}=12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H\gtrsim2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- $\mathcal{M}(\sigma)$: The mean field of the self-interaction in the Kohn-Sham equation of the gas: the 16×16 matrix $\lambda_\Psi(n_\mathrm{s}\mathbf{1}-\mathcal{N})$ at each σ, which adds to the mass m_Ψ; positive (repulsive) for $\lambda_\Psi>0$. Unit: GeV. Status: DERIVED.
- G_KS: The dimensionless strength of the self-interaction of the gas of particles χ in the Kohn-Sham states: the self-coupling λ_Ψ times the number density n_χ of the particles on the fold divided by the volume V_* of the extra times, $G_\mathrm{KS}=\lambda_\Psi n_\chi/V_*$ (in units of m_Ψ when numbers are given). Unit: dimensionless. Value: 10^{-6} to 0.5 in the solutions. Status: DERIVED.
- $\hat\rho(\sigma)$: The scalar density of the occupied orbitals, $\hat\rho = \frac14\sum_o\bar\chi_o\chi_o$ (per particle); the scalar density of the gas is $n_\mathrm{s} = (n_\chi/V_*)\,\mathrm{e}^{7H|\sigma|}\hat\rho$. Unit: GeV. Status: DERIVED.
- $\mathbf{1}$, $\mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.

- $\hat{\mathcal{N}}(\sigma)$: The one-body density matrix of the occupied orbitals, $\hat{\mathcal{N}} = \frac{1}{4}\sum_o \chi_o\bar{\chi}_o$ (a 16×16 matrix at each σ, per particle); the density matrix of the gas is $\mathcal{N} = (n_\chi/V_*)\,\mathrm{e}^{7H|\sigma|}\hat{\mathcal{N}}$. Unit: GeV. Status: DERIVED.

The Kohn-Sham equations of dirac16complex coupled to itself, to gravity and to every Standard-Model field: the quantised field is described by single-particle orbitals φ_o with the occupations n_o (Fermi-Dirac with the temperature T_Ψ and the chemical potential μ_Ψ of dirac16complex; $\mathsf{n}_o = 0$ or 1 in a degenerate state). Each orbital obeys the field equation with the mean fields: gravity through D_M and γ^M (the background metric and its spin connection), the self-interaction through the Hartree term $\lambda_\Psi n_\mathrm{s}$ (the scalar density) and the exact exchange term $-\lambda_\Psi\mathcal{N}$ (the one-body density matrix, a 16×16 matrix at each point), and every Standard-Model field through the fold operator with the mean fields of the Standard Model, $\langle\mathbb{M}\rangle$ (FE-13, FE-39). The mean-field decomposition of $-\frac{\lambda_\Psi}{2}(\bar\Psi\Psi)^2$ is $-\lambda_\Psi n_\mathrm{s}\bar\Psi\Psi + \lambda_\Psi\bar\Psi\mathcal{N}\Psi$ (the second term with the sign of fermion exchange); correlation beyond exchange is neglected and the negative-energy sea is not included (no-sea approximation). The third line is the reduction to the hidden direction for a homogeneous gas of the particles χ at rest on the static block: the four occupied orbitals χ_o of the level of FE-40 (Krein charge -1 each, normalised with the measure of FE-43, $\int \mathrm{d}\sigma\,\mathrm{e}^{H|\sigma|}\chi_o^\dagger\mathfrak{B}\chi_o = -1$, with $\Psi = \mathrm{e}^{(7/2)H|\sigma|}\chi$ as in FE-36), the mean field $\mathcal{M}(\sigma) = \lambda_\Psi(n_\mathrm{s} - \mathcal{N})$ written with the scalar density $\hat\rho$ and the density matrix $\hat{\mathcal{N}}$ per particle and the dimensionless strength $G_\mathrm{KS} = \lambda_\Psi n_\chi/V_*$ (n_χ the number density of the particles on the fold); the proper densities carry the factor $\mathrm{e}^{7H|\sigma|}$ of $\bar\Psi\Psi = \mathrm{e}^{7H|\sigma|}\bar\chi\chi$ (UM-3: $\mathrm{e}^{6H|\sigma|}$), and the frequency carries the warp $\mathrm{e}^{H|\sigma|}$. The solutions (DERIVED, `um4_kohnsham.py`, K1 to K9, for $\mu_\mathrm{L} = -0.6$, $\mu_\mathrm{R} = 0$, $H = 10^{-4}m_\Psi$, solved in the box $|\sigma| \le 40/m_\Psi$ well inside the turning point, with the author's matrices, reduced exactly by the rotations of ordinary space and of the extra times and lifted back to sixteen components for the checks): the self-consistent shift of the level equals first-order perturbation theory in the dilute gas to 5×10^{-4} for $G_\mathrm{KS} \le 10^{-3}$, $\delta\omega \propto G_\mathrm{KS}$; the residuals of FE-41 and of the matching are at most 10^{-4} for $G_\mathrm{KS} = 10^{-6}$ to 0.5; the proper density decays away from the fold (the gas stays at the fold, also for an attractive self-interaction); the Higgs-portal loop through FE-49 converges and lowers the level; the pressure of the gas is positive and vanishes in the dilute limit (cold dark matter); and the repulsive self-interaction raises the level towards m_Ψ, which shrinks the barrier of FE-86, so that at $m_\Psi/H = 10^4$ the gas lives longer than the universe only for $G_\mathrm{KS} < 0.147$. Status: RECORD (the Kohn-Sham treatment), DERIVED (the exchange term, the reduction and the solutions), ASSUMED (no correlation, no sea).

$$\langle\mathcal{T}_{\Psi\,MN}\rangle = -\frac{1}{4}\sum_o \mathsf{n}_o\Big(\bar\varphi_o\gamma_M D_N\varphi_o + \bar\varphi_o\gamma_N D_M\varphi_o - (D_M\bar\varphi_o)\gamma_N\varphi_o - (D_N\bar\varphi_o)\gamma_M\varphi_o\Big) + \frac{\lambda_\Psi}{2}\,g_{MN}\big(n_\mathrm{s}^2 - \mathrm{tr}\,\mathcal{N}^2\big) \quad \text{(FE-42)}$$

Symbols in (FE-42).

- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $\mathcal{T}_\Psi$: The energy-momentum tensor of dirac16complex: $-2/\sqrt{|g|}$ times the variation of the action S_Ψ of dirac16complex with respect to g^{MN}. Unit: GeV^8. Status: STANDARD.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- o: The label of a single-particle state of dirac16complex: a Kohn-Sham orbital, or a mode of the quantised field; a sum over o runs over the orbitals or the modes. Status: STANDARD.
- n_o: The occupation of the Kohn-Sham orbital o (between 0 and 1). Unit: dimensionless. Status: RECORD.
- φ_o, $\bar\varphi_o$: A Kohn-Sham orbital of dirac16complex (a single-particle solution of the Kohn-Sham equation: the Dirac equation of dirac16complex with the mean field of the gas added to its mass) and its Dirac adjoint. Unit: GeV$^{7/2}$. Status: RECORD.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu}D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC}G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2\epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- λ_Ψ: The coupling of the contact self-interaction of dirac16complex. Unit: GeV^{-6}. Status: RECORD.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order

$x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- n_s: The scalar density of dirac16complex in the Kohn-Sham state, $n_\mathrm{s} = \langle\bar\Psi\Psi\rangle = \mathrm{tr}\,\mathcal{N}$. Unit: GeV^7. Status: DERIVED.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- $\mathcal{N}$: The one-body density matrix of dirac16complex, $\mathcal{N} = \sum_o \mathsf{n}_o \varphi_o \bar\varphi_o$ (a 16×16 matrix at each point). Unit: GeV^7. Status: DERIVED.

The energy-momentum tensor of the Kohn-Sham state, the source of dirac16complex in FE-1 and, through the portals, in FE-5: the sum over the occupied orbitals of FE-6, and the mean value of the self-interaction (on solutions of FE-41, $\langle\mathcal{L}_\Psi\rangle = \frac{\lambda_\Psi}{2}(n_\mathrm{s}^2 - \mathrm{tr}\,\mathcal{N}^2)$). It closes the loop: the orbitals determine the densities, the densities the mean fields and the metric, the metric and the mean fields the orbitals. For the gas of FE-41 the energy per particle is the orbital energy minus the interaction energy per particle, and the pressure along ordinary space, equal to that along the extra times, is the interaction energy per particle times n_χ: the equation of state is $w_\chi = 5.4 \times 10^{-8}$ to 5.0×10^{-3} for $G_\mathrm{KS} = 10^{-6}$ to 0.1 (DERIVED, `um4_kohnsham.py`, K7). Status: DERIVED.

$$
\begin{aligned}
&Q_\Psi = \int \mathrm{d}\sigma\,\mathrm{d}^3x\,\sqrt{|g|}\;\mathrm{e}^{H|\sigma|}\;\Psi^\dagger \mathfrak{B}\Psi = b_4^3 \int \mathrm{d}\sigma\,\mathrm{d}^3x\;\mathrm{e}^{H|\sigma|}\;\chi^\dagger\mathfrak{B}\chi,\\
&\mathfrak{B} = \mathrm{i}\,\sigma_{16}\,\gamma^{(4)}, \qquad \mathfrak{B}^\dagger = \mathfrak{B}, \qquad \mathfrak{B}^2 = \mathbf{1}, \qquad E_o = -\omega_o\,\mathcal{Q}_o
\end{aligned}
\tag{FE-43}
$$

Symbols in (FE-43).

- Q_Ψ: The conserved charge of dirac16complex (the Noether charge of a change of the phase of Ψ by a constant), a Krein form (indefinite); as written it is the charge per unit coordinate volume of the extra times, on which no field depends. Unit: GeV^3. Status: DERIVED.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- x: The coordinates of the fold: x_1, x_2, x_3 of ordinary space, x_4 the time, x_5, x_6, x_7 the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $\mathfrak{B}$: The Krein operator, $\mathfrak{B} = \mathrm{i}\sigma_{16}\gamma^{(4)}$: a Hermitian involution with the signature (8, 8); $\Psi^\dagger\mathfrak{B}\Psi$ is the charge density. Unit: dimensionless. Status: DERIVED.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.
- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{\frac{7}{2}H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x} - \omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of the level of dirac16complex at the fold (a long-lived resonance) and $\bar\chi$ its antiparticle. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- σ_{16}: The matrix of the Dirac adjoint, $\sigma_{16} = T_0T_1T_2T_3$, real and symmetric; $\sigma_{16}T_a$ is real and antisymmetric. Unit: dimensionless. Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1}$, $\mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- o: The label of a single-particle state of dirac16complex: a Kohn-Sham orbital, or a mode of the quantised field; a sum over o runs over the orbitals or the modes. Status: STANDARD.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- $\mathcal{Q}_o$: The charge Q_Ψ (the Krein charge) of the single-particle state o of dirac16complex, a Kohn-Sham orbital or a mode: $+1$ or -1 for a normalised state. Unit: dimensionless. Status: DERIVED.

The conserved charge of dirac16complex (the Noether charge of a constant change of the phase of Ψ, whose density is $\mathrm{i}\bar\Psi\gamma^{x_4}\Psi$, with $\gamma^{x_4} = \mathrm{e}^{H|\sigma|}\gamma^{(4)}$ in the metric of UM-4; per unit coordinate volume of the extra times) is a Krein form: $\mathfrak{B}$ is a Hermitian involution with eight eigenvalues $+1$ and eight eigenvalues -1, so the charge, and with it the norm of the quantised field, is indefinite (the consequence of the signature (4,4); DERIVED, `um4_fields.py`, F3). $\mathfrak{B}$ commutes with $\gamma^{(0)}, \ldots, \gamma^{(4)}$ and anticommutes with $\gamma^{(5)}, \gamma^{(6)}, \gamma^{(7)}$ and with Γ_8: every term that treats the two halves differently (here: the fold terms with $\mu_\mathrm{L} \neq \mu_\mathrm{R}$) mixes the two Krein sectors. The energy of a stationary orbital of frequency ω_o and charge $\mathcal{Q}_o$ is $E_o = -\omega_o\mathcal{Q}_o$ (from FE-6), positive when the frequency and the charge have opposite signs. Of the eight resonances at $+\omega_\mathrm{r}$, four have positive and four negative charge (integrated over the region of the fold inside the turning point; no null states): four carry positive and four negative energy; the mirror $\chi(\sigma) \to T_0T_1T_2T_3\,\chi(-\sigma)$ of the fold (with that of ordinary space) maps the eight onto themselves, the four of one Krein sign even, the four of the other odd. The quantisation of this charge is FE-67. Status: DERIVED (`um4_spectra.py`, U7 and U8; `um4_krein.py`, KR1 and KR3).

$$\left\{\mathsf{a}_o, \mathsf{a}_o^\dagger\right\} = -\mathcal{Q}_o, \qquad \mathsf{H}_\Psi = \sum_o \left(-\mathcal{Q}_o\,\omega_o\right)\mathsf{a}_o^\dagger\mathsf{a}_o, \qquad \left[\mathsf{H}_\Psi, \mathsf{a}_o^\dagger\right] = \omega_o\,\mathsf{a}_o^\dagger \tag{FE-67}$$

Symbols in (FE-67).

- a_o, $\mathsf{a}_o^\dagger$: The annihilation and the creation operator of the mode o of the quantised dirac16complex (frequency ω_o, Krein charge $\mathcal{Q}_o$); the dagger is the adjoint with respect to the indefinite (Krein) scalar product. Unit: dimensionless. Status: STANDARD.
- o: The label of a single-particle state of dirac16complex: a Kohn-Sham orbital, or a mode of the quantised field; a sum over o runs over the orbitals or the modes. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $\mathcal{Q}_o$: The charge Q_Ψ (the Krein charge) of the single-particle state o of dirac16complex, a Kohn-Sham orbital or a mode: $+1$ or -1 for a normalised state. Unit: dimensionless. Status: DERIVED.
- H_Ψ: The Hamiltonian (the energy operator) of the quantised dirac16complex. Unit: GeV. Status: DERIVED.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.

The quantised dirac16complex. The canonical anticommutator of the field is $\{\Psi, \Psi^\dagger\} = -\mathfrak{B}$ (times the delta function of the positions), the overall sign chosen so that the occupied states of charge -1, the particles χ, have a positive norm. A mode o of the frequency ω_o and the Krein charge $\mathcal{Q}_o = \pm 1$ then has the anticommutator $-\mathcal{Q}_o$, the field has the Hamiltonian H_Ψ (with FE-43, the energy of the mode is $-\mathcal{Q}_o\omega_o$), and every quantum raises the energy by its frequency. In the positive-energy quantisation (the vacuum is annihilated by a_o for $\omega_o > 0$ and by $\mathsf{a}_o^\dagger$ for $\omega_o < 0$) every particle and every hole carries a positive energy and the norm $-\mathcal{Q}_o$ of its mode: the particles and the holes of the modes of charge -1 are physical, those of the modes of charge $+1$ are ghosts (negative norm). A fermion cannot trade its norm for its energy ($\{\mathsf{a}, \mathsf{a}^\dagger\} = -1$ has no representation with a positive norm), so no choice of the vacuum removes the ghosts. The consequences (DERIVED, `um4_krein.py`, KR1 to KR7, on the static block of UM-4 with the warped time): the vacuum cannot decay, because every quantum has a positive energy. Whether the

gas of the particles χ is classically stable is not decided: the boosts between ordinary space and the extra times, which mix the two charges, are exact flat directions of the self-consistent gas; the linear response truncated to the occupied level has frequencies far below the splitting of the two multiplets (0.06 of it), two of them imaginary, but of the size of its truncation error; and in the self-consistent mean field the partner level of charge +1 moves away from the occupied level as the self-interaction grows (the two have opposite parity under the mirror of the fold) and stays real and at the fold up to $G_{\mathrm{KS}} = 0.3$ (in UM-3 it turned complex in the denser gases). The couplings at the fold decide unitarity: the chirality-blind fold operators (products of $\gamma^{(1)}, \ldots, \gamma^{(4)}$) never connect a physical state with a ghost at rest, the chiral ones (the same times Γ_8) always do, and the left-handed Higgs portal of the author creates a physical particle χ with a ghost antiparticle with 0.94 times its elastic amplitude; for moving particles (whose energies are $\sqrt{\omega_0^2 + \vec{k}^2}$ exactly, FE-78) the magnetic dipole of the fold connects the two sectors as well, in proportion to the momentum. With chiral couplings at the fold UM-4 is therefore not a unitary quantum theory, and chirality-blind couplings are necessary for unitarity but not shown to be sufficient. Status: DERIVED, OPEN (the classical stability of the gas, a unitary set of couplings).

9 The Standard Model on the fold

9.1 The four-dimensional couplings

$$
\begin{gathered}
g_1^2 = \frac{3H\mathtt{g}_1^2}{2V_*}, \qquad g_2^2 = \frac{3H\mathtt{g}_2^2}{2V_*}, \qquad g_3^2 = \frac{3H\mathtt{g}_3^2}{2V_*}, \qquad 8\pi G_{\mathrm{N}} = \frac{\kappa_7}{V_*}, \\
g_{1,\mathrm{eff}} = \frac{g_1}{\sqrt{\mathcal{Z}}}, \quad g_{2,\mathrm{eff}} = \frac{g_2}{\sqrt{\mathcal{Z}}}, \quad g_{3,\mathrm{eff}} = \frac{g_3}{\sqrt{\mathcal{Z}}}, \qquad \lambda_{\mathrm{eff}} = \frac{\lambda}{\mathcal{Z}}, \qquad \mathcal{Y}_{\mathrm{eff}} = \frac{\mathcal{Y}}{\sqrt{\mathcal{Z}}}, \qquad G_{\mathrm{N,eff}} = \frac{G_{\mathrm{N}}}{\mathcal{Z}}, \qquad \text{(FE-44)} \\
\mathcal{Z} = \frac{V_{\mathrm{t}}\left(\mathrm{e}^{-a_4}b_4\right)^3}{V_*}
\end{gathered}
$$

Symbols in (FE-44).

- g_1: The $U(1)_Y$ gauge coupling of the four-dimensional world (usually g'; not normalised for grand unification): the coupling of the massless mode of the hypercharge field of the bulk, $g_1^2 = 3H\mathtt{g}_1^2/(2V_*)$. Unit: dimensionless. Value: $g_1(m_Z) = 0.3574$. Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling of the four-dimensional world (usually g): the coupling of the massless mode of the $SU(2)_L$ field of the bulk, $g_2^2 = 3H\mathtt{g}_2^2/(2V_*)$. Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- g_3: The $SU(3)_C$ gauge coupling of the four-dimensional world (the strong coupling g_s): the coupling of the massless mode of the gluon field of the bulk, $g_3^2 = 3H\mathtt{g}_3^2/(2V_*)$. Unit: dimensionless. Value: $g_3(m_Z) = 1.2177$ ($g_3^2/(4\pi) = 0.1180$). Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_{\mathrm{F}} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- $\mathtt{g}_1$, $\mathtt{g}_2$, $\mathtt{g}_3$: The gauge couplings of eight dimensions of $U(1)_Y$, $SU(2)_L$ and $SU(3)_C$: the coefficients of the kinetic terms of the gauge fields of the bulk in their geometric normalisation; the four-dimensional couplings are $g_k^2 = 3H\mathtt{g}_k^2/(2V_*)$. Unit: GeV^{-2} ($\mathtt{g}^2$: length^4). Status: ASSUMED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_{\mathrm{t}} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_{N}: Newton's constant of four-dimensional gravity (the gravity of our world, on the fold), $G_{\mathrm{N}} = 1/M_{\mathrm{Pl}}^2$ with the Planck mass $M_{\mathrm{Pl}} = 1.221 \times 10^{19}$ GeV. Unit: GeV^{-2}. Value: 6.708×10^{-39} GeV^{-2}. Status: STANDARD.
- κ_7: The gravitational coupling of the fold: to the first order in its curvature over H^2 the fold obeys the Einstein equations of seven dimensions $\hat{G}^a{}_b = \kappa_7 \mathtt{t}^a{}_b$ with $\kappa_7 = 5\kappa H/2$, the coupling of the massless graviton of the bulk bound to the fold; four-dimensional gravity on the fold has $8\pi G_{\mathrm{N}} = \kappa_7/V_*$ (plus the induced term, which is negligible). Unit: GeV^{-5}. Value: $5\kappa H/2$. Status: DERIVED.
- $g_{k,\mathrm{eff}}$: The effective (canonically normalised) gauge couplings of the fold when the fold factor is $\mathcal{Z}$, $g_{k,\mathrm{eff}} = g_k/\sqrt{\mathcal{Z}}$ ($k = 1, 2, 3$): the kinetic terms of the massless modes of the gauge fields of the bulk carry the volume of the

extra times as the terms of the fold do. Unit: dimensionless. Status: DERIVED.

- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3/V_*$: the factor by which the size of the extra times multiplies every term of the fold and the kinetic terms of the massless modes of the fields of the bulk; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- λ_eff: The effective quartic Higgs coupling, $\lambda/\mathcal{Z}$. Unit: dimensionless. Status: DERIVED.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- $\mathcal{Y}_\mathrm{eff}$: The effective Yukawa matrices, $\mathcal{Y}/\sqrt{\mathcal{Z}}$ (for each of $\mathcal{Y}^u, \mathcal{Y}^d, \mathcal{Y}^e, \mathcal{Y}^\nu$). Unit: dimensionless. Status: DERIVED.
- $\mathcal{Y}^u$, $\mathcal{Y}^d$, $\mathcal{Y}^e$, $\mathcal{Y}^\nu$: The Yukawa matrices (3×3 complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- $G_\mathrm{N,eff}$: The effective Newton constant of the fold, $G_\mathrm{N}/\mathcal{Z}$. Unit: GeV^{-2}. Status: DERIVED.
- V_t: The coordinate volume of the (compact) extra times, $V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$. Unit: length^3. Status: ASSUMED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor (the factor of the lengths) of the extra times is $\mathrm{e}^{-a_4}b_4$ on the fold and $\mathrm{e}^{-a_4}b_4\,\mathrm{e}^{-H|\sigma|}$ at the distance σ from it (the square root of the warp factor times $\mathrm{e}^{-a_4}b_4$). Unit: dimensionless. Status: AUTHOR.

The gauge couplings of the Standard Model are those of the massless modes of the gauge fields of the bulk (FE-74): the integral of the kinetic weight $\mathrm{e}^{-3H|\sigma|}$ over the hidden direction, $2/(3H)$, and the volume V_* of the extra times turn the couplings $\mathtt{g}_k$ of eight dimensions into the four-dimensional couplings g_k (first line, $k = 1, 2, 3$); Newton's constant is that of the graviton of the fold (FE-71; the induced term IN-9 adds $1/\kappa_\mathrm{ind}$ to $1/(8\pi G_\mathrm{N})$, FE-59). When the extra times have the size $\mathrm{e}^{-a_4}b_4$, every term of the fold and the kinetic terms of the zero modes of the bulk carry the factor $\mathcal{Z}$ (IN-8); rescaling every field by $\sqrt{\mathcal{Z}}$ restores the canonical normalisation and leaves the effective couplings of the second line, while the masses (products of a coupling and a vacuum value) are unchanged. With frozen extra times $\mathcal{Z} = 1$ (FE-72), and the couplings are constant; if the volume of the extra times changed, the gauge and Yukawa couplings would change with $\mathcal{Z}^{-1/2}$, the quartic Higgs coupling and Newton's constant with $\mathcal{Z}^{-1}$, which observation excludes (FE-72): the fold fluid holds the volume fixed (FE-65). Status: DERIVED (`um4_bulk.py`, B2; `um4_background.py`, BG4).

9.2 The electroweak vacuum with the ad hoc component

$$m_\mathrm{W}^2 = \frac{g_2^2\,(v^2+\mathcal{A}^2)}{4}, \qquad m_\mathrm{Z}^2 = \frac{(g_1^2+g_2^2)(v^2+\mathcal{A}^2)}{4}, \qquad \frac{G_\mathrm{F}}{\sqrt{2}} = \frac{g_2^2}{8m_\mathrm{W}^2} = \frac{1}{2(v^2+\mathcal{A}^2)}, \qquad m_h^2 = 2\lambda v^2,$$
$$A_\mu = \frac{g_2\,B_\mu - g_1\,n^I W_\mu^I}{\sqrt{g_1^2+g_2^2}}, \qquad n^I = \frac{\left(2\mathcal{A}v,\ 0,\ \mathcal{A}^2-v^2\right)}{\mathcal{A}^2+v^2} \tag{FE-45}$$

Symbols in (FE-45).

- m_W: The mass of the W bosons. Unit: GeV. Value: 80.37 GeV ($\mathcal{A} = 0$). Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling of the four-dimensional world (usually g): the coupling of the massless mode of the $SU(2)_L$ field of the bulk, $g_2^2 = 3H\mathtt{g}_2^2/(2V_*)$. Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_\mathrm{F})^{-1/2}$ with the Fermi constant $G_\mathrm{F} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_\mathcal{A}\mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- m_Z: The mass of the Z boson. Unit: GeV. Value: 91.19 GeV ($\mathcal{A} = 0$). Status: STANDARD.
- g_1: The $U(1)_Y$ gauge coupling of the four-dimensional world (usually g'; not normalised for grand unification): the coupling of the massless mode of the hypercharge field of the bulk, $g_1^2 = 3H\mathtt{g}_1^2/(2V_*)$. Unit: dimensionless. Value: $g_1(m_Z) = 0.3574$. Status: STANDARD.

- G_F: The Fermi constant, $G_\mathrm{F} = 1/(\sqrt{2}\,v^2)$ in the Standard-Model vacuum (its measured value); in the equations of the neutrino flavours it stands for the local Fermi constant of the ad hoc vacuum. Unit: GeV^{-2}. Value: $1.1664 \times 10^{-5}\ \mathrm{GeV}^{-2}$. Status: STANDARD.
- m_h: The mass of the Higgs boson. Unit: GeV. Value: 125.25 GeV (Particle Data Group 2022). Status: STANDARD.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- A_μ: The photon field of the fold (canonically normalised): the massless combination of the massless modes of W^1, W^2, W^3 and B that the Higgs vacuum leaves massless; a field of the bulk whose zero mode travels at the speed of light. Unit: GeV. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- $B_M,\ B_\mu,\ B_{MN}$: The hypercharge gauge field of $U(1)_Y$: in the bulk B_M (geometric normalisation), on the fold the canonically normalised four-dimensional field $B_\mu(x)$ of its massless mode, $g_1 B_\mu(x) = B_\mu(\sigma = 0, x)$; B_{MN}, $B_{\mu\nu}$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- n^I: The unit isovector of the ad hoc vacuum, $n = (2\mathcal{A}v, 0, \mathcal{A}^2 - v^2)/(\mathcal{A}^2 + v^2)$, which fixes which combination of W^1, W^2, W^3 joins B in the photon. Unit: dimensionless. Value: $(0, 0, -1)$ for $\mathcal{A} = 0$. Status: DERIVED.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $W^I_M,\ W^I_\mu,\ W^I_{MN}$: The gauge field of $SU(2)_L$ ($I = 1, 2, 3$): in the bulk W^I_M (geometric normalisation), on the fold the canonically normalised four-dimensional field $W^I_\mu(x)$ of its massless mode, $g_2 W^I_\mu(x) = W^I_\mu(\sigma = 0, x)$; W^I_{MN}, $W^I_{\mu\nu}$ their field strengths; $W^\pm_\mu = (W^1_\mu \mp \mathrm{i}W^2_\mu)/\sqrt{2}$. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.

The masses of the gauge bosons and the photon in the author's ad hoc vacuum (the eigenvalues and eigenvectors of the mass matrix of the canonical fields W^1, W^2, W^3, B of the fold from $|D_\mu \Phi|^2$ with $\Phi = (\mathcal{A}, v)/\sqrt{2}$; DERIVED, `um4_fields.py`, F5): the vacuum value that breaks the electroweak symmetry is $\sqrt{v^2 + \mathcal{A}^2}$, and the unbroken direction, the photon A_μ, is rotated from the usual combination by the unit isovector n^I (for $\mathcal{A} = 0$: $n = (0, 0, -1)$ and $A_\mu = (g_2 B_\mu + g_1 W^3_\mu)/\sqrt{g_1^2 + g_2^2}$). The Fermi constant measured in muon decay is that of the total vacuum value. As $\mathcal{A}$ decays (rate $2a_{40}$), the electroweak vacuum becomes the Standard-Model vacuum. The Higgs mass is that of the Standard-Model vacuum ($\mathcal{A} \to 0$). These are the masses of the zero modes of the gauge fields of the bulk; their massive modes lower the squares of the masses of the W and the Z by $m_\mathrm{W}^4/(3H^2)$ and $m_\mathrm{Z}^4/(3H^2)$ and give them the widths $m(m/H)^3/3$ into the bulk (FE-82), while the photon and the gluons stay massless. Status: DERIVED.

9.3 The masses and the mixing of quarks and leptons

$$
\begin{aligned}
&\frac{v}{\sqrt{2}}\,\mathcal{Y}^u = U_{u\mathrm{L}}\,\mathrm{diag}(m_u, m_c, m_t)\,U^\dagger_{u\mathrm{R}}, \qquad \frac{v}{\sqrt{2}}\,\mathcal{Y}^d = U_{d\mathrm{L}}\,\mathrm{diag}(m_d, m_s, m_b)\,U^\dagger_{d\mathrm{R}},\\
&\frac{v}{\sqrt{2}}\,\mathcal{Y}^e = U_{e\mathrm{L}}\,\mathrm{diag}(m_e, m_\mu, m_\tau)\,U^\dagger_{e\mathrm{R}}, \qquad \frac{v}{\sqrt{2}}\,\mathcal{Y}^\nu = U_{\nu\mathrm{L}}\,\mathrm{diag}(m_1, m_2, m_3)\,U^\dagger_{\nu\mathrm{R}},\\
&V_\mathrm{CKM} = U^\dagger_{u\mathrm{L}} U_{d\mathrm{L}}, \qquad U_\mathrm{PMNS} = U^\dagger_{e\mathrm{L}} U_{\nu\mathrm{L}}, \qquad \nu_\alpha = \sum_{i=1}^{3} \big(U_\mathrm{PMNS}\big)_{\alpha i}\,\nu_i,\\
&\mathcal{L}_\mathrm{CC} = \frac{g_2}{\sqrt{2}}\,W^+_\mu \Big(\bar{\hat{u}}_\mathrm{L}\,\hat{\gamma}^\mu V_\mathrm{CKM}\,\hat{d}_\mathrm{L} + \sum_\alpha \bar{\nu}_\alpha\,\hat{\gamma}^\mu \hat{e}_{\mathrm{L}\alpha}\Big) + \mathrm{h.c.}, \qquad W^\pm_\mu = \frac{W^1_\mu \mp \mathrm{i}W^2_\mu}{\sqrt{2}}
\end{aligned}
\tag{FE-46}
$$

Symbols in (FE-46).

- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_\mathrm{F})^{-1/2}$ with the Fermi constant $G_\mathrm{F} = 1.1663788 \times 10^{-5}\ \mathrm{GeV}^{-2}$ of the muon lifetime (Particle Data Group). Status: STANDARD.
- $\mathcal{Y}^u,\ \mathcal{Y}^d,\ \mathcal{Y}^e,\ \mathcal{Y}^\nu$: The Yukawa matrices ($3 \times 3$ complex matrices in generation space) of the up quarks, the down quarks, the charged leptons and the neutrinos. Unit: dimensionless. Status: STANDARD.
- $U_{f\mathrm{L}},\ U_{f\mathrm{R}}$: The unitary 3×3 matrices that diagonalise the mass matrices of the up quarks, the down quarks, the charged leptons and the neutrinos (left-handed and right-handed rotations). Unit: dimensionless. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor).

Status: STANDARD.

- $m_u, \ldots, m_3$: The masses of the quarks (m_u, m_c, m_t; m_d, m_s, m_b), of the charged leptons (m_e, m_μ, m_τ) and of the three neutrino mass states (m_1, m_2, m_3, Dirac masses). Unit: GeV. Value: measured values; $\Delta m^2_{21} = 7.41\times10^{-5}$ eV^2, $|\Delta m^2_{31}| = 2.511 \times 10^{-3}$ eV^2 in the computations of UM-4 (the global fit NuFIT 5.2 of the oscillation data, November 2022), 7.49×10^{-5} and 2.513×10^{-3} eV^2 in NuFIT 6.0 (2024); the lightest mass $m_1 = 0$ in the computations, below 23 meV by DESI DR2 with the CMB and NuFIT 6.0. Status: STANDARD.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $V_{\rm CKM}$: The quark mixing matrix (Cabibbo-Kobayashi-Maskawa). Unit: dimensionless. Status: STANDARD.
- $U_{\rm PMNS}$: The lepton mixing matrix (Pontecorvo-Maki-Nakagawa-Sakata) of the three Dirac neutrinos; its entries $(U_{\rm PMNS})_{\alpha i}$ connect the flavour state ν_α with the mass state ν_i. Unit: dimensionless. Value: $\theta_{12} = 33.40°$, $\theta_{23} = 49.14°$, $\theta_{13} = 8.54°$ (NuFIT 5.2, November 2022, normal ordering, the fit without the Super-Kamiokande atmospheric data); the Dirac phase is set to zero (ASSUMED; the fit prefers 197°). Status: STANDARD, ASSUMED (the phase).
- ν_α, ν_i: A neutrino field: ν_α the flavour states (the flavour index runs over the electron, muon and tau flavours), ν_i the mass states ($i = 1, 2, 3$). Unit: GeV$^{3/2}$. Status: STANDARD.
- α: A lepton-flavour index, $\alpha \in \{e, \mu, \tau\}$. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the three neutrino mass states; in a frame component such as $\mathcal{S}^{(i)}{}_{(i)}$ one of the three directions of ordinary space (the three components are equal), and in a tensor of ordinary space such as h_{ij} one of the directions x_1, x_2, x_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $\mathcal{L}_{\rm CC}$: The charged-current interaction of the quarks and leptons with the W bosons. Unit: GeV^4. Status: STANDARD.
- g_2: The $SU(2)_L$ gauge coupling of the four-dimensional world (usually g): the coupling of the massless mode of the $SU(2)_L$ field of the bulk, $g_2^2 = 3H\mathfrak{g}_2^2/(2V_*)$. Unit: dimensionless. Value: $g_2(m_Z) = 0.6517$. Status: STANDARD.
- W^I_M, W^I_μ, W^I_{MN}: The gauge field of $SU(2)_L$ ($I = 1, 2, 3$): in the bulk W^I_M (geometric normalisation), on the fold the canonically normalised four-dimensional field $W^I_\mu(x)$ of its massless mode, $g_2 W^I_\mu(x) = W^I_\mu(\sigma = 0, x)$; W^I_{MN}, $W^I_{\mu\nu}$ their field strengths; $W^\pm_\mu = (W^1_\mu \mp \mathrm{i}W^2_\mu)/\sqrt{2}$. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- $\hat{u}_{\rm L}$, $\hat{d}_{\rm L}$, $\hat{e}_{{\rm L}\alpha}$: The left-handed quarks and charged leptons in the mass basis ($\hat{e}_{{\rm L}\alpha}$ the charged lepton of flavour α). Unit: GeV$^{3/2}$. Status: STANDARD.
- $\hat{\gamma}^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold: $\hat{\gamma}^{(a)}$ along the frame directions of the fold and $\hat{\gamma}^\mu$ along its coordinates (the frame matrices contracted with the inverse frame field of the fold), $\{\hat{\gamma}^\mu, \hat{\gamma}^\nu\} = -2\hat{h}^{\mu\nu}\mathbf{1}$; $\hat{\gamma}_\mu = \hat{h}_{\mu\nu}\hat{\gamma}^\nu$. Unit: dimensionless. Status: STANDARD.
- h.c.: The Hermitian conjugate of the preceding terms. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.

The Dirac masses of the quarks, the charged leptons and the three neutrinos (in the vacuum with $\mathcal{A} = 0$; with $\mathcal{A} \neq 0$ the same with the rotated vacuum), the unitary matrices that diagonalise the Yukawa matrices, the quark mixing matrix $V_{\rm CKM}$ and the lepton mixing matrix $U_{\rm PMNS}$, and the charged-current interaction in the mass basis (fields with a hat; the neutrino flavour states ν_α are the partners of the charged leptons of definite mass). This is the flavour structure of the Standard Model with three Dirac neutrinos; the portals of dirac16complex add nothing to it (they are flavour blind, IN-16). The values of the masses and mixing angles are the measured ones (STANDARD inputs). Status: STANDARD.

9.4 The four-dimensional couplings of the particles of dirac16complex

$$\mathfrak{g}_\Phi = \frac{\partial\omega_{\rm b}}{\partial\mu_{\rm L}}\,\frac{c_\Phi}{V_*}, \qquad \mathfrak{g}_G = \sum_X \frac{\partial\omega_{\rm b}}{\partial\mu_X}\,\frac{c^G_X}{4V_*}, \qquad \mathfrak{g}_f = \sum_X \frac{\partial\omega_{\rm b}}{\partial\mu_X}\,\frac{2\omega_{\rm b}\,c^f_X}{V_*}, \qquad \frac{\partial\omega_{\rm b}}{\partial\mu_X} = -\frac{q_{\rm b}}{\omega_{\rm b}}\,\frac{\partial q_{\rm b}}{\partial\mu_X},$$
$$\Delta\omega_{\rm b} = \mathfrak{g}_\Phi\,\Delta\langle\Phi^\dagger\Phi\rangle + \mathfrak{g}_G\,\Delta\langle G^2\rangle + \sum_f \mathfrak{g}_f\,\Delta n_f, \qquad \mathfrak{g}_{h\chi} = \mathfrak{g}_\Phi\,v = -\frac{m^2_\Psi\,\mu^2_{\rm L}}{2\,\omega_{\rm b}\,v} \quad (\mu_{\rm R} = 0) \tag{FE-47}$$

Symbols in (FE-47).

- $\mathfrak{g}_\Phi$, $\mathfrak{g}_G$, $\mathfrak{g}_f$, $\mathfrak{g}_{h\chi}$: The four-dimensional couplings of the particles χ of dirac16complex to the Standard Model: to $\Phi^\dagger\Phi$, to the gluon kinetic density, to the number densities of the fermions of kind f, and the Yukawa coupling of the Higgs boson to χ. Unit: $\mathfrak{g}_\Phi$: GeV^{-1}; $\mathfrak{g}_G$: GeV^{-3}; $\mathfrak{g}_f$: GeV^{-2}; $\mathfrak{g}_{h\chi}$: dimensionless. Status: DERIVED.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- ω_b: The energy at zero momentum of the state of dirac16complex at the fold in the limit $H \to 0$ (the bound state of UM-3): the particle χ of UM-4 has the mass $\omega_\mathrm{r} = \omega_\mathrm{b}(1 - H/(2q_\mathrm{b})) + \ldots$, equal to ω_b up to the relative correction $H/(2q_\mathrm{b})$. Unit: GeV. Status: DERIVED.
- μ_X, μ_L, μ_R: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex, set by the mean fields of the Standard Model at the fold: the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_\mathrm{L}P_\mathrm{L} + \mu_\mathrm{R}P_\mathrm{R})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_\mathrm{t} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- q_b: The inverse width of a state of dirac16complex at the fold: inside the turning point its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.
- Δ: A change, difference or small variation of the quantity that follows (for example an interval of the time, a perturbation of a density, a difference of two squared masses, or an addition to an effective number of species). Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $\langle G^2\rangle$: The mean square of the gluon field strength, $\langle G^A_{\mu\nu}G^{A\mu\nu}\rangle$ (the gluon condensate in the QCD vacuum, a thermal value in the plasma). Unit: GeV^4. Status: STANDARD.
- G^A_M, G^A_μ, G^A_{MN}: The gluon field of $SU(3)_C$ ($A = 1, \ldots, 8$): in the bulk G^A_M, a field of eight dimensions in its geometric normalisation (the covariant derivative $\partial - \mathrm{i}G$), and on the fold the canonically normalised four-dimensional field $G^A_\mu(x)$ of its massless mode, $g_3 G^A_\mu(x) = G^A_\mu(\sigma = 0, x)$; G^A_{MN} and $G^A_{\mu\nu}$ their field strengths. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- n_f: The number density of the Standard-Model fermions of kind f (particles minus antiparticles, summed over generations and gauge components), the time component of the mean current $\langle J^\mu_f\rangle$ in the rest frame of the medium. Unit: GeV^3. Status: STANDARD.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_\mathrm{F})^{-1/2}$ with the Fermi constant $G_\mathrm{F} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.

The couplings of the particles χ (the resonances of FE-40 of the mass ω_r, equal to ω_b up to the relative correction $H/(2q_\mathrm{b})$) to the Standard Model, by the theorem of Feynmann and Hellmann: a slowly varying Standard-Model

field shifts the strengths μ_X of FE-39 and with them the energy of the state, so the coupling of the particle to a Standard-Model operator is the derivative of its energy with respect to its mean value (valid for momentum transfers small compared with q_b). $\mathfrak{g}_{h\chi}$ is the Yukawa coupling of the Higgs boson h to the particle ($\Delta\langle\Phi^\dagger\Phi\rangle = vh$), here for the Higgs portal alone; it is fixed by the mass of the particle and the bulk mass, $\mathfrak{g}_{h\chi} = -2q_\mathrm{b}^2/(\omega_\mathrm{b} v)$, and it grows with the mass at a fixed ratio $q_\mathrm{b}/\omega_\mathrm{b}$: the heavy particles χ that UM-4 requires (FE-86) are strongly coupled to the Higgs boson unless they are very weakly bound (FE-61). The couplings to W and B follow like $\mathfrak{g}_G$; the dipole portal gives the particles magnetic and electric dipole moments. These couplings govern the annihilation (FE-61), the scattering on nuclei and the production of the particles χ. Status: DERIVED (the theorem checked with the resonance level of UM-4: the left-handed density at the fold equals $\partial\omega/\partial\mu_\mathrm{L}$ to 3×10^{-5}, `um4_kohnsham.py`, K6).

10 The three neutrino flavours

The neutrinos are the upper components ν_L of the three lepton doublets on the fold; with the three right-handed neutrinos they have Dirac masses and the lepton mixing of FE-46. Their flavour conversion follows from FE-22 and FE-24 for neutrinos that propagate through a medium of electrons, nucleons, other neutrinos and dirac16complex, by the standard reduction of the Dirac equations of ultrarelativistic neutrinos to equations for their flavour content (the mean-field quantum kinetic equations of Sigl and Raffelt, 1993). The author asks for correct flavour conversion because it shapes the fate of core-collapse supernovae: Gogilashvili and Tamborra ("Neutrino Flavor Conversion Shapes the Rate of Failed Core-collapse Supernovae", arXiv:2605.16504v3) simulate the collapse of 195 stars of 9 to 120 solar masses and find that flavour equipartition below a critical baryon density raises the fraction of failed explosions from 25.6% (without flavour conversion) to 50.8%, 77.4%, 93.3% and 96.4% for critical densities of 10^9, 10^{10}, 10^{11} and 10^{13} g/cm^3 (from 27.0% to between 47.6% and 88.3% when the stars are weighted with the Salpeter initial mass function). This section writes the equations of motion of the flavour content and shows exactly which terms UM-4 adds to those of the Standard Model.

10.1 The equations of motion of the flavour content

$$
\begin{aligned}
&\mathrm{i}\big(\partial_{x_4} + \hat v\cdot\vec\nabla\big)\,\varrho_{\vec p} = \big[\mathsf{H}_{\vec p},\ \varrho_{\vec p}\big] + \mathrm{i}\,\mathcal{C}_{\vec p}, \qquad \mathrm{i}\big(\partial_{x_4} + \hat v\cdot\vec\nabla\big)\,\bar\varrho_{\vec p} = \big[\bar{\mathsf{H}}_{\vec p},\ \bar\varrho_{\vec p}\big] + \mathrm{i}\,\bar{\mathcal{C}}_{\vec p},\\
&\mathsf{H}_{\vec p} = \frac{v_\mathrm{loc}^2}{v^2}\,\frac{U_\mathrm{PMNS}\,\mathsf{M}^2\,U_\mathrm{PMNS}^\dagger}{2E} + \mathsf{V}_\mathrm{m} + \mathsf{V}_{\nu\nu} + V_\Psi\mathbf{1}, \qquad \mathsf{M}^2 = \mathrm{diag}\big(m_1^2,\ m_2^2,\ m_3^2\big),\\
&\bar{\mathsf{H}}_{\vec p} = -\frac{v_\mathrm{loc}^2}{v^2}\,\frac{U_\mathrm{PMNS}\,\mathsf{M}^2\,U_\mathrm{PMNS}^\dagger}{2E} + \mathsf{V}_\mathrm{m} + \mathsf{V}_{\nu\nu} + V_\Psi\mathbf{1}
\end{aligned}
\qquad \text{(FE-48)}
$$

Symbols in (FE-48).

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\hat v$, $\hat v^\mu$: The direction of motion of a neutrino: the unit vector $\hat v = \vec p/|\vec p|$ ($\hat v_{\vec p}$, $\hat v_{\vec q}$ for the momenta $\vec p$, $\vec q$), and the four-vector $\hat v^\mu = (1, \hat v)$ in the order $(x_4; x_1, x_2, x_3)$; $\mathrm{d}^2\hat v$ is the element of solid angle of the directions. Unit: dimensionless. Status: STANDARD.
- $\vec\nabla$: The gradient along ordinary space, $\vec\nabla = (\partial_{x_1}, \partial_{x_2}, \partial_{x_3})$ (in the local frame of the medium). Unit: 1/length. Status: STANDARD.
- $\varrho_{\vec p}$, $\bar\varrho_{\vec p}$: The 3×3 flavour density matrices of the neutrinos and of the antineutrinos of momentum $\vec p$ in the flavour basis (electron, muon, tau): the diagonal entries are the occupation numbers of the three flavours, the off-diagonal entries the coherence between the flavours; $\bar\varrho$ is defined with the flavour indices in the reversed order, so that it obeys an equation of the same form. Unit: dimensionless. Status: STANDARD.
- $\vec p$: The momentum of a neutrino (three components along ordinary space, in the rest frame of the medium); $|\vec p| = E$ for a neutrino of negligible mass. Unit: GeV. Status: STANDARD.
- $\mathsf{H}_{\vec p}$, $\bar{\mathsf{H}}_{\vec p}$: The flavour Hamiltonians (3×3 Hermitian matrices) of the neutrinos and of the antineutrinos of momentum $\vec p$. Unit: GeV. Status: STANDARD.
- $\mathcal{C}_{\vec p}$, $\bar{\mathcal{C}}_{\vec p}$: The collision terms ($3\times3$ matrices) of the neutrinos and the antineutrinos of momentum $\vec p$: the rates of change of $\varrho_{\vec p}$ and $\bar\varrho_{\vec p}$ by emission, absorption, scattering and pair processes with the matter. Unit: GeV. Status: STANDARD.

- v_{loc}: The local value of the electroweak vacuum, $v_{\mathrm{loc}}^2 = 2\langle\Phi^\dagger\Phi\rangle$: it sets the local masses of the quarks and leptons (each mass times v_{loc}/v) and the local Fermi constant; $v_{\mathrm{loc}} = v$ in the Standard-Model vacuum. Unit: GeV. Value: v in the late universe outside dense dirac16complex. Status: DERIVED.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_{\mathrm{F}})^{-1/2}$ with the Fermi constant $G_{\mathrm{F}} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- U_{PMNS}: The lepton mixing matrix (Pontecorvo-Maki-Nakagawa-Sakata) of the three Dirac neutrinos; its entries $(U_{\mathrm{PMNS}})_{\alpha i}$ connect the flavour state ν_α with the mass state ν_i. Unit: dimensionless. Value: $\theta_{12} = 33.40°$, $\theta_{23} = 49.14°$, $\theta_{13} = 8.54°$ (NuFIT 5.2, November 2022, normal ordering, the fit without the Super-Kamiokande atmospheric data); the Dirac phase is set to zero (ASSUMED; the fit prefers 197°). Status: STANDARD, ASSUMED (the phase).
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- M^2: The diagonal matrix of the squared masses of the three neutrino mass states, $\mathsf{M}^2 = \mathrm{diag}(m_1^2, m_2^2, m_3^2)$ (M appears only squared). Unit: GeV^2 (of M^2). Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- V_{m}: The matter potential of the neutrinos (a 3×3 matrix in flavour space) from their forward scattering on the electrons, positrons and neutrons of the medium. Unit: GeV. Status: STANDARD.
- $\mathsf{V}_{\nu\nu}$: The neutrino-neutrino potential (a 3×3 matrix in flavour space) from the forward scattering of a neutrino on the other neutrinos and antineutrinos of the medium; it depends on their flavour content and their directions. Unit: GeV. Status: STANDARD.
- V_Ψ: The potential energy that dirac16complex gives every neutrino through the current portal of the lepton doublets, $V_\Psi = \mathcal{V}^\ell_{x_4}$ (the same for the three flavours). Unit: GeV. Status: DERIVED.
- $\mathbf{1}$, $\mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- $m_u, \ldots, m_3$: The masses of the quarks (m_u, m_c, m_t; m_d, m_s, m_b), of the charged leptons (m_e, m_μ, m_τ) and of the three neutrino mass states (m_1, m_2, m_3, Dirac masses). Unit: GeV. Value: measured values; $\Delta m^2_{21} = 7.41 \times 10^{-5}$ eV^2, $|\Delta m^2_{31}| = 2.511 \times 10^{-3}$ eV^2 in the computations of UM-4 (the global fit NuFIT 5.2 of the oscillation data, November 2022), 7.49×10^{-5} and 2.513×10^{-3} eV^2 in NuFIT 6.0 (2024); the lightest mass $m_1 = 0$ in the computations, below 23 meV by DESI DR2 with the CMB and NuFIT 6.0. Status: STANDARD.

The equations of motion of the 3×3 flavour density matrices of the neutrinos ($\varrho_{\vec{p}}$) and of the antineutrinos ($\bar{\varrho}_{\vec{p}}$, defined with the flavour indices in the reversed order) of momentum $\vec{p}$, energy $E = |\vec{p}|$ and direction $\hat{v}$; the diagonal entries are the occupation numbers of the three flavours, the off-diagonal entries the coherence between them. Free streaming stands on the left, the commutator with the flavour Hamiltonian and the collisions $\mathcal{C}$ (emission, absorption, scattering and pair processes) on the right. The Hamiltonian contains the vacuum term of the three Dirac masses (with the opposite sign for antineutrinos in this convention), multiplied by v_{loc}^2/v^2 because the masses are the Yukawa couplings times the local value of the Higgs field (FE-49); the matter potential; the neutrino-neutrino potential; and the potential V_Ψ of dirac16complex. Status: STANDARD (UM-4 adds V_Ψ and the factor v_{loc}^2/v^2).

$$
\begin{aligned}
&\mathsf{V}_{\mathrm{m}} = \sqrt{2}\,G_{\mathrm{F,loc}}\,\mathrm{diag}\big(n_e - n_{\bar{e}},\ 0,\ 0\big) - \frac{G_{\mathrm{F,loc}}}{\sqrt{2}}\,n_n\mathbf{1}, \qquad \mathsf{V}_{\nu\nu} = \sqrt{2}\,G_{\mathrm{F,loc}}\int \frac{\mathrm{d}^3\vec{q}}{(2\pi)^3}\big(1 - \hat{v}_{\vec{p}}\cdot\hat{v}_{\vec{q}}\big)\big(\varrho_{\vec{q}} - \bar{\varrho}_{\vec{q}}\big),\\
&V_\Psi = \mathcal{V}^\ell_{x_4} = \sum_X c^\ell_X\, j_{X\,x_4}, \qquad G_{\mathrm{F,loc}} = \frac{1}{\sqrt{2}\,v_{\mathrm{loc}}^2},\\
&v_{\mathrm{loc}}^2 = 2\langle\Phi^\dagger\Phi\rangle, \qquad v_{\mathrm{loc}}^2 = v^2 + \mathcal{A}^2 \ \text{ (the ad hoc vacuum)},\\
&v_{\mathrm{loc}}^2 = v^2 - \frac{c_\Phi}{\lambda}\,\langle\bar{\Psi}P_{\mathrm{L}}\Psi\rangle \ \text{ (the relaxed vacuum with dirac16complex)}
\end{aligned}
\tag{FE-49}
$$

Symbols in (FE-49).

- V_m: The matter potential of the neutrinos (a 3×3 matrix in flavour space) from their forward scattering on the electrons, positrons and neutrons of the medium. Unit: GeV. Status: STANDARD.
- $G_{\mathrm{F,loc}}$: The local Fermi constant, $G_{\mathrm{F,loc}} = 1/(\sqrt{2}\, v_\mathrm{loc}^2)$ ($= G_\mathrm{F}$ where $v_\mathrm{loc} = v$). Unit: GeV^{-2}. Status: DERIVED.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- n_e: The number density of electrons (bound and free; in a supernova all are free, in the recombining universe $x_\mathrm{e} n_e$ is the density of the free electrons). Unit: GeV^3 (cm^{-3}). Status: STANDARD.
- $n_{\bar{e}}$: The number density of positrons. Unit: GeV^3. Status: STANDARD.
- n_n: The number density of neutrons. Unit: GeV^3. Status: STANDARD.
- $\mathbf{1}$, $\mathbf{1}_8$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space); $\mathbf{1}_8$ is the 8×8 identity matrix. Unit: dimensionless. Status: STANDARD.
- $\mathsf{V}_{\nu\nu}$: The neutrino-neutrino potential (a 3×3 matrix in flavour space) from the forward scattering of a neutrino on the other neutrinos and antineutrinos of the medium; it depends on their flavour content and their directions. Unit: GeV. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $\vec{q}$: The momentum of another neutrino of the medium (the integration variable of the neutrino-neutrino potential); $\mathrm{d}^3\vec{q}$ its volume element. Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\hat{v}$, $\hat{v}^\mu$: The direction of motion of a neutrino: the unit vector $\hat{v} = \vec{p}/|\vec{p}|$ ($\hat{v}_{\vec{p}}$, $\hat{v}_{\vec{q}}$ for the momenta $\vec{p}$, $\vec{q}$), and the four-vector $\hat{v}^\mu = (1, \hat{v})$ in the order $(x_4; x_1, x_2, x_3)$; $\mathrm{d}^2\hat{v}$ is the element of solid angle of the directions. Unit: dimensionless. Status: STANDARD.
- $\vec{p}$: The momentum of a neutrino (three components along ordinary space, in the rest frame of the medium); $|\vec{p}| = E$ for a neutrino of negligible mass. Unit: GeV. Status: STANDARD.
- $\varrho_{\vec{p}}$, $\bar{\varrho}_{\vec{p}}$: The 3×3 flavour density matrices of the neutrinos and of the antineutrinos of momentum $\vec{p}$ in the flavour basis (electron, muon, tau): the diagonal entries are the occupation numbers of the three flavours, the off-diagonal entries the coherence between the flavours; $\bar{\varrho}$ is defined with the flavour indices in the reversed order, so that it obeys an equation of the same form. Unit: dimensionless. Status: STANDARD.
- V_Ψ: The potential energy that dirac16complex gives every neutrino through the current portal of the lepton doublets, $V_\Psi = \mathcal{V}^\ell_{x_4}$ (the same for the three flavours). Unit: GeV. Status: DERIVED.
- $\mathcal{V}^f_\mu$: The vector potential that dirac16complex exerts on the Standard-Model fermions of kind f through the current portals, $\mathcal{V}^f_\mu = \sum_X c^f_X j_{X\mu}$ ($\mathcal{V}^Q, \mathcal{V}^u, \ldots, \mathcal{V}^\nu$ for the six kinds); flavour blind. Unit: GeV. Status: DERIVED.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c^f_X: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c^ℓ_X and c^ν_X are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- j^μ_X: The convective current of the half X of dirac16complex, $j^\mu_X = \mathrm{i}(\bar{\Psi} P_X D^\mu \Psi - (D^\mu \bar{\Psi}) P_X \Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^8. Status: DERIVED.
- v_loc: The local value of the electroweak vacuum, $v^2_\mathrm{loc} = 2\langle \Phi^\dagger \Phi \rangle$: it sets the local masses of the quarks and leptons (each mass times v_loc/v) and the local Fermi constant; $v_\mathrm{loc} = v$ in the Standard-Model vacuum. Unit: GeV. Value: v in the late universe outside dense dirac16complex. Status: DERIVED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor; for an operator on the space of states, its adjoint with respect to the scalar product of that space (for the quantised dirac16complex the indefinite Krein scalar product). Status: STANDARD.
- $\langle \cdots \rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.

- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_{\mathrm{F}})^{-1/2}$ with the Fermi constant $G_{\mathrm{F}} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- $\mathcal{A}$: The author's ad hoc upper component of the Higgs vacuum, almightyAdHoc, $\mathcal{A}(x_4) = v_{\mathcal{A}} \mathrm{e}^{-2a_{40}x_4}$. Unit: GeV. Status: AUTHOR.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- λ: The quartic coupling of the Higgs potential, $\lambda > 0$. Unit: dimensionless. Value: $\lambda = m_h^2/(2v^2) = 0.129$. Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: GeV$^{7/2}$. Status: AUTHOR.
- P_{L}: The author's left-handed chiral projector, $P_{\mathrm{L}} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.

The potentials and the local electroweak vacuum. Forward scattering on electrons and positrons through W exchange gives the electron flavour the potential $\sqrt{2}G_{\mathrm{F,loc}}(n_e - n_{\bar e})$; neutral currents give all flavours the same potential, here from the neutrons (the neutral-current potentials of electrons and protons cancel in neutral matter for any weak mixing angle); forward scattering on the other neutrinos gives the neutrino-neutrino potential, which depends on the flavour content and the directions of all other neutrinos and makes the equations non-linear. dirac16complex acts on the neutrinos in exactly two ways, both fixed by the action (IN-16):

- through the current portal of the lepton doublets: V_Ψ is the time component of the vector potential $\mathcal{V}_\mu^\ell$ of FE-22, the same for the three flavours because the portals are flavour blind. A multiple of the unit matrix drops out of every commutator: V_Ψ shifts the energies of the neutrinos but does not change their flavour;
- through the Higgs portal: the scalar density of the left-handed half shifts the mass parameter of the Higgs field (FE-15). Where the Higgs field relaxes to the minimum of its potential (a homogeneous, static vacuum with $\mathcal{A} = 0$ in FE-15), its local value is $v_{\mathrm{loc}}^2 = v^2 - c_\Phi\langle\bar\Psi P_{\mathrm{L}}\Psi\rangle/\lambda$; in the author's ad hoc vacuum $v_{\mathrm{loc}}^2 = v^2 + \mathcal{A}^2$ (FE-45). All masses scale with v_{loc}/v and the Fermi constant with v^2/v_{loc}^2 (at tree level $G_{\mathrm{F}} = 1/(\sqrt{2}v^2)$ does not depend on the gauge couplings, so the gauge-kinetic portals do not change it), while the Yukawa matrices, and with them U_{PMNS}, are unchanged.

So dirac16complex changes neither the mixing angles nor the flavour pattern of the conversion: it rescales the vacuum oscillation frequencies by v_{loc}^2/v^2 and the potentials by v^2/v_{loc}^2 (the resonance density of FE-50 by v_{loc}^4/v^4). In the late universe $\mathcal{A} = 0$, and the dark matter is far too dilute to move v_{loc}: with FE-47 the shift is $\Delta v^2/v^2 = -\mathfrak{g}_{h\chi}\rho_\chi/(\lambda v^3\omega_{\mathrm{b}})$, about $10^{-50}\,\mathfrak{g}_{h\chi}$ for the local dark-matter density 0.3 GeV/cm^3 and $\omega_{\mathrm{b}} = 100$ GeV, and less for the heavier particles χ that UM-4 requires (FE-86). The flavour conversion of UM-4 is therefore that of the Standard Model with three Dirac neutrinos wherever $v_{\mathrm{loc}} = v$, which includes every supernova and the Sun (DERIVED). The Dirac neutrinos of UM-4 have the magnetic moments of the Standard Model, about 3.2×10^{-19} Bohr magnetons times the mass in eV, too small to turn left-handed into right-handed neutrinos in the magnetic fields of supernovae; the dipole portal does not act on neutrinos directly. Status: STANDARD (the potentials), DERIVED (V_Ψ, v_{loc} and their consequences; `um4_neutrino.py`, N1).

10.2 Matter-enhanced conversion

$$\tan 2\theta_{\mathrm{m}} = \frac{\sin 2\theta}{\cos 2\theta - 2EV_{\mathrm{CC}}/\Delta m^2}, \qquad V_{\mathrm{CC}} = \sqrt{2}\,G_{\mathrm{F,loc}}\,n_e, \qquad n_{e,\mathrm{res}} = \frac{\Delta m^2\cos 2\theta}{2\sqrt{2}\,G_{\mathrm{F,loc}}\,E},$$
$$P_{\mathrm{LZ}} = \exp\Big(-\frac{\pi}{2}\,\frac{\Delta m^2\sin^2 2\theta}{2E\cos 2\theta}\,\Big|\frac{\mathrm{d}\ln n_e}{\mathrm{d}r}\Big|_{\mathrm{res}}^{-1}\Big) \tag{FE-50}$$

Symbols in (FE-50).

- θ_{m}: The mixing angle of two neutrino flavours in matter. Unit: dimensionless. Status: STANDARD.
- θ: The vacuum mixing angle of two neutrino flavours (in the two-flavour description of a resonance, one of the three angles $\theta_{12}, \theta_{13}, \theta_{23}$ of U_{PMNS}). Unit: dimensionless. Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- V_{CC}: The charged-current matter potential of the electron neutrinos (forward scattering on electrons through W exchange). Unit: GeV. Status: STANDARD.
- Δ: A change, difference or small variation of the quantity that follows (for example an interval of the time, a perturbation of a density, a difference of two squared masses, or an addition to an effective number of species).

Status: STANDARD.

- m: A mass: m_i the mass of the neutrino mass state i; $\Delta m^2_{ij} = m^2_i - m^2_j$ the squared-mass difference of the states i and j (Δm^2_{21}, Δm^2_{31}, Δm^2_{32}); Δm^2 without labels the difference of the two states of a resonance, $m^2_2 - m^2_1$, multiplied by $v^2_{\rm loc}/v^2$ where this differs from 1; m without a label the mass of a mode of a field of the bulk as the fold sees it, $m^2 = \omega^2 - k^2$; for the W and Z bosons their complex pole mass, $m^2 = m^2_0 - {\rm i}m_0\Gamma_{\rm bulk}$ to the leading order; in the relic abundance of the particles χ their mass. Unit: GeV. Status: STANDARD.
- $G_{\rm F,loc}$: The local Fermi constant, $G_{\rm F,loc} = 1/(\sqrt{2}\,v^2_{\rm loc})$ ($= G_{\rm F}$ where $v_{\rm loc} = v$). Unit: GeV^{-2}. Status: DERIVED.
- n_e: The number density of electrons (bound and free; in a supernova all are free, in the recombining universe $x_e n_e$ is the density of the free electrons). Unit: GeV^3 (cm^{-3}). Status: STANDARD.
- $n_{e,\rm res}$: The electron density at which the flavour conversion in matter is resonant (the Mikheyev-Smirnov-Wolfenstein resonance). Unit: GeV^3. Status: STANDARD.
- $P_{\rm LZ}$: The probability that a neutrino jumps between the eigenstates of matter at the resonance (Landau-Zener); $P_{\rm LZ} = 0$ is an adiabatic resonance. Unit: dimensionless. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as ${\rm d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- r: A distance from a centre: the distance from the centre of the star along the path of a neutrino; in Newton's and Coulomb's laws the distance between two point sources on the fold. Unit: length. Status: STANDARD.
- res: The label of a quantity evaluated at the resonance, where $n_e = n_{e,\rm res}$. Status: STANDARD.

The resonant flavour conversion in matter (Mikheyev-Smirnov-Wolfenstein) for two flavours with the vacuum mixing angle θ and the squared-mass difference Δm^2 (which includes the factor $v^2_{\rm loc}/v^2$ of FE-48): the mixing angle in matter $\theta_{\rm m}$ (FE-48 with the diagonal potential $V_{\rm CC}$ in the electron flavour), the electron density of the resonance, where $\theta_{\rm m} = 45°$, and the probability $P_{\rm LZ}$ that the neutrino jumps between the eigenstates of matter at the resonance for the density profile $n_e(r)$ along its path (the Landau-Zener formula, exact for a linear profile). With the measured masses and mixing angles this gives the survival probability of solar neutrinos and the conversion in the outer layers of supernovae (the resonances of Δm^2_{31} and Δm^2_{21}). Status: STANDARD (`um4_neutrino.py`, N2).

10.3 The fast flavour instability of dense neutrino gases

$$\mathcal{G}_{\rm ELN}(\hat{v}) = \sqrt{2}\,G_{\rm F,loc}\int_0^\infty \frac{E^2\,{\rm d}E}{(2\pi)^3}\Big[\mathcal{F}_e - \bar{\mathcal{F}}_e - \big(\mathcal{F}_x - \bar{\mathcal{F}}_x\big)\Big],$$
$$\det\Big[\eta^{\mu\nu}_{\rm M} + \int {\rm d}^2\hat{v}\;\mathcal{G}_{\rm ELN}(\hat{v})\,\frac{\hat{v}^\mu\hat{v}^\nu}{\Omega_{\rm fw} - \vec{K}\cdot\hat{v}}\Big] = 0, \qquad \hat{v}^\mu = (1,\ \hat{v}) \tag{FE-51}$$

Symbols in (FE-51).

- $\mathcal{G}_{\rm ELN}(\hat{v})$: The angular distribution of the electron lepton number carried by the neutrinos minus that carried by the heavy-lepton neutrinos, per unit solid angle of the direction $\hat{v}$, multiplied by $\sqrt{2}G_{\rm F,loc}$. Unit: GeV. Status: STANDARD.
- $\hat{v}$, $\hat{v}^\mu$: The direction of motion of a neutrino: the unit vector $\hat{v} = \vec{p}/|\vec{p}|$ ($\hat{v}_{\vec{p}}$, $\hat{v}_{\vec{q}}$ for the momenta $\vec{p}$, $\vec{q}$), and the four-vector $\hat{v}^\mu = (1, \hat{v})$ in the order $(x_4; x_1, x_2, x_3)$; ${\rm d}^2\hat{v}$ is the element of solid angle of the directions. Unit: dimensionless. Status: STANDARD.
- $G_{\rm F,loc}$: The local Fermi constant, $G_{\rm F,loc} = 1/(\sqrt{2}\,v^2_{\rm loc})$ ($= G_{\rm F}$ where $v_{\rm loc} = v$). Unit: GeV^{-2}. Status: DERIVED.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no bound. Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as ${\rm d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\mathcal{F}_e$, $\bar{\mathcal{F}}_e$, $\mathcal{F}_x$, $\bar{\mathcal{F}}_x$: The occupation numbers of the electron neutrinos, of the electron antineutrinos, and of one heavy-lepton neutrino and antineutrino ($x = \mu$ or τ), as functions of the energy E and the direction $\hat{v}$ (the

diagonal entries of $\varrho_{\vec{p}}$ and $\bar{\varrho}_{\vec{p}}$); in a transport code, the same for the number of particles in an energy bin. Unit: dimensionless. Status: STANDARD.
- $\eta_{\mathrm{M}}^{\mu\nu}$: The Minkowski metric of the four-vectors $\hat{v}^{\mu}$, $\eta_{\mathrm{M}} = \mathrm{diag}(1,-1,-1,-1)$ in the order $(x_4; x_1, x_2, x_3)$. Unit: dimensionless. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- Ω_{fw}: The (complex) frequency of a flavour wave of the neutrino gas, measured from the shift caused by the potentials; an imaginary part greater than zero is an instability. Unit: GeV. Status: STANDARD.
- $\vec{K}$: The (real) wave vector of a flavour wave of the neutrino gas, measured from the shift caused by the potentials. Unit: GeV. Status: STANDARD.

The fast flavour instability (STANDARD: FE-48 linearised in the flavour coherence, without the vacuum term, which is negligible in the dense gas; Izaguirre, Raffelt and Tamborra 2017). $\mathcal{G}_{\mathrm{ELN}}$ is the angular distribution of the electron lepton number carried by the neutrinos minus that of the heavy-lepton flavours ($\mathcal{F}$ the occupation numbers of ν_e, $\bar{\nu}_e$ and of one heavy flavour ν_x, $\bar{\nu}_x$, as functions of the energy and the direction; $\mathrm{d}^2\hat{v}$ the element of solid angle). A small flavour coherence $\propto \mathrm{e}^{-\mathrm{i}(\Omega_{\mathrm{fw}} x_4 - \vec{K}\cdot\vec{x})}$ grows exponentially if the dispersion relation (the second line, written with the Minkowski metric η_{M} of the four-vectors $\hat{v}^{\mu}$) has a solution with an imaginary part of Ω_{fw} greater than zero for a real wave vector $\vec{K}$ (the frequency and the wave vector are measured from the shifts that the matter and the neutrino-neutrino potentials cause, which drop out). This happens if and only if $\mathcal{G}_{\mathrm{ELN}}$ changes sign as a function of the direction, an angular crossing (Morinaga 2022). The instability then grows on lengths of the order of $1/(\sqrt{2}G_{\mathrm{F}}n_{\nu})$, centimetres in a supernova core, much shorter than the scales of the explosion, and drives the flavours toward equipartition; transport codes therefore model its outcome by a prescription such as FE-52. Status: STANDARD (`um4_neutrino.py`, N3).

$$
\begin{aligned}
&\mathcal{F}_e \to \mathcal{F}_e - \delta_{\mathrm{FC}}, \qquad \bar{\mathcal{F}}_e \to \bar{\mathcal{F}}_e - \delta_{\mathrm{FC}}, \qquad \mathcal{F}_x = \bar{\mathcal{F}}_x \to \mathcal{F}_x + \frac{\delta_{\mathrm{FC}}}{2}, \\
&\delta_{\mathrm{FC}} = \max\Big(\frac{2}{3}\Big(\min\left(\mathcal{F}_e,\ \bar{\mathcal{F}}_e\right) - \mathcal{F}_x\Big),\ \max\left(\mathcal{F}_e,\ \bar{\mathcal{F}}_e\right) - 1\Big), \\
&\text{applied where}\ \ \rho_{\mathrm{b}} < \rho_{\mathrm{FC}}\ \ \text{and}\ \ x_4 - x_{4,\mathrm{b}} > t_{\mathrm{FC}}
\end{aligned}
\tag{FE-52}
$$

Symbols in (FE-52).

- $\mathcal{F}_e,\ \bar{\mathcal{F}}_e,\ \mathcal{F}_x,\ \bar{\mathcal{F}}_x$: The occupation numbers of the electron neutrinos, of the electron antineutrinos, and of one heavy-lepton neutrino and antineutrino ($x = \mu$ or τ), as functions of the energy E and the direction $\hat{v}$ (the diagonal entries of $\varrho_{\vec{p}}$ and $\bar{\varrho}_{\vec{p}}$); in a transport code, the same for the number of particles in an energy bin. Unit: dimensionless. Status: STANDARD.
- δ_{FC}: The occupation (or number per energy bin) of electron neutrino-antineutrino pairs converted into heavy-lepton pairs by the flavour equipartition (the conversion that equalises the occupations of the three flavours where the fast flavour instability acts); a negative value converts heavy-lepton pairs into electron pairs. Unit: dimensionless. Status: ASSUMED.
- ρ_{b}: The (rest-mass) energy density of the baryons: in a supernova the local density of the matter (quoted in g/cm^3), in cosmology the mean density of the baryons ($\rho_{\mathrm{b}}^{(0)}$ today). Unit: GeV^4. Status: STANDARD.
- ρ_{FC}: The baryon density below which the flavour equipartition is applied (the critical density ρ_{c} of Gogilashvili and Tamborra). Unit: GeV^4 (g/cm^3). Value: 10^9, 10^{10}, 10^{11} or 10^{13} g/cm^3. Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $x_{4,\mathrm{b}}$: The time of the bounce of the collapsing stellar core. Unit: length (= time). Status: STANDARD.
- t_{FC}: The time after the bounce from which the flavour equipartition is applied. Unit: time. Value: 0.02 s. Status: STANDARD.

The flavour equipartition that models the outcome of the fast instability in the transport of a core-collapse supernova, written in the form that satisfies the constraints stated by Gogilashvili and Tamborra for their scheme: in every energy bin (and in the same way for the flux of every energy bin) electron neutrino-antineutrino pairs are converted into heavy-lepton pairs, which conserves the electron lepton number $\mathcal{F}_e - \bar{\mathcal{F}}_e$, the number of neutrinos $\mathcal{F}_e + 2\mathcal{F}_x$, the number of antineutrinos $\bar{\mathcal{F}}_e + 2\mathcal{F}_x$ and the momentum, with the four heavy-lepton species equal as in the transport ($\mathcal{F}_x = \bar{\mathcal{F}}_x$). The amount δ_{FC} brings the less abundant of ν_e and $\bar{\nu}_e$ into equipartition with the heavy-lepton flavours (complete equipartition of all six species is impossible when the electron lepton number is not zero, because flavour conversion conserves it); the second argument of the maximum keeps every occupation number at most 1 (the Pauli

principle) when heavy-lepton pairs are converted into electron pairs, and every occupation number stays between 0 and 1. The map is applied where the baryon density is below ρ_{FC} (their ρ_{c}, varied from 10^9 to 10^{13} g/cm^3) and from $t_{\mathrm{FC}} = 0.02$ s after the bounce of the core at $x_{4,\mathrm{b}}$. In UM-4, FE-48 and FE-49 are those of the Standard Model wherever $v_{\mathrm{loc}} = v$, so the prescription and its consequences carry over unchanged: the supernova physics of UM-4 is that of the Standard Model with three Dirac neutrinos, and the failed-explosion fractions of the paper apply (DERIVED from FE-49). Status: STANDARD (the constraints), ASSUMED (this explicit form), DERIVED (its validity in UM-4; `um4_neutrino.py`, N4).

10.4 Neutrinos in the expanding universe

$$\Big(\partial_{x_4} - a_4' \,|\vec{p}|\, \frac{\partial}{\partial|\vec{p}|}\Big)\varrho_{\vec{p}} = -\mathrm{i}\big[\mathsf{H}_{\vec{p}},\ \varrho_{\vec{p}}\big] + \mathcal{C}_{\vec{p}},$$
$$N_{\mathrm{eff}} = \frac{8}{7}\Big(\frac{11}{4}\Big)^{4/3}\frac{\rho_\nu}{\rho_\gamma}, \qquad \mathsf{T}_\nu^{(0)} = \Big(\frac{4}{11}\Big)^{1/3}\mathsf{T}_\gamma^{(0)}, \qquad n_\nu^{(0)} = \frac{3\zeta(3)}{2\pi^2}\big(\mathsf{T}_\nu^{(0)}\big)^3 \tag{FE-53}$$

Symbols in (FE-53).

- ∂: A partial derivative: the derivative with respect to one variable at fixed values of the others (∂_σ with respect to σ; ∂_M and ∂_μ with respect to the coordinate that carries the same index, of the eight dimensions or of the fold). Status: STANDARD.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $\vec{p}$: The momentum of a neutrino (three components along ordinary space, in the rest frame of the medium); $|\vec{p}| = E$ for a neutrino of negligible mass. Unit: GeV. Status: STANDARD.
- $\varrho_{\vec{p}}$, $\bar{\varrho}_{\vec{p}}$: The 3×3 flavour density matrices of the neutrinos and of the antineutrinos of momentum $\vec{p}$ in the flavour basis (electron, muon, tau): the diagonal entries are the occupation numbers of the three flavours, the off-diagonal entries the coherence between the flavours; $\bar{\varrho}$ is defined with the flavour indices in the reversed order, so that it obeys an equation of the same form. Unit: dimensionless. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\mathsf{H}_{\vec{p}}$, $\bar{\mathsf{H}}_{\vec{p}}$: The flavour Hamiltonians (3×3 Hermitian matrices) of the neutrinos and of the antineutrinos of momentum $\vec{p}$. Unit: GeV. Status: STANDARD.
- $\mathcal{C}_{\vec{p}}$, $\bar{\mathcal{C}}_{\vec{p}}$: The collision terms ($3 \times 3$ matrices) of the neutrinos and the antineutrinos of momentum $\vec{p}$: the rates of change of $\varrho_{\vec{p}}$ and $\bar{\varrho}_{\vec{p}}$ by emission, absorption, scattering and pair processes with the matter. Unit: GeV. Status: STANDARD.
- N_{eff}: The effective number of neutrino species: the energy density of the relativistic particles other than photons in units of that of one neutrino species whose temperature is $(4/11)^{1/3}$ times that of the photons. Unit: dimensionless. Value: 3.044 (Standard Model); measured 2.99 ± 0.17 (Planck 2018). Status: STANDARD.
- ρ_ν: The energy density of the neutrinos and antineutrinos of all flavours. Unit: GeV^4. Status: STANDARD.
- ρ_γ: The energy density of the photons. Unit: GeV^4. Status: STANDARD.
- T_ν: The temperature of the cosmic neutrino background; $\mathsf{T}_\nu^{(0)}$ its value today. Unit: GeV (or K). Value: 1.945 K today. Status: STANDARD.
- T_γ: The temperature of the photons (of the CMB); $\mathsf{T}_\gamma^{(0)}$ its value today. Unit: GeV (or K). Value: 2.7255 K today. Status: STANDARD.
- n_ν: The number density of the cosmic neutrino background per flavour (neutrinos plus antineutrinos, one helicity each); $n_\nu^{(0)}$ its value today. Unit: GeV^3. Value: 112 cm^{-3} today. Status: STANDARD.
- ζ: The Riemann zeta function; $\zeta(3) = 1.2020569$. Unit: dimensionless. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

The neutrinos in the homogeneous, isotropic, expanding fold (the momentum redshifts with the expansion rate a_4') and the cosmic neutrino background (CNB); the superscript (0) marks a value today. Until the temperature falls to about 1 MeV the collisions keep the neutrinos in equilibrium with the plasma; they then decouple, the annihilation of electrons and positrons heats the photons, and the effective number of neutrino species is $N_{\mathrm{eff}} = 3.044$ in the Standard Model (with the small heating of the neutrinos and the flavour conversion during decoupling included).

Today the CNB has the temperature $\mathsf{T}_\nu^{(0)} = 1.945$ K and $n_\nu^{(0)} = 112$ neutrinos and antineutrinos per cm^3 and flavour (one helicity each: the right-handed states of the Dirac neutrinos are not populated by the Standard-Model collisions). The right-handed neutrinos add to N_{eff} only if something else brings them into equilibrium (FE-62). Status: STANDARD (`um4_neutrino.py`, N5).

11 The cosmological perturbations

The CMB, the CNB and the BAO are computed on the fold, where four-dimensional gravity is general relativity with G_{N} (FE-59: the fold obeys the Einstein equations of seven dimensions to the first order in its curvature over H^2, FE-71, and with the volume of the extra times held fixed these are the equations of four-dimensional general relativity), the matter is that of the Standard Model, the dark matter is the particles χ of dirac16complex (the long-lived resonances of FE-40 and FE-86), and the dark energy is the vacuum energy ρ_Λ of the fold (FE-59). The metric of the fold is the background plus the first-order cosmological perturbation (of the order ϵ_{c}, the size of the perturbations); the equations are those of Ma and Bertschinger (1995) in the conformal Newtonian gauge, written out here.

$$
\begin{aligned}
&\hat{h}_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu = a^2\Big[-(1+2\psi_{\mathrm{B}})\,\mathrm{d}\tau_{\mathrm{c}}^2 + (1-2\phi_{\mathrm{B}})\big(\mathrm{d}x_1^2+\mathrm{d}x_2^2+\mathrm{d}x_3^2\big)\Big],\\
&a = \mathrm{e}^{a_4}, \qquad \mathrm{d}\tau_{\mathrm{c}} = \frac{\mathrm{d}x_4}{a}, \qquad \mathcal{H} = \frac{\dot{a}}{a},\\
&k^2\phi_{\mathrm{B}} + 3\mathcal{H}\big(\dot{\phi}_{\mathrm{B}} + \mathcal{H}\psi_{\mathrm{B}}\big) = -4\pi G_{\mathrm{N}} a^2\,\Delta\rho, \qquad k^2\big(\dot{\phi}_{\mathrm{B}} + \mathcal{H}\psi_{\mathrm{B}}\big) = 4\pi G_{\mathrm{N}} a^2(\rho+\mathcal{P})\,\vartheta,\\
&\ddot{\phi}_{\mathrm{B}} + \mathcal{H}\big(\dot{\psi}_{\mathrm{B}} + 2\dot{\phi}_{\mathrm{B}}\big) + \Big(2\frac{\ddot{a}}{a} - \mathcal{H}^2\Big)\psi_{\mathrm{B}} + \frac{k^2}{3}\big(\phi_{\mathrm{B}} - \psi_{\mathrm{B}}\big) = 4\pi G_{\mathrm{N}} a^2\,\Delta\mathcal{P},\\
&k^2\big(\phi_{\mathrm{B}} - \psi_{\mathrm{B}}\big) = 12\pi G_{\mathrm{N}} a^2(\rho+\mathcal{P})\,\Sigma_{\mathrm{s}}
\end{aligned}
\qquad \text{(FE-54)}
$$

Symbols in (FE-54).

- $\hat{h}_{\mu\nu}$: The four-dimensional metric of the fold, $\hat{h}_{\mu\nu} = \mathrm{diag}(\mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, \mathrm{e}^{2a_4}, -1)$ in the coordinates $x_1, \ldots, x_4$ (the induced metric at $\sigma = 0$ without the extra times); $\hat{h}^{\mu\nu}$ its inverse, $\hat{h}$ its determinant, $\sqrt{-\hat{h}} = \mathrm{e}^{3a_4}$; $\hat{h}^{(1)}_{\mu\nu}$ its first-order cosmological perturbation. Unit: dimensionless. Status: DERIVED.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- x: The coordinates of the fold: x_1, x_2, x_3 of ordinary space, x_4 the time, x_5, x_6, x_7 the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- a, $\dot{a}$, $\ddot{a}$: The scale factor of the fold, $a = \mathrm{e}^{a_4}$ (normalised to 1 today by the choice of the comoving coordinates x_1, x_2, x_3), and its first and second derivatives with respect to the conformal time τ_{c}. Unit: dimensionless. Status: STANDARD.
- ψ_{B}, $\dot{\psi}_{\mathrm{B}}$: The Bardeen potential of the time-time part of the perturbed metric of the fold (the Newtonian potential) in the conformal Newtonian gauge, a function of τ_{c} and of the wave number k, and its derivative with respect to τ_{c}. Unit: dimensionless. Status: STANDARD.
- τ_{c}: The conformal time of the fold, $\mathrm{d}\tau_{\mathrm{c}} = \mathrm{e}^{-a_4}\mathrm{d}x_4$ (the time divided by the scale factor e^{a_4} of the fold); a dot on a quantity of the cosmological perturbations is $\mathrm{d}/\mathrm{d}\tau_{\mathrm{c}}$. Unit: length (= time). Status: STANDARD.
- ϕ_{B}, $\dot{\phi}_{\mathrm{B}}$, $\ddot{\phi}_{\mathrm{B}}$: The Bardeen potential of the space-space part of the perturbed metric of the fold (the curvature potential) in the conformal Newtonian gauge, and its first and second derivatives with respect to τ_{c}. Unit: dimensionless. Status: STANDARD.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.

- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\mathcal{H}$: The conformal expansion rate, $\mathcal{H} = \dot{a}/a = a\,a_4'$. Unit: 1/length. Status: STANDARD.
- k: A wave number: the momentum of a plane wave along x_3, $\mathrm{e}^{\mathrm{i}(kx_3 - \omega x_4)}$; in the cosmological perturbations the comoving wave number of a Fourier mode, $k = |\vec{k}|$. Unit: GeV (1/length). Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_N: Newton's constant of four-dimensional gravity (the gravity of our world, on the fold), $G_\mathrm{N} = 1/M_\mathrm{Pl}^2$ with the Planck mass $M_\mathrm{Pl} = 1.221 \times 10^{19}$ GeV. Unit: GeV^{-2}. Value: 6.708×10^{-39} GeV^{-2}. Status: STANDARD.
- Δ: A change, difference or small variation of the quantity that follows (for example an interval of the time, a perturbation of a density, a difference of two squared masses, or an addition to an effective number of species). Status: STANDARD.
- ρ: The total energy density of the matter of the fold (all species); $\Delta\rho$ its perturbation. Unit: GeV^4. Status: STANDARD.
- $\mathcal{P}$: The total pressure of the matter of the fold; $\Delta\mathcal{P}$ its perturbation. Unit: GeV^4. Status: STANDARD.
- ϑ: The divergence of the velocity of the total matter: $(\rho + \mathcal{P})\vartheta$ is the sum over the species of their energy density plus pressure times the divergence of their velocity. Unit: 1/length. Status: STANDARD.
- Σ_s: The anisotropic stress (shear) of the total matter: $(\rho + \mathcal{P})\Sigma_\mathrm{s}$ is the sum over the species (photons and neutrinos) of their energy density plus pressure times their shear. Unit: dimensionless. Status: STANDARD.

The metric of the fold with the scalar perturbations in the conformal Newtonian gauge (ψ_B and ϕ_B the two Bardeen potentials, functions of the conformal time τ_c and of the comoving wave number k of a Fourier mode; a dot is $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$; the comoving coordinates are chosen so that $a = 1$ today), and the Einstein equations of the fold (FE-59) to first order: the energy constraint, the momentum constraint, the trace of the space-space equations and their traceless part. $\Delta\rho$ and $\Delta\mathcal{P}$ are the perturbations of the total energy density and pressure; $(\rho + \mathcal{P})\vartheta$ and $(\rho + \mathcal{P})\Sigma_\mathrm{s}$ are the sums over the species of (energy density plus pressure) times the divergence of their velocity and times their anisotropic stress. By FE-34 and FE-71 the bulk metric of any metric of the fold, perturbed or not, is fixed to the first order in the curvature of the fold over H^2, and the fold then obeys the Einstein equations of seven dimensions: with the volume of the extra times held fixed by the fold fluid (for the perturbations as well, ASSUMED) the first-order perturbations of the fold are those of general relativity, with corrections of the relative orders $(H_4/H)^2$ and $(k/(aH))^2$ (k/a the physical wave number) from the massive modes of the bulk, below 10^{-78} at the scales of the CMB for $H > 2.5$ TeV (ESTIMATE), and its gravitational waves are those of general relativity (FE-63). The Weyl term $\mathfrak{c}_{ab}$ of the bulk in FE-34 (dark radiation) is taken to be zero (ASSUMED). Status: STANDARD (DERIVED: the equations of the fold are those of general relativity to the first order in its curvature over H^2, FE-71; ASSUMED: the volume held fixed for the perturbations, no dark radiation).

$$
\begin{aligned}
&\dot\delta_\chi = -\vartheta_\chi + 3\dot\phi_\mathrm{B}, \qquad \dot\vartheta_\chi = -\mathcal{H}\vartheta_\chi + k^2\psi_\mathrm{B} + \Gamma_{\chi\mathrm{b}}\big(\vartheta_\mathrm{b} - \vartheta_\chi\big),\\
&\dot\delta_\mathrm{b} = -\vartheta_\mathrm{b} + 3\dot\phi_\mathrm{B}, \qquad \dot\vartheta_\mathrm{b} = -\mathcal{H}\vartheta_\mathrm{b} + c_\mathrm{s}^2k^2\delta_\mathrm{b} + k^2\psi_\mathrm{B} + \frac{a\,x_\mathrm{e}n_e\sigma_\mathrm{T}}{\mathsf{R}}\big(\vartheta_\gamma - \vartheta_\mathrm{b}\big) + \frac{\rho_\chi}{\rho_\mathrm{b}}\,\Gamma_{\chi\mathrm{b}}\big(\vartheta_\chi - \vartheta_\mathrm{b}\big), \qquad \mathsf{R} = \frac{3\rho_\mathrm{b}}{4\rho_\gamma},\\
&\dot\delta_\gamma = -\frac{4}{3}\vartheta_\gamma + 4\dot\phi_\mathrm{B}, \qquad \dot\vartheta_\gamma = k^2\Big(\frac{\delta_\gamma}{4} - \Sigma_\gamma\Big) + k^2\psi_\mathrm{B} + a\,x_\mathrm{e}n_e\sigma_\mathrm{T}\big(\vartheta_\mathrm{b} - \vartheta_\gamma\big),\\
&\dot{\mathsf{F}}_l = \frac{k}{2l+1}\big(l\,\mathsf{F}_{l-1} - (l+1)\,\mathsf{F}_{l+1}\big) - a\,x_\mathrm{e}n_e\sigma_\mathrm{T}\Big(\mathsf{F}_l - \frac{\mathsf{G}_0 + \mathsf{F}_2 + \mathsf{G}_2}{10}\,\delta_{l2}\Big) \quad (l \geq 2),\\
&\dot{\mathsf{G}}_l = \frac{k}{2l+1}\big(l\,\mathsf{G}_{l-1} - (l+1)\,\mathsf{G}_{l+1}\big) - a\,x_\mathrm{e}n_e\sigma_\mathrm{T}\Big(\mathsf{G}_l - \frac{\mathsf{G}_0 + \mathsf{F}_2 + \mathsf{G}_2}{2}\Big(\delta_{l0} + \frac{\delta_{l2}}{5}\Big)\Big) \quad (l \geq 0),\\
&\mathsf{F}_0 = \delta_\gamma, \qquad \mathsf{F}_1 = \frac{4\vartheta_\gamma}{3k}, \qquad \mathsf{F}_2 = 2\Sigma_\gamma,\\
&\dot\delta_\nu = -\frac{4}{3}\vartheta_\nu + 4\dot\phi_\mathrm{B}, \qquad \dot\vartheta_\nu = k^2\Big(\frac{\delta_\nu}{4} - \Sigma_\nu\Big) + k^2\psi_\mathrm{B}, \qquad \dot{\mathsf{N}}_l = \frac{k}{2l+1}\big(l\,\mathsf{N}_{l-1} - (l+1)\,\mathsf{N}_{l+1}\big) \quad (l \geq 2),\\
&\mathsf{N}_1 = \frac{4\vartheta_\nu}{3k}, \qquad \mathsf{N}_2 = 2\Sigma_\nu,\\
&\dot\Upsilon_0 = -\frac{\mathsf{q}k}{\mathsf{E}}\Upsilon_1 - \dot\phi_\mathrm{B}\frac{\mathrm{d}\ln\mathcal{F}_0}{\mathrm{d}\ln\mathsf{q}}, \qquad \dot\Upsilon_1 = \frac{\mathsf{q}k}{3\mathsf{E}}\big(\Upsilon_0 - 2\Upsilon_2\big) - \frac{\mathsf{E}k}{3\mathsf{q}}\,\psi_\mathrm{B}\frac{\mathrm{d}\ln\mathcal{F}_0}{\mathrm{d}\ln\mathsf{q}},\\
&\dot\Upsilon_l = \frac{\mathsf{q}k}{(2l+1)\mathsf{E}}\big(l\,\Upsilon_{l-1} - (l+1)\,\Upsilon_{l+1}\big) \quad (l \geq 2), \qquad \mathsf{E} = \sqrt{\mathsf{q}^2 + a^2m_i^2}, \qquad \mathcal{F}_0 = \frac{1}{\mathrm{e}^{\mathsf{q}/\mathsf{T}_\nu^{(0)}} + 1}
\end{aligned}
\tag{FE-55}
$$

Symbols in (FE-55).

- δ_χ, $\dot\delta_\chi$: The density contrast of the dark matter (the particles χ of dirac16complex), the perturbation of its energy density divided by its mean, and its derivative with respect to τ_c. Unit: dimensionless. Status: STANDARD.
- ϑ_χ, $\dot\vartheta_\chi$: The divergence of the velocity of the dark matter (for a Fourier mode, ik times its velocity along the wave vector), and its derivative. Unit: 1/length. Status: STANDARD.
- ϕ_B, $\dot\phi_\mathrm{B}$, $\ddot\phi_\mathrm{B}$: The Bardeen potential of the space-space part of the perturbed metric of the fold (the curvature potential) in the conformal Newtonian gauge, and its first and second derivatives with respect to τ_c. Unit: dimensionless. Status: STANDARD.
- $\mathcal{H}$: The conformal expansion rate, $\mathcal{H} = \dot a/a = a\,a_4'$. Unit: 1/length. Status: STANDARD.
- k: A wave number: the momentum of a plane wave along x_3, $\mathrm{e}^{\mathrm{i}(kx_3 - \omega x_4)}$; in the cosmological perturbations the comoving wave number of a Fourier mode, $k = |\vec{k}|$. Unit: GeV (1/length). Status: STANDARD.
- ψ_B, $\dot\psi_\mathrm{B}$: The Bardeen potential of the time-time part of the perturbed metric of the fold (the Newtonian potential) in the conformal Newtonian gauge, a function of τ_c and of the wave number k, and its derivative with respect to τ_c. Unit: dimensionless. Status: STANDARD.
- $\Gamma_{\chi\mathrm{b}}$: The rate (per conformal time) at which the dark matter exchanges momentum with the baryons through the portal couplings (elastic scattering of χ on nuclei and electrons). Unit: 1/length. Value: negligible for the couplings that direct detection allows. Status: DERIVED.
- ϑ_b, $\dot\vartheta_\mathrm{b}$: The divergence of the velocity of the baryons and its derivative. Unit: 1/length. Status: STANDARD.
- δ_b, $\dot\delta_\mathrm{b}$: The density contrast of the baryons and its derivative. Unit: dimensionless. Status: STANDARD.
- c_s: The sound speed of the baryon gas: its square is the temperature of the gas divided by the mean mass per particle, times one minus a third of the logarithmic derivative of that temperature with respect to a. Unit: dimensionless. Status: STANDARD.
- a, $\dot a$, $\ddot a$: The scale factor of the fold, $a = \mathrm{e}^{a_4}$ (normalised to 1 today by the choice of the comoving coordinates x_1, x_2, x_3), and its first and second derivatives with respect to the conformal time τ_c. Unit: dimensionless. Status: STANDARD.
- x_e: The ionisation fraction: the fraction of the electrons that are free. Unit: dimensionless. Status: STANDARD.
- n_e: The number density of electrons (bound and free; in a supernova all are free, in the recombining universe $x_\mathrm{e}n_e$ is the density of the free electrons). Unit: GeV^3 (cm^{-3}). Status: STANDARD.
- σ_T: The Thomson cross-section of the scattering of photons on free electrons. Unit: GeV^{-2}. Value: $6.6524587 \times 10^{-25}$ cm^2. Status: STANDARD.
- R: The ratio of the inertia of the baryons to that of the photons, $\mathsf{R} = 3\rho_\mathrm{b}/(4\rho_\gamma)$. Unit: dimensionless. Status:

STANDARD.

- ϑ_γ, $\dot{\vartheta}_\gamma$: The divergence of the velocity of the photons and its derivative. Unit: 1/length. Status: STANDARD.
- ρ_χ: The energy density of the dark matter, the particles χ of dirac16complex; $\rho_\chi^{(0)}$ its value today. Unit: GeV^4. Status: DERIVED.
- ρ_b: The (rest-mass) energy density of the baryons: in a supernova the local density of the matter (quoted in $\mathrm{g/cm}^3$), in cosmology the mean density of the baryons ($\rho_\mathrm{b}^{(0)}$ today). Unit: GeV^4. Status: STANDARD.
- ρ_γ: The energy density of the photons. Unit: GeV^4. Status: STANDARD.
- δ_γ, $\dot{\delta}_\gamma$: The density contrast of the photons and its derivative. Unit: dimensionless. Status: STANDARD.
- Σ_γ: The anisotropic stress (shear) of the photons. Unit: dimensionless. Status: STANDARD.
- F_l, $\dot{\mathsf{F}}_l$: The multipoles of the temperature perturbation of the photons ($\mathsf{F}_0 = \delta_\gamma$, $\mathsf{F}_1 = 4\vartheta_\gamma/(3k)$, $\mathsf{F}_2 = 2\Sigma_\gamma$; the photon multipoles of Ma and Bertschinger) and their derivatives with respect to τ_c. Unit: dimensionless. Status: STANDARD.
- l: The multipole index and number ($l = 0, 1, 2, \ldots$) of an expansion in Legendre polynomials of the direction of a photon or a neutrino relative to the wave vector, or of the CMB sky in spherical harmonics. Status: STANDARD.
- G_l, $\dot{\mathsf{G}}_l$: The multipoles of the polarisation perturbation of the photons (the $G_{\gamma l}$ of Ma and Bertschinger) and their derivatives. Unit: dimensionless. Status: STANDARD.
- δ: The Kronecker delta: with two indices $\delta_N^M = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{X\mathrm{L}}$ is 1 if the label X of a half of dirac16complex is L and 0 otherwise); with two index strings of equal length the generalised Kronecker delta $\delta_{N_1 N_2 \cdots}^{M_1 M_2 \cdots}$, the determinant of the square matrix of the Kronecker deltas $\delta_{N_b}^{M_a}$ (+1 or −1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- δ_ν, $\dot{\delta}_\nu$: The density contrast of the relativistic neutrinos and its derivative. Unit: dimensionless. Status: STANDARD.
- ϑ_ν, $\dot{\vartheta}_\nu$: The divergence of the velocity of the relativistic neutrinos and its derivative. Unit: 1/length. Status: STANDARD.
- Σ_ν: The anisotropic stress (shear) of the relativistic neutrinos. Unit: dimensionless. Status: STANDARD.
- N_l, $\dot{\mathsf{N}}_l$: The multipoles of the perturbation of the relativistic neutrinos ($\mathsf{N}_0 = \delta_\nu$, $\mathsf{N}_1 = 4\vartheta_\nu/(3k)$, $\mathsf{N}_2 = 2\Sigma_\nu$) and their derivatives. Unit: dimensionless. Status: STANDARD.
- $\Upsilon_l(\mathsf{q})$, $\dot{\Upsilon}_l$: The multipoles of the relative perturbation of the distribution of a massive neutrino state at the comoving momentum q (the Ψ_l of Ma and Bertschinger) and their derivatives. Unit: dimensionless. Status: STANDARD.
- q: The comoving momentum of a neutrino, $a|\vec{p}|$. Unit: GeV. Status: STANDARD.
- E: The comoving energy of a massive neutrino (a times its energy). Unit: GeV. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $\mathcal{F}_0(\mathsf{q})$: The unperturbed (Fermi-Dirac) occupation number of the cosmic neutrino background at the comoving momentum q. Unit: dimensionless. Status: STANDARD.
- m: A mass: m_i the mass of the neutrino mass state i; $\Delta m_{ij}^2 = m_i^2 - m_j^2$ the squared-mass difference of the states i and j (Δm_{21}^2, Δm_{31}^2, Δm_{32}^2); Δm^2 without labels the difference of the two states of a resonance, $m_2^2 - m_1^2$, multiplied by v_loc^2/v^2 where this differs from 1; m without a label the mass of a mode of a field of the bulk as the fold sees it, $m^2 = \omega^2 - k^2$; for the W and Z bosons their complex pole mass, $m^2 = m_0^2 - \mathrm{i}m_0\Gamma_\mathrm{bulk}$ to the leading order; in the relic abundance of the particles χ their mass. Unit: GeV. Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the three neutrino mass states; in a frame component such as $\mathcal{S}^{(i)}{}_{(i)}$ one of the three directions of ordinary space (the three components are equal), and in a tensor of ordinary space such as h_{ij} one of the directions x_1, x_2, x_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- T_ν: The temperature of the cosmic neutrino background; $\mathsf{T}_\nu^{(0)}$ its value today. Unit: GeV (or K). Value: 1.945 K today. Status: STANDARD.

The Boltzmann equations of the species in the same gauge (Ma and Bertschinger 1995, their equations 42, 66, 63, 49 and 57): the dark matter χ (the long-lived resonances of dirac16complex at the fold, FE-86: pressureless, FE-42, with the drag $\Gamma_{\chi\mathrm{b}}$ on the baryons from its portal couplings, FE-47, which is negligible for the couplings that direct detection allows); the baryons (with the sound speed c_s of the gas and the Thomson coupling to the

photons through the free electrons, x_e the ionisation fraction); the photons (the multipoles F_l of the temperature and G_l of the polarisation, l the multipole index; δ_{l0}, δ_{l2} are Kronecker deltas, and the term $l\,\mathsf{G}_{l-1}$ vanishes for $l=0$); the neutrinos while they are relativistic (the multipoles N_l); and each massive neutrino state i of mass m_i (the multipoles Υ_l of the relative perturbation of its distribution at the comoving momentum q and energy E, with the Fermi-Dirac occupation $\mathcal{F}_0$ of the CNB). The hierarchies are truncated at a large l in a numerical solution. Status: STANDARD (UM-4: the dark matter is χ, with the drag term).

$$
\begin{aligned}
&r_\mathrm{s} = \int_0^{\tau_\mathrm{d}} c_{\gamma\mathrm{b}}\,\mathrm{d}\tau_\mathrm{c}, \qquad c_{\gamma\mathrm{b}} = \frac{1}{\sqrt{3(1+\mathsf{R})}},\\
&D_\mathrm{M}(\mathsf{z}) = \int_0^{\mathsf{z}} \frac{\mathrm{d}\mathsf{z}'}{H_4(\mathsf{z}')}, \qquad D_\mathrm{H}(\mathsf{z}) = \frac{1}{H_4(\mathsf{z})}, \qquad D_\mathrm{V}(\mathsf{z}) = \big(\mathsf{z}\,D_\mathrm{M}^2(\mathsf{z})\,D_\mathrm{H}(\mathsf{z})\big)^{1/3}
\end{aligned}
\tag{FE-56}
$$

Symbols in (FE-56).

- r_s: The comoving sound horizon at the drag epoch: the comoving distance that a sound wave of the photon-baryon fluid travels until the baryons are released from the photons. Unit: length. Value: about 147 Mpc (Planck 2018). Status: STANDARD.
- τ_d: The conformal time of the drag epoch, when the baryons are released from the photons (redshift about 1060). Unit: length (= time). Status: STANDARD.
- $c_{\gamma\mathrm{b}}$: The sound speed of the tightly coupled photon-baryon fluid. Unit: dimensionless. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- τ_c: The conformal time of the fold, $\mathrm{d}\tau_\mathrm{c} = \mathrm{e}^{-a_4}\mathrm{d}x_4$ (the time divided by the scale factor e^{a_4} of the fold); a dot on a quantity of the cosmological perturbations is $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$. Unit: length (= time). Status: STANDARD.
- R: The ratio of the inertia of the baryons to that of the photons, $\mathsf{R} = 3\rho_\mathrm{b}/(4\rho_\gamma)$. Unit: dimensionless. Status: STANDARD.
- $D_\mathrm{M}(\mathsf{z})$: The transverse comoving distance (comoving angular-diameter distance) to the redshift z, for a spatially flat fold. Unit: length. Status: STANDARD.
- z: The redshift, $1+\mathsf{z} = 1/a$; z' an integration variable. Unit: dimensionless. Status: STANDARD.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc with $h_{100} = 0.674$ (Planck 2018): 67.4 km/s/Mpc $= 1.44\times10^{-42}$ GeV. Status: STANDARD.
- $D_\mathrm{H}(\mathsf{z})$: The Hubble distance at the redshift z. Unit: length. Status: STANDARD.
- $D_\mathrm{V}(\mathsf{z})$: The volume-averaged distance to the redshift z. Unit: length. Status: STANDARD.

The baryon acoustic oscillations: the sound horizon r_s at the drag epoch τ_d (the comoving distance that a sound wave of the photon-baryon fluid, of speed $c_{\gamma\mathrm{b}}$, travels before the baryons are released from the photons), and the distances to the redshift z at which galaxy surveys measure it: the transverse comoving distance D_M (for a spatially flat fold), the Hubble distance D_H and their volume average D_V ($H_4 = a_4'$ the expansion rate of the fold as a function of redshift). The surveys measure $D_\mathrm{M}/r_\mathrm{s}$, $D_\mathrm{H}/r_\mathrm{s}$ and $D_\mathrm{V}/r_\mathrm{s}$. Status: STANDARD.

$$
\begin{aligned}
&C_l^\mathrm{TT} = 4\pi\int\frac{\mathrm{d}k}{k}\,\mathcal{P}_\mathcal{R}(k)\,\big(\mathcal{I}_l^\mathrm{T}(k)\big)^2, \qquad C_l^\mathrm{TE} = 4\pi\int\frac{\mathrm{d}k}{k}\,\mathcal{P}_\mathcal{R}(k)\,\mathcal{I}_l^\mathrm{T}(k)\,\mathcal{I}_l^\mathrm{E}(k),\\
&C_l^\mathrm{EE} = 4\pi\int\frac{\mathrm{d}k}{k}\,\mathcal{P}_\mathcal{R}(k)\,\big(\mathcal{I}_l^\mathrm{E}(k)\big)^2, \qquad \mathcal{P}_\mathcal{R}(k) = A_\mathrm{s}\Big(\frac{k}{k_*}\Big)^{n_\mathrm{S}-1}
\end{aligned}
\tag{FE-57}
$$

Symbols in (FE-57).

- C_l^TT, C_l^TE, C_l^EE: The angular power spectra of the CMB at the multipole l: temperature with temperature, temperature with E-mode polarisation, E-mode polarisation with itself. Unit: dimensionless. Status: STANDARD.
- l: The multipole index and number ($l = 0, 1, 2, \ldots$) of an expansion in Legendre polynomials of the direction of a photon or a neutrino relative to the wave vector, or of the CMB sky in spherical harmonics. Status: STANDARD.

- T, E: The labels of the CMB observables: T the temperature anisotropy, E the E-mode (gradient) polarisation; TT, TE, EE label the spectra of their products. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- k: A wave number: the momentum of a plane wave along x_3, $\mathrm{e}^{\mathrm{i}(kx_3-\omega x_4)}$; in the cosmological perturbations the comoving wave number of a Fourier mode, $k=|\vec{k}|$. Unit: GeV (1/length). Status: STANDARD.
- $\mathcal{P}_\mathcal{R}(k)$: The primordial power spectrum of the comoving curvature perturbation, per logarithmic interval of k. Unit: dimensionless. Status: ASSUMED.
- $\mathcal{I}_l^{\mathrm{T}}(k)$, $\mathcal{I}_l^{\mathrm{E}}(k)$: The transfer functions of the CMB: the multipole l of the temperature or E-mode anisotropy today produced by a Fourier mode k of unit primordial curvature perturbation, computed by the line-of-sight integral. Unit: dimensionless. Status: STANDARD.
- A_s: The amplitude of the primordial curvature spectrum at the pivot k_*. Unit: dimensionless. Value: 2.10×10^{-9} (Planck 2018). Status: ASSUMED.
- k_*: The pivot wave number of the primordial spectrum. Unit: GeV (1/length). Value: 0.05 Mpc^{-1}. Status: STANDARD.
- n_S: The spectral index of the primordial curvature spectrum. Unit: dimensionless. Value: 0.965 (Planck 2018). Status: ASSUMED.

The angular power spectra of the CMB temperature (T) and E-mode polarisation (E) at the multipole l: the transfer functions $\mathcal{I}_l^{\mathrm{T}}$, $\mathcal{I}_l^{\mathrm{E}}$ follow from FE-54 and FE-55, integrated from the early universe to today (the line-of-sight integral), and are weighted with the primordial spectrum $\mathcal{P}_\mathcal{R}$ of the comoving curvature perturbation (amplitude A_s, spectral index n_S, pivot k_*). The primordial spectrum is an input of UM-4 (ASSUMED): the primordial era can be an inflation of the fold driven by a vacuum energy of the fold with the volume of the extra times held fixed (FE-88: with the holding pressure of the fold fluid, whose origin is OPEN, and a Weyl tensor of the bulk that the exact equations of the fold require, OPEN), in the regime of general relativity only if its rate is below $H/10$, and UM-4 does not fix its spectrum (the Higgs field could drive it only with a large coupling to the curvature of the fold, which the action of UM-4 does not contain, IN-11, and then needs $H>1.6\times10^{14}$ GeV, FE-88). Status: STANDARD (ASSUMED: the primordial spectrum).

$$\Omega_\chi=\frac{\rho_\chi^{(0)}}{\rho_\mathrm{c}^{(0)}},\qquad \Omega_\mathrm{b}=\frac{\rho_\mathrm{b}^{(0)}}{\rho_\mathrm{c}^{(0)}},\qquad \Omega_\Lambda=\frac{\rho_\Lambda}{\rho_\mathrm{c}^{(0)}},\qquad \Omega_\nu h_{100}^2=\frac{\sum_i m_i}{93.14\ \mathrm{eV}},$$
$$\rho_\mathrm{c}^{(0)}=\frac{3\left(H_4^{(0)}\right)^2}{8\pi G_\mathrm{N}},\qquad H_4^{(0)}=100\,h_{100}\ \mathrm{km/s/Mpc} \tag{FE-58}$$

Symbols in (FE-58).

- Ω_χ: The density parameter of the dark matter today. Unit: dimensionless. Value: $\Omega_\chi h_{100}^2=0.120$ (Planck 2018). Status: DERIVED.
- ρ_χ: The energy density of the dark matter, the particles χ of dirac16complex; $\rho_\chi^{(0)}$ its value today. Unit: GeV^4. Status: DERIVED.
- ρ_c: The critical density, the energy density of a spatially flat fold at the expansion rate H_4; $\rho_\mathrm{c}^{(0)}$ its value today. Unit: GeV^4. Value: $\rho_\mathrm{c}^{(0)}=1.0537\times10^{-5}\,h_{100}^2$ $\mathrm{GeV/cm}^3$. Status: STANDARD.
- Ω_b: The density parameter of the baryons today. Unit: dimensionless. Value: $\Omega_\mathrm{b}h_{100}^2=0.0224$ (Planck 2018). Status: STANDARD.
- ρ_b: The (rest-mass) energy density of the baryons: in a supernova the local density of the matter (quoted in $\mathrm{g/cm}^3$), in cosmology the mean density of the baryons ($\rho_\mathrm{b}^{(0)}$ today). Unit: GeV^4. Status: STANDARD.
- Ω_Λ: The density parameter of the dark energy (the vacuum energy of the fold) today. Unit: dimensionless. Value: 0.685 (Planck 2018). Status: DERIVED.
- ρ_Λ: The vacuum energy density of the fold above its static tension, the dark energy of UM-4: $\rho_\Lambda=\rho_\mathrm{ff}+\rho_\mathrm{vac}-\lambda_\mathrm{F}V_*$ (four-dimensional normalisation). Unit: GeV^4. Value: $0.685\,\rho_\mathrm{c}^{(0)}=2.5\times10^{-47}$ GeV^4 (Planck 2018). Status: DERIVED.
- Ω_ν, $\Omega_\nu^{(i)}$: The density parameter of the neutrinos of the CNB: Ω_ν that of the massive states today; $\Omega_\nu^{(i)}(\mathsf{z})$ the energy density of the mass state i at the redshift z in units of the critical density today. Unit: dimensionless.

Status: STANDARD.

- h_{100}: The present expansion rate of the fold in units of 100 km/s/Mpc. Unit: dimensionless. Value: 0.674 (Planck 2018). Status: STANDARD.
- i: A generation index ($i = 1, 2, 3$), which also labels the three neutrino mass states; in a frame component such as $\mathcal{S}^{(i)}{}_{(i)}$ one of the three directions of ordinary space (the three components are equal), and in a tensor of ordinary space such as h_{ij} one of the directions x_1, x_2, x_3; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- m: A mass: m_i the mass of the neutrino mass state i; $\Delta m^2_{ij} = m_i^2 - m_j^2$ the squared-mass difference of the states i and j (Δm^2_{21}, Δm^2_{31}, Δm^2_{32}); Δm^2 without labels the difference of the two states of a resonance, $m_2^2 - m_1^2$, multiplied by v_{loc}^2/v^2 where this differs from 1; m without a label the mass of a mode of a field of the bulk as the fold sees it, $m^2 = \omega^2 - k^2$; for the W and Z bosons their complex pole mass, $m^2 = m_0^2 - \mathrm{i}m_0\Gamma_{\mathrm{bulk}}$ to the leading order; in the relic abundance of the particles χ their mass. Unit: GeV. Status: STANDARD.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc with $h_{100} = 0.674$ (Planck 2018): 67.4 km/s/Mpc $= 1.44 \times 10^{-42}$ GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_{N}: Newton's constant of four-dimensional gravity (the gravity of our world, on the fold), $G_{\mathrm{N}} = 1/M_{\mathrm{Pl}}^2$ with the Planck mass $M_{\mathrm{Pl}} = 1.221 \times 10^{19}$ GeV. Unit: GeV^{-2}. Value: 6.708×10^{-39} GeV^{-2}. Status: STANDARD.

The density parameters today: the dark matter (the particles χ, whose origin is OPEN, FE-61), the baryons, the dark energy (the vacuum energy of the fold, FE-59) and the massive neutrinos of the CNB (for the three left-handed states with the temperature and the small non-thermal distortion of the Standard Model; the right-handed states are not populated). Together with A_{s}, n_{S} and the optical depth of reionisation they fix the CMB, the CNB and the BAO. Status: STANDARD.

12 The thermal history

$$H_4^2 = \frac{8\pi G_{\mathrm{N}}}{3}\big(\rho_{\mathrm{SM}} + \rho_\chi + \rho_\Lambda\big), \qquad 2H_4' + 3H_4^2 = -8\pi G_{\mathrm{N}}\,\mathcal{P}, \qquad \frac{1}{8\pi G_{\mathrm{N}}} = \frac{V_*}{\kappa_7} + \frac{1}{\kappa_{\mathrm{ind}}}, \qquad \kappa_7 = \frac{5}{2}\,\kappa H,$$
$$\rho_\Lambda = \rho_{\mathrm{ff}} + \rho_{\mathrm{vac}} - \lambda_{\mathrm{F}} V_*, \qquad q_{\mathrm{SM}} + q_\chi + q_{\mathrm{P}} + q_{\mathrm{ff}} + \lambda_{\mathrm{F}} V_* + \frac{\rho - 3\mathcal{P}}{2} = 0, \tag{FE-59}$$
$$q_{\mathrm{SM}} = \big\langle \mathcal{L}_{\mathrm{H}} + \mathcal{L}_{\mathrm{f}} + \mathcal{L}_{\mathrm{Y}} \big\rangle, \qquad q_{\mathrm{P}} = \langle \mathcal{L}_{\mathrm{P}} \rangle$$

Symbols in (FE-59).

- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc with $h_{100} = 0.674$ (Planck 2018): 67.4 km/s/Mpc $= 1.44 \times 10^{-42}$ GeV. Status: STANDARD.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_{N}: Newton's constant of four-dimensional gravity (the gravity of our world, on the fold), $G_{\mathrm{N}} = 1/M_{\mathrm{Pl}}^2$ with the Planck mass $M_{\mathrm{Pl}} = 1.221 \times 10^{19}$ GeV. Unit: GeV^{-2}. Value: 6.708×10^{-39} GeV^{-2}. Status: STANDARD.
- ρ_{SM}: The energy density of the Standard-Model particles. Unit: GeV^4. Status: STANDARD.
- ρ_χ: The energy density of the dark matter, the particles χ of dirac16complex; $\rho_\chi^{(0)}$ its value today. Unit: GeV^4. Status: DERIVED.
- ρ_Λ: The vacuum energy density of the fold above its static tension, the dark energy of UM-4: $\rho_\Lambda = \rho_{\mathrm{ff}} + \rho_{\mathrm{vac}} - \lambda_{\mathrm{F}} V_*$ (four-dimensional normalisation). Unit: GeV^4. Value: $0.685\,\rho_{\mathrm{c}}^{(0)} = 2.5 \times 10^{-47}$ GeV^4 (Planck 2018). Status: DERIVED.
- $\mathcal{P}$: The total pressure of the matter of the fold; $\Delta\mathcal{P}$ its perturbation. Unit: GeV^4. Status: STANDARD.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3$ (with $V_{\mathrm{t}} = \int \mathrm{d}x_5\,\mathrm{d}x_6\,\mathrm{d}x_7$ the coordinate volume of the extra times) at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- κ_7: The gravitational coupling of the fold: to the first order in its curvature over H^2 the fold obeys the Einstein

equations of seven dimensions $\hat{G}^a{}_b = \kappa_7 \mathrm{t}^a{}_b$ with $\kappa_7 = 5\kappa H/2$, the coupling of the massless graviton of the bulk bound to the fold; four-dimensional gravity on the fold has $8\pi G_\mathrm{N} = \kappa_7/V_*$ (plus the induced term, which is negligible). Unit: GeV^{-5}. Value: $5\kappa H/2$. Status: DERIVED.

- κ_ind: The coupling of the Einstein-Hilbert term $\hat{R}/(2\kappa_\mathrm{ind})$ that the quantum fluctuations of the fields of the fold may induce, $\kappa_\mathrm{ind} = 8\pi G_\mathrm{ind}$; it adds to the gravity of the fold, $1/(8\pi G_\mathrm{N}) = V_*/\kappa_7 + 1/\kappa_\mathrm{ind}$. Unit: GeV^{-2}. Value: not fixed by UM-4; negligible for a cutoff of the order of H. Status: ESTIMATE.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- ρ_ff: The energy density of the fold fluid. Unit: GeV^4. Value: $\lambda_\mathrm{F} V_* + \rho_\Lambda - \rho_\mathrm{vac}$: together with the vacuum energy ρ_vac of the fields of the fold it makes up the tension $\lambda_\mathrm{F} V_*$ that the static block requires plus the dark energy ρ_Λ. Status: ASSUMED (the fluid), DERIVED (the value).
- ρ_vac: The vacuum energy of the fields of the fold: the minimum of their potential (for the Standard Model the minimum of the Higgs potential with its loop corrections), a tension of the seven-dimensional fold. Unit: GeV^4. Status: STANDARD.
- λ_F: The tension that the static block requires of the fold, $\kappa\lambda_\mathrm{F} = 12H$: the total vacuum energy of the seven-dimensional fold, made of the tension of the fold fluid and the vacuum energy of the fields of the fold (both have the pressure minus their energy density along ordinary space and along the extra times), per unit volume of the six other directions of the fold (eight-dimensional normalisation); in the four-dimensional normalisation it is $\lambda_\mathrm{F} V_*$, and the excess of the total over $\lambda_\mathrm{F} V_*$ is the dark energy ρ_Λ. Unit: GeV^7. Value: $12H/\kappa$; $\lambda_\mathrm{F} V_* = 30H^2 M_\mathrm{red}^2 = 1.1 \times 10^{45}$ GeV^4 for $H = 2.5$ TeV. Status: DERIVED.
- q_SM: The pressure of the Standard Model along the extra times at the fold: the mean on-shell Lagrangian density of the Standard-Model fields (the extra-time component of the energy-momentum tensor of every term of the fold is its Lagrangian, because the fields of the fold do not depend on the extra times). Unit: GeV^4. Status: DERIVED (for radiation and free fields), ASSUMED (for the baryons).
- q_χ: The pressure of the particles χ of dirac16complex along the extra times, integrated across the fold (their on-shell Lagrangian, the self-interaction energy per volume; zero for a dilute gas). Unit: GeV^4. Status: DERIVED.
- q_P: The pressure of the portals along the extra times: their mean on-shell Lagrangian density. Unit: GeV^4. Status: DERIVED.
- q_ff: The pressure of the fold fluid along the extra times: minus its energy density (its tension) plus the pressure q_hold that holds the volume of the extra times fixed. Unit: GeV^4. Value: $-\rho_\mathrm{ff} + q_\mathrm{hold}$. Status: ASSUMED (the fluid), DERIVED (the value).
- ρ: The total energy density of the matter of the fold (all species); $\Delta\rho$ its perturbation. Unit: GeV^4. Status: STANDARD.
- $\langle \cdots \rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $\mathcal{L}_\mathrm{H}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.
- $\mathcal{L}_\mathrm{f}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_\mathrm{Y}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_\mathrm{P}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.

The Einstein equations of the fold with frozen extra times: FE-71 and FE-72 with the sources of FE-5, written in the four-dimensional normalisation (IN-8; $H_4 = a_4'$ the expansion rate of the fold). The first line is general relativity: the Friedmann equation and the acceleration equation of a flat universe with Newton's constant $8\pi G_\mathrm{N} = \kappa_7/V_*$ of the graviton of the fold (FE-63, FE-71), plus the induced term IN-9, which is negligible (FE-64), with corrections of the relative order $(12/25)(H_4/H)^2$ from the high-energy term of FE-35, below 10^{-50} after nucleosynthesis for $H > 2.5$ TeV. ρ_SM is the energy density of the Standard-Model particles (the gauge fields included: their zero modes are the radiation of the fold, and their massive modes are not excited at temperatures far below H), ρ_χ that of the particles χ of dirac16complex, $\mathcal{P}$ the total pressure along ordinary space and $\rho = \rho_\mathrm{SM} + \rho_\chi + \rho_\Lambda$ the total energy density above the tension of the static block. ρ_Λ is the vacuum energy of the fold above that tension, the dark energy of UM-4 ($w = -1$ exactly): the tension of the fold fluid plus the vacuum energy ρ_vac of the fields of the fold, minus the tension $\lambda_\mathrm{F} V_* = 30H^2/(8\pi G_\mathrm{N})$ that the static block requires (FE-65). UM-4 does not explain why the two nearly cancel (for $H = 2.5$ TeV, $\lambda_\mathrm{F} V_* = 1.1 \times 10^{45}$ GeV^4 against $\rho_\Lambda = 2.5 \times 10^{-47}$ GeV^4, one part in 10^{92}): the

cosmological-constant problem in the form it takes for a brane of Randall and Sundrum (OPEN). The second line, last equation, is the component along the extra times (the pressures q_{SM}, q_χ, q_{P} and q_{ff} along the extra times; for every term of the fold its on-shell Lagrangian density by FE-5, the mean $\langle\ \rangle$ over the state of the fold; the gauge fields live in the bulk and their pressure along the extra times enters through the bulk, where their zero modes carry, like radiation, none on average): it is the radion condition FE-72, the condition that the volume of the extra times stays fixed, and with the fold fluid holding it fixed it determines q_{hold} in $q_{\mathrm{ff}} = -\rho_{\mathrm{ff}} + q_{\mathrm{hold}}$ (FE-65). Status: DERIVED (`um4_fold.py`, F2, F4 and F5; `um4_background.py`, BG2 to BG4), ASSUMED (that the fold fluid holds the volume of the extra times fixed), OPEN (the tuning of the tension).

$$\rho_{\mathrm{SM}} = \frac{\pi^2}{30}\, g_\rho(\mathsf{T})\, \mathsf{T}^4, \qquad s = \frac{2\pi^2}{45}\, g_{\mathrm{s}}(\mathsf{T})\, \mathsf{T}^3, \qquad H_4 = \sqrt{\frac{8\pi^3 g_\rho}{90}}\, \frac{\mathsf{T}^2}{M_{\mathrm{Pl}}}, \qquad M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2} \tag{FE-60}$$

Symbols in (FE-60).

- ρ_{SM}: The energy density of the Standard-Model particles. Unit: GeV^4. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- g_ρ: The effective number of relativistic degrees of freedom of the energy density. Unit: dimensionless. Value: 106.75 in the Standard Model above the electroweak scale. Status: STANDARD.
- T: The temperature of the Standard-Model plasma (of the photons after the annihilation of electrons and positrons). Unit: GeV. Status: STANDARD.
- s: The entropy density of the Standard-Model plasma. Unit: GeV^3. Status: STANDARD.
- g_{s}: The effective number of relativistic degrees of freedom of the entropy density. Unit: dimensionless. Value: 3.909 today. Status: STANDARD.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\, h_{100}$ km/s/Mpc with $h_{100} = 0.674$ (Planck 2018): 67.4 km/s/Mpc $= 1.44 \times 10^{-42}$ GeV. Status: STANDARD.
- M_{Pl}: The Planck mass, $M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2}$. Unit: GeV. Value: 1.221×10^{19} GeV. Status: STANDARD.
- G_{N}: Newton's constant of four-dimensional gravity (the gravity of our world, on the fold), $G_{\mathrm{N}} = 1/M_{\mathrm{Pl}}^2$ with the Planck mass $M_{\mathrm{Pl}} = 1.221 \times 10^{19}$ GeV. Unit: GeV^{-2}. Value: 6.708×10^{-39} GeV^{-2}. Status: STANDARD.

The radiation of the Standard Model in thermal equilibrium at the temperature T: the energy density and the entropy density with the effective numbers of relativistic degrees of freedom g_ρ and g_{s} (106.75 above the electroweak scale in the Standard Model; three right-handed neutrinos in equilibrium would add $\frac{7}{8} \times 6 = 5.25$; the particles χ add theirs while they are relativistic), and the expansion rate of the radiation era. The gauge fields count with the two polarisations of their zero modes for $\mathsf{T} \ll H$. Status: STANDARD (`um4_thermal.py`, T1).

$$\frac{\mathrm{d}n_\chi}{\mathrm{d}x_4} + 3H_4\, n_\chi = -\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle\big(n_\chi^2 - n_{\chi,\mathrm{eq}}^2\big), \qquad n_{\chi,\mathrm{eq}} = g_\chi\Big(\frac{\omega_{\mathrm{b}}\mathsf{T}}{2\pi}\Big)^{3/2} \mathrm{e}^{-\omega_{\mathrm{b}}/\mathsf{T}}, \qquad \Omega_\chi h_{100}^2 \simeq \frac{1.07 \times 10^9\ \mathrm{GeV}^{-1}\, x_{\mathrm{f}}}{\sqrt{g_*}\, M_{\mathrm{Pl}}\, \langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle} \tag{FE-61}$$

Symbols in (FE-61).

- d: The differential: d in front of a quantity is an infinitesimal change of it (as dσ, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- n_χ: The number density of the dark-matter particles χ (particles plus antiparticles). Unit: GeV^3. Status: DERIVED.
- x_4: The time; a prime on a function of x_4 alone is d/dx_4. Unit: length (= time, $c = 1$). Status: AUTHOR.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\, h_{100}$ km/s/Mpc with $h_{100} = 0.674$ (Planck 2018): 67.4 km/s/Mpc $= 1.44 \times 10^{-42}$ GeV. Status: STANDARD.
- σ_{ann}: The annihilation cross-section of the particles χ into Standard-Model particles (through the portals) in the normalisation of the relic abundance, where n_χ counts the four particle and the four antiparticle states of the level at the fold: $\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle$ is a quarter of the annihilation rate $(\sigma\mathsf{v})_{\chi\bar\chi}$ of one particle χ_a with one antiparticle $\bar\chi_a$. Unit: GeV^{-2}. Status: DERIVED.
- v: The relative velocity of two annihilating particles; $\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle$ is the thermal average of the product. Unit: dimensionless. Status: STANDARD.

- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $n_{\chi,\mathrm{eq}}$: The number density of χ in thermal equilibrium with the plasma (non-relativistic). Unit: GeV^3. Status: STANDARD.
- g_χ: The number of internal states of the level of χ at the fold that are in thermal equilibrium. Unit: dimensionless. Value: 8 (four particles and four antiparticles). Status: DERIVED.
- ω_b: The energy at zero momentum of the state of dirac16complex at the fold in the limit $H \to 0$ (the bound state of UM-3): the particle χ of UM-4 has the mass $\omega_\mathrm{r} = \omega_\mathrm{b}(1 - H/(2q_\mathrm{b})) + \ldots$, equal to ω_b up to the relative correction $H/(2q_\mathrm{b})$. Unit: GeV. Status: DERIVED.
- T: The temperature of the Standard-Model plasma (of the photons after the annihilation of electrons and positrons). Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- Ω_χ: The density parameter of the dark matter today. Unit: dimensionless. Value: $\Omega_\chi h_{100}^2 = 0.120$ (Planck 2018). Status: DERIVED.
- h_{100}: The present expansion rate of the fold in units of 100 km/s/Mpc. Unit: dimensionless. Value: 0.674 (Planck 2018). Status: STANDARD.
- x_f: The ratio of the mass of χ to the temperature at which its annihilations stop (freeze-out). Unit: dimensionless. Value: 23 to 30 (for $\omega_\mathrm{b} = 10$ GeV to 10 TeV). Status: STANDARD.
- g_*: The effective number of relativistic degrees of freedom at the freeze-out (g_ρ there). Unit: dimensionless. Status: STANDARD.
- M_Pl: The Planck mass, $M_\mathrm{Pl} = G_\mathrm{N}^{-1/2}$. Unit: GeV. Value: 1.221×10^{19} GeV. Status: STANDARD.

The relic density of the particles χ of dirac16complex if they were once in thermal equilibrium: their number density n_χ (particles plus antiparticles) follows the Boltzmann equation with the thermally averaged annihilation cross-section $\langle\sigma_\mathrm{ann}\mathsf{v}\rangle$ into Standard-Model particles through the portal couplings of FE-47 (Higgs exchange, pairs of gluons and photons, pairs of fermions); g_χ is the number of states of the level that are in equilibrium (8: four particles and four antiparticles). The freeze-out at $x_\mathrm{f} = \omega_\mathrm{b}/\mathsf{T}_\mathrm{f} \approx 20$ to 30 (T_f the temperature at which the annihilations stop) gives the standard estimate of the relic density for an annihilation that does not depend on the velocity (g_* the degrees of freedom at freeze-out); the numerical solution gives $\Omega_\chi h_{100}^2 = 0.12$ for $\langle\sigma_\mathrm{ann}\mathsf{v}\rangle = 2.2$ to 2.3×10^{-26} cm^3/s from 10 GeV to 10 TeV. The particles χ of UM-4 cannot be such a thermal relic: the largest annihilation that partial-wave unitarity allows, $(\sigma\mathsf{v})_{\chi\bar\chi} = 4\pi/(m^2\mathsf{v})$ for the s-wave, gives $\Omega_\chi h_{100}^2 = 0.12$ already at $\omega_\mathrm{b} = 8.9 \times 10^4$ GeV, and heavier particles would be overproduced, while their lifetime requires $\omega_\mathrm{b} > 2.4 \times 10^6$ GeV (FE-86), 27 times more. Their abundance therefore needs a non-thermal origin (OPEN). Moreover, if the fold binds them through the Higgs portal (FE-47), the limit of LZ on their scattering on nuclei (Phys. Rev. Lett. 135 (2025) 011802, extrapolated in proportion to the mass above 40 GeV) allows at most $q_\mathrm{b}/\omega_\mathrm{b} = 0.019$ at 10 TeV, falling like $\omega_\mathrm{b}^{-1/4}$; with that binding the width of FE-86 ($F \simeq (q_\mathrm{b}/\omega_\mathrm{b})^3/3$) is below $1/t_0$ only for H below 1.8×10^{-3} GeV at 2.4×10^6 GeV and below 2.1 GeV at the Planck mass, never for $H > 2.5$ TeV: the particles χ bound by the Higgs portal cannot be the dark matter of UM-4, which needs a binding of χ to the fold that does not couple it to nucleons (OPEN). For the record of the masses of UM-3 (10 GeV to 10 TeV): a relic through the quark current portals is excluded by direct detection by factors 2×10^7 (100 GeV) to 20 (10 TeV), one through the lepton portals is allowed above about 1.5 TeV, and the Higgs portal requires $q_\mathrm{b}/\omega_\mathrm{b} < 0.062$ to 0.019. Status: STANDARD (the equation; UM-4: the cross-section from the portals), DERIVED (`um4_thermal.py`, T2 and T7), ESTIMATE (T3, T4 and T8: the limit of LZ and its extrapolation, the loop cut-off of the lepton portals, the prefactor of the width), OPEN (the origin of the particles χ and a binding that escapes direct detection).

$$\frac{\mathrm{d}\rho_{\nu_\mathrm{R}}}{\mathrm{d}x_4} + 4H_4\,\rho_{\nu_\mathrm{R}} = \mathcal{C}_{\nu_\mathrm{R}}, \qquad \Delta N_\mathrm{eff} = 3\Big(\frac{10.75}{g_\mathrm{s}(\mathsf{T}_\mathrm{dec})}\Big)^{4/3} \quad \text{(three right-handed neutrinos decoupling at } \mathsf{T}_\mathrm{dec}) \quad \text{(FE-62)}$$

Symbols in (FE-62).

- d: The differential: d in front of a quantity is an infinitesimal change of it (as dσ, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- ρ_{ν_R}: The energy density of the three right-handed neutrinos. Unit: GeV^4. Status: STANDARD.

- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc with $h_{100} = 0.674$ (Planck 2018): 67.4 km/s/Mpc $= 1.44 \times 10^{-42}$ GeV. Status: STANDARD.
- $\mathcal{C}_{\nu_\mathrm{R}}$: The collision term of the right-handed neutrinos: the rate at which collisions produce (or remove) their energy density. Unit: GeV^5. Status: STANDARD.
- Δ: A change, difference or small variation of the quantity that follows (for example an interval of the time, a perturbation of a density, a difference of two squared masses, or an addition to an effective number of species). Status: STANDARD.
- N_eff: The effective number of neutrino species: the energy density of the relativistic particles other than photons in units of that of one neutrino species whose temperature is $(4/11)^{1/3}$ times that of the photons. Unit: dimensionless. Value: 3.044 (Standard Model); measured 2.99 ± 0.17 (Planck 2018). Status: STANDARD.
- g_s: The effective number of relativistic degrees of freedom of the entropy density. Unit: dimensionless. Value: 3.909 today. Status: STANDARD.
- T_dec: The temperature at which the right-handed neutrinos decouple from the plasma. Unit: GeV. Status: STANDARD.

The right-handed neutrinos in the early universe: their energy density grows by the collision term $\mathcal{C}_{\nu_\mathrm{R}}$ (production by the neutrino Yukawa couplings, negligible for Dirac masses below 0.1 eV because it is suppressed by the square of the mass over the temperature, and by the current portal c_X^ν with dirac16complex), and if they were in equilibrium until the temperature T_dec they add ΔN_eff to the effective number of neutrinos (from the conservation of entropy after their decoupling). The measured $N_\mathrm{eff} = 2.99 \pm 0.17$ (Planck 2018) and 2.89 ± 0.11 (ACT DR6, the sixth data release of the Atacama Cosmology Telescope, with Planck, lensing, BAO and nucleosynthesis, 2025) bound ΔN_eff and with it the portal couplings c_X^ν: if the portal kept the right-handed neutrinos in equilibrium with χ they would decouple at $\mathsf{T}_\mathrm{dec} \approx \omega_\mathrm{b}/13$ to $\omega_\mathrm{b}/17$ and add $\Delta N_\mathrm{eff} = 0.14$ to 0.24 (for $\omega_\mathrm{b} = 10$ TeV to 10 GeV; at the masses of UM-4 the decoupling lies far above the electroweak scale, $\Delta N_\mathrm{eff} = 0.14$), 2.7 to 3.5 standard deviations above the ACT value; the right-handed neutrinos of UM-4 must therefore never thermalise (c_X^ν much smaller than the lepton-doublet couplings), and then $N_\mathrm{eff} = 3.044$. Status: STANDARD (the equations), DERIVED (the bound; `um4_thermal.py`, T5).

12.1 The primordial era

$$g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N = \mathrm{d}\sigma^2 + f_\mathrm{dS}^2\Big[\mathrm{e}^{2H_\mathrm{i}x_4}\big(\mathrm{d}\vec{x}^2 - \mathrm{d}\vec{x}_5^2\big) - \mathrm{d}x_4^2\Big], \qquad f_\mathrm{dS} = \frac{\sinh\big(H(\sigma_0 - |\sigma|)\big)}{\sinh(H\sigma_0)}, \qquad H_\mathrm{i} = \frac{H}{\sinh(H\sigma_0)},$$
$$\kappa\big(\lambda_\mathrm{F} + \rho_\mathrm{i}\big) = 12H\coth(H\sigma_0), \qquad H_\mathrm{i}^2 = \Big(\frac{\kappa(\lambda_\mathrm{F} + \rho_\mathrm{i})}{12}\Big)^2 - H^2 = \frac{\kappa_7\rho_\mathrm{i}}{15}\Big(1 + \frac{\rho_\mathrm{i}}{2\lambda_\mathrm{F}}\Big) \tag{FE-87}$$

Symbols in (FE-87).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: d in front of a quantity is an infinitesimal change of it (as $\mathrm{d}\sigma$, a small step of σ); the quotient of two such changes is a derivative; d with an exponent, in front of several variables, is their volume element, the product of their differentials. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

- f_{dS}: The warp of the de Sitter fold: the factor of the lengths of the fold at the depth σ. Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H_{i}: A constant expansion rate of ordinary space: of the de Sitter fold (whose extra times expand at the same rate) or of an inflation of the fold with frozen extra times. Unit: GeV. Status: DERIVED (FE-87), ASSUMED (FE-88).
- x_4: The time; a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- $\vec{x}$: The position in ordinary space, $\vec{x} = (x_1, x_2, x_3)$. Unit: length. Status: AUTHOR.
- $\vec{x}_5$: The position along the extra times, $\vec{x}_5 = (x_5, x_6, x_7)$; $\mathrm{d}\vec{x}_5^2 = \mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2$. Unit: length. Status: AUTHOR.
- σ_0: The depth at which the warp of the de Sitter fold vanishes (a horizon of the bulk), $\sinh(H\sigma_0) = H/H_{\mathrm{i}}$. Unit: length. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_{\mathrm{F}} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- λ_{F}: The tension that the static block requires of the fold, $\kappa\lambda_{\mathrm{F}} = 12H$: the total vacuum energy of the seven-dimensional fold, made of the tension of the fold fluid and the vacuum energy of the fields of the fold (both have the pressure minus their energy density along ordinary space and along the extra times), per unit volume of the six other directions of the fold (eight-dimensional normalisation); in the four-dimensional normalisation it is $\lambda_{\mathrm{F}}V_*$, and the excess of the total over $\lambda_{\mathrm{F}}V_*$ is the dark energy ρ_Λ. Unit: GeV^7. Value: $12H/\kappa$; $\lambda_{\mathrm{F}}V_* = 30H^2M_{\mathrm{red}}^2 = 1.1 \times 10^{45}$ GeV^4 for $H = 2.5$ TeV. Status: DERIVED.
- ρ_{i}: The vacuum energy of the fold above the critical tension λ_{F} that expands the de Sitter fold (eight-dimensional normalisation). Unit: GeV^7. Status: DERIVED.
- κ_7: The gravitational coupling of the fold: to the first order in its curvature over H^2 the fold obeys the Einstein equations of seven dimensions $\hat{G}^a{}_b = \kappa_7 \mathsf{t}^a{}_b$ with $\kappa_7 = 5\kappa H/2$, the coupling of the massless graviton of the bulk bound to the fold; four-dimensional gravity on the fold has $8\pi G_{\mathrm{N}} = \kappa_7/V_*$ (plus the induced term, which is negligible). Unit: GeV^{-5}. Value: $5\kappa H/2$. Status: DERIVED.

An exact solution of UM-4 with an inflating fold: a fold with the tension $\lambda_{\mathrm{F}} + \rho_{\mathrm{i}}$ above the critical one (the vacuum energy ρ_{i} of the fold, seven-dimensional normalisation) in a bulk of the constant curvature $-H^2$ (the static block's bulk, sliced differently). The metric of the fold is a de Sitter space of seven dimensions with the expansion rate H_{i} in all its space-like and time-like directions ($\mathrm{d}\vec{x}^2$ and $\mathrm{d}\vec{x}_5^2$ the squared steps along ordinary space and along the extra times), and the warp factor f_{dS}^2 vanishes at the depth σ_0 (a horizon of the bulk). The bulk equations hold exactly ($G^M{}_N = 21H^2\delta^M_N$ in all eight directions), Israel's condition gives the tension of the second line, and the expansion rate obeys the Friedmann equation of seven dimensions with the correction of high energies $\rho_{\mathrm{i}}/(2\lambda_{\mathrm{F}})$, as for the brane of Randall and Sundrum (DERIVED, `um4_primordial.py`, PR1, exactly). It is not the primordial era of our universe: the extra times inflate together with ordinary space, so the factor $\mathcal{Z}$ of the couplings of the fold grows like e^{3N} after N e-folds and every coupling of the fold falls like e^{-3N}, by 7×10^{-79} after 60 e-folds (PR5). Status: DERIVED.

$$\mathsf{c}' = 0, \quad a_4' = H_{\mathrm{i}} : \qquad \kappa_7\rho_7 = 3H_{\mathrm{i}}^2, \qquad P_7 = -\rho_7, \qquad q_7 = -2\rho_7, \qquad \frac{H_{\mathrm{i}}}{H} < \frac{1}{10} \tag{FE-88}$$

Symbols in (FE-88).

- c: The logarithm of the scale factor of the extra times on the fold, $\mathsf{c} = \ln(\mathrm{e}^{-a_4}b_4)$; $\mathsf{c}' = \mathsf{V}H$; frozen extra times have $\mathsf{c}' = 0$. Unit: dimensionless. Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- H_{i}: A constant expansion rate of ordinary space: of the de Sitter fold (whose extra times expand at the same rate) or of an inflation of the fold with frozen extra times. Unit: GeV. Status: DERIVED (FE-87), ASSUMED (FE-88).
- κ_7: The gravitational coupling of the fold: to the first order in its curvature over H^2 the fold obeys the Einstein

equations of seven dimensions $\hat{G}^a{}_b = \kappa_7 \mathsf{t}^a{}_b$ with $\kappa_7 = 5\kappa H/2$, the coupling of the massless graviton of the bulk bound to the fold; four-dimensional gravity on the fold has $8\pi G_\mathrm{N} = \kappa_7/V_*$ (plus the induced term, which is negligible). Unit: GeV^{-5}. Value: $5\kappa H/2$. Status: DERIVED.

- ρ_7: The energy density of the matter of the fold above its tension in the seven-dimensional normalisation (minus the component $\mathsf{t}^{(4)}{}_{(4)}$: an energy per unit volume of the six other directions of the fold); $V_*\rho_7$ in the four-dimensional normalisation. Unit: GeV^7. Status: DERIVED.
- P_7: The pressure along ordinary space of the matter of the fold above its tension in the seven-dimensional normalisation (the component $\mathsf{t}^{(i)}{}_{(i)}$). Unit: GeV^7. Status: DERIVED.
- q_7: The pressure along the extra times of the matter of the fold above its tension in the seven-dimensional normalisation (the component $\mathsf{t}^{(r)}{}_{(r)}$). Unit: GeV^7. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the warp factor $\mathrm{e}^{-2H|\sigma|}$ of the metric, the cosmological constant $\Lambda = -21H^2$ and the tension $\kappa\lambda_\mathrm{F} = 12H$ of the fold; in UM-4 a free parameter, bounded from below by the propagation of the gauge fields in the bulk. Unit: GeV (1/length). Value: $H \gtrsim 2.5$ TeV (from the invisible width of the Z boson). Status: AUTHOR.

The inflation of the fold with frozen extra times (the primordial era that UM-4 assumes): a vacuum energy ρ_7 of the fold expands ordinary space at the constant rate H_i by the Friedmann equation of FE-72, and the radion condition requires the pressure $q_7 = -2\rho_7$ along the extra times, twice that of a vacuum energy: the fold fluid adds $q_\mathrm{hold} = -\rho_7$ (FE-65). These are the equations of the first order; the exact equations FE-35 of a fold with this extrinsic curvature have no solution in a bulk of constant curvature for $H_\mathrm{i} \neq 0$, so the inflation with frozen extra times needs a Weyl tensor of the bulk, whose solution is not computed (OPEN). The regime of general relativity, $(12/25)(H_\mathrm{i}/H)^2 \ll 1$, needs $H_\mathrm{i} < H/10$ (third relation). The bound $r_\mathrm{T} < 0.032$ on the ratio of tensor to scalar perturbations limits any inflation to $H_\mathrm{i} < 4.4 \times 10^{13}$ GeV, so an inflation at that bound needs $H > 4.4 \times 10^{14}$ GeV; with H at its lower bound of 2.5 TeV only an inflation with $H_\mathrm{i} < 250$ GeV (an energy scale $\rho^{1/4} < 3.2 \times 10^{10}$ GeV) is in the regime of general relativity. The Higgs field could drive the inflation (Higgs inflation, with the coupling $\xi_\mathrm{H} = 1.6 \times 10^4$ of the Higgs doublet to the curvature of the fold, which the action of UM-4 does not contain: $n_\mathrm{S} = 0.964$, $r_\mathrm{T} = 0.004$, and $H_\mathrm{i} = 1.6 \times 10^{13}$ GeV, so UM-4 would need $H > 1.6 \times 10^{14}$ GeV). Status: DERIVED (`um4_primordial.py`, PR2 and PR3), ESTIMATE (PR4, Higgs inflation: an extension outside the action), OPEN (the Weyl tensor of the bulk, the origin of the holding pressure).

13 The coupled system

13.1 Which field appears in which equation

equation	the fields in it	how they enter
bulk gravity FE-1, FE-29, FE-32, FE-34	metric, dirac16complex, gauge fields	the Einstein tensor of the metric; $\mathcal{T}_\Psi$ and $\mathcal{T}_\mathrm{gauge}$
fold junction FE-4, FE-5, FE-30, FE-35, FE-59, FE-71, FE-72	metric, every field of the fold, dirac16complex at the fold	the jump of the extrinsic curvature (to the first order the Einstein tensor of the fold divided by κ_7); $\mathcal{T}_\mathrm{H}$, $\mathcal{T}_\mathrm{f}$, $\mathcal{T}_\mathrm{P}$, $\mathcal{T}_\mathrm{ff}$; the induced gravity $\hat{G}/\kappa_\mathrm{ind}$
dirac16complex FE-12, FE-13, FE-36 to FE-43, FE-85, FE-86	metric, itself, every Standard-Model field	the spin connection and the warped frequency; the self-interaction; the fold operator $\mathbb{M}$ of the portals (the Higgs portal for Ψ_L only)
gauge fields FE-16 to FE-18, FE-74, FE-79 to FE-82	metric, dirac16complex, Higgs, quarks and leptons	the metric of the bulk; at the fold the currents of the fold, the portal factors $\mathcal{K}_G$, $\mathcal{K}_W$, $\mathcal{K}_B$, the dipole density $\mathcal{D}_X$ and the mass term of the Higgs vacuum
Higgs FE-15	metric, dirac16complex, the gauge fields at the fold, quarks and leptons	$\hat{h}$ and $\mathcal{Z}$; $c_\Phi \bar{\Psi} P_\mathrm{L} \Psi$; D_μ; the Yukawa source Θ
quarks and leptons FE-19 to FE-24	metric, the gauge fields at the fold, Higgs, dirac16complex	$\hat{h}$; D_μ; the Yukawa matrices; the potentials $\mathcal{V}^f$
neutrino flavours FE-48 to FE-53	expansion, Higgs, dirac16complex, matter, other neutrinos	a_4'; v_loc and $G_\mathrm{F,loc}$; V_Ψ; V_m and $\mathsf{V}_{\nu\nu}$

equation	the fields in it	how they enter
cosmology FE-54 to FE-58	metric perturbations, dark matter χ, baryons, photons, neutrinos	ϕ_{B} and ψ_{B}; $\Gamma_{\chi\mathrm{b}}$; Thomson scattering; ρ_Λ
thermal history FE-59 to FE-62, FE-87, FE-88	metric, plasma, dark matter χ, right-handed neutrinos, fold fluid	H_4, G_{N} and ρ_Λ; $\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle$; $\mathcal{C}_{\nu_{\mathrm{R}}}$; q_{hold}

13.2 How the system is solved

The equations are coupled in a chain that UM-4 solves in steps, each with the results of the earlier ones:

1. The background: the static block (FE-32, exact: a space of constant curvature $-H^2$ on each side of a fold of pure tension) and the expanding fold with frozen extra times to the first order in its curvature over H^2 (FE-34, FE-71, FE-72: the Friedmann equations of general relativity), from an inflation of the fold (FE-88) to the late universe; the fold fluid holds the volume of the extra times fixed (FE-65; ASSUMED, its origin OPEN). The exact de Sitter fold FE-87 checks the first-order equations and shows why the volume must be held.
2. The propagation in the bulk (FE-73 to FE-84): the zero modes of gravity and of the gauge fields at the speed of light, the photon and the gravitational wave as waves of spin 1 and 2, the propagators from one point of the fold to another, and the bounds on H that the massive modes give (FE-81, FE-82). At energies far below H the zero modes of the graviton and of the gauge fields form a four-dimensional effective theory on the fold, in which the radiation of the gauge fields enters the Friedmann equations as four-dimensional radiation (the low-energy limit of FE-74 and FE-79; the source term of the bulk in the exact equations FE-35 is not computed).
3. The states of dirac16complex at the fold on the background: the resonances (FE-36 to FE-40, FE-85, FE-86) with the mean fields of the Standard-Model vacuum (FE-39), and their lifetime.
4. The Kohn-Sham states (FE-41, FE-42): the level of the resonances filled to a given density, with the self-interaction and the mean fields, to self-consistency; their energy-momentum tensor enters the bulk equations and the fold (their density integrated across the fold counts as matter of the fold, and their decay into the bulk as the flux of FE-11).
5. The Standard Model on the fold (FE-44 to FE-47): the couplings of the zero modes, the masses with the ad hoc vacuum, the residuals FE-26, the widths of the W and Z into the bulk (FE-82), the couplings of the particles χ.
6. The neutrino flavours (FE-48 to FE-53), the thermal history (FE-59 to FE-62) and the abundance of χ (whose origin is OPEN).
7. The cosmological perturbations (FE-54 to FE-58): the CMB, the CNB and the BAO.

14 What was verified

Every statement labelled DERIVED in this document rests on a computation in the folder `universe_model_4/code/` of the repository, which checks itself and writes a report to `universe_model_4/reports/` (the equations and results themselves are all written out above). The 185 checks of the 20 modules, all passed:

- `um4_dirac_matrices.py` (55 checks): the author's matrices, their reality, the Clifford relations, the chiral projectors, Pin(4,4) (`um4_dirac_matrices_doc.py` writes them out in the document "dirac matrices").
- `um4_geometry.py` (10 checks): the proper chart with the warped time (G1), the exact curvature of the metric of UM-4 for any a_4, b_4 on both halves of the period (G2), the static block of constant curvature (G3), its Lovelock tensors (G4), the exact bulk equations FE-29 (G5), the Ricci-flat folds FE-33 (G6), the junction FE-30 of a pure tension (G7), the light cone of every slice (G8), the spin connection FE-14 and the vielbein postulate with the author's matrices (G9), the static warps of the author's form, whose only vacua of Einstein gravity have equal rates in all seven directions of the fold (G10).
- `um4_gravity.py` (4 checks): the vacua of the static blocks for any weights FE-70 (L1), the graviton around the static block for any weights FE-68 (L2), the light cone of the gravitons on every warped background FE-69 (L3), the choice of Einstein gravity (L4).
- `um4_fold.py` (6 checks): the bulk metric of an expanding fold to the first order FE-34 (F1), the fold as a seven-dimensional Einstein brane FE-71 (F2), the exact equations of the fold FE-35 (F3), the Friedmann

equations and the radion condition FE-72 (F4), the size of the corrections (F5), the Weyl term that dust requires (F6).

- `um4_foldsources.py` (5 checks): the tension of a fold in n dimensions FE-30 (FS1), the gravitational wave of the fold FE-63 (FS2), the induced gravity FE-64 (FS3), the tuning of the tension (FS4), the fold fluid FE-65 (FS5).
- `um4_background.py` (6 checks): the vacua of the author's form (BG1), the expanding fold FE-66 and FE-72 (BG2), the free volume of the extra times excluded by lunar laser ranging (BG3), the factor $\mathcal{Z}$ of the couplings (BG4), the gauge equivalence of the ad hoc vacuum and its radial residual (BG5), with numbers for three values of a_{40} (BG6).
- `um4_bulk.py` (10 checks): the modes of a scalar FE-73 (B1) and of a gauge field FE-74 (B2) of the bulk, the photon as a transverse wave of spin 1 FE-75 (B3), the graviton of the bulk (B4) and the gravitational wave with its plus and cross polarisations FE-76 (B5), the bulk fermion FE-77 (B6), the boost invariance of dirac16complex FE-78 (B7), the summary of the speeds (B8), the speeds of UM-3 recomputed (B9), and the massive modes of FE-73 and FE-74, a continuum from zero mass whose spectral density at the fold reproduces the propagators FE-79 (B10).
- `um4_propagators.py` (5 checks): the propagators FE-79 (P1), their rational forms and expansions FE-80 (P2), Newton's and Coulomb's laws FE-81 (P3), the W and Z resonances FE-82 (P4), the bounds on H (P5).
- `um4_causality.py` (4 checks): the static block conformally flat FE-83 (C1), the light cone of an expanding fold and the largest advance FE-84 (C2, C3), the closed time-like curves of the four time-like directions (C4).
- `um4_fields.py` (16 checks): the reality and chirality of the bilinears of dirac16complex, the Krein operator, the four-dimensional chirality, the electroweak masses and the photon of the ad hoc vacuum, its isospin charge and residual, the Yukawa invariance, the anomalies with three right-handed neutrinos, the electric charges.
- `um4_spectra.py` (8 checks): the reduction to eight copies (U1), the exact solutions FE-85 (U2), the resonances FE-40 with the 16×16 matching (U3), their widths FE-86 (U4), their dispersion (U5), the lifetime condition (U6), their Krein form (U7) and their mirror (U8).
- `um4_kohnsham.py` (9 checks): the Kohn-Sham equations FE-41 for a homogeneous gas of the particles χ, solved self-consistently with the author's matrices (reduced exactly by the rotations of ordinary space and of the extra times, and lifted back to sixteen components for the checks): the free level FE-40 and the box method against the exact resonance (K1), first-order perturbation theory in the dilute limit (K2), the residuals of FE-41 and of the matching (K3), the exponent of the dilute limit (K4), the gas at the fold and the attractive case (K5), the Higgs-portal loop through FE-49 (K6), the equation of state from FE-42 (K7), the width that grows with the self-interaction (K8), a weakly bound point (K9), and how dilute the gas of the universe is (K10).
- `um4_krein.py` (7 checks): the Krein form and its quantisation FE-67 (KR1), the symmetries at rest and in motion (KR2), the mirrors of the fold (KR3), the flat directions of the dilute gas (KR4), the partner level of the gas (KR5), the selection rules at the fold (KR6), the consequences for stability and unitarity (KR7).
- `um4_extratimes.py` (5 checks): the dispersion with a momentum along the extra times on the fold (EX1), the states of dirac16complex with such a momentum FE-38 (EX2), the growth with frozen extra times (EX3), compact extra times in the late universe: no size helps (EX4), the truncation to fields without that momentum (EX5).
- `um4_primordial.py` (6 checks): the author's vacuum no longer a solution FE-31 (PR0), the exact de Sitter fold FE-87 (PR1), the inflation with frozen extra times FE-88 (PR2), its scales (PR3), Higgs inflation, an extension outside the action (PR4), the factor $\mathcal{Z}$ during inflation (PR5).
- `um4_neutrino.py` (5 checks): the flavour blindness of V_Ψ and the invariance of the mixing under v_{loc} (N1), the resonance formulas FE-50 against an exact integration (N2), the crossing criterion of FE-51 (N3), the conservation laws of FE-52 (N4), the CNB numbers of FE-53 (N5).
- `um4_flavour.py` (4 checks): the survival probability of solar neutrinos by exact integration of FE-48 and FE-49, against the adiabatic formula of FE-50 and the measurements of Borexino; the three-flavour conversion in the envelope of a supernova for both mass orderings; the fast instability near a neutrinosphere and the dispersion relation FE-51 at its solutions; the equipartition FE-52 (its conservation laws, its fixed point, the bound $0 \leq \mathcal{F} \leq 1$) and its effect on the heating behind the shock.
- `um4_numass.py` (5 checks): the neutrino masses of both orderings (NM1), the bounds of DESI DR2 and KATRIN (NM2), the Dirac phase in the results of UM-4 (NM3), what the lightest mass changes (NM4), the record against NuFIT 6.0 (NM5).
- `um4_cosmology.py` (6 checks): the expansion rate of FE-59 and the ruler FE-56 computed independently and by the Boltzmann code CAMB; the derived quantities of the CMB against Planck 2018; the CMB spectra (FE-57) against the Planck 2018 high-l data, with a refit for thermal right-handed neutrinos; the BAO distances

against DESI DR2 (the second data release of the Dark Energy Spectroscopic Instrument); the CNB numbers of FE-53 and FE-58.
- `um4_thermal.py` (8 checks): g_ρ and g_s of FE-60 (T1), the relic density FE-61 by numerical solution (T2), the current portals against the limits of direct detection (T3), the Higgs-portal coupling of FE-47 and the binding of χ (T4), the right-handed neutrinos and N_eff FE-62 (T5), the ad hoc vacuum before nucleosynthesis (T6), no thermal relic (T7), the binding that direct detection allows and the lifetime that it gives (T8).

The helper modules `um4_common.py` (the report), `um4_gammas.py` (the author's matrices) and `um4_lovelock.py` (the Lovelock tensors from the frame curvature) have no checks of their own. The Kohn-Sham module runs in parts (`um4_kohnsham.py part NAME` for NAME = K1, I, K6, K9, then `um4_kohnsham.py assemble`), so that no single calculation takes longer than 423 seconds; the Krein module reads the snapshots of the parts, the spectra module the bounds of the propagators, and the thermal module the lifetimes of the spectra.

15 Further symbols

Further symbols. These symbols appear in the definitions of the symbols listed after the equations of this document.

- M_red: The reduced Planck mass, $M_\mathrm{red} = (8\pi G_\mathrm{N})^{-1/2}$. Unit: GeV. Value: 2.435×10^{18} GeV. Status: STANDARD.
- θ_{12}, θ_{13}, θ_{23}: The three mixing angles of U_PMNS (its fourth parameter is the Dirac phase, which does not enter the electron-flavour results of UM-4). Unit: dimensionless. Value: 33.40°, 8.54°, 49.14° in the computations of UM-4 ($\sin^2\theta_{12} = 0.303$, $\sin^2\theta_{13} = 0.02203$, $\sin^2\theta_{23} = 0.572$: NuFIT 5.2, November 2022, normal ordering, the fit without the Super-Kamiokande atmospheric data); 33.71°, 8.56°, 43.28° in NuFIT 6.0 (2024, with the Super-Kamiokande atmospheric data; $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$). Status: STANDARD.
- L_t: The physical period of the extra times, if they are compact. Unit: length. Status: ASSUMED.
- $\mathcal{W}$: The warp factor: in a warped metric, the function of the position along one direction that multiplies the squared steps of other directions (its square root multiplies their lengths); in UM-4 the function of σ that multiplies all seven directions of the fold, ordinary space, the time and the extra times, $\mathcal{W} = \mathrm{e}^{-2H|\sigma|} = (\sin^2 z)^{1/6}$, equal to 1 on the fold. Unit: dimensionless. Status: DERIVED.
- c_b: The speed coefficient of the particles χ of dirac16complex along ordinary space, in units of the speed of light on the fold: at small momentum $\omega^2 = \omega_0^2 + c_\mathrm{b}^2\vec{k}^2$. In UM-3, whose time was not warped, $c_\mathrm{b}^2 = 1.19$ for $H = 0.05\,m_\Psi$ and $q_\mathrm{b} = 0.3\,m_\Psi$ (faster than light); in UM-4 the boosts of the fold are a symmetry and $c_\mathrm{b} = 1$ exactly. Unit: dimensionless. Value: 1 in UM-4; 1.09 in UM-3. Status: DERIVED.
- ϵ_c: The bookkeeping parameter of the cosmological perturbations of the metric of the fold (the size of the CMB anisotropies). Unit: dimensionless. Value: $\sim 10^{-5}$. Status: STANDARD.
- r_T: The ratio of the amplitudes of the primordial spectra of gravitational waves (tensor) and of curvature (scalar) at the pivot scale. Unit: dimensionless. Value: below 0.032 (Planck PR4 with BICEP/Keck 2018 and BAO, M. Tristram et al. 2022, 95 per cent). Status: STANDARD.
- w_χ: The equation of state of the gas of the particles χ: its pressure divided by its energy density, $w_\chi = P_\mathrm{KS}/(n_\chi \epsilon_\mathrm{p})$ (0 for cold dark matter, -1 for a vacuum energy). Unit: dimensionless. Status: DERIVED.
- P_KS: The pressure of the gas of χ along ordinary space in the Kohn-Sham state: the particles are at rest, so it is the mean self-interaction, $P_\mathrm{KS} = n_\chi \mathsf{E}_\mathrm{int}$. Unit: GeV^4. Status: DERIVED.
- ϵ_p: The energy per particle of the gas of χ: the orbital energy minus the interaction energy per particle, which the orbital energy counts twice. Unit: GeV. Status: DERIVED.
- E_int: The interaction energy per particle of the gas of χ: the mean self-interaction energy $\frac{\lambda_\Psi}{2}(n_\mathrm{s}^2 - \mathrm{tr}\,\mathcal{N}^2)$ integrated across the hidden direction, per particle. Unit: GeV. Status: DERIVED.